

The canonical system approach to scaling limits of β -ensembles

Vincent Painchaud



McGill

Department of Mathematics and Statistics
McGill University, Montreal

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Abstract

The general goal of this thesis is to argue that canonical systems theory can be used to improve our understanding of scaling limits of β -ensembles, in the sense that it provides an appropriate framework under which to describe all of the possible scaling limits, and therefore is especially suited to studying the relationships between them.

Over the last twenty years, important progress has been made in understanding scaling limits of β -ensembles. Scaling limits in the three main regimes, the soft edge, the hard edge, and the bulk, have been described as the spectra of three random differential operators called the stochastic Airy, Bessel and sine operators. The former two are random Sturm–Liouville operators, while the latter is a random Dirac operator, a type of two-dimensional first-order differential operator. As these operators belong to different classes, much of the analysis of their properties has, so far, been done using different methods. However, all of them—and in fact even the tridiagonal random matrix models for β -ensembles—can also be realized as random canonical systems. This allows for a consistent analysis of the relationships between all of these models.

A first contribution of this thesis is a proof that the canonical system representations of certain tridiagonal random matrix models converge in law to the canonical system representations of the stochastic Airy and Bessel operators. The classes of random matrices considered in these two cases include what are arguably the most important examples: the tridiagonal models for the Hermite and Laguerre β -ensembles scaled at the soft edge, and that for the Laguerre β -ensemble scaled at the hard edge. The convergence considered is first that of the coefficient matrices of the canonical systems under their vague topology, which is then upgraded to the compact convergence of the associated Weyl–Titchmarsh functions and thus to the vague convergence of the spectral measures. This provides an alternative operator level picture of the convergence of tridiagonal random matrix models of β -ensembles to the stochastic Airy and Bessel operators to the existing theory.

Another contribution is a description of soft and hard edge to bulk transitions, at the operator level, again using canonical systems theory. More precisely, it is shown that the canonical system for the stochastic sine operator can be recovered as a limit of the canonical system for a suitably scaled version of either the stochastic Airy or Bessel operator. Again, the convergence is proven in the vague topology of the coefficient matrices, and then upgraded to the compact convergence of the Weyl–Titchmarsh functions and therefore to the vague convergence of the spectral measures.

Résumé

L'objectif général de cette thèse est de montrer que la théorie des systèmes canoniques peut être exploitée pour améliorer notre compréhension des limites d'échelle des β -ensembles, au sens où elle donne un cadre théorique cohérent dans lequel les différentes limites d'échelle peuvent toutes être décrites, et est donc particulièrement appropriée pour étudier les relations entre elles.

Au cours des vingt dernières années, d'importants progrès ont été réalisés dans l'étude des limites d'échelle des β -ensembles. Les limites d'échelle dans les trois principaux régimes, c'est-à-dire le bord mou, le bord ferme et l'intérieur du spectre, ont été décrites comme les spectres de trois opérateurs différentiels aléatoires appelés les opérateurs d'Airy, de Bessel et sinus stochastiques. Les deux premiers sont des opérateurs de Sturm–Liouville aléatoires, mais l'autre est plutôt un opérateur de Dirac aléatoire, qui est un type d'opérateur différentiel de premier ordre, mais bi-dimensionnel. Comme ces opérateurs appartiennent à des classes distinctes, l'analyse de plusieurs de leurs propriétés repose sur des méthodes distinctes. Cependant, ces trois opérateurs peuvent être représentés comme des systèmes canoniques aléatoires, de même que les modèles de matrices aléatoires tridiagonales qui décrivent les β -ensembles. Ces représentations permettent ainsi une analyse plus cohérente des relations entre ces différents modèles.

Une première contribution de cette thèse est une preuve que les systèmes canoniques associés à certains modèles de matrices aléatoires tridiagonales convergent en loi vers les systèmes canoniques associés aux opérateurs d'Airy et de Bessel stochastiques. Les modèles de matrices aléatoires considérés dans les deux cas incluent certains des exemples les plus importants : les modèles de matrices tridiagonales pour les β -ensembles d'Hermite et de Laguerre au bord mou, ainsi que celui pour le β -ensemble de Laguerre au bord ferme. La convergence étudiée est d'abord celle des matrices coefficient des systèmes canoniques dans leur topologie vague. Elle est ensuite renforcée en la convergence localement uniforme des fonctions de Weyl–Titchmarsh des systèmes canoniques, et donc en la convergence vague des mesures spectrales associées. Par rapport aux descriptions déjà présentes dans la littérature, ceci donne une façon alternative, au niveau des opérateurs, de comprendre la convergence de modèles de matrices aléatoires tridiagonales vers les opérateurs d'Airy et de Bessel stochastiques.

Une autre contribution importante est une nouvelle description des transitions du bord mou et du bord ferme vers l'intérieur du spectre des β -ensembles, au niveau des systèmes canoniques. Plus

précisément, il est démontré que le système canonique associé à l'opérateur sinus stochastique peut être obtenu comme une limite du système canonique associé à une version à l'échelle de l'opérateur d'Airy ou de Bessel stochastique. À nouveau, la convergence est d'abord prouvée sous la topologie vague des matrices coefficient des systèmes canoniques, puis renforcée en la convergence localement uniforme des fonctions de Weyl–Titchmarsh, et donc en la convergence vague des mesures spectrales.

*Mais c'est quelquefois au moment où tout
nous semble perdu que l'avertissement
arrive qui peut nous sauver : on a frappé à
toutes les portes qui ne donnent sur rien, et
la seule par où on peut entrer et qu'on
aurait cherchée en vain pendant cent ans,
on y heurte sans le savoir et elle s'ouvre.*

Marcel Proust, *Le Temps retrouvé*

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Contributions

With a few exceptions, the material in this thesis is either review from previous work or based on two preprints [54, 55]. I am the sole author of [54], but [55] was written in collaboration with my advisor Elliot Paquette. I wrote the proofs of all of the results taken from [55] that appear in the thesis, but Elliot provided significant input on proof strategies.

The thesis is split in three main chapters.

Chapter 2 is essentially review of existing theory, with the exceptions of §2.2.3 and §2.3 which are (to my knowledge) original, although rather elementary.

Chapter 3 is partly review, and partly original. The material from §3.1.2 is taken from [55]. Then, Propositions 3.15 and 3.17 and Lemma 3.16 as well as the content of §3.2.2 are taken from [54]. The results proven in §3.1.3 and §3.2.3 are original in their present formulation, but have important similarities with celebrated results from other authors [7, 58, 59, 60].

Chapter 4 is based on [54, 55], and slightly extends some of the results.

Chapter 1

Introduction

1.1 β -ensembles

This thesis is concerned with random canonical systems that arise in random matrix theory as scaling limits of tridiagonal matrix models of β -ensembles. A β -ensemble on a domain $D \subseteq \mathbb{R}$ is a point process with an explicit density of the form

$$\lambda = (\lambda_1, \dots, \lambda_n) \mapsto \frac{1}{Z_{n,V,\beta}} \exp\left(-\beta \sum_{j=1}^n V(\lambda_j)\right) \prod_{1 \leq j < k \leq n} |\lambda_j - \lambda_k|^\beta \quad (1.1)$$

where $n \in \mathbb{N}$, $\beta > 0$ is a parameter usual called the *inverse temperature*, and $V: D \rightarrow \mathbb{R}$ is a constraining potential. These point processes originate from the work of Dyson in the 1960's [25, 26, 27, 28], and have attracted a lot of attention in random matrix theory [3, 51] and statistical physics [35, 66] ever since.

1.1.1 The Hermite β -ensemble

Perhaps the most important example of a β -ensemble is the *Hermite* or *Gaussian* β -ensemble, which is obtained in (1.1) with a quadratic potential $V(\lambda) := \lambda^2/4$. One of the reasons this particular example is interesting is that it generalizes the eigenvalue distribution of the Gaussian orthogonal, unitary and symplectic ensembles from random matrix theory, which we now introduce briefly. A random matrix is said to follow a *Gaussian orthogonal* ($\beta = 1$), *unitary* ($\beta = 2$) or *symplectic* ($\beta = 4$) *ensemble* distribution if it is self-dual with $\mathcal{N}(0, 2/\beta)$ diagonal entries and standard real, complex or quaternionic normal off-diagonal entries, with all entries independent up to the symmetry constraint. Here, by *standard complex* and *quaternionic* normal distributions, we mean the distributions of

$$\frac{1}{\sqrt{2}}(\xi_1 + i\xi_2) \quad \text{and} \quad \frac{1}{2}(\xi_1 + \mathbf{i}\xi_2 + \mathbf{j}\xi_3 + \mathbf{k}\xi_4)$$

respectively, where all ξ_j 's are independent standard normals. It can be shown that these three random matrix models have eigenvalue point processes with density (1.1) for $V(\lambda) = \lambda^2/4$ and $\beta = 1, 2$ and 4 for the orthogonal, unitary and symplectic case respectively [3, §4.1].

While these three classical random matrix models were introduced by Dyson in the 1960's [28], it was unknown for a long time whether it was possible to build a random matrix model that would have the Hermite β -ensemble as its eigenvalue point process for a general $\beta > 0$. This question was answered positively by Dumitriu and Edelman in 2002 [24]. Expanding on the work of Trotter [71], they extrapolated the result of tridiagonalizing the Gaussian orthogonal and unitary ensembles, and showed that for any $\beta > 0$, the tridiagonal random matrix

$$H_{\beta,n} := \frac{1}{\sqrt{\beta}} \begin{pmatrix} \sqrt{2}\xi_{n,1} & X_{n,1} & & \\ X_{n,1} & \ddots & \ddots & \\ & \ddots & \ddots & X_{n,n-1} \\ & & X_{n,n-1} & \sqrt{2}\xi_{n,n} \end{pmatrix} \quad \text{where} \quad \begin{aligned} \xi_{n,j} &\sim \mathcal{N}(0, 1) \\ X_{n,j} &\sim \chi_{(n-j)\beta}, \end{aligned} \quad (1.2)$$

with all $\xi_{n,j}$'s and $X_{n,j}$'s independent, has eigenvalue point process with density (1.1) for $V(\lambda) = \lambda^2/4$ [24, Theorem 2.12]. An important property of the Hermite β -ensemble is that as the number of points increases, their density on the real line takes the shape of a semicircle. Indeed, if λ is the Hermite β -ensemble with n points and $d\mu_{\text{sc}}(x) = \frac{1}{2\pi} \sqrt{4 - x^2} \mathbb{1}_{[-2,2]}(x) dx$ is a semicircle measure,

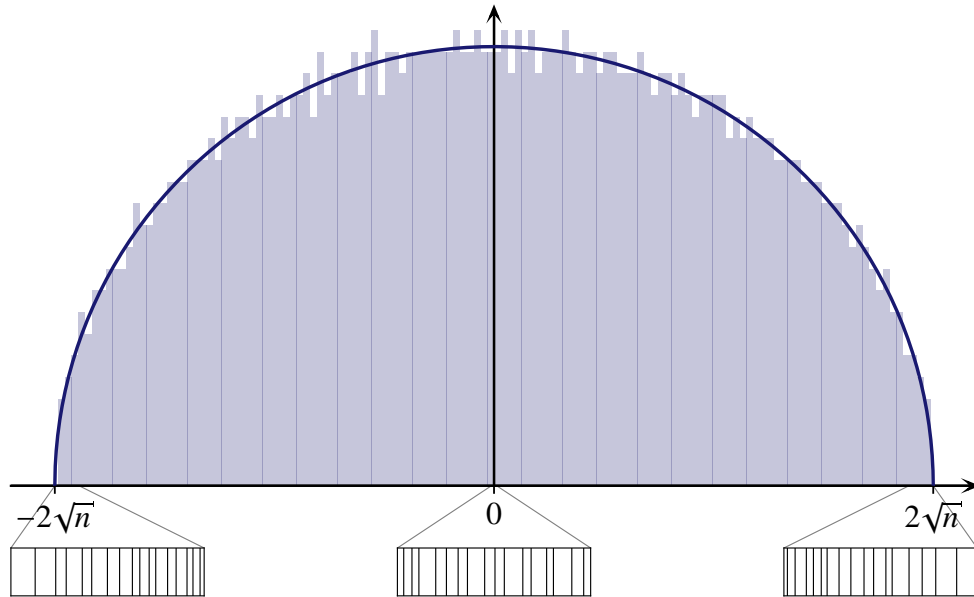


Figure 1.1: Empirical spectral distribution of a Hermite β -ensemble matrix $H_{\beta,n}$ as given in (1.2) with $\beta = 8$ and $n = 2048$. The solid line is a semicircle density between $-2\sqrt{n}$ and $2\sqrt{n}$.

then

$$\frac{1}{n} \sum_{j=1}^n \delta_{\lambda_j/\sqrt{n}} \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \mu_{\text{sc}} \quad (1.3)$$

vaguely (see [42] or [23, Theorem 6.4.1]). Intuitively, this means that when n is large, the density of points in the Hermite β -ensemble roughly forms a semicircle between $-2\sqrt{n}$ and $2\sqrt{n}$, as illustrated on Figure 1.1. The figure pictures the *empirical spectral distribution* of a tridiagonal Hermite β -ensemble matrix, which is the measure $\sum_{j=1}^n \delta_{\lambda_j}$ placing a Dirac mass at each eigenvalue λ_j of the matrix. In a sense, this result extends the classical Wigner semicircle law for Wigner matrices [3, Theorem 2.1.1], a class to which Gaussian orthogonal, unitary and symplectic ensemble matrices all belong.

1.1.2 The Laguerre β -ensemble

Another important example of a β -ensemble is the *Laguerre β -ensemble*, which is obtained in (1.1) with the potential $V(\lambda) := \frac{\lambda}{2} - (\frac{m-n+1}{2} - \frac{1}{\beta}) \log \lambda$ where $m > n - 1$ is an extra parameter. This β -ensemble generalizes the eigenvalue distribution of real ($\beta = 1$), complex ($\beta = 2$) and quaternionic ($\beta = 4$) Wishart matrices, which are classical random matrix models of the form XX^* where X is an $n \times m$ matrix with independent standard real, complex or quaternionic normal entries [3, §4.1].

Like for the Hermite case, there is a tridiagonal random matrix model that interpolates between the classical values of β . Indeed, Dumitriu and Edelman have shown that for any $\beta > 0$, if

$$L_{\beta,n} := \frac{1}{\sqrt{\beta}} \begin{pmatrix} Y_{n,1} & X_{n,1} & & \\ & \ddots & \ddots & \\ & & \ddots & X_{n,n-1} \\ & & & Y_{n,n} \end{pmatrix} \quad \text{where} \quad \begin{aligned} Y_{n,j} &\sim \mathcal{X}_{(m-j+1)\beta} \\ X_{n,j} &\sim \mathcal{X}_{(n-j)\beta} \end{aligned} \quad (1.4)$$

with all $X_{n,j}$'s and $Y_{n,j}$'s independent, then $L_{\beta,n} L_{\beta,n}^\top$ has eigenvalue point process with density (1.1) for $V(\lambda) := \frac{\lambda}{2} - (\frac{m-n+1}{2} - \frac{1}{\beta}) \log \lambda$ [24, Theorem 3.4].

The macroscopic behavior of the Laguerre β -ensemble has a similar (but distinct) description as that of the Hermite β -ensemble, which is illustrated on Figure 1.2. If λ is the Laguerre β -ensemble with n points and if $n/m \rightarrow \gamma \in (0, 1]$, then in the vague topology of measures,

$$\frac{1}{n} \sum_{j=1}^n \delta_{\lambda_j/n} \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \mu_{\text{MP}} \quad \text{where} \quad d\mu_{\text{MP}}(x) := \frac{1}{2\pi\gamma x} \sqrt{(\gamma_+ - x)(x - \gamma_-)} dx \quad (1.5)$$

with $\gamma_{\pm} := (1 \pm \sqrt{\gamma})^2$ [23, Theorem 6.5.1]. The measure μ_{MP} is called a *Marčenko–Pastur distribution* [50].

1.1.3 Rank-1 perturbations of matrix models of β -ensembles

It can also be natural to consider perturbations of the random matrix models for β -ensembles.

An example of this is the so-called *spiked* version of the Laguerre β -ensemble, which originates from the work on Johnstone [43] and has important applications in high-dimensional statistics. In the classical cases where $\beta \in \{1, 2, 4\}$, this is the eigenvalue point process of random matrices of the form $X\Sigma X^*$ where again X is $n \times m$ with independent standard real ($\beta = 1$), complex ($\beta = 2$) or quaternionic ($\beta = 4$) normal entries, but $\Sigma := \text{diag}(\ell_n, 1, \dots, 1)$ for an $\ell_n > 0$. The description of this point process has garnered a lot of interest in the random matrix theory community [5, 31, 34]. In particular, Baik, Ben Arous and P      famously showed in [5] that a phase transition occurs in the complex model: if $n/m \rightarrow \gamma \in (0, 1]$, then the largest eigenvalue has the same behavior as in the unspiked case when $\ell_n \rightarrow \ell < 1 + \sqrt{\gamma}$, but it separates from the bulk when $\ell_n \rightarrow \ell > 1 + \sqrt{\gamma}$.

It is also possible to extend the Dumitriu–Edelman method to build tridiagonal matrix models for these spiked ensembles [7]. The resulting tridiagonal matrix has the form $L_{\beta,n,\ell_n} L_{\beta,n,\ell_n}^\top$ where

$$L_{\beta,n,\ell_n} := \frac{1}{\sqrt{\beta}} \begin{pmatrix} \sqrt{\ell_n} Y_{n,1} & X_{n,1} & & \\ & \ddots & \ddots & \\ & & \ddots & X_{n,n-1} \\ & & & Y_{n,n} \end{pmatrix} \quad \text{and again} \quad \begin{aligned} Y_{n,j} &\sim \chi_{(m-j+1)\beta} \\ X_{n,j} &\sim \chi_{(n-j)\beta} \end{aligned} \quad (1.6)$$

with all $X_{n,j}$'s and $Y_{n,j}$'s independent.

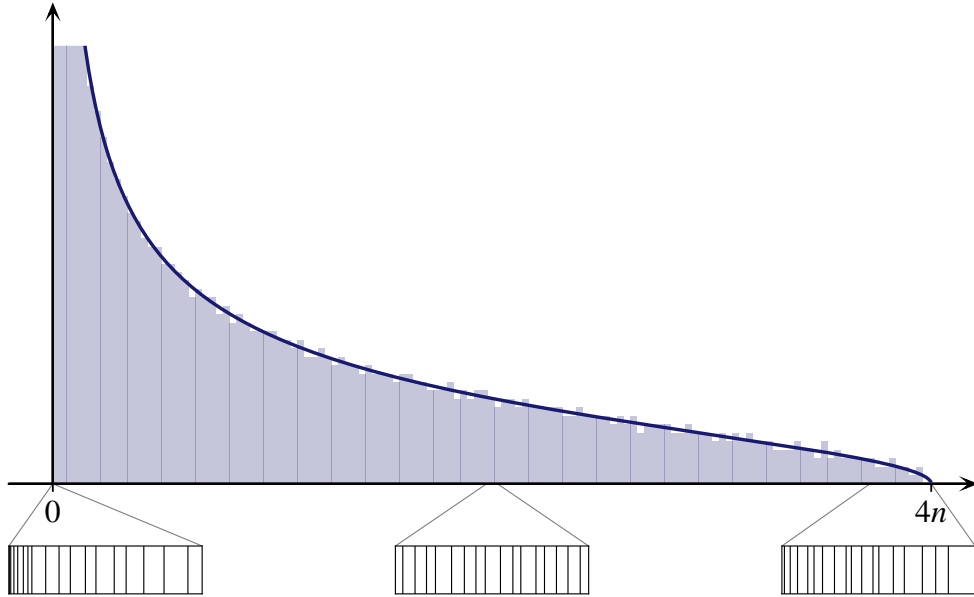


Figure 1.2: Empirical spectral distribution of a Laguerre β -ensemble matrix $L_{\beta,n} L_{\beta,n}^\top$ with $L_{\beta,n}$ as in (1.2) with $\beta = 8$ and $n = m = 2048$. The solid line is a Mar     –Pastur distribution rescaled between 0 and $4n$.

Another example of a perturbed matrix model of a β -ensemble comes from perturbed Gaussian orthogonal, unitary or symplectic matrices where the expectation of every entry has been shifted by the same constant. These are sometimes called *deformed* ensembles, and have been studied before by several authors—see e.g. [33, 56]. As a specific example, one might shift every entry of a Gaussian orthogonal ensemble matrix by α/n for some $\alpha \in \mathbb{R}$ which, by orthogonal invariance, has the same effect on the spectrum as shifting only the first entry of the matrix by α . Using the method of Dumitriu and Edelman, this matrix can still be tridiagonalized [7], and the resulting matrix simply has the form $H_{\beta,n} + \alpha \mathbf{e}_1 \mathbf{e}_1^\top$ where $H_{\beta,n}$ is as in (1.2) and $\mathbf{e}_1 := (1, 0, \dots, 0)$.

1.2 Local statistics of β -ensembles

Convergence results like those from (1.3) and (1.5) describe the *global* statistics of β -ensembles in the asymptotic regime where the number of points diverges. To get a more complete description of the point process, one can then try to understand its *local* statistics in the same asymptotic regime. This is usually done by taking scaling limits of the point process, the idea being to “zoom in” somewhere in the spectrum to see how points are spaced in that region, asymptotically.

It is far from obvious to understand what happens to the point process under a scaling limit, but Edelman and Sutton introduced in [30] a groundbreaking idea to do so: to describe these scaling limits as the spectra of stochastic differential operators. It turns out that for large classes of β -ensembles, only a few (families of) point processes suffice to describe all possible scaling limits.

1.2.1 The soft edge

A first scaling limit is the *soft edge* limit, which corresponds to the behavior of points in β -ensembles near the edges of the spectrum when the density vanishes like a square root—this is the behavior at both ends of the semicircle in the Hermite β -ensemble (see Figure 1.1), and at the top end of the spectrum in the Laguerre β -ensemble (see Figure 1.2). The soft edge limit is described by the spectrum of the *stochastic Airy operator* $\mathcal{H}_\beta$, whose action on a function $f: [0, \infty) \rightarrow \mathbb{R}$ is heuristically given by

$$\mathcal{H}_\beta f(t) = -f''(t) + \left(t + \frac{2}{\sqrt{\beta}} B'(t)\right) f(t) \quad (1.7)$$

where B is a standard Brownian motion. Of course, the white noise B' is not actually defined as a function, so it is not obvious that one can even define such an operator $\mathcal{H}_\beta$, and if so how it can be done rigorously—we will come back to this in detail in §3.1. The idea of this operator comes from Edelman and Sutton [30], but it was first defined in a rigorous way by Ramírez, Rider and Virág in [60]. As we will see later, this operator can be shown to be almost surely self-adjoint with real

eigenvalues on the appropriate domain under a boundary condition at 0. Its precise relationship with the Hermite and Laguerre β -ensembles is given by the following.

Theorem 1.1 [60, Theorems 1.1 and 1.4]. *Let $\Lambda_1 < \Lambda_2 < \dots$ denote the eigenvalues of $\mathcal{H}_\beta$ for the boundary condition $f(0) = 0$, and let $\lambda_{\beta,1}^H \geq \dots \geq \lambda_{\beta,n}^H$ and $\lambda_{\beta,1}^L \geq \dots \geq \lambda_{\beta,n}^L$ denote the points of the Hermite and Laguerre β -ensembles, assuming that $m > n - 1$ in the latter. Write $\mu_{n,m} := (\sqrt{n} + \sqrt{m})^2$ and $\sigma_{n,m} := \mu_{n,m}^{-2/3} n^{1/6} m^{1/6}$. Then for any fixed $k \in \mathbb{N}$, both*

$$\left\{ n^{1/6} (2\sqrt{n} - \lambda_{\beta,j}^H) \right\}_{j=1}^k \xrightarrow[n \rightarrow \infty]{\text{law}} \{\Lambda_j\}_{j=1}^k \quad \text{and} \quad \left\{ \sigma_{n,m} (\mu_{n,m} - \lambda_{\beta,j}^L) \right\}_{j=1}^k \xrightarrow[n \rightarrow \infty]{\text{law}} \{\Lambda_j\}_{j=1}^k.$$

The negatives of the eigenvalues Λ_j of the stochastic Airy operator are the points of the Airy_β point process. This point process has more generally been shown to arise as a universal scaling limit in the soft edge regime of β -ensembles for large classes of potentials [6, 10, 48].

The results of Theorem 1.1 were extended by Bloemendal and Virág in [7] to perturbed models of β -ensembles, such as the spiked Laguerre β -ensemble and the deformed Hermite β -ensemble mentioned before. They showed that, for suitably scaled perturbations, these point processes converge to the eigenvalues of the stochastic Airy operator for a different boundary condition at 0.

1.2.2 The hard edge

Another important scaling limit is the *hard edge* limit, which corresponds to the behavior of points in β -ensembles near the edges of the spectrum where the density diverges like $x \mapsto 1/\sqrt{x}$. This regime never appears in the Hermite β -ensemble, but it corresponds to points near the left edge of the Laguerre β -ensemble when $m \simeq n + a$ for some constant $a > -1$, as one can see on Figure 1.2. The hard edge limit is described by another stochastic differential operator.

Definition 1.1. Let $\beta > 0$ and $a > -1$, and let $B = \{B(t)\}_{t \geq 0}$ be a standard Brownian motion. The *stochastic Bessel operator* is the second-order differential operator $\mathfrak{G}_{\beta,a}$ that maps $f: [0, \infty) \rightarrow \mathbb{R}$ to

$$\mathfrak{G}_{\beta,a} f = -\frac{1}{w_{\beta,a}} (p_{\beta,a} f')' \tag{1.8}$$

where

$$p_{\beta,a}(t) := \exp\left(-at - \frac{2}{\sqrt{\beta}} B(t)\right) \quad \text{and} \quad w_{\beta,a}(t) := \exp\left(-(a+1)t - \frac{2}{\sqrt{\beta}} B(t)\right). \tag{1.9}$$

The idea of this operator, again, comes from Edelman and Sutton [30], but the above definition is that of Ramírez and Rider [58], who were the first to relate it in a rigorous way to the Laguerre β -ensemble.

Theorem 1.2 [58, Theorem 1]. *With probability one, with boundary conditions $f(0) = 0$ and $\lim_{t \rightarrow \infty} p_{\beta,a}(t)f'(t) = 0$, $\mathfrak{G}_{\beta,a}$ has discrete spectrum comprised of simple eigenvalues $0 < \Lambda_1 < \Lambda_2 < \dots \uparrow \infty$. Moreover, if $\lambda_{\beta,1}^L \leq \dots \leq \lambda_{\beta,n}^L$ denote the points of the Laguerre β -ensemble with $m = n + a$, then for any $k \in \mathbb{N}$,*

$$\{n\lambda_{\beta,j}^L\}_{j=1}^k \xrightarrow[n \rightarrow \infty]{\text{law}} \{\Lambda_j\}_{j=1}^k.$$

These eigenvalues Λ_j are the points of the Bessel $_{\beta,a}$ point process, and again, this scaling limit has been shown to be the universal description of the hard edge regime for a wide class of potentials [63]. Ramírez and Rider also extended their results from Theorem 1.2 to show that for a suitably scaled perturbation, the spiked Laguerre β -ensemble converges to the eigenvalues of the stochastic Bessel operator for a different boundary condition at 0 [59].

1.2.3 The bulk

Finally, another important scaling limit is the *bulk* limit, which corresponds to the behavior of points in the middle of β -ensembles when the density is bounded from below. In particular, this is the case anywhere away from the edges in the Hermite and Laguerre β -ensembles, as can be seen in Figures 1.1 and 1.2. The point process associated with this scaling limit, the sine $_{\beta}$ point process, is a symmetric, translation invariant point process that was first constructed independently by Valkó and Virág in [72] and by Killip and Stoiciu in [47], and was later realized as the spectrum of a stochastic differential operator in [76]. This operator is defined in terms of a *hyperbolic Brownian motion*, which is a complex-valued stochastic process $\mathcal{B}$ defined as a strong solution to the stochastic differential equation

$$d\mathcal{B}(t) = \sigma \operatorname{Im} \mathcal{B}(t) dW(t) \quad (1.10)$$

where $\sigma^2 > 0$ is referred to as the *variance* of $\mathcal{B}$, and where W is a standard complex Brownian motion, that is, a complex-valued process whose real and imaginary parts are independent standard (real) Brownian motions. Having defined this process, we can introduce the following.

Definition 1.2. With $\beta > 0$, let $\mathcal{B}_{\beta}$ be a hyperbolic Brownian motion with variance $4/\beta$ started at i in the upper half-plane, and let

$$R_{\beta} := \frac{1}{2 \det X_{\beta}} (X_{\beta})^{\top} X_{\beta} \quad \text{with} \quad X_{\beta} := \begin{pmatrix} 1 & -\operatorname{Re} \mathcal{B}_{\beta} \\ 0 & \operatorname{Im} \mathcal{B}_{\beta} \end{pmatrix}.$$

The *stochastic sine operator* is the operator acting on $u: (0, 1) \rightarrow \mathbb{C}^2$ as $u \mapsto (R_{\beta}^{-1} \circ \nu)Ju'$ where $\nu(t) := -\log(1-t)$ and $J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

The stochastic sine operator was introduced by Valkó and Virág in [76], where they proved that

it is almost surely self-adjoint on a domain with appropriate boundary conditions, with a discrete spectrum distributed as the sine_β point process. Connecting with their earlier work [72], this yields the following link with the Hermite β -ensemble.

Theorem 1.3 [72, Theorem 1]. *For $\beta > 0$, let Λ_n denote the Hermite β -ensemble with n points, and let $\{\mu_n\}_{n \in \mathbb{N}} \subset \mathbb{R}$ be a sequence satisfying $n^{1/6}(2\sqrt{n} - |\mu_n|) \rightarrow \infty$ as $n \rightarrow \infty$. Then*

$$\sqrt{4n - \mu_n^2}(\Lambda_n - \mu_n) \xrightarrow[n \rightarrow \infty]{\text{law}} \text{sine}_\beta$$

with respect to the vague topology for the counting measure of the point processes.

As in the other cases, it has also been shown that this limit object is the universal description of the bulk regime of β -ensembles [6, 11, 12, 67].

1.3 Canonical systems for scaling limits of β -ensembles

1.3.1 Motivation

The stochastic Airy, Bessel and sine operators that we introduced in the last section allow us to describe the local behavior of β -ensembles in the soft edge, hard edge and bulk asymptotic regimes. However, these three operators are not all of the same type: the stochastic Airy and Bessel operators are both random (generalized) Sturm–Liouville operators, whereas the stochastic sine operator is a random two-dimensional first-order differential operator.

This discrepancy complicates the study of the relationships that might exist between these operators. An example of this is that one could expect that it is possible, in some sense, to retrieve the stochastic sine operator as a scaling limit of the stochastic Airy or Bessel operator. The reason for this is that the Airy_β and $\text{Bessel}_{\beta,a}$ point processes both represent an infinite number of points on a half-line, with the density of points increasing or decreasing when going away from the edge. The sine_β point process, on the other hand, is translation invariant, and has points on the whole real line, equally spaced on average. Since all of these occur as scaling limits of the same β -ensembles, one might expect that, starting from the Airy_β or $\text{Bessel}_{\beta,a}$ point process, one can shift the process in order to zoom in the direction where there are infinitely many points, rescale the shifted process to make the density of points constant, and retrieve the sine_β process. Heuristically, looking at the pictures of Figures 1.1 and 1.2, the idea is to start from the edge process and zoom back in the bulk. In fact, this was proven by Valkó and Virág when they introduced the sine_β point process.

Theorem 1.4 [72, Corollary 3]. $2\sqrt{E}(\text{Airy}_\beta + E) \rightarrow \text{sine}_\beta$ vaguely in law as $E \rightarrow \infty$.

Because these point processes are the spectra of the stochastic Airy and sine operators, one might expect to be able to upgrade this convergence statement to the associated operators. However, because they are of different types, it is not obvious to see how to even formulate a convergence statement that would make sense.

Another reason to seek a unifying framework is that because the types of operators are different, the ways to relate them to β -ensembles (and to prove for example Theorems 1.1, 1.2 and 1.3) are different, and would not necessarily be obvious to generalize in order to derive other scaling limits of β -ensembles.

Fortunately, there exists a type of object that can be used to represent all of these three operators: the canonical system.

Definition 1.3. A *canonical system* on a possibly infinite interval $(a, b) \subseteq \mathbb{R}$ is a differential equation of the form

$$Ju' = -zHu \quad \text{with} \quad J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

where $z \in \mathbb{C}$, $u: (a, b) \rightarrow \mathbb{C}^2$ and $H: (a, b) \rightarrow \mathbb{C}^{2 \times 2}$. The matrix H is called the *coefficient matrix* of the system.

As we will see in detail in Chapter 2, a canonical system should be thought of as an eigenvalue equation. This is particularly obvious if the coefficient matrix H is invertible on the whole domain, in which case the canonical system can be written as the eigenvalue equation $-H^{-1}Ju' = zu$ for the differential operator $u \mapsto -H^{-1}Ju'$, which we will call a *Dirac operator*, following [76]. Canonical systems offer a very general framework for second-order differential operators. In addition to the Dirac operators mentioned above, they can also represent for example Sturm–Liouville operators and Jacobi difference operators.

The fact that the stochastic Airy, Bessel and sine operators can all be represented as canonical systems was already noticed by Valkó and Virág in [76]. In this work, they developed a theory of a certain class of Dirac operators, and they applied it to describe the stochastic sine operator as well as another Dirac operator equivalent to the stochastic Bessel operator.

One of the main points of this thesis is to argue that the canonical system representation, in its full generality, appears to be the right framework to study scaling limits of β -ensembles. Indeed, this framework is general enough to encompass all of the three operators introduced in §1.2, and because canonical systems can reproduce Jacobi difference operators, there are even canonical system representations of the tridiagonal random matrix models for β -ensembles! Hence, it can be used not only to describe the relationships between the different operators that occur as scaling limits, but also those between the operators and the random matrix models.

1.3.2 Canonical system convergence of tridiagonal random matrix models

The observation made at the end of the last subsection leads to an interesting point: using canonical systems, one can formalize the idea that the operators introduced in §1.2 are scaling limits of the tridiagonal matrix models in a consistent way, for all scaling limits. A contribution of this thesis is to prove, at the canonical system level, the convergence of suitably scaled tridiagonal random matrix models for β -ensembles to the stochastic Airy and Bessel operators. The precise results are given in §3.1.3 for the Airy case and in §3.2.3 for the Bessel case, and we give a glimpse of the main consequences here.

As we will see in detail in §2.4, using a suitable change of variables, it is possible to represent as a canonical system the eigenvalue equation for a (generalized) Sturm–Liouville operator. For instance, taking two solutions f_β and g_β of $\mathcal{H}_\beta f = 0$ with initial conditions $g_\beta(0) = f'_\beta(0) = 0$ and $f_\beta(0) = g'_\beta(0) = 1$, it can be shown that if

$$A_\beta := \begin{pmatrix} g'_\beta - \frac{2}{\sqrt{\beta}} B g_\beta & f'_\beta - \frac{2}{\sqrt{\beta}} B f_\beta \\ g_\beta & f_\beta \end{pmatrix},$$

then a function $f: [0, \infty) \rightarrow \mathbb{R}$ solves the eigenvalue equation $\mathcal{H}_\beta f = z f$ if and only if $u := A_\beta^{-1} \begin{pmatrix} f' - \frac{2}{\sqrt{\beta}} B f \\ f \end{pmatrix}$ solves the canonical system

$$J u' = -z H_\beta u \quad \text{on} \quad (0, \infty) \quad \text{where} \quad H_\beta := \begin{pmatrix} g_\beta^2 & f_\beta g_\beta \\ f_\beta g_\beta & f_\beta^2 \end{pmatrix}. \quad (1.11)$$

An eigenvalue problem for $\mathcal{H}_\beta$ transfers to an eigenvalue problem for this canonical system. As we will see in detail in §2.5, there is a similar way to represent the eigenvalue equation for a tridiagonal matrix as a canonical system.

There are various ways to describe how a sequence of canonical systems can converge. The simplest one is probably to use their coefficient matrices, by topologizing them through a *vague topology*. Essentially, this is done by thinking of coefficient matrices as 2×2 matrix-valued measures, and testing them against compactly supported continuous functions—we will come back to this in detail in §2.2. So, a first way to write the results from Theorem 1.1 at the canonical system level is the following.

Theorem 1.5. *Let $H_{\beta,n}$ be the coefficient matrix built from the tridiagonal matrix $n^{1/6}(2\sqrt{n} - H_{\beta,n})$ or $\sigma_{n,m}(\mu_{n,m} - L_{\beta,n} L_{\beta,n}^\top)$ where $\mu_{n,m} := (\sqrt{n} + \sqrt{m})^2$ and $\sigma_{n,m} := \mu_{n,m}^{-2/3} n^{1/6} m^{1/6}$, with $H_{\beta,n}$ as in (1.2) and $L_{\beta,n}$ as in (1.4). Then $H_{\beta,n} \rightarrow H_\beta$ vaguely in law as $n \rightarrow \infty$, where H_β is as in (1.11).*

We will prove a more general version of this in Theorem 3.8, in §3.1.3, with weaker conditions

on the tridiagonal random matrix model and allowing for perturbations, so that this will include spiked versions of the Laguerre β -ensemble as well as deformed Hermite β -ensembles.

In the same way, we will see in §3.2.1 that the eigenvalue equation for the stochastic Bessel operator is equivalent to a canonical system on $(0, \infty)$ with a coefficient matrix $G_{\beta,a}$. We will then prove the following.

Theorem 1.6. *Let $G_{\beta,a,n}$ denote the coefficient matrix of the canonical system built from the tridiagonal matrix $nL_{\beta,n}L_{\beta,n}^T$ where $L_{\beta,n}$ is as in (1.4) for $m = n + a$ with $a > -1$ fixed. Then $G_{\beta,a,n} \rightarrow G_{\beta,a}$ vaguely in law as $n \rightarrow \infty$.*

We will prove a more general version of this in Theorem 3.23, in §3.2.3, again with weaker conditions on the tridiagonal matrix model and allowing for a spiked version of the model.

Now, the vague convergence of coefficient matrices doesn't say anything, a priori, about the spectral information of the system. This motivates to improve the results of Theorems 1.5 and 1.6 to obtain a stronger result. To do so, we use the canonical systems' *Weyl–Titchmarsh functions*.

As we will see in §2.1, given suitable boundary conditions depending on the system's behavior near the endpoints of its domain (a, b) , a canonical system $Ju' = -zHu$ can be associated with a holomorphic function $m_H: \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ from the upper half-plane $\mathbb{H} := \{z \in \mathbb{C} : \text{Im } z > 0\}$ to its closure $\overline{\mathbb{H}}_\infty$ into the Riemann sphere. This function m_H is called the *Weyl–Titchmarsh function* of the system, and is especially useful from a spectral theoretic point of view because it actually extends to a meromorphic function on $\mathbb{C}$, with poles on the real line placed at the system's eigenvalues. It also has an integral representation with respect to a measure μ_H on $\mathbb{R}$, called the system's *spectral measure*.

Building on Theorem 1.5, we will also show the following in §3.1.3.

Theorem 1.7. *Let $H_{\beta,n}$ be as in Theorem 1.5, and let $m_{H_{\beta,n}}$ and $\mu_{H_{\beta,n}}$ be the system's Weyl–Titchmarsh function and spectral measure. Likewise, let m_{H_β} and μ_{H_β} be the stochastic Airy system's Weyl–Titchmarsh function and spectral measure. Then $m_{H_{\beta,n}} \rightarrow m_{H_\beta}$ compactly on $\mathbb{H}$ in law and $\mu_{H_{\beta,n}} \rightarrow \mu_{H_\beta}$ vaguely in law as $n \rightarrow \infty$. This holds jointly, and jointly with the convergence of coefficient matrices in Theorem 1.5.*

In §3.2.3 we will also extend the convergence to the stochastic Bessel operator from Theorem 1.6 to the following.

Theorem 1.8. *Let $G_{\beta,a,n}$ be as in Theorem 1.6, and let $m_{G_{\beta,a,n}}$ and $\mu_{G_{\beta,a,n}}$ be the system's Weyl–Titchmarsh function and spectral measure. Likewise, let $m_{G_{\beta,a}}$ and $\mu_{G_{\beta,a}}$ be the stochastic Bessel system's Weyl–Titchmarsh function and spectral measure. Then $m_{G_{\beta,a,n}} \rightarrow m_{G_{\beta,a}}$ compactly on $\mathbb{H}$ in law and $\mu_{G_{\beta,a,n}} \rightarrow \mu_{G_{\beta,a}}$ vaguely in law as $n \rightarrow \infty$. This holds jointly, and jointly with the convergence of coefficient matrices in Theorem 1.6.*

As in the cases of Theorems 1.5 and 1.6, we will in fact prove this for more general tridiagonal models, and possibly allowing for perturbations in the first entry of the matrix. All of the precise results are given in §3.1.3 for the Airy case and in §3.2.3 for the Bessel case.

1.3.3 Edge to bulk transitions via canonical systems

As mentioned earlier, one of the main motivations to study canonical system representations of the stochastic Airy, Bessel and sine operators is that it allows one to describe relationships between them, which is otherwise not obvious to do since the stochastic sine operator is of a different type than the other two.

An example of such a relationship is the operator level version of the soft edge to bulk transition from Theorem 1.4. Indeed, since $2\sqrt{E}(\text{Airy}_\beta + E) \rightarrow \text{sine}_\beta$ vaguely in law as $E \rightarrow \infty$, we can expect that a similar result should hold at the operator level: the operator $\mathcal{H}_{\beta,E} := 2\sqrt{E}(\mathcal{H}_\beta - E)$ should be expected to converge, in some sense, to the stochastic sine operator. Of course, this question doesn't make sense when talking about the operators themselves, but once they are represented as canonical systems, the convergence can be given a precise meaning. In the same way, one should also expect an operator level *hard edge* to bulk transition to be possible, although with a different scaling. In that case, an analysis of the deterministic limit case $\beta = \infty$ suggests that $\mathfrak{G}_{\beta,a,E} := \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E)$ should converge to the stochastic sine operator as $E \rightarrow \infty$.

These two edge to bulk transitions, at the canonical system level, are the main subject of Chapter 4. These two problems were addressed separately in the two preprints [54, 55], and the results are reproduced here.

In the same way as we described earlier for $\mathcal{H}_\beta$, the two operators $\mathcal{H}_{\beta,E}$ and $\mathfrak{G}_{\beta,a,E}$ can be associated with canonical systems on $(0, \infty)$ with coefficient matrices $H_{\beta,E}$ and $G_{\beta,a,E}$ that are built in terms of solutions to the eigenvalue equations $\mathcal{H}_{\beta,E}f = 0$ and $\mathfrak{G}_{\beta,a,E}f = 0$. The first canonical system convergence that we prove is the vague convergence of the coefficient matrices.

Theorem 1.9 [54, 55, Theorems 1]. *Let $\mathcal{I}_\beta := [0, 1)$ if $\beta \leq 2$ and $\mathcal{I}_\beta := [0, 1]$ if $\beta > 2$. For any diverging sequence $\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$, there are $\mathcal{C}^1$ bijections $\eta_{A,E_n} : [0, 1 + \varepsilon_{A,E_n}) \rightarrow [0, \infty)$ where $\varepsilon_{A,E_n} \rightarrow 0$ such that*

$$\tilde{H}_{\beta,E_n} := \eta'_{A,E_n}(H_{\beta,E_n} \circ \eta_{A,E_n}) \xrightarrow[n \rightarrow \infty]{\text{law}} \tilde{R}_\beta := R_\beta \circ \nu$$

vaguely on $\mathcal{I}_\beta$, where R_β and ν are as in Definition 1.2. Likewise, there are $\mathcal{C}^1$ bijections $\eta_{B,E_n} : [0, 1 + \varepsilon_{B,E_n}) \rightarrow [0, \infty)$ where $\varepsilon_{B,E_n} \rightarrow 0$ such that

$$\tilde{G}_{\beta,a,E_n} := \eta'_{B,E_n}(G_{\beta,a,E_n} \circ \eta_{B,E_n}) \xrightarrow[n \rightarrow \infty]{\text{law}} \tilde{R}_\beta$$

vaguely on $\mathcal{I}_\beta$.

To retrieve information about the canonical systems' spectra, this result is then extended to the convergence of the Weyl–Titchmarsh functions.

Theorem 1.10 [54, 55, Theorems 2]. *Let $\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$ be a diverging sequence, and let $m_{\tilde{H}_{\beta, E_n}}, m_{\tilde{G}_{\beta, a, E_n}}, m_{\tilde{R}_{\beta}} : \mathbb{H} \rightarrow \overline{\mathbb{H}}_{\infty}$ be the Weyl–Titchmarsh functions of the shifted and scaled Airy and Bessel systems and of the sine system. Then*

$$m_{\tilde{H}_{\beta, E_n}} \xrightarrow[n \rightarrow \infty]{\text{law}} m_{\tilde{R}_{\beta}} \quad \text{and} \quad m_{\tilde{G}_{\beta, a, E_n}} \xrightarrow[n \rightarrow \infty]{\text{law}} m_{\tilde{R}_{\beta}}$$

compactly on $\mathbb{H}$, jointly with the respective coefficient and transfer matrices. In particular, the spectral measures of the corresponding systems converge vaguely in law, also jointly with the other objects.

1.3.4 Remarks on spectral measures and eigenvalue point processes

Consider a canonical system $Ju' = -zHu$ with appropriate boundary conditions, and suppose that it has purely discrete spectrum. The spectral measure μ_H of such a system is discrete, and has the form

$$\mu_H = \sum_{\lambda \in \text{spec } S_H} \mu_H(\{\lambda\}) \delta_{\lambda}$$

where $\text{spec } S_H$ denotes the spectrum for this boundary value problem, which is in this case the set of eigenvalues. The number $\mu_H(\{\lambda\})$ is called the *spectral weight* for the eigenvalue $\lambda \in \text{spec } S_H$. This spectral measure is distinct from the eigenvalue point process ν_H of the same spectral problem, which is simply

$$\nu_H = \sum_{\lambda \in \text{spec } S_H} \delta_{\lambda}.$$

Although these two measures are obviously related, they are distinct objects. In particular, given a sequence of boundary value problems for canonical systems with purely discrete spectra, the convergence of their spectral measures is distinct from the convergence of their eigenvalue point processes: neither implies the other one. Indeed, the atoms of a sequence of spectral measures can merge or disappear in the vague limit, but these two phenomena are impossible for the associated point processes, since they require the spectral weights to combine or vanish in the limit, whereas the “weights” of the point process are always 1. Conversely, the convergence of the eigenvalue point processes does not imply the convergence of the spectral measures either, because the point processes carry no information at all about the spectral weights.

This distinction means, in particular, that the vague convergences of spectral measures that we prove in Theorems 1.7, 1.8 and 1.10 from canonical systems theory are not equivalent to the

corresponding results at the eigenvalue level, which are already known from [58, 60, 72] as pointed out earlier.

While the tools from canonical systems theory are very well adapted to study the convergence of spectral problems up to the convergence of the spectral measures, it is less natural to deduce the convergence of the associated eigenvalue point processes. We will say a few words about this problem in a general setting in §2.2.2.2, but we won't tackle it in the context of problems from β -ensembles theory in this thesis.

Nevertheless, the convergence of the eigenvalues themselves is, in a sense, a more natural goal from the point of view of random matrix theory, and for that reason it would be valuable to be able to strengthen the results from Theorems 1.7, 1.8 and 1.10 in that sense, in order to complete the canonical system theoretic picture of these scaling limits. This could possibly be done through an analysis of the spectral weights. As we will see later, the spectral weights can be described from norms of eigenfunctions, or from residues of the Weyl–Titchmarsh functions. Maybe these tools could allow us to get a sufficient understanding of their asymptotic behavior to be able to deduce the convergence of eigenvalues.

1.4 Organization of the thesis

The rest of the thesis is organized in three chapters as follows. Each chapter also starts with a more detailed overview of its contents.

In Chapter 2, we review important concepts of the theory of canonical systems that will be needed in the sequel. We review the classical theory in §2.1, and then we present topologies on canonical systems and associated objects in §2.2. Some extensions and adaptations to random canonical systems are given in §2.3. Finally, we present in §2.4 and §2.5 the methods used to represent generalized Sturm–Liouville operators and tridiagonal matrices or Jacobi operators as canonical systems.

Then, Chapter 3 is split in three sections in which are successively presented in more detail the stochastic Airy (§3.1), Bessel (§3.2) and sine (§3.3) operators. For the stochastic Airy and Bessel operators, we review important properties, derive asymptotics of solutions to the eigenvalue equation on the negative real line, and prove convergence from tridiagonal random matrix models at the canonical system level. In particular, Theorems 1.5 and 1.7 are proven in §3.1.3, and Theorems 1.6 and 1.8 are proven in §3.2.3. For the stochastic sine operator, we only review some important properties.

Finally, Chapter 4 is about the operator level soft and hard edge to bulk transitions described earlier. The goals of this chapter are essentially to prove Theorems 1.9 and 1.10.

1.5 Notation

Below is a list of symbols used repeatedly throughout the thesis.

$\text{AC}_{\text{loc}}(a, b)$	Set of locally absolutely continuous functions on (a, b) , i.e., functions $f: (a, b) \rightarrow \mathbb{C}$ such that $f(t) = f(t_0) + \int_{t_0}^t g(t) dt$ for a $t_0 \in (a, b)$ and a $g \in L_{\text{loc}}^1(a, b)$.
$\mathcal{C}(X, Y)$	Set of continuous functions $X \rightarrow Y$.
$\mathcal{C}_c(X, Y)$	Set of compactly supported continuous functions $X \rightarrow Y$.
$\mathbb{C}_{\infty}$	Riemann sphere.
e_{ϕ}	Vector $(\cos \phi, \sin \phi) \in \mathbb{R}^2$ for $\phi \in [0, 2\pi)$.
$\mathbf{e}_j$	Basis vector in $\mathbb{R}^n$ or $\mathbb{C}^n$, with a 1 in j th position and zeros elsewhere.
$\mathbb{H}$	Upper half-plane $\{z \in \mathbb{C} : \text{Im } z > 0\}$.
$\overline{\mathbb{H}}_{\infty}$	Closure of $\mathbb{H}$ in the Riemann sphere.
I_n	$n \times n$ identity matrix.
$\mathcal{L}$	Lebesgue measure.
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2 .
O	Big-o notation. We say that $f(t) = O(g(t))$ as $t \rightarrow \infty$ if there is a $C > 0$ such that for all t large enough, $f(t) \leq Cg(t)$. Sometimes we add subscripts to O to indicate parameters on which C might depend.
$O_{\mathbb{P}}$	Probabilistic version of the big-o notation. We say that $f(t) = O_{\mathbb{P}}(g(t))$ as $t \rightarrow \infty$ if for all $\varepsilon > 0$, there is a $C_{\varepsilon} > 0$ such that $f(t) \leq C_{\varepsilon}g(t)$ with probability at least $1 - \varepsilon$.
o	Little-o notation. We say that $f(t) = o(g(t))$ as $t \rightarrow \infty$ if $f(t)/g(t) \rightarrow 0$ as $t \rightarrow \infty$. Sometimes we add subscripts to o to refer to a set of parameters over which the convergence $f(t)/g(t) \rightarrow 0$ is uniform.
$o_{\mathbb{P}}$	Probabilistic version of the little-o notation. We say that $f(t) = o_{\mathbb{P}}(g(t))$ as $t \rightarrow \infty$ if $f(t)/g(t) \rightarrow 0$ in probability as $t \rightarrow \infty$.
$\mathcal{P}$	Mapping $\mathbb{C}^2 \rightarrow \mathbb{C}_{\infty}$ defined by $\mathcal{P}\left(\frac{z}{w}\right) := z/w$.
$\mathbb{S}^1$	Unit circle $\{z \in \mathbb{C} : z = 1\}$.
Unif S	Uniform distribution on a set S .
$\Gamma(\alpha, \theta)$	Γ distribution with shape α and scale θ , hence with mean $\alpha\theta$ and variance $\alpha\theta^2$.
$\lesssim$	We use $f \lesssim g$ or $f(t) \lesssim g(t)$ as an alternative notation for $f(t) = O(g(t))$.

Chapter 2

Canonical systems theory

In this chapter, we introduce the concepts from the theory of canonical systems that will be used in the sequel. In §2.1, we recall what canonical systems are, and we introduce the main objects that are used to describe their spectral theory. In §2.2, we introduce topologies that are used to characterize the convergence of a sequence of canonical systems. Some of these ideas are then adapted to random canonical systems in §2.3. Finally, §2.4 and §2.5 present methods to represent generalized Sturm–Liouville operators and tridiagonal matrices or Jacobi operators as canonical systems.

The classical concepts of canonical systems theory are due to de Branges [14, 15, 16, 17, 18]. The presentation given here is essentially taken or extended from Remling’s book [61], with the exceptions of §2.2.3 and §2.3. More specific references are given throughout the chapter.

2.1 Classical theory

2.1.1 Canonical systems and their solutions

Recall the definition of a canonical system from the introduction.

Definition 1.3. A *canonical system* on a possibly infinite interval $(a, b) \subseteq \mathbb{R}$ is a differential equation of the form

$$Ju' = -zHu \quad \text{with} \quad J := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

where $z \in \mathbb{C}$, $u: (a, b) \rightarrow \mathbb{C}^2$ and $H: (a, b) \rightarrow \mathbb{C}^{2 \times 2}$. The matrix H is called the *coefficient matrix* of the system.

Here, we make a few hypotheses on the coefficient matrix that will be assumed to hold unless specified otherwise.

Hypothesis 2.1. The coefficient matrix H of a canonical system on an interval (a, b) has the following properties:

- i) $H: (a, b) \rightarrow \mathbb{R}^{2 \times 2}$;
- ii) $H \in L^1_{\text{loc}}(a, b)$ entrywise;
- iii) $H(t) \succeq 0$ for almost every $t \in (a, b)$.
- iv) $H(t) \neq 0$ for almost every $t \in (a, b)$.

The coefficient matrix does not have to be invertible. The most dramatic way in which this can happen is that H could be a projector onto a particular direction, which leads to the following classification of times in the domain.

Definition 2.1. A point $t \in (a, b)$ of the domain of a canonical system $Ju' = -zHu$ is called *singular* if there is an $\varepsilon > 0$ and a $v \in \mathbb{R}^2 \setminus \{0\}$ such that $H(s)v = 0$ for almost all $s \in (t - \varepsilon, t + \varepsilon)$. The connected components of the set of singular points are called *singular intervals*, and points that are not singular are called *regular*.

Based on the above, we will sometimes add another hypothesis to the list from Hypothesis 2.1.

Hypothesis 2.2. The domain of a canonical system is not a single singular interval.

We now turn to the description of a canonical system's solutions. Since the coefficient matrix does not have to be very regular, solutions are understood in the following weak sense.

Definition 2.2. A function $u: (a, b) \rightarrow \mathbb{C}^2$ is said to solve a canonical system $Ju' = -zHu$ on an interval (a, b) if $u \in AC_{\text{loc}}(a, b)$ and if $Ju' = -zHu$ holds a.e. on (a, b) .

With this definition of solutions, it can be shown that for any initial condition $v \in \mathbb{C}^2$ at some $c \in (a, b)$, a canonical system has a unique solution u such that $u(c) = v$ [61, Theorem 1.1]. Moreover, since the equation $Ju' = -zHu$ only has to hold almost everywhere on the interval for u to be a solution, two coefficient matrices that are equal almost everywhere yield two canonical systems with the exact same solutions. Therefore, we never distinguish between two such canonical systems.

In Definition 1.3, we defined a canonical system on an open interval, but sometimes it is possible to extend its solutions to the endpoints. Whether this is possible depends on the system's behavior near the endpoints.

Definition 2.3. Consider a canonical system $Ju' = -zHu$ on an interval (a, b) . The endpoint a is said to be *limit circle* if $H \in L^1(a, a + \varepsilon)$ for some $\varepsilon > 0$, and *limit point* otherwise. In the same way, b is limit circle if $H \in L^1(b - \varepsilon, b)$ for some $\varepsilon > 0$ and limit point otherwise.

Remark. The names “limit point” and “limit circle” originate from Weyl theory, where they have

a clear geometrical interpretation. In fact, the above definition is not the usual way to define these terms when building the theory, and the fact that this characterization is equivalent is an important result. We refer the interested reader to [61, Definition 2.6] for the usual definition, and to [61, §3.4] for the theory leading to the geometrical interpretation.

This classification of endpoints plays a major role in the spectral theory of canonical systems, but for now we only point out the following.

Lemma 2.1 [61, Lemma 2.6]. *Consider a canonical system $Ju' = -zHu$ that satisfies Hypothesis 2.2 on an interval (a, b) . If a is limit circle, then any solution to $Ju' = -zHu$ has a continuous extension to $[a, b)$, and in fact $u \in AC_{\text{loc}}[a, b)$. The analogous result holds at b if it is limit circle.*

Many important features of canonical systems are preserved by monotone changes of variables. Indeed, the following result is easily verified by direct calculation (see also [61, Theorem 1.5]).

Theorem 2.2. *Consider a canonical system $Ju' = -zHu$ on an interval (a, b) . Let $\eta: (A, B) \rightarrow (a, b)$ be an absolutely continuous bijection, where (A, B) is another real interval. Then the initial canonical system is equivalent to*

$$Jv' = -z\tilde{H}v \quad \text{on} \quad (A, B) \quad \text{with} \quad \tilde{H} := \eta' \cdot H \circ \eta$$

in the sense that an absolutely continuous function $u: (a, b) \rightarrow \mathbb{C}^2$ solves $Ju' = -zHu$ if and only if $v := u \circ \eta$ solves $Jv' = -z\tilde{H}v$.

An important example of such an equivalence is the case where η satisfies $\eta^{-1'} = \text{tr } H$ a.e., which can always be defined since $\text{tr } H$ is locally integrable. In that case, $\eta' = \frac{1}{\text{tr } H \circ \eta}$, and it follows that $\text{tr } \tilde{H} \equiv 1$. The resulting canonical system $Jv' = -z\tilde{H}v$ is said to be *trace-normed*. Such systems have a special theoretical importance in that they give a natural way to choose a particular representative of a class of systems that are equivalent up to a time change, but we will not need to use them much here.

2.1.2 Transfer matrices

A useful tool in the study of a canonical system's solutions is a specific matrix solution defined as follows.

Definition 2.4. The *transfer matrix* associated with a canonical system $Ju' = -zHu$ on an interval (a, b) is the map $T_H: (a, b)^2 \times \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ such that for each $s \in (a, b)$, $T_H(\cdot, s, z)$ is the (matrix) solution to $JT' = -zHT$ with $T(s) = I_2$.

Because $\partial_1 T_H = zJHT_H$ and H is symmetric, the determinant of T is constant. Indeed, it is

easy to check by direct computation that $\partial_1 \det T_H = -\operatorname{tr}(JT_H^\top J \partial_1 T_H) = z \operatorname{tr}(JT_H^\top H T_H)$, but this also equals $z \operatorname{tr}(T_H^\top H^\top T_H J^\top) = -z \operatorname{tr}(JT_H^\top H T_H)$ by the cyclic property of the trace, and it follows that $\partial_1 \det T_H \equiv 0$. As $T_H(t, t, z) = I_2$ for any t , this shows that $\det T_H \equiv 1$.

The existence and uniqueness of canonical system solutions has rather immediate and very useful consequences on transfer matrices. If $v \in \mathbb{C}^2$, then the unique solution with initial value $u(s) = v$ is $u = T_H(\cdot, s, z)v$. Moreover, for any times t, r, s in the canonical system's domain, $T_H(t, s, z) = T_H(t, r, z)T_H(r, s, z)$, which also implies that $T_H^{-1}(t, s, z) = T_H(s, t, z)$. These properties lead to the important intuition that the transfer matrix $T_H(t, s, z)$ updates the system's solutions from time s to time t . They also imply that there is no loss of generality in considering the transfer matrix as a function of two arguments, in the sense that the second one can be taken to be constant. Indeed, having fixed some time r , we can always write $T_H(t, s, z) = T_H(t, r, z)T_H^{-1}(s, r, z)$. When the left endpoint a of the domain is limit circle, by Lemma 2.1 the transfer matrix extends to a , so we will often base it at that point and drop the dependence on the second argument in the notation.

As transfer matrices are $\operatorname{SL}(2, \mathbb{C})$ transforms, they act on the Riemann sphere $\mathbb{C}_\infty$ by Möbius transforms.

Definition 2.5. Given $T := \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \operatorname{SL}(2, \mathbb{C})$ and $z \in \mathbb{C}_\infty$, we define $Tz := \frac{az+b}{cz+d}$.

Remark. The identity acts as $I_2 z = z$, and it is easy to verify by direct calculation that if $T, S \in \operatorname{SL}(2, \mathbb{C})$, then $T(Sz) = (TS)z$. Hence, the above does define a group action of $\operatorname{SL}(2, \mathbb{C})$ on $\mathbb{C}_\infty$.

2.1.3 Associated L^2 spaces

A canonical system $Ju' = -zHu$ on an interval (a, b) can naturally be associated with a specific Hilbert space. To build this space, given a measurable $u: (a, b) \rightarrow \mathbb{C}^2$, we define

$$\|u\|_H^2 := \int_a^b u^*(t)H(t)u(t) dt. \quad (2.1)$$

Then, we define the family of functions

$$\mathcal{L} := \left\{ u: (a, b) \rightarrow \mathbb{C}^2 : u \text{ is measurable, } \|u\|_H < \infty \right\},$$

and $\|\cdot\|_H$ is a seminorm on $\mathcal{L}$, as in the case of usual L^2 spaces. To get a normed space, we use the usual method and define

$$L_H^2(a, b) := \mathcal{L} / \{u \in \mathcal{L} : \|u\|_H = 0\}. \quad (2.2)$$

This is a separable, infinite-dimensional Hilbert space [61, §1.4].

This construction is, for the most part, entirely analogous to the classical one. However, it should be noted that it is not always the case that two functions u and v that are equivalent in $L_H^2(a, b)$ are

equal almost everywhere. Indeed, the equivalence of u and v in $L_H^2(a, b)$ means that $\|u - v\|_H = 0$, but this only means that $(u - v)^* H(u - v)$ is zero a.e., which only requires $Hu = Hv$ a.e. Since H can have a kernel, it is therefore possible for u and v to be two distinct continuous functions.

2.1.4 Relations on Hilbert spaces

A canonical system has the structure of an eigenvalue equation: whenever the coefficient matrix H is invertible, the equation $Ju' = -zHu$ can be written as $-H^{-1}Ju' = zu$, which is an eigenvalue equation for a differential operator $u \mapsto -H^{-1}Ju'$. As in general H may fail to be invertible, in order to develop this point of view we have to use the more general theory of *relations* rather than operators. We start this section by introducing important concepts from the theory of relations.

Definition 2.6. A *relation* on a Hilbert space $\mathcal{H}$ is a linear subspace of $\mathcal{H} \oplus \mathcal{H}$.

In particular, an operator on a Hilbert space can be seen as a relation by identifying it with its graph.

The following objects associated to a relation will be useful in the sequel.

Definition 2.7. Let $\mathcal{T}$ be a relation on a Hilbert space $\mathcal{H}$.

- i) The *domain* of $\mathcal{T}$ is $\text{dom } \mathcal{T} := \{x \in \mathcal{H} : \exists y \in \mathcal{H}, (x, y) \in \mathcal{T}\}$.
- ii) The *kernel* of $\mathcal{T}$ is $\ker \mathcal{T} := \{x \in \mathcal{H} : (x, 0) \in \mathcal{T}\}$.
- iii) The *multi-valued part* of $\mathcal{T}$ is $\mathcal{T}(0) := \{y \in \mathcal{H} : (0, y) \in \mathcal{T}\}$.
- iv) The *inverse* of $\mathcal{T}$ is the relation $\mathcal{T}^{-1} := \{(y, x) \in \mathcal{H} \oplus \mathcal{H} : (x, y) \in \mathcal{T}\}$.
- v) The *closure* $\bar{\mathcal{T}}$ of $\mathcal{T}$ is its closure in $\mathcal{H} \oplus \mathcal{H}$, and $\mathcal{T}$ is *closed* if $\mathcal{T} = \bar{\mathcal{T}}$.
- vi) The *adjoint* of $\mathcal{T}$ is the relation $\mathcal{T}^* := \{(z, w) \in \mathcal{H} \oplus \mathcal{H} : \forall (x, y) \in \mathcal{T}, \langle z, y \rangle = \langle w, x \rangle\}$.
- vii) $\mathcal{T}$ is *symmetric* if $\mathcal{T} \subseteq \mathcal{T}^*$ and *self-adjoint* if $\mathcal{T} = \mathcal{T}^*$.

The spectral theory of self-adjoint relations can be studied in a similar fashion as that of (unbounded) operators. In fact, a self-adjoint relation contains (the graph of) a self-adjoint operator on a possibly smaller space, to which is added the multi-valued part of the relation as the orthogonal complement.

Theorem 2.3 [61, Theorem 2.11]. Let S be a self-adjoint relation from a Hilbert space $\mathcal{H}$ to itself. Let $\mathcal{H}_1 := \overline{\text{dom } S}$ and $\mathcal{H}_2 := S(0)$. Then $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2$, and this decomposition reduces S in the sense that $S = S_1 \oplus S_2$ where $S_j := S \cap (\mathcal{H}_j \oplus \mathcal{H}_j)$. Moreover, S_1 is a self-adjoint operator on $\mathcal{H}_1$, and $S_2 = \{(0, y) \in \mathcal{H} \oplus \mathcal{H} : y \in \mathcal{H}_2\}$.

This naturally leads to defining the spectrum of a self-adjoint relation from that of the corresponding self-adjoint operator.

Definition 2.8. Let S be a self-adjoint relation from a Hilbert space $\mathcal{H}$ to itself, and let $S := S \cap (\overline{\text{dom } S \oplus \text{dom } S})$ be its operator part. The *spectrum* $\text{spec } S$ of S is the spectrum of S if $\text{dom } S \neq 0$, or $\emptyset$ if $\text{dom } S = 0$. The *discrete spectrum* $\text{spec}_d S$ of S is the set of isolated eigenvalues with finite multiplicity, and its *essential spectrum* is $\text{spec}_{\text{ess}} S := \text{spec } S \setminus \text{spec}_d S$.

Furthermore, a complete spectral representation can be given, like for a self-adjoint operator.

Definition 2.9. Let S be a self-adjoint relation from a Hilbert space $\mathcal{H}$ to itself. A *spectral representation* of S is a collection $\{\mu_j\}_{j \in A}$, where A is some index set, of Borel measures on $\mathbb{R}$, together with a partial isometry $U: \mathcal{H} \rightarrow \bigoplus_{j \in A} L^2(\mathbb{R}, \mu_j)$, such that

- i) $\ker U = S(0)$;
- ii) $U: \overline{\text{dom } S} \rightarrow \bigoplus_{j \in A} L^2(\mathbb{R}, \mu_j)$ is unitary;
- iii) $USU^* = M$, the operator of multiplication by the variable in $\bigoplus_{j \in A} L^2(\mathbb{R}, \mu_j)$.

The measures μ_j are called the *spectral measures* of S .

More generally, if one does not have a self-adjoint relation but rather a closed and symmetric relation $\mathcal{T}$, one can still study the spectral theory of $\mathcal{T}$ by considering a self-adjoint *extension* S , that is, a self-adjoint relation such that $\mathcal{T} \subseteq S \subseteq \mathcal{T}^*$. It is possible to find conditions for the existence of such an extension (as done in [61, Chapter 2]) by generalizing von Neumann's theory of symmetric operators, which uses Cayley transforms and their relations to deficiency spaces (see e.g. [13, Chapter X] or [64, Chapter 13] for short introductions and [65] for a more comprehensive treatment). We will not dive into the details of this theory here, and will rather settle for stating the results for relations that arise from canonical systems.

2.1.5 Self-adjoint relations associated with canonical systems

A canonical system $Ju' = -zHu$ on an interval (a, b) can be naturally associated with a relation.

Definition 2.10. Consider a canonical system $Ju' = -zHu$ on an interval (a, b) . The *maximal relation* associated with this canonical system is

$$\begin{aligned} \mathcal{T}_H := \{ (u, v) \in L_H^2(a, b) \oplus L_H^2(a, b) : u \text{ has a representative } u_0 \in \text{AC}_{\text{loc}}(a, b) \\ \text{s.t. } Ju'_0 = -Hv \text{ a.e.} \}. \end{aligned} \quad (2.3)$$

The *minimal relation* of the canonical system is $\mathcal{T}_H^*$.

Remark. To make this definition a bit more intuitive, note that when H is invertible on all of (a, b) , $\mathcal{T}_H$ is exactly the graph of the operator $u \mapsto -H^{-1}Ju'$.

It can be shown that when Hypothesis 2.2 holds, a canonical system's maximal relation is closed, and its minimal relation is closed and symmetric (see e.g. [61, Theorem 2.5]). Hence, there is hope

in finding self-adjoint extensions of the minimal relation, which we simply refer to as self-adjoint *realizations* of the canonical system's relations. Their description can be obtained from Cayley transforms through von Neumann's theory, and this leads to the following classification.

Theorem 2.4 [61, Theorem 2.25]. *Consider a canonical system $Ju' = -zHu$ that satisfies Hypothesis 2.2 on an interval (a, b) .*

- i) If both endpoints are limit point, then $\mathcal{T}_H$ is self-adjoint.*
- ii) If a is limit circle and b is limit point, then for $\phi \in [0, \pi)$, the restriction of $\mathcal{T}_H$ by the boundary condition $e_\phi^*Ju(a) = 0$ with $e_\phi := (\cos \phi, \sin \phi)$ is a self-adjoint relation. All self-adjoint realizations of $\mathcal{T}_H$ are obtained in this way.*
- iii) If both endpoints are limit circle, then for $\phi, \theta \in [0, \pi)$, the restriction of $\mathcal{T}_H$ by the boundary conditions $e_\phi^*Ju(a) = e_\theta^*Ju(b) = 0$ is a self-adjoint relation. Other self-adjoint realizations exist, with coupled boundary conditions.*

In other words, to obtain a self-adjoint realization of a canonical system, one adds boundary conditions at limit circle endpoints. At limit point endpoints, the integrability condition essentially plays the role of a boundary condition: it can be shown that if $(u, v), (w, x) \in \mathcal{T}_H$, then representatives $u_0, w_0 \in \text{AC}_{\text{loc}}(a, b)$ of u and w have the property that $u_0^*(t)Jw_0(t) \rightarrow 0$ as t converges to a limit point endpoint (this is contained e.g. in [61, Theorem 2.24]).

It can also be shown that, given a self-adjoint realization, there is always a unique (up to multiplicative constant) solution that has the correct behavior near an endpoint.

Proposition 2.5 [61, Lemma 3.2]. *Consider a canonical system $Ju' = -zHu$ that satisfies Hypothesis 2.2 on an interval (a, b) , and take a self-adjoint realization of $\mathcal{T}_H$ of the appropriate form from Theorem 2.4. Then for any $z \in \mathbb{C} \setminus \mathbb{R}$, at each endpoint $t \in \{a, b\}$, there is a unique (up to multiplicative constant) non-trivial solution u to $Ju' = -zHu$ that is $L_H^2(I_t)$ for a neighborhood I_t of t and that satisfies the boundary condition at t if there is one (i.e., if t is limit circle).*

2.1.6 Weyl–Titchmarsh functions and spectral representations

An object that is especially useful in the study of the spectral theory of canonical systems is the Weyl–Titchmarsh function, which is a holomorphic function on the upper half-plane $\mathbb{H} := \{z \in \mathbb{C} : \text{Im } z > 0\}$ with values in its closure $\overline{\mathbb{H}}_\infty$ in the Riemann sphere. We introduce this function only for canonical systems that are limit circle at their left endpoint, as this simplifies the treatment and suffices for our purposes. However, as we impose no condition on the right endpoint, technically this also covers systems that are limit point on the left but limit circle on the right, by time reversal. The interested reader is referred to [61, §3.7] for the generalization of the theory to systems that are limit point at both endpoints.

To simplify notation below, we write $R_\phi \in \mathrm{SL}(2, \mathbb{R})$ for the rotation matrix with columns e_ϕ and Je_ϕ . We also write $\mathcal{P}: \mathbb{C}^2 \rightarrow \mathbb{C}_\infty$ for the mapping $\mathcal{P}w := w_1/w_2$. We point out right away that this mapping has the property that if $T \in \mathrm{SL}(2, \mathbb{C})$ and $w \in \mathbb{C}^2$, then $\mathcal{P}Tw = T\mathcal{P}w$, where the latter is understood as the action of T on $\mathcal{P}w \in \mathbb{C}_\infty$.

Definition 2.11. Consider a canonical system $Ju' = -zHu$ on (a, b) that is limit circle at a with boundary condition $e_\phi^*Ju(a) = 0$ for a $\phi \in [0, \pi)$. Suppose either that b is limit point, or that b is limit circle with boundary condition $e_\theta^*Ju(b) = 0$ for a $\theta \in [0, \pi)$. The *Weyl–Titchmarsh function* for this problem is the function $m_H^\phi: \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ defined as $m_H^\phi(z) := \mathcal{P}R_{-\phi}u(a, z)$ where $u: [a, b) \times \mathbb{H} \rightarrow \mathbb{C}^2$ is defined so that $u(\cdot, z) \in L_H^2(a, b)$ solves $J\partial_1 u(\cdot, z) = -zHu(\cdot, z)$ for any $z \in \mathbb{H}$ and satisfies the boundary condition at b (if it is limit circle).

Remark. By Proposition 2.5, the function u used above is well defined up to a multiplicative constant (which might even depend on z), and this suffices for m_H^ϕ to be well defined since the multiplicative constant cancels out in the ratio.

Remark. The Weyl–Titchmarsh function m_H^ϕ can be written in terms of m_H^0 (as long as the right boundary condition, if any, is kept the same). Indeed, $R_0 = I_2$ so $m_H^0 = \mathcal{P}u(a, \cdot)$, and therefore $m_H^\phi = R_{-\phi}m_H^0$. This can be useful to deduce properties of m_H^ϕ from those of m_H^0 .

The Weyl–Titchmarsh function can be related to the transfer matrix T_H of the system. In the case where b is limit circle, this is especially explicit, since the boundary condition actually forces $u(\cdot, z)$ to be $T_H^{-1}(b, \cdot, z)e_\theta$ (up to a multiplicative constant), so that $m_H^\phi(z) = \mathcal{P}R_{-\phi}T_H^{-1}(b, a, z)e_\theta = R_{-\phi}T_H^{-1}(b, a, z)\cot\theta$. When b is limit point, such a relation does not hold since the solutions cannot be continued up to the endpoint, but Weyl theory in fact still shows that for any $\theta_t \in [0, \pi)$, $m_H^\phi(z) = \lim_{t \rightarrow b} R_{-\phi}T_H^{-1}(t, a, z)\cot\theta_t$ (see e.g. [61, §3.4]).

Weyl–Titchmarsh functions belong to a specific class of holomorphic function: the *generalized Herglotz functions*.

Definition 2.12. A *Herglotz function* is a holomorphic map $\mathbb{H} \rightarrow \mathbb{H}$. A *generalized Herglotz function* is a holomorphic map $\mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$.

Remark. In fact, by the open mapping theorem, a generalized Herglotz function is either a genuine Herglotz function or an extended real constant.

It can be shown that the Weyl–Titchmarsh function of a canonical system that satisfies Hypothesis 2.2 is a Herglotz function (this is contained in [61, Theorems 3.10 and 3.15]), while that of a canonical system that does not satisfy Hypothesis 2.2 can be computed explicitly and shown to be a Herglotz function when b is limit circle and an extended real constant when b is limit point.

For us, the most important property of Herglotz functions is the following representation.

Theorem 2.6 (Herglotz representation theorem). *Let $m: \mathbb{H} \rightarrow \mathbb{H}$ be a Herglotz function. Then, there are $A \in \mathbb{R}$, $B \geq 0$, and a positive regular Borel measure μ on $\mathbb{R}$ with $\int_{\mathbb{R}} \frac{1}{1+t^2} d\mu(t) < \infty$ such that*

$$m(z) = A + Bz + \int_{\mathbb{R}} \left(\frac{1}{\lambda - z} - \frac{\lambda}{1 + \lambda^2} \right) d\mu(\lambda), \quad (2.4)$$

and A , B and μ are uniquely determined by m . Conversely, any function F defined as such is a Herglotz function.

As one can guess from the above representation, Herglotz functions can often be seen as Stieltjes transforms of finite positive Borel measures on $\mathbb{R}$. Thus, the measure μ associated with a Herglotz function m by the Herglotz representation theorem can often be recovered by Stieltjes inversion. In particular, the absolutely continuous part μ_{ac} of μ satisfies $d\mu_{ac}(\lambda) = \frac{1}{\pi} \operatorname{Im} m(\lambda) d\lambda$, and its singular part can be obtained through the relation

$$\mu(\{\lambda\}) = \lim_{\varepsilon \downarrow 0} -i\varepsilon m(\lambda + i\varepsilon),$$

which is valid for all $\lambda \in \mathbb{R}$. We refer to [65, Appendix F] for a survey of the important properties of these functions and their relations to Stieltjes transforms. Classical references include [1, 20].

The Weyl–Titchmarsh function actually leads to a spectral representation of the canonical system in the sense of Definition 2.9.

Theorem 2.7. *Consider a canonical system $Ju' = -zHu$ on (a, b) that satisfies Hypothesis 2.2. Suppose that a is limit circle with boundary condition $e_{\phi}^*Ju(a) = 0$, and suppose either that b is limit point or that it is limit circle with a boundary condition. Let S_H^{ϕ} denote the corresponding self-adjoint realization of the system's relations, and let μ_H^{ϕ} denote the measure induced by the Weyl–Titchmarsh function m_H^{ϕ} from its Herglotz representation. If b is limit circle, define $U: L_H^2(a, b) \rightarrow L^2(\mathbb{R}, \mu_H^{\phi})$ as*

$$U(v)(\lambda) := \int_a^b u_{\phi}^*(t, \lambda) H(t) v(t) dt$$

where $u_{\phi}: (a, b) \times \mathbb{C} \rightarrow \mathbb{C}^2$ is defined so that for each z , $u_{\phi}(\cdot, z)$ solves the canonical system with $u_{\phi}(a, z) = e_{\phi}$. If b is limit point, define $U: L_H^2(a, b) \rightarrow L^2(\mathbb{R}, \mu_H^{\phi})$ as above for $v \in \bigcup_{t \in (a, b)} L_H^2(a, t)$, or as $U(v) := \lim_{t \rightarrow b} U(\mathbb{1}_{(a, t)} v)$ for other $v \in L_H^2(a, b)$. In both cases, U and μ_H^{ϕ} set up a spectral representation of S_H^{ϕ} .

Proof. The case $\phi = 0$ is contained in [61, Theorems 3.16 and 3.17], and this easily extends to the general case by rotating the system, using the idea from [61, Theorem 3.20].

To do so, let $\tilde{H} := R_{-\phi} H R_{\phi}$, and define $\mathcal{R}_{\phi}: L_{\tilde{H}}^2(a, b) \rightarrow L_H^2(a, b)$ by $\mathcal{R}_{\phi} u := R_{\phi} u$. A direct

computation readily shows that the adjoint of $\mathcal{R}_\phi$ is the mapping $\mathcal{R}_{-\phi}: L^2_H(a, b) \rightarrow L^2_{\tilde{H}}(a, b)$ that sends $v \mapsto \mathcal{R}_{-\phi}v$, and it follows that $\mathcal{R}_\phi$ is unitary.

This unitary mapping rotates the spectral problem: the boundary condition $e_\phi^*Ju(a) = 0$ of H , which asks for $u(a)$ to be parallel to e_ϕ , is equivalent to asking for a rotated solution to be parallel to e_0 at a , and if b is limit circle then its boundary condition is rotated similarly. Thus, the rotation induces a self-adjoint realization $S_{\tilde{H}}^0$ of $\tilde{H}$'s maximal relation. Applying the theorem (for $\phi = 0$) shows the existence of a mapping $\tilde{U}: L^2_{\tilde{H}}(a, b) \rightarrow L^2(\mathbb{R}, \mu_{\tilde{H}}^0)$ that, together with the spectral measure $\mu_{\tilde{H}}^0$ of the Weyl–Titchmarsh function $m_{\tilde{H}}^0$, sets up a spectral representation of $S_{\tilde{H}}^0$.

Rotating back the system, $U := \tilde{U} \circ \mathcal{R}_{-\phi}: L^2_H(a, b) \rightarrow L^2(\mathbb{R}, \mu_H^0)$ now sets up a spectral representation of S_H^ϕ , still with the spectral measure μ_H^0 . By definition, μ_H^0 is the spectral measure of the Weyl–Titchmarsh function $m_H^0 := \mathcal{P}\tilde{u}(a, \cdot)$, where for each z , $\tilde{u}(\cdot, z) \in L^2_{\tilde{H}}(a, b)$ solves the canonical system for $\tilde{H}$, and satisfies the boundary condition at b if any. But

$$J\partial_1\tilde{u}(t, z) = -z\tilde{H}(t)\tilde{u}(t, z) = -z\mathcal{R}_{-\phi}H(t)\mathcal{R}_\phi\tilde{u}(t, z) \iff J\partial_1(R_\phi\tilde{u})(t, z) = -zH(t)R_\phi\tilde{u}(t, z),$$

so $u := R_\phi\tilde{u}$ solves the canonical system for H and has the appropriate boundary behavior at b . Therefore, $m_H^0 = \mathcal{P}\tilde{u}(a, \cdot) = \mathcal{P}R_\phi u(a, \cdot) = m_H^\phi$, and $\mu_H^\phi = \mu_H^0$. To conclude, it only remains to note that $U = \tilde{U} \circ \mathcal{R}_{-\phi}$ has exactly the form announced in the theorem under the identification $u_\phi := R_\phi\tilde{u}_0$ where $\tilde{u}_0$ is the function that appears in the expression of $\tilde{U}$. But in the same way as above, each $u_\phi(\cdot, z)$ is now a solution to the canonical system for H with $u_\phi(a, z) = e_\phi$, so U is indeed the desired mapping. ■

It is sometimes possible to obtain a more explicit description of the spectral measure μ_H^ϕ from above. In particular, this is the case when the spectrum of the self-adjoint realization S_H^ϕ is a discrete set and comprises only eigenvalues. The system then also has a well-defined *eigenvalue point process*

$$\nu_H^\phi := \sum_{\lambda \in \text{spec } S_H^\phi} \delta_\lambda. \quad (2.5)$$

The spectral measure is then a weighted version of this point process, and the following result gives a representation of the spectral weights.

Theorem 2.8. *In the setting of Theorem 2.7, suppose that the spectrum of S_H^ϕ is purely discrete. Then*

$$\mu_H^\phi = \sum_{\lambda \in \text{spec } S_H^\phi} \frac{1}{\|u_\phi(\cdot, \lambda)\|_H^2} \delta_\lambda$$

where again $u_\phi: (a, b) \times \mathbb{C} \rightarrow \mathbb{C}^2$ is defined so that for each z , $u_\phi(\cdot, z)$ is a canonical system solution with $u_\phi(a, z) = e_\phi$.

Remark. An important case in which the hypotheses are satisfied is that where the operator part of S_H^ϕ is densely defined and has a compact resolvent, for example when the canonical system (with appropriate boundary conditions) has a Hilbert–Schmidt resolvent. In particular, this is the case for systems that are limit circle at both endpoints (see [61, Theorem 3.4]).

Proof. This argument is adapted from the analogous argument for generalized Sturm–Liouville operators in [29, Lemma 9.5]. See also [61, Theorem 3.16] for a proof using Weyl theory, for canonical systems that are limit circle at both endpoints.

Because of the boundary condition at a , the eigenfunction at any $\lambda \in \text{spec } S_H^\phi$ is (proportional to) the solution $u_\phi(\cdot, \lambda)$, which means in particular that $u_\phi(\cdot, \lambda) \in L_H^2(a, b)$. This means that regardless of the system’s behavior at b , as long as $\lambda, \lambda' \in \text{spec } S_H^\phi$,

$$U(u_\phi(\cdot, \lambda))(\lambda') = \int_a^b u_\phi^*(t, \lambda') H(t) u_\phi(t, \lambda) dt.$$

However, as S_H^ϕ is self-adjoint, eigenfunctions associated with distinct eigenvalues are orthogonal, so in fact the above shows that $U(u_\phi(\cdot, \lambda))(\lambda') = 0$ if $\lambda \neq \lambda'$ and $U(u_\phi(\cdot, \lambda))(\lambda) = \|u_\phi(\cdot, \lambda)\|_H^2$. As U is an isometry, it follows that if $\|\cdot\|_{\mu_H^\phi}$ denotes the norm on $L^2(\mathbb{R}, \mu_H^\phi)$, then for $\lambda \in \text{spec } S_H^\phi$,

$$\|u_\phi(\cdot, \lambda)\|_H^2 = \|U(u_\phi(\cdot, \lambda))\|_{\mu_H^\phi}^2 = \sum_{\lambda' \in \text{spec } S_H^\phi} |U(u_\phi(\cdot, \lambda))(\lambda')|^2 \mu_H^\phi(\{\lambda'\}) = \|u_\phi(\cdot, \lambda)\|_{H\mu_H^\phi}^4(\{\lambda\}),$$

and the result follows. ■

2.1.7 Resolvents

Another way to study the spectral theory of a canonical system is through a resolvent.

Definition 2.13. The *resolvent* of a self-adjoint relation S at a $z \in \mathbb{C}$ is the relation $(S - z)^{-1}$.

Because relations always have inverses, the resolvent always exists. It can moreover be shown that away from the spectrum, a relation’s resolvent is in fact an operator [61, Theorem 3.1].

In a similar way as for self-adjoint operators on Hilbert spaces, resolvents of self-adjoint realizations of canonical systems’ relations can be described explicitly using integral representations.

Theorem 2.9 [61, Theorem 3.3]. Consider a canonical system $Ju' = -zHu$ on (a, b) , and let S_H be a self-adjoint realization of the system’s relations as in Theorem 2.4. Fix $z \in \mathbb{C} \setminus \mathbb{R}$, and for each endpoint $t \in \{a, b\}$, let u_t denote a non-trivial solution of $Ju' = -zHu$ that satisfies the boundary or integrability condition at t , normalized so that $u_a^\top J u_b \equiv -1$. Then the resolvent $(S_H - z)^{-1}$ is

an operator $L_H^2(a, b) \rightarrow L_H^2(a, b)$ given by

$$((S_H - z)^{-1}v)(t) = \int_a^b G(t, s, z)H(s)v(s) ds \quad \text{where} \quad G(t, s, z) := \begin{cases} u_b(t)u_a^\top(s) & \text{if } s < t, \\ u_a(t)u_b^\top(s) & \text{if } s \geq t. \end{cases}$$

Remark. The solutions u_a and u_b exist and are unique up to a multiplicative constant by Proposition 2.5. Moreover, while the normalization condition $u_a^\top J u_b \equiv -1$ might look very strong at first glance, it only concerns the ambiguity in these multiplicative constants. Indeed, the quantity $u_a^\top J u_b$, which can be seen as the Wronskian of u_a and u_b , is already constant:

$$(u_a^\top J u_b)' = -(J u_a')^\top u_b + u_a^\top J u_b' = z(H u_a)^\top u_b - z u_a^\top H u_b = 0$$

since H is symmetric.

When the system is limit circle at a and $z \in \mathbb{H}$, the integral kernel G of the resolvent can be expressed in terms of the transfer matrix T_H and of its Weyl–Titchmarsh function m_H . For simplicity, we consider here only the case where the left boundary condition is $e_0^* J u(a) = 0$, but the more general result could be obtained by rotating the system, using the same ideas as in the proof of Theorem 2.7.

By definition of the transfer matrix, $u_a := T_H(\cdot, a, z)e_0$ and $u_b := T_H(\cdot, a, z)\begin{pmatrix} m_H^0(z) \\ 1 \end{pmatrix}$ are solutions to the canonical system, and u_a clearly satisfies the boundary condition at a . If b is limit circle, u_b satisfies the boundary condition at b by definition of the Weyl–Titchmarsh function, and it can be shown from Weyl theory that $u_b \in L_H^2(a, b)$ if b is limit point—see [61, Theorem 3.14(a)]. A direct computation then shows that $u_a^\top J u_b \equiv -1$, so these two solutions can indeed be used as those appearing in the kernel G from Theorem 2.9. This yields the representation

$$G(t, s, z) = T_H(t, a, z) \begin{pmatrix} m_H^0(z) & \mathbb{1}_{[t, b)}(s) \\ \mathbb{1}_{(a, t)}(s) & 0 \end{pmatrix} T_H^\top(s, a, z). \quad (2.6)$$

2.2 Topologies on canonical systems

The goal of this section is to give a precise meaning to the idea of a sequence of canonical systems converging. As a canonical system is really defined from a coefficient matrix, we start by introducing a topology on such matrices in §2.2.1. Then, we describe what this topology can and cannot capture about canonical systems' spectral information, and we see how to relate the convergence of a sequence of coefficient matrices to that of the associated transfer matrices and Weyl–Titchmarsh functions in §2.2.2. We will see that the behavior of canonical systems at the endpoints of their domains plays an important role in this. Finally, §2.2.3 presents a method to describe the convergence

of a sequence of canonical systems when the endpoint behavior changes from limit point to limit circle in the limit.

The material presented in §2.2.1 and §2.2.2 is essentially extended from [61, §5.2], which presents the theory for trace-normed canonical systems, and from standard results from functional analysis that can be found for example in [13, 64]. The material from §2.2.3, however, is my own understanding of the problem of a sequence of canonical systems whose endpoint behavior changes in the limit. Indeed, after having failed at finding an approach to this problem in the literature, this is what I came up with, but I do not know whether something similar exists elsewhere.

2.2.1 Coefficient matrices

We start by studying spaces of coefficient matrices. First, we set up some notation.

Definition 2.14. Let $I \subseteq \mathbb{R}$ be an interval that could be open, closed or half-open. We denote by $\text{CS } I$ the set of coefficient matrices of canonical systems defined on the interior of I and for which the endpoints included in I (if any) are limit circle, and where coefficient matrices that are equal a.e. are identified. Given $f \in L^1_{\text{loc}}(I)$, we write $\text{CS}_f I$ for the *trace-dominated* subset $\text{CS}_f I := \{H \in \text{CS } I : \text{tr } H \leq f \text{ a.e.}\}$.

By definition, a coefficient matrix $H \in \text{CS } I$ is always locally integrable, and therefore it always defines a 2×2 matrix-valued measure $A \mapsto \int_A H \, d\mathcal{L}$ on Borel sets $A \subseteq I$, where $\mathcal{L}$ denotes the Lebesgue measure. Here, by “ 2×2 matrix-valued measure”, we simply mean a countably additive function from a σ -algebra to $\mathbb{R}^{2 \times 2}$ that is zero at $\emptyset$. Such a measure can be understood as a 2×2 array of signed measures on $\mathbb{R}$, so that integration with respect to a matrix-valued measure is simply understood as entrywise integration with respect to the measures of the entries, using the usual conventions of matrix products.

Given this correspondence, we can use a topology on measures, such as the vague topology, to describe a convergence of coefficient matrices.

Definition 2.15. Let $I \subseteq \mathbb{R}$ be an interval, and let $\mathcal{C}_c(I, \mathbb{C})$ be the set of continuous compactly supported functions $I \rightarrow \mathbb{C}$. Let $\mathcal{M}(I)$ be the set of locally finite signed measures on I . The *vague topology* on $\mathcal{M}(I)$ is the coarsest topology on $\mathcal{M}(I)$ under which all maps $\mu \mapsto \int_I \varphi \, d\mu$ for $\varphi \in \mathcal{C}_c(I, \mathbb{C})$ are continuous.

From this, we can topologize $\text{CS } I$ with the topology of vague convergence, using it entrywise. The following result, which extends part of [61, Theorem 5.3], gives criteria for the vague convergence of coefficient matrices in terms of different kinds of test functions.

Theorem 2.10. *Let $I \subseteq \mathbb{R}$ be an interval, and let $m \in \mathbb{N}$. Let $\{\mu_n\}_{n \in \mathbb{N}}$ be a sequence of locally finite $m \times m$ matrix-valued measures on I , and let μ be another such measure. Then, the following statements are equivalent.*

- i) $\mu_n \rightarrow \mu$ vaguely entrywise;
- ii) For each $\varphi \in \mathcal{C}_c(I, \mathbb{C}^m)$, $\int_I d\mu_n \varphi \rightarrow \int_I d\mu \varphi$ as $n \rightarrow \infty$.
- iii) For each $\varphi, \psi \in \mathcal{C}_c(I, \mathbb{C}^m)$, $\int_I \psi^* d\mu_n \varphi \rightarrow \int_I \psi^* d\mu \varphi$ as $n \rightarrow \infty$.

If in addition the measures μ_n and μ are symmetric, then both statements are also equivalent to:

- iv) For each $\varphi \in \mathcal{C}_c(I, \mathbb{C}^m)$, $\int_I \varphi^* d\mu_n \varphi \rightarrow \int_I \varphi^* d\mu \varphi$ as $n \rightarrow \infty$.

Proof. It is clear that i) implies both ii) and iii) by linearity. To see that ii) implies i), let $\varphi \in \mathcal{C}_c(I, \mathbb{C})$, and notice that for $k \in \{1, \dots, m\}$, applying ii) to $\varphi \mathbf{e}_k \in \mathcal{C}_c(I, \mathbb{C}^m)$ shows that for each $j \in \{1, \dots, m\}$, $\int_I \varphi d\mu_{n,jk} \rightarrow \int_I \varphi d\mu_{jk}$. Similarly, we can see that iii) implies i) by fixing $\varphi \in \mathcal{C}_c(I, \mathbb{C})$ first with $\varphi \geq 0$, and applying iii) with $\sqrt{\varphi} \mathbf{e}_j$ and $\sqrt{\varphi} \mathbf{e}_k$. The hypothesis that $\varphi \geq 0$ can then be lifted by writing $\varphi = (\operatorname{Re} \varphi)^+ - (\operatorname{Re} \varphi)^- + i(\operatorname{Im} \varphi)^+ - i(\operatorname{Im} \varphi)^-$ and using the result for nonnegative test functions.

To prove the equivalence of iv) with the other statements when the measures are symmetric, note that the symmetry of μ_n implies that $\langle \varphi, \psi \rangle_n := \int_I \psi^* d\mu_n \varphi$ is conjugate symmetric, so it is a semi-inner product. Therefore, the polarization identity

$$\langle \varphi, \psi \rangle_n = \frac{1}{4} \left(\langle \varphi + \psi, \varphi + \psi \rangle_n - \langle \varphi - \psi, \varphi - \psi \rangle_n + i \langle \varphi + i\psi, \varphi + i\psi \rangle_n - i \langle \varphi - i\psi, \varphi - i\psi \rangle_n \right)$$

holds for any $\varphi, \psi \in \mathcal{C}_c(I, \mathbb{C}^2)$, and as the same construction also works with μ , it follows that iv) implies iii). As iii) trivially implies iv), this completes the proof. ■

The matrix-valued measures induced by coefficient matrices of canonical systems are always locally finite and symmetric, since coefficient matrices are locally integrable and positive semi-definite. Therefore, Theorem 2.10 shows that $H_n \rightarrow H$ vaguely on I if and only if for each $\varphi \in \mathcal{C}_c(I, \mathbb{C}^2)$, $\int_I \varphi^* H_n \varphi d\mathcal{L} \rightarrow \int_I \varphi^* H \varphi d\mathcal{L}$. This leads to introduce on CS I the pseudometrics

$$d_\varphi(H, H') := \left| \int_I \varphi^* (H - H') \varphi d\mathcal{L} \right| \quad (2.7)$$

for $\varphi \in \mathcal{C}_c(I, \mathbb{C}^2)$, where by “pseudometric”, we mean a function that satisfies all properties of a metric with the exception that $d_\varphi(H, H') = 0$ does not necessarily imply that $H = H'$. Indeed, it could be that $\varphi \in \ker H = \ker H'$ a.e. on I , in which case $d_\varphi(H, H') = 0$. These pseudometrics can be turned into an actual metric on CS I using standard techniques from functional analysis (c.f. [64, §8.2] or [13, §IV.2]; see also [45, Theorem 4.2] which shows that the space of locally finite measures on a complete separable metric space is Polish under the vague topology).

Theorem 2.11. *Let $I \subseteq \mathbb{R}$ be an interval. There is a countable collection of functions $\{\varphi_m\}_{m \in \mathbb{N}}$ that is dense in $\mathcal{C}_c(I, \mathbb{C}^2)$ with respect to the topology of compact convergence, and such that $d_{\text{CS } I}: \text{CS } I \times \text{CS } I \rightarrow [0, \infty)$ given by*

$$d_{\text{CS } I}(H, H') := \sum_{m=1}^{\infty} \frac{1}{2^m} \frac{d_{\varphi_m}(H, H')}{1 + d_{\varphi_m}(H, H')}$$

is a metric on $\text{CS } I$ that induces the vague topology. Moreover, the resulting space is separable.

Proof. Let $\{\varphi_m\}_{m \in \mathbb{N}}$ be dense in $\mathcal{C}_c(I, \mathbb{C}^2)$ with respect to the topology of compact convergence. It is clear that $d_{\text{CS } I}$ is at least a pseudometric on $\text{CS } I$ since every d_{φ_m} is a pseudometric. To see that it is an actual metric, take $H, H' \in \text{CS } I$ with $d_{\text{CS } I}(H, H') = 0$, and let λ denote an eigenvalue of $H - H'$ with corresponding normalized eigenvector v . Suppose that λ is positive on a set $A \subseteq I$ of positive measure. Taking a subset of A if necessary, we may assume that it is compact so that $\mathbb{1}_A v$ is integrable. By density of $\mathcal{C}_c(I, \mathbb{C}^2)$ in $L^2(I)$, there are $\psi_n \in \mathcal{C}_c(I, \mathbb{C}^2)$ such that $\|\psi_n - \mathbb{1}_A v\|_2 \rightarrow 0$ as $n \rightarrow \infty$. Then, by density of the φ_m 's in $\mathcal{C}_c(I, \mathbb{C}^2)$, for each n there are $\varphi_{m_n, \ell}$ such that $\varphi_{m_n, \ell} \rightarrow \psi_n$ compactly as $\ell \rightarrow \infty$. Since all of these are compactly supported, we can recursively pick $\ell_n \geq \ell_{n-1}$ so that $\|\varphi_{m_n, \ell_n} - \psi_n\|_{\infty} < 1/n$, and $\|\varphi_{m_n, \ell_n} - \mathbb{1}_A v\|_2 \rightarrow 0$ as $n \rightarrow \infty$. Now,

$$\begin{aligned} d_{\varphi_{m_n, \ell_n}}(H, H') &\geq \left| \int_A v^*(H - H')v \, d\mathcal{L} \right| - \left| \int_I (\varphi_{m_n, \ell_n} - \mathbb{1}_A v)^*(H - H')(\varphi_{m_n, \ell_n} - \mathbb{1}_A v) \, d\mathcal{L} \right| \\ &\quad - \left| \int_A v^*(H - H')(\varphi_{m_n, \ell_n} - \mathbb{1}_A v) \, d\mathcal{L} \right| - \left| \int_A (\varphi_{m_n, \ell_n} - \mathbb{1}_A v)^*(H - H')v \, d\mathcal{L} \right|. \end{aligned} \quad (2.8)$$

Here, by hypothesis, the first term is some constant $c := \int_A v^*(H - H')v \, d\mathcal{L} = \int_A \lambda \, d\mathcal{L} > 0$. Because $H - H'$ is locally integrable and $\|\varphi_{m_n, \ell_n} - \mathbb{1}_A v\|_2 \rightarrow 0$, the other three terms in (2.8) vanish as $n \rightarrow \infty$. It follows that $d_{\varphi_{m_n, \ell_n}}(H, H') > 0$ for n large enough. As this contradicts the fact that $d_{\text{CS } I}(H, H') = 0$, we conclude that $H - H'$ cannot have an eigenvalue that is positive on a set of positive measure. Applying this argument to $H' - H$, we conclude that $H - H'$ can also not have an eigenvalue that is negative on a set of positive measure, and by symmetry of H and H' this implies that $H = H'$ a.e. on I , so that $H = H'$ in $\text{CS } I$. This proves that $d_{\text{CS } I}$ is indeed a metric on $\text{CS } I$.

We now show that the φ_m 's can be taken so that $d_{\text{CS } I}$ induces the vague topology. To do so, it suffices to show that given $H_n, H \in \text{CS } I$, $d_{\text{CS } I}(H_n, H) \rightarrow 0$ if and only if $H_n \rightarrow H$ vaguely on I . It is clear by definition of $d_{\text{CS } I}$ that the “if” part holds. To see that the other implication is also true, note that if $\varphi \in \mathcal{C}_c(I, \mathbb{C}^2)$ then by density there are φ_{m_k} 's such that $\varphi_{m_k} \rightarrow \varphi$ compactly as $k \rightarrow \infty$. Because φ is compactly supported, we can assume without loss of generality that the supports of all φ_{m_k} 's are contained in a compact set $K \subseteq I$, so that actually $\|\varphi - \varphi_{m_k}\|_{\infty} \rightarrow 0$. Now,

if $d_{\text{CS } \mathcal{I}}(H_n, H) \rightarrow 0$, then for any n and m_k ,

$$\begin{aligned} d_\varphi(H_n, H) &\leq d_{\varphi_{m_k}}(H_n, H) + \left| \int_{\mathcal{I}} (\varphi - \varphi_{m_k})^* (H_n - H) \varphi \, d\mathcal{L} \right| + \left| \int_{\mathcal{I}} \varphi_{m_k}^* (H_n - H) (\varphi - \varphi_{m_k}) \, d\mathcal{L} \right| \\ &\leq d_{\varphi_{m_k}}(H_n, H) + \|\varphi - \varphi_{m_k}\|_\infty (\|\varphi\|_\infty + \|\varphi_{m_k}\|_\infty) \left| \int_K \text{tr}(H_n - H) \, d\mathcal{L} \right|. \end{aligned} \quad (2.9)$$

Because $H \in L^1_{\text{loc}}(\mathcal{I})$ entrywise, it is clear that $\int_K \text{tr } H \, d\mathcal{L} < \infty$, and we can easily verify that $\int_K \text{tr } H_n \, d\mathcal{L}$ is bounded uniformly in n . Indeed, by density there is certainly a φ_{m_0} with $\varphi_{m_0,1} \geq 1$ on K , and adding $(\varphi_{m_0}, 0)$ to the list if necessary, we may assume that $\varphi_{m_0,2} \equiv 0$. For this φ_{m_0} ,

$$\int_{\mathcal{I}} \varphi_{m_0}^* H_n \varphi_{m_0} \, d\mathcal{L} \geq \int_K H_{n,11} \, d\mathcal{L} + \int_{\mathcal{I} \setminus K} \varphi_{m_0}^* H_n \varphi_{m_0} \, d\mathcal{L}.$$

Because all terms here are nonnegative and the left-hand side converges to $\int_{\mathcal{I}} \varphi_{m_0}^* H \varphi_{m_0} \, d\mathcal{L} < \infty$ by hypothesis, both terms in the right-hand side must be bounded in n . Using the same idea with φ_{m_0} such that $\varphi_{m_0,2} \geq 1$ on K and $\varphi_{m_0,1} \equiv 0$ (where, again, we might have to add $(0, \varphi_{m_0,2})$ to the list), we conclude that $\int_K \text{tr } H_n \, d\mathcal{L}$ is bounded in n . Therefore, for any $\varepsilon > 0$, we can take k large enough so that the second term of (2.9) is bounded by $\varepsilon/2$ for all n . With this k fixed, $d_{\varphi_{m_k}}(H_n, H) < \varepsilon/2$ for all n large enough, and it follows that $d_\varphi(H_n, H) \rightarrow 0$ as $n \rightarrow \infty$. Because φ was arbitrary in $\mathcal{C}_c(\mathcal{I}, \mathbb{C}^2)$, this shows that $H_n \rightarrow H$ vaguely on $\mathcal{I}$.

We finally turn to the separability of the resulting space. Let $\mathbb{Q}_{\geq 0}^{2 \times 2}[x]$ denote the set of polynomials $P: \mathcal{I} \rightarrow \mathbb{R}^{2 \times 2}$ with rational coefficients, modified so that the value $P(x)$ is replaced with the identity matrix at all x such that otherwise $P(x)$ would not be positive semi-definite. We will show that $\mathbb{Q}_{\geq 0}^{2 \times 2}[x]$ is dense in $\text{CS } \mathcal{I}$, which implies separability as this set is countable.

Let $H \in \text{CS } \mathcal{I}$, and take an increasing sequence of compacts $K_n \subseteq \mathcal{I}$ such that $\bigcup_{n \in \mathbb{N}} K_n = \mathcal{I}$. For each n , $H|_{K_n}$ is integrable, so by density of $\mathcal{C}(K_n, \mathbb{R})$ in $L^1(K_n)$ there are $H_{n,k} \in \mathcal{C}(K_n, \mathbb{R}^{2 \times 2})$ such that $\|H_{n,k} - H|_{K_n}\|_1 \rightarrow 0$ entrywise as $k \rightarrow \infty$. By the Stone–Weierstrass theorem and by density of $\mathbb{Q}$ in $\mathbb{R}$, there are 2×2 matrix polynomials $P_{n,k,\ell}$ with rational coefficients such that $\|P_{n,k,\ell}|_{K_n} - H_{n,k}\|_\infty \rightarrow 0$ as $\ell \rightarrow \infty$. Because K_n is compact, this means that $\|P_{n,k,\ell}|_{K_n} - H_{n,k}\|_1 \rightarrow 0$ as $\ell \rightarrow \infty$, and thus we can choose k_n and ℓ_n such that $\|(P_{n,k_n,\ell_n} - H)|_{K_n}\|_1 < 1/n$. Note that it is clear that all $P_{n,k,\ell}$'s can be chosen to be symmetric, and because the eigenvalues are continuous functions of the matrix entries, they can also be taken to be positive semi-definite for k and ℓ large enough so that $P_{n,k_n,\ell_n} \in \mathbb{Q}_{\geq 0}^{2 \times 2}[x]$. Now, let $\varphi \in \mathcal{C}_c(\mathcal{I}, \mathbb{C}^2)$. Then for all n large enough that $\text{supp } \varphi \in K_n$,

$$\left| \int_{\mathcal{I}} \varphi^* (P_{n,k_n,\ell_n} - H) \varphi \, d\mathcal{L} \right| \leq \int_{K_n} |\varphi|^2 \|P_{n,k_n,\ell_n} - H\|_{\text{op}} \, d\mathcal{L} \leq \frac{4\|\varphi\|_\infty^2}{n} \rightarrow 0.$$

As this holds for any φ , it follows that $P_{n,k_n,\ell_n} \rightarrow H$ vaguely on $\mathcal{I}$, and we can conclude that $\mathbb{Q}_{\geq 0}^{2 \times 2}[x]$ is indeed dense in $\text{CS } \mathcal{I}$. ■

2.2.2 Transfer matrices, Weyl–Titchmarsh functions and associated objects

2.2.2.1 Transfer matrices and Weyl–Titchmarsh functions

It is more straightforward to introduce relevant topologies to describe the convergence of transfer matrices or Weyl–Titchmarsh functions: all of these are continuous functions, which can be topologized using the compact-open topology. Since all codomains here are metric, the compact-open topology is the topology of compact convergence.

We now introduce a notation for spaces of transfer matrices. To simplify the discussion and to be able to use Weyl–Titchmarsh functions as they were defined in §2.1.6, we restrict ourselves to systems defined on intervals that are closed on the left.

Definition 2.16. Let $I = [a, b)$ or $[a, b]$ be a real interval. We denote by $\text{TM } I$ the set of transfer matrices $T: I \times \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ of canonical systems with coefficient matrices from $\text{CS } I$ and that are fixed at a , that is, with $T(a, z) = I_2$ for any $z \in \mathbb{C}$. We also denote by $\text{TM}_{\text{LP}} I$ and $\text{TM}_{\text{LC}} I$ the subsets of $\text{TM } I$ corresponding to canonical systems that are respectively limit point and limit circle at b .

Remark. Here, we consider transfer matrices to be functions of only two arguments, rather than three as in Definition 2.4, by fixing them at a . More precisely, if $T: I^2 \times \mathbb{C}$ is a transfer matrix in the notation of Definition 2.4, if a is limit circle then $T(\cdot, a, \cdot) \in \text{TM } I$. Note that this is allowed by Lemma 2.1 since the canonical systems are limit circle at a .

A space $\text{TM } I$, like the space $\text{Hol}(\mathbb{H}, \overline{\mathbb{H}}_\infty)$ of generalized Herglotz functions, is a space of continuous functions $X \rightarrow Y$ between locally compact metric spaces X and Y . Because in both cases the domain X is also hemicompact (that is, it contains a sequence of compact sets $K_n \subseteq X$ with the property that any compact $K \subseteq X$ of the space is contained in one of the K_n 's), the topology of compact convergence is metrizable—see e.g. [13, Proposition IV.2.1 and Example IV.2.2] for an explicit construction of a metric. Moreover, in both cases the spaces X and Y are second-countable Hausdorff spaces, so $\mathcal{C}(X, Y)$ is second-countable under the compact-open topology (see e.g. [21, Theorem XII.5.2]). As second-countability and separability are equivalent in metric spaces, it follows that any space $\text{TM } I$ and the space $\text{Hol}(\mathbb{H}, \overline{\mathbb{H}}_\infty)$ of generalized Herglotz functions are separable metric spaces.

Now, a coefficient matrix always defines a transfer matrix and a Weyl–Titchmarsh function. This raises the question as to whether the maps that relate these objects are continuous. The following result answers this question positively. It is a simple generalization of [61, Theorem 5.7(a)], which restricts to trace-normed coefficient matrices. The proof is heavily based on the proof given in [61], and essentially replaces the trace-norming condition by a weaker trace-domination condition.

Theorem 2.12. *Let $\mathcal{I} = [a, b)$ or $[a, b]$. For any $f \in L^1_{\text{loc}}(\mathcal{I})$, the mapping $\text{CS}_f \mathcal{I} \rightarrow \text{TM } \mathcal{I}$ sending a coefficient matrix to the corresponding transfer matrix is continuous.*

Remark. Recall that the transfer matrix acts as a fundamental basis of solutions, in the sense that the solution u to $Ju' = -zHu$ with $u(a) = v$ for a $v \in \mathbb{C}^2$ is $u = T_H(\cdot, z)v$. Hence, this result also shows that canonical system solutions converge in a more general sense: if $H_n, H \in \text{CS}_f \mathcal{I}$ and $v_n, v \in \mathbb{C}^2$ are such that $H_n \rightarrow H$ vaguely and $v_n \rightarrow v$, then the solution u_n to $Ju'_n = -zH_n u_n$ with $u_n(a) = v_n$ converges compactly to the solution u to $Ju' = -zHu$ with $u(a) = v$.

Proof. Let $\{H_n\}_{n \in \mathbb{N}} \subseteq \text{CS}_f \mathcal{I}$ be a sequence of coefficient matrices that converges vaguely to some $H \in \text{CS}_f \mathcal{I}$, and let $T_{H_n}, T_H \in \text{TM } \mathcal{I}$ denote the corresponding transfer matrices. As the spaces $\text{CS}_f \mathcal{I}$ and $\text{TM } \mathcal{I}$ are metric, it suffices to prove that $T_{H_n} \rightarrow T_H$ compactly.

Given $\tau \in \mathcal{I}$, we first prove uniform convergence on $[a, \tau]$ for fixed $z \in \mathbb{C}$. By definition of the transfer matrix, $\partial_1 T_{H_n}(\cdot, z) = zJH_n T_{H_n}(\cdot, z)$ and $T_{H_n}(a, z) = I_2$, so for $t \in [a, \tau]$,

$$T_{H_n}(t, z) = I_2 + zJ \int_a^t H_n(s) T_{H_n}(s, z) \, ds. \quad (2.10)$$

Note that $\|H_n\|_{\text{op}} \leq f$ on (a, τ) for all n since $\|H_n\|_{\text{op}}$, being the maximal eigenvalue of H_n , is bounded by its trace. Therefore, taking norms in (2.10) shows that

$$\|T_{H_n}(t, z)\|_{\text{op}} \leq 1 + |z| \int_a^t f(s) \|T_{H_n}(s, z)\|_{\text{op}} \, ds.$$

Gronwall's lemma then shows that for any $t \in [a, \tau]$, $\|T_{H_n}(t, z)\|_{\text{op}} \leq e^{|z| \|f\|_{[a, \tau]}}$, which also implies that $\|\partial_1 T_{H_n}(\cdot, z)\|_{\text{op}} \leq |z| e^{|z| \|f\|_{[a, \tau]}} f$. It follows that for $t, s \in [a, \tau]$,

$$\|T_{H_n}(t, z) - T_{H_n}(s, z)\|_{\text{op}} = \left\| \int_s^t \partial_1 T_{H_n}(u, z) \, du \right\|_{\text{op}} \leq |z| e^{|z| \|f\|_{[a, \tau]}} \left| \int_s^t f(u) \, du \right|.$$

Since f is integrable on $[a, \tau]$, the map $t \mapsto \int_a^t f(u) \, du$ is uniformly continuous on $[a, \tau]$, and therefore $\|T_{H_n}(t, z) - T_{H_n}(s, z)\|_{\text{op}}$ can be made arbitrarily small by taking t and s sufficiently close to each other. We thus conclude that the family $\{T_{H_n}(\cdot, z)\}_{n \in \mathbb{N}}$ is equicontinuous on $[a, \tau]$, and it follows by Arzelà–Ascoli's theorem that there is a subsequence $\{T_{H_{n_k}}(\cdot, z)\}_{k \in \mathbb{N}}$ that converges uniformly to some continuous $S: [a, \tau] \rightarrow \mathbb{C}^{2 \times 2}$.

To prove that this limit is in fact $T_H(\cdot, z)$, we fix $t \in [a, \tau]$ and define for each $\ell \in \mathbb{N}$ a function $S_\ell \in \mathcal{C}_c(\mathcal{I}, \mathbb{C}^{2 \times 2})$ to be a continuous approximation of $S\mathbb{1}_{[a, t]}$. Specifically, if $t = b$ we can directly take $S_\ell = S$, and otherwise we define S_ℓ for each $\ell \in \mathbb{N}$ large enough so that $t + 1/\ell < b$ as

$$S_\ell(s) := \begin{cases} S(s) & \text{if } a \leq s \leq t, \\ S(t) \exp\left(-\frac{\ell^2(s-t)}{1-\ell(s-t)}\right) & \text{if } t < s < t + 1/\ell, \\ 0 & \text{if } s \geq t + 1/\ell. \end{cases}$$

In any case, $S_\ell = S$ on $[a, t]$ and the support of S_ℓ shrinks to $[a, t]$ as ℓ grows. Then for any ℓ ,

$$\begin{aligned} \int_a^t H_{n_k}(s) T_{H_{n_k}}(s, z) \, ds &= \int_a^t H(s) S(s) \, ds + \int_a^t (H_{n_k}(s) - H(s)) S(s) \, ds \\ &\quad + \int_a^t H_{n_k}(s) (T_{H_{n_k}}(s, z) - S(s)) \, ds \\ &= \int_a^t H(s) S(s) \, ds + \int_a^b (H_{n_k}(s) - H(s)) S_\ell(s) \, ds \\ &\quad - \int_t^b (H_{n_k}(s) - H(s)) S_\ell(s) \, ds + \int_a^t H_{n_k}(s) (T_{H_{n_k}}(s, z) - S(s)) \, ds. \end{aligned}$$

Let $\varepsilon > 0$. By definition of S_ℓ and by integrability of f on $[a, \tau]$, it is clear that it is possible to take ℓ large enough so that for any k ,

$$\left\| \int_t^b (H_{n_k}(s) - H(s)) S_\ell(s) \, ds \right\|_{\text{op}} \leq 2 \|S(t)\|_{\text{op}} \int_t^{t+1/\ell} f(s) \, ds < \frac{\varepsilon}{3}.$$

(Here, if $t = b$, the first inequality of course does not make sense, strictly speaking, but this has no incidence on the argument since the left-hand side is zero in that case). Then, since $S_\ell \in \mathcal{C}_c(\mathcal{I}, \mathbb{C}^{2 \times 2})$ the fact that $H_{n_k} \rightarrow H$ vaguely on $\mathcal{I}$ implies that for k large enough,

$$\left\| \int_a^b (H_{n_k}(s) - H(s)) S_\ell(s) \, ds \right\|_{\text{op}} < \frac{\varepsilon}{3}.$$

Finally, by definition of S , enlarging k if needed,

$$\left\| \int_a^t H_{n_k}(s) (T_{H_{n_k}}(s, z) - S(s)) \, ds \right\|_{\text{op}} \leq \|f \mathbb{1}_{[a, \tau]}\|_1 \sup_{s \in [a, t]} \|T_{H_{n_k}}(s, z) - S(s)\|_{\text{op}} < \frac{\varepsilon}{3}.$$

It follows that

$$\left\| \int_a^t H_{n_k}(s) T_{H_{n_k}}(s, z) \, ds - \int_a^t H(s) S(s) \, ds \right\|_{\text{op}} < \varepsilon$$

for any k large enough, and $\int_a^t H_{n_k}(s) T_{H_{n_k}}(s, z) \, ds \rightarrow \int_a^t H(s) S(s) \, ds$. But this means that for any $t \in [a, \tau]$,

$$S(t) = \lim_{k \rightarrow \infty} T_{H_{n_k}}(t, z) = \lim_{k \rightarrow \infty} \left(I_2 + zJ \int_a^t H_{n_k}(s) T_{H_{n_k}}(s, z) \, ds \right) = I_2 + zJ \int_a^t H(s) S(s) \, ds,$$

which implies that $S = T_H(\cdot, z)$ by uniqueness of the transfer matrix.

The above argument shows that every uniformly convergent subsequence of $\{T_{H_n}(\cdot, z)\}_{n \in \mathbb{N}}$ on $[a, \tau]$ converges to $T_H(\cdot, z)$. If the whole sequence was not uniformly convergent, there would be an $\varepsilon > 0$ and a subsequence $\{T_{H_{n_k}}(\cdot, z)\}_{k \in \mathbb{N}}$ such that $\|T_{H_{n_k}}(\cdot, z) - T_H(\cdot, z)\|_{\infty} \geq \varepsilon$ for each k , but

then by Arlezá–Ascoli’s theorem we could still extract a uniformly convergent subsequence from $\{T_{H_{n_k}}(\cdot, z)\}_{k \in \mathbb{N}}$. This contradiction shows that $T_{H_n}(\cdot, z) \rightarrow T_H(\cdot, z)$ uniformly on $[a, \tau]$.

To prove that this convergence is uniform on compacts in z , note that for any $t \in [a, \tau]$ and $z \in \mathbb{C}$, $\partial_2 \partial_1 T_{H_n}(t, z) = JH_n(t)T_{H_n}(t, z) + zJH_n(t)\partial_2 T_{H_n}(t, z)$, so

$$\partial_2 T_{H_n}(t, z) = J \int_a^t H_n(s) (T_{H_n}(s, z) + z\partial_2 T_{H_n}(s, z)) ds.$$

Taking norms and using the bound $\|T_n(s, z)\|_{\text{op}} \leq e^{|z|\|f\|_{[a, \tau]}\|_1}$, it follows that

$$\|\partial_2 T_{H_n}(t, z)\|_{\text{op}} \leq \|f\|_{[a, \tau]}\|_1 e^{|z|\|f\|_{[a, \tau]}\|_1} + |z| \int_a^t f(s) \|\partial_2 T_{H_n}(s, z)\|_{\text{op}} ds,$$

and Gronwall’s lemma now shows that $\|\partial_2 T_{H_n}(t, z)\|_{\text{op}} \leq \|f\|_{[a, \tau]}\|_1 e^{2|z|\|f\|_{[a, \tau]}\|_1}$. Since this holds for any $z \in \mathbb{C}$, then for any compact $K \subset \mathbb{C}$ and any $t \in [a, \tau]$, the sequence $\{T_{H_n}(t, \cdot)\}_{n \in \mathbb{N}}$ is equicontinuous when restricted to K . Thus, Arzelá–Ascoli’s theorem gives again a subsequence that converges uniformly, and as the pointwise convergence holds, any uniformly convergent subsequence has to converge to $T_H(t, \cdot)$. As before, if the whole sequence does not converge, we can build a subsequence that cannot converge but from which we can still extract a convergent subsequence, and this contradiction shows that the whole sequence $\{T_{H_n}(t, \cdot)\}_{n \in \mathbb{N}}$ must converge uniformly to $T_H(t, \cdot)$ when restricted to K . ■

Theorem 2.12 relates the convergence of coefficient matrices to that of transfer matrices, and the convergence of transfer matrices can then be related to that of Weyl–Titchmarsh functions. To simplify the discussion, we restrict our attention to systems that are subject to the boundary condition $e_0^* J u(a) = 0$ on the left. The results are easily generalized to more general left boundary conditions using the fact that the Weyl–Titchmarsh function for another boundary condition e_ϕ is simply the action of the Möbius transform $R_{-\phi}$ on that for the boundary condition e_0 , as seen in the remark following Definition 2.11. We split the treatment in two different cases, depending on the right endpoint behavior of the system to which the sequence converges.

The first case is the following, which can be proven like [61, Theorem 5.7(b)].

Theorem 2.13. *Let $I = [a, b)$ or $[a, b]$. Let $T_n \in \text{TM } I$ and $T \in \text{TM}_{\text{LP}} I$ be transfer matrices. Suppose that the system for T_n is given a boundary condition at b if necessary, and let $m_n^0, m^0 \in \text{Hol}(\mathbb{H}, \overline{\mathbb{H}}_\infty)$ be the Weyl–Titchmarsh functions of the corresponding canonical systems with boundary condition $e_0^* J u(a) = 0$. If $T_n \rightarrow T$ compactly on $I \times \mathbb{C}$, then $m_n^0 \rightarrow m^0$ compactly on $\mathbb{H}$. In particular, the corresponding map $\text{TM}_{\text{LP}} I \rightarrow \text{Hol}(\mathbb{H}, \overline{\mathbb{H}}_\infty)$ is continuous.*

When the sequence converges to a system that is limit circle on the right, its Weyl–Titchmarsh function also depends on the right boundary condition. In the following, we allow for extra flexibility on what qualifies as the “right boundary condition” for reasons that will become clear later.

Theorem 2.14. Define $M: \text{TM}_{\text{LC}}[a, b] \times \mathcal{C}(\mathbb{H}, \mathbb{C}^2) \rightarrow \mathbb{C}_{\infty}^{\mathbb{H}}$ by $M(T, w)(z) := \mathcal{P}T^{-1}(b, z)w(z)$. Then M is continuous on $M^{-1}(\text{Hol}(\mathbb{H}, \overline{\mathbb{H}}_{\infty}))$ under the topology of compact convergence.

Remark. If w is a constant in $\alpha\mathbb{R}^2 \setminus \{0\}$ for some $\alpha \in \mathbb{C}$, then $M(T, w)$ is the Weyl–Titchmarsh function of the canonical system associated to T with boundary conditions e_0 at a and w at b , as defined in Definition 2.11. However, M also maps other points to generalized Herglotz functions, which motivates to enlarge its domain. An example of this arises when T is the transfer matrix of a canonical system on $[a, c)$ with $c > b$. This will be discussed in detail in §2.2.3.

Proof. It suffices to prove that $M(T_n, w_n) \rightarrow M(T, w)$ compactly if $(T_n, w_n) \rightarrow (T, w)$ compactly in $M^{-1}(\text{Hol}(\mathbb{H}, \overline{\mathbb{H}}_{\infty}))$, and to do so, it suffices to prove that $M(T_n, w_n)(z) \rightarrow M(T, w)(z)$ for every $z \in \mathbb{H}$, since the space of generalized Herglotz functions is sequentially compact (see [61, Theorem 5.6(a)] for a proof of this).

Fix $z \in \mathbb{H}$. As $T_n \rightarrow T$ compactly, $\begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} := T_n^{-1}(b, z) \rightarrow T^{-1}(b, z) =: \begin{pmatrix} A & B \\ C & D \end{pmatrix}$. If both $|M(T, w)(z)| < \infty$ and $|\mathcal{P}w(z)| < \infty$, then $C\mathcal{P}w(z) + D$ is bounded away from zero, so $C_n\mathcal{P}w_n(z) + D_n$ is bounded away from zero for n large enough, and

$$M(T_n, w_n)(z) = \frac{A_n\mathcal{P}w_n(z) + B_n}{C_n\mathcal{P}w_n(z) + D_n} \rightarrow \frac{A\mathcal{P}w(z) + B}{C\mathcal{P}w(z) + D} = M(T, w)(z).$$

If $|M(T, w)(z)| < \infty$ but $\mathcal{P}w(z) = \infty$, then $M(T_n, w_n)(z) \rightarrow A/C = M(T, w)(z)$. Finally, if $M(T, w)(z) = \infty$, then $C\mathcal{P}w(z) + D = 0$, and it must be that $\mathcal{P}w(z) < \infty$ and that $A\mathcal{P}w(z) + B \neq 0$ because $\det T(b, z) = 1$, so $M(T_n, w_n)(z) \rightarrow \infty$. ■

2.2.2.2 Spectral measures and eigenvalue point processes

As we have seen before, it is possible to extract from a canonical system's Weyl–Titchmarsh function a spectral measure through the Herglotz representation (see Theorem 2.6). This spectral measure can then be used in the description of the canonical system's spectral representation, as in Theorem 2.7. It is therefore natural to ask what does the convergence of a sequence of Weyl–Titchmarsh functions imply about the associated spectral measures.

The following result is contained in [61, Theorem 7.3(a)]. Below, we write $\mathcal{M}_+(\mathbb{R})$ for the space of locally finite (nonnegative) measures on $\mathbb{R}$ equipped with the vague topology.

Theorem 2.15. The mapping $\text{Hol}(\mathbb{H}, \mathbb{H}) \rightarrow \mathcal{M}_+(\mathbb{R})$ sending a Herglotz function to the spectral measure μ from its Herglotz representation is continuous.

The vague convergence of the spectral measures provides, of course, important information about the convergence of the spectrum of a sequence of canonical systems. However, this information can be insufficient to provide a clear picture of the asymptotic behavior of the spectrum. For

example, if all spectral measures involved are discrete, with only eigenvalues, then the vague convergence of the spectral measures does not imply the convergence of the eigenvalues themselves; heuristically, this is because atoms could merge or disappear in the limit. Thus, it can be desirable to upgrade the vague convergence of the spectral measures to that of the eigenvalue point processes.

To clarify what the vague convergence implies about the eigenvalues, a very simple first step is the following.

Lemma 2.16. *Let $\mu_n, \mu \in \mathcal{M}_+(\mathbb{R})$ be discrete measures supported on discrete sets such that $\mu_n \rightarrow \mu$ vaguely as $n \rightarrow \infty$. If λ is an atom of μ , then there is a sequence of atoms λ_n of μ_n such that $\lambda_n \rightarrow \lambda$ as $n \rightarrow \infty$.*

Proof. Take an atom λ of μ and set $\lambda_n := \arg \min\{|\lambda - \lambda'| : \lambda' \in \text{supp } \mu_n\}$. Given $\varepsilon > 0$ small enough, there is a $\varphi_\varepsilon \in \mathcal{C}_c(\mathbb{R})$ supported on $[\lambda - \varepsilon, \lambda + \varepsilon]$ such that $\varphi_\varepsilon(\lambda) = 1$ and $\varphi_\varepsilon(\lambda') = 0$ for all other $\lambda' \in \text{supp } \mu$. But then $\int_{\mathbb{R}} \varphi_\varepsilon d\mu = \mu(\{\lambda\}) > 0$, so for all n large enough, μ_n must have at least one atom in $(\lambda - \varepsilon, \lambda + \varepsilon)$, and in particular $|\lambda - \lambda_n| < \varepsilon$. ■

From this, it is easy to find sufficient conditions to be able to deduce from the convergence of discrete measures the convergence of the associated point processes.

Proposition 2.17. *Let $\mu_n, \mu \in \mathcal{M}_+(\mathbb{R})$ be discrete measures supported on discrete sets such that $\mu_n \rightarrow \mu$ vaguely as $n \rightarrow \infty$. Suppose that*

- i) *The spectral weights of μ are limits of those of μ_n , in the sense that for any $\lambda \in \text{supp } \mu$, if $\lambda_n \in \text{supp } \mu_n$ satisfy $\lambda_n \rightarrow \lambda$ then $\mu_n(\{\lambda_n\}) \rightarrow \mu(\{\lambda\})$;*
- ii) *For any compact $K \subset \mathbb{R}$, there is an $\varepsilon > 0$ such that $\limsup_{n \rightarrow \infty} \mu_n(\{\lambda_n\}) \geq \varepsilon$ whenever $\lambda_n \in K \cap \text{supp } \mu_n$.*

Then $\nu_n := \sum_{\lambda \in \text{supp } \mu_n} \delta_\lambda \rightarrow \nu := \sum_{\lambda \in \text{supp } \mu} \delta_\lambda$ vaguely as $n \rightarrow \infty$.

Intuitively, the first condition prevents atoms from merging in the limit whereas the second one prevents them from disappearing.

Proof. Let $\varphi \in \mathcal{C}_c(\mathbb{R}, [0, \infty))$, and to simplify notation, enumerate $\text{supp } \varphi \cap \text{supp } \mu = \{\lambda^k\}_{k=1}^N$ for $N \in \mathbb{N}$. By Lemma 2.16, for all n large enough, there are $\lambda_n^k \in \text{supp } \varphi \cap \text{supp } \mu_n$ such that $\lambda_n^k \rightarrow \lambda^k$, and we may assume that this correspondence between $\{\lambda_n^k\}_{k=1}^N$ and $\{\lambda^k\}_{k=1}^N$ is one-to-one (because the λ^k 's are distinct, so the sequences that converge to them must eventually be distinct). Now,

$$\int_{\mathbb{R}} \varphi d\mu = \sum_{k=1}^N \varphi(\lambda^k) \mu(\{\lambda^k\}) \quad \text{and} \quad \int_{\mathbb{R}} \varphi d\mu_n = \sum_{k=1}^N \varphi(\lambda_n^k) \mu_n(\{\lambda_n^k\}) + \sum_{\lambda \in S_n} \varphi(\lambda) \mu_n(\{\lambda\})$$

where $S_n := (\text{supp } \varphi \cap \text{supp } \mu_n) \setminus \{\lambda_n^k\}_{k=1}^N$. A priori, S_n could be nonempty, but $\varphi(\lambda_n^k) \rightarrow \varphi(\lambda^k)$ and hypothesis i) on spectral weights ensures that $\mu_n(\{\lambda_n^k\}) \rightarrow \mu(\{\lambda^k\})$ for each k . By hypothesis ii) on

the spectral weights of the μ_n 's, $\sum_{\lambda \in S_n} \varphi(\lambda) \mu_n(\{\lambda\})$ cannot vanish as $n \rightarrow \infty$ unless S_n is eventually empty (or unless the $\lambda \in S_n$ evolve in a specific way depending on the details of φ , but this can be ruled out by repeating the argument for a slightly different φ). Therefore, S_n is empty for all n large enough, and

$$\int_{\mathbb{R}} \varphi \, d\nu_n = \sum_{k=1}^N \varphi(\lambda_n^k) \rightarrow \sum_{k=1}^N \varphi(\lambda^k) = \int_{\mathbb{R}} \varphi \, d\nu$$

so $\nu_n \rightarrow \nu$ vaguely. ■

For spectral measures of canonical systems that are limit circle at both endpoints, it is easy to show that the spectral weights converge once the coefficient matrices and boundary conditions also converge.

Proposition 2.18. *Let $H_n, H \in \text{CS}_f[a, b]$ with $f \in L^1[a, b]$ have boundary conditions $e_{\phi_n}^* J u(a) = e_{\theta_n}^* J u(b) = 0$ and $e_{\phi}^* J u(a) = e_{\theta}^* J u(b) = 0$ respectively for $\phi_n, \phi, \theta_n, \theta \in [0, \pi]$, and let $\mu_{H_n}^{\phi_n}, \mu_H^{\phi} \in \mathcal{M}_+(\mathbb{R})$ be the spectral measures induced from the corresponding Weyl–Titchmarsh functions. If $H_n \rightarrow H$ vaguely on $[a, b]$, $\phi_n \rightarrow \phi$ and $\theta_n \rightarrow \theta$, then the spectral weights of μ_H^{ϕ} are limits of those of $\mu_{H_n}^{\phi_n}$ in the sense of hypothesis i) from Proposition 2.17.*

Proof. By Theorem 2.8, the spectral weights have the form

$$\mu_{H_n}^{\phi_n}(\{\lambda\}) = \frac{1}{\|u_{n,\phi_n}(\cdot, \lambda)\|_{H_n}^2} \quad \text{and} \quad \mu_H^{\phi}(\{\lambda\}) = \frac{1}{\|u_{\phi}(\cdot, \lambda)\|_H^2} \quad (2.11)$$

for $u_{n,\phi_n} := T_{H_n} e_{\phi_n}$ and $u_{\phi} := T_H e_{\phi}$, where T_{H_n} and T_H are the transfer matrices of the two systems. By Theorem 2.12, $T_{H_n} \rightarrow T_H$ compactly on $[a, b] \times \mathbb{C}$, which also means that $u_{n,\phi_n} \rightarrow u_{\phi}$ compactly on $[a, b] \times \mathbb{C}$, and then by Theorems 2.14 and 2.15, $\mu_{H_n}^{\phi_n} \rightarrow \mu_H^{\phi}$ vaguely.

Now, fix a $\lambda \in \text{supp } \mu_H^{\phi}$, and take $\lambda_n \in \text{supp } \mu_{H_n}^{\phi_n}$ such that $\lambda_n \rightarrow \lambda$. Then

$$\begin{aligned} \|u_{n,\phi_n}(\cdot, \lambda_n)\|_{H_n}^2 - \|u_{\phi}(\cdot, \lambda)\|_H^2 &= \int_a^b u_{n,\phi_n}^*(t, \lambda_n) H_n(t) (u_{n,\phi_n}(t, \lambda_n) - u_{\phi}(t, \lambda)) \, dt \\ &\quad + \int_a^b (u_{n,\phi_n}(t, \lambda_n) - u_{\phi}(t, \lambda))^* H_n(t) u_{\phi}(t, \lambda) \, dt \\ &\quad + \int_a^b u_{\phi}^*(t, \lambda) (H_n(t) - H(t)) u_{\phi}(t, \lambda) \, dt. \end{aligned} \quad (2.12)$$

The first two lines vanish as $n \rightarrow \infty$ by dominated convergence, since $u_{n,\phi_n} \rightarrow u_{\phi}$ compactly on $[a, b] \times \mathbb{C}$ and $\text{tr } H_n \leq f \in L^1[a, b]$ by hypothesis. The last line also vanishes as $n \rightarrow \infty$, because $u_{\phi}(\cdot, \lambda) \in \mathcal{C}_c([a, b], \mathbb{C}^2)$ and $H_n \rightarrow H$ vaguely on $[a, b]$. Thus, $\|u_{n,\phi_n}(\cdot, \lambda_n)\|_{H_n} \rightarrow \|u_{\phi}(\cdot, \lambda)\|_H$ and the associated spectral weights also converge. ■

Hence, for a sequence of canonical systems that are limit circle at both endpoints, the convergence of the eigenvalue point processes can be deduced from Propositions 2.17 and 2.18 once the spectral weights are shown not to vanish in the limit (e.g., hypothesis *ii*) of Proposition 2.17 is shown to be satisfied by the spectral measures).

For a sequence of canonical systems on $[a, b)$ that are limit point at b instead of being limit circle, when the spectra are still discrete with only eigenvalues, it can be possible to prove the convergence of the spectral weights using a similar analysis as in the proof of Proposition 2.18, but this requires a better control on the eigenfunctions. Indeed, in that case Theorem 2.8 still applies so the spectral weights are as in (2.11), where the solutions u_ϕ still have the same representations in terms of transfer matrices and boundary conditions, but now only converge compactly on $[a, b) \times \mathbb{C}$. Hence, the decomposition (2.12) is not sufficient to show the convergence of the spectral weights, and a better uniform control on the decay of the u_{n, ϕ_n} near b is needed to localize the argument on a compact time interval in $[a, b)$.

2.2.2.3 Remarks on resolvents

We have seen in §2.1.7 that a self-adjoint realization of a canonical system's relations can be given a resolvent, which can possibly be expressed exclusively in terms of the system's transfer matrix and Weyl–Titchmarsh function. Indeed, consider a canonical system on $\mathcal{I} = [a, b)$ or $[a, b]$ with coefficient matrix H , subject to the boundary condition $e_0^* J u(a) = 0$ at a and to some other boundary condition at b if it is limit circle. This yields a self-adjoint realization $\mathcal{S}_H$ of its relations, and combining Theorem 2.9 with the expression of the integral kernel given in (2.6), for $z \in \mathbb{C} \setminus \mathbb{R}$ the resolvent $(\mathcal{S}_H - z)^{-1}: L_H^2(a, b) \rightarrow L_H^2(a, b)$ acts as

$$((\mathcal{S}_H - z)^{-1}v)(t) = \int_a^b T_H(t, z) \begin{pmatrix} m_H^0(z) & \mathbb{1}_{[t, b)}(s) \\ \mathbb{1}_{(a, t)}(s) & 0 \end{pmatrix} T_H^\top(s, z) H(s) v(s) \, ds$$

where T_H and m_H^0 are the associated transfer matrix and Weyl–Titchmarsh function.

Now, it is clear from the above that given a sequence of systems $H_n \in \text{CS } \mathcal{I}$, if $H_n \rightarrow H$ vaguely, $T_{H_n} \rightarrow T_H$ compactly on $\mathcal{I} \times \mathbb{C}$ and $m_{H_n}^0 \rightarrow m_H^0$ compactly on $\mathbb{H}$, then $(\mathcal{S}_{H_n} - z)^{-1}\varphi \rightarrow (\mathcal{S}_H - z)^{-1}\varphi$ whenever $z \in \mathbb{H}$ and $\varphi \in \mathcal{C}_c(\mathcal{I}, \mathbb{C}^2)$. This type of convergence is similar to a strong resolvent convergence (i.e., the convergence of resolvents in the strong operator topology, see [69, §6.6] for precise definitions and more details), and it is natural to ask whether it can be upgraded to an actual operator level convergence. To do so, one would need to overcome the problem that these resolvents act on a priori distinct Hilbert spaces (namely the spaces $L_{H_n}^2(a, b)$). It is possible that this could lead to a *generalized* strong resolvent convergence—where the Hilbert spaces on which the operators act are allowed to be different—following or extending the methods from [57, 70], for example, but we won't investigate this further here.

2.2.3 Change of behavior at the right endpoint

Theorems 2.13 and 2.14 allow us to deduce the convergence of a sequence of Weyl–Titchmarsh function from that of the associated transfer matrices in several cases, but they a priori leave out an important one: that in which a sequence of systems that are limit point on the right converges to a system which is limit circle on the right. As far as I know, this phenomenon appears not to be considered in the standard theory—for example, I have not seen it in [61]—perhaps because it cannot happen for trace-normed systems: whether a trace-normed system on an interval $[a, b)$ is limit circle or limit point at b entirely depends on whether b is finite or not.

The purpose of this section is to present a method to tackle this problem. Let $H_n, H \in \text{CS}[a, b)$ with associated transfer matrices $T_{H_n} \in \text{TM}_{\text{LP}}[a, b)$ and $T_H \in \text{TM}_{\text{LC}}[a, b]$. Suppose that both systems are subject to the usual boundary condition $e_0^* J u(a) = 0$ on the left, and add a boundary condition $e_\theta^* J u(b) = 0$ at b for H . Let $m_{H_n}^0$ and m_H^0 denote the Weyl–Titchmarsh functions of the two systems. By definition, these Weyl–Titchmarsh functions both depend on the transfer matrices T_{H_n} and T_H , but m_H^0 also depends on the boundary condition at b . This suggests that the convergence of the transfer matrices $T_{H_n} \rightarrow T_H$ cannot be enough for the systems to converge: the integrability conditions of the systems in the sequences have to converge in some sense to the boundary condition of the limit system.

To see how this can happen, recall that by definition, $m_H^0 = T_H^{-1}(b, \cdot) \cot \theta$, while for any $t \in [a, b)$, we can write $m_{H_n}^0 = T_{H_n}^{-1}(t, \cdot) \mathcal{P}u_n(t, \cdot)$ where for each $z \in \mathbb{C}$, $u_n(\cdot, z)$ is an integrable solution to $Ju' = -zH_n u$. Comparing these two expressions leads to the following observation: for any $z \in \mathbb{H}$, a sufficient condition for $m_{H_n}^0(z) \rightarrow m_H^0(z)$ is the existence of a sequence $\{t_n\}_{n \in \mathbb{N}} \subset [a, b)$ such that $T_{H_n}^{-1}(t_n, z) \mathcal{P}u_n(t_n, z) \rightarrow T_H^{-1}(b, z) \cot \theta$. If $t_n \uparrow b$, we can then think of this convergence as that of the systems restricted to $[a, t_n]$ (which effectively become limit circle at t_n) to the limit system, along with the convergence of the “boundary conditions” at t_n (this being $\mathcal{P}u_n(t_n, z)$) to that at b . In that sense, we can think of the convergence $H_n \rightarrow H$ as a convergence of systems that are all limit circle at both endpoints. However, here the “boundary conditions” at t_n of the systems in the sequence are not actual boundary conditions, as they depend on z .

The above interpretation also has the important drawback that the domains $[a, t_n]$ grow larger with n , so there is no direct way to study the vague convergence of the coefficient matrices on all of these domains at the same time. Fortunately, one can always rescale time, at least when $b < \infty$: given $t_n \uparrow b$ it is always possible to define a map $\eta_n: [a, b_n] \rightarrow [a, b]$ by

$$\eta_n(t) := a + \frac{t_n - a}{b - a}(t - a) \quad \text{with} \quad b_n := a + \frac{(b - a)^2}{t_n - a}. \quad (2.13)$$

This is a $\mathcal{C}^1$ bijection: indeed it satisfies $\eta_n' \equiv \frac{t_n - a}{b - a} > 0$, so it is increasing from $\eta_n(a) = a$ to $\eta_n(b_n) = b$. As $\eta_n(b) = t_n$, η_n is also a bijection $[a, b] \rightarrow [a, t_n]$, so it can be taken as a time

change to make the system on $[a, t_n]$ act on $[a, b]$. Moreover, $\eta_n^{-1}(t) = a + \frac{b-a}{t_n-a}(t-a)$ satisfies

$$\|\eta_n^{-1} - \text{id}_{[a,b]}\|_\infty = \sup_{t \in [a,b]} \left| a + \frac{b-a}{t_n-a}(t-a) - t \right| = \sup_{t \in [a,b]} \frac{(t-a)(b-t_n)}{t_n-a} = \frac{(b-a)(b-t_n)}{t_n-a},$$

so $\eta_n^{-1} \rightarrow \text{id}_{[a,b]}$ uniformly on $[a, b]$ since $t_n \uparrow b$.

Using such time changes η_n , we arrive at the following plan to work out the convergence of the canonical systems H_n to H and deduce the convergence of their Weyl–Titchmarsh functions.

- i) Find a sequence of times $t_n \in [a, b)$ such that $t_n \uparrow b$, which would be expected to have the property that $T_{H_n}(t_n, \cdot) \rightarrow T_H(b, \cdot)$.
- ii) Define $\mathcal{C}^1$ bijections $\eta_n: [a, b_n] \rightarrow [a, b]$ with $\eta_n(b) = t_n$ and $b_n \downarrow b$ (e.g. as in (2.13)), and prove that $\eta'_n(H_n \circ \eta_n) \rightarrow H$ vaguely on the *closed* interval $[a, b]$.
- iii) Deduce using Theorem 2.12 that $T_{H_n}(\eta_n, \cdot) \rightarrow T_H$ compactly on $[a, b] \times \mathbb{C}$. Note that this proves in particular that $T_{H_n}(\eta_n(b), \cdot) = T_{H_n}(t_n, \cdot) \rightarrow T_H(b, \cdot)$.
- iv) Prove the convergence of the z -dependent “boundary conditions” $u_n(\eta_n(b), z)$ to e_θ , and deduce the convergence of the Weyl–Titchmarsh functions from Theorem 2.14.

It can be verified that in a sense, this plan always describes what happens if the systems converge at all. More precisely, if the transfer matrices converge, then it is always possible to build such a sequence of times t_n (this will be shown in Proposition 2.19). Moreover, if both the transfer matrices and the Weyl–Titchmarsh functions converge, then the z -dependent “boundary conditions” do converge to the boundary condition of the limit system (this will be shown in Proposition 2.20).

Proposition 2.19. *Let $T_n \in \text{TM}[a, b)$ and $T \in \text{TM}_{\text{LC}}[a, b)$ with $b < \infty$. If $T_n \rightarrow T$ compactly on $[a, b) \times \mathbb{C}$, then there is a sequence $t_n \uparrow b$ such that for all $z \in \mathbb{C}$, $T_n(t_n, z) \rightarrow T(b, z)$ as $n \rightarrow \infty$.*

Proof. Let $\{K_k\}_{k \in \mathbb{N}} \subset \mathbb{C}$ be an increasing sequence of compacts with $\bigcup_{k \in \mathbb{N}} K_k = \mathbb{C}$. It suffices to build two sequences $\{t_n\}_{n \in \mathbb{N}}$ and $\{\varepsilon_n\}_{n \in \mathbb{N}}$ as well as a subsequence $\{n_k\}_{k \in \mathbb{N}}$ such that

- i) $\{t_n\}_{n \in \mathbb{N}}$ is nondecreasing, and for all k , $|t_{n_k} - b| < 1/k$ and $t_n = t_{n_k}$ if $n_k \leq n < n_{k+1}$,
- ii) $\{\varepsilon_n\}_{n \in \mathbb{N}}$ is nonincreasing, and for all k , $\varepsilon_{n_{k+1}} = \varepsilon_{n_k}/n_k$ and $\varepsilon_n = \varepsilon_{n_k}$ if $n_k \leq n < n_{k+1}$,
- iii) For all k , all $n \geq n_k$ and all $z \in K_k$, $\|T_n(t_n, z) - T(b, z)\|_{\text{op}} < \varepsilon_{n_k}$.

Having done so, the proof is completed by noting that the sequence $\{t_n\}_{n \in \mathbb{N}}$ has the desired properties. By construction, it is nondecreasing and satisfies $|t_{n_k} - b| < 1/k$ for all k , so $t_{n_k} \uparrow b$. Moreover, because the sequence $\{\varepsilon_n\}_{n \in \mathbb{N}}$ is constant over $n \in \{n_k, \dots, n_{k+1} - 1\}$ and because the K_k ’s form an increasing sequence, the last condition actually shows that if $z \in K_k$, then for all $n \geq n_k$, $\|T_n(t_n, z) - T(b, z)\|_{\text{op}} < \varepsilon_n$. But the sequence $\{\varepsilon_n\}_{n \in \mathbb{N}}$ is nonincreasing and satisfies $\varepsilon_{n_{k+1}} = \varepsilon_{n_k}/n_k$ for all k , so it is clear that $\varepsilon_n \downarrow 0$. As for any $z \in \mathbb{C}$ there must be an ℓ such that $z \in K_k$ for all $k \geq \ell$, it follows that $T_n(t_n, z) \rightarrow T(b, z)$ for any $z \in \mathbb{C}$.

We now construct these sequences recursively. Since K_1 is compact, T is uniformly continuous on $[a, b] \times K_1$, and therefore there is a $t \in [a, b)$ with $|t - b| < 1$ such that for all $z \in K_1$,

$\|T(t, z) - T(b, z)\|_{\text{op}} < 1/2$. Then, since $T_n \rightarrow T$ compactly, there is an n_1 such that for all $z \in K_1$ and all $n \geq n_1$, $\|T_n(t, z) - T(t, z)\|_{\text{op}} < 1/2$. Set $t_n := t$ and $\varepsilon_n := 1$ for all $n \leq n_1$. Then for all $n \geq n_1$ and all $z \in K_1$, $\|T_n(t_n, z) - T(b, z)\|_{\text{op}} < 1$, and all of the desired properties are satisfied for $\{t_n\}_{n=1}^{n_1}$, $\{\varepsilon_n\}_{n=1}^{n_1}$ and $\{n_1\}$.

Now, suppose that for some k we have built $\{t_n\}_{n=1}^{n_k}$, $\{\varepsilon_n\}_{n=1}^{n_k}$ and $\{n_\ell\}_{\ell=1}^k$ that satisfy the three conditions up to k . As T is uniformly continuous on $[a, b] \times K_{k+1}$, there is again a $t > t_{n_k}$ with $|t - b| < 1/(k+1)$ such that for all $z \in K_{k+1}$, $\|T(t, z) - T(b, z)\|_{\text{op}} < \varepsilon_{n_k}/2n_k$. Since $T_n \rightarrow T$ compactly, there is an m such that for all $n \geq m$ and all $z \in K_{k+1}$, $\|T_n(t, z) - T(t, z)\|_{\text{op}} < \varepsilon_k/2n_k$. Here, there are two possibilities:

- i) If $m \leq n_k + 1$, we set $n_{k+1} := n_k + 1$ and define $t_{n_{k+1}} := t$ and $\varepsilon_{n_{k+1}} := \varepsilon_{n_k}/n_k$.
- ii) If $m > n_k + 1$, we set $n_{k+1} := m$ and define $t_{n_{k+1}} := t$ and $\varepsilon_{n_{k+1}} := \varepsilon_{n_k}/n_k$, and we fill in the rest by setting $t_n := t$ and $\varepsilon_n := \varepsilon_{n_k}$ for $n_k < n < n_{k+1}$.

In both cases, the updated sequences $\{t_n\}_{n=1}^{n_{k+1}}$, $\{\varepsilon_n\}_{n=1}^{n_{k+1}}$ and $\{n_\ell\}_{\ell=1}^{k+1}$ now satisfy the three conditions up to $k + 1$. ■

Proposition 2.20. *Let $H_n, H \in \text{CS}[a, b]$ satisfy Hypothesis 2.2 and have transfer matrices $T_{H_n} \in \text{TM}_{\text{LP}}[a, b]$ and $T_H \in \text{TM}_{\text{LC}}[a, b]$. Let $m_{H_n}^0$ and m_H^0 be their Weyl–Titchmarsh functions for the boundary condition $e_0^*Ju(a) = 0$, adding $e_\theta^*Ju(b) = 0$ for some $\theta \in [0, \pi)$ for H . Suppose that the sequence $\{t_n\}_{n \in \mathbb{N}} \subset [a, b]$ satisfies $t_n \uparrow b$ and $T_{H_n}(t_n, z) \rightarrow T_H(b, z)$ for all $z \in \mathbb{H}$. If $m_{H_n}^0 \rightarrow m_H^0$ compactly, then $\mathcal{P}u_n(t_n, z) \rightarrow \cot \theta$ for all $z \in \mathbb{H}$ where $u_n: [a, b] \times \mathbb{H} \rightarrow \mathbb{C}^2$ is defined so that for each $z \in \mathbb{H}$, $u_n(\cdot, z)$ is an integrable solution to $Ju' = -zH_nu$.*

Proof. Let $z \in \mathbb{H}$. By definition of the Weyl–Titchmarsh function, $\mathcal{P}u_n(t_n, z) = T_{H_n}(t_n, z)m_{H_n}^0(z)$ and $\cot \theta = T_H(b, z)m_H^0(z)$. If $T_H(b, z)m_H^0(z) < \infty$, then

$$|\mathcal{P}u_n(t_n, z) - \cot \theta| \leq |T_{H_n}(t_n, z)m_{H_n}^0(z) - T_H(b, z)m_{H_n}^0(z)| + |T_H(b, z)m_{H_n}^0(z) - T_H(b, z)m_H^0(z)|$$

and both terms vanish in the limit. It remains to see what happens when $T_H(b, z)m_H^0(z) = \infty$. By Hypothesis 2.2, it is impossible that $m_H^0(z) = \infty$, so this can happen only if $m_H^0(z) = -D/C$ with $C \neq 0$ where $\begin{pmatrix} A & B \\ C & D \end{pmatrix} := T_H(b, z)$. In that case, writing $\begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} := T_{H_n}(b, z)$,

$$C_n m_{H_n}^0(z) + D_n = \frac{1}{C} (CD_n - C_n D) + C_n (m_{H_n}^0(z) - m_H^0(z)),$$

and both terms vanish in the limit. Because $A_n m_{H_n}^0(z) + B_n \rightarrow A m_H^0(z) + B \neq 0$ (this being equal to zero would imply that $\det T_H(b, z) = 0$, but this determinant is always equal to 1), this shows that $T_{H_n}(t_n, z)m_{H_n}^0(z) \rightarrow \infty$. ■

We finish this section by pointing out a simple result. When working out the vague convergence of the time-changed systems $\eta'_n(H_n \circ \eta_n) \rightarrow H$ on $[a, b]$, one might worry that information is lost

since the time intervals (b, b_n) , which correspond to (t_n, b) for the original systems H_n , are removed from the analysis. We now verify that this is not the case.

Proposition 2.21. *Let $H_n, H \in \text{CS}[a, b)$ with $b < \infty$. Let $\eta_n: [a, b_n] \rightarrow [a, b]$ be $\mathcal{C}^1$ increasing bijections with $b_n \downarrow b$, and suppose that $\|\eta_n^{-1} - \text{id}_{[a,b]}\|_\infty \rightarrow 0$. Then $H_n \rightarrow H$ vaguely on $[a, b)$ if and only if $\eta'_n(H_n \circ \eta_n) \rightarrow H$ vaguely on $[a, b)$.*

Remark. Thus, the vague convergence $\eta'_n(H_n \circ \eta_n) \rightarrow H$ on $[a, b]$ is in fact a *stronger* condition than the vague convergence $H_n \rightarrow H$ on $[a, b)$, since the latter is equivalent to the vague convergence $\eta'_n(H_n \circ \eta_n) \rightarrow H$ on $[a, b)$, with b excluded.

Proof. Let $\varphi \in \mathcal{C}_c([a, b), \mathbb{C}^2)$. If $t = b$ or $t = b_n$, then

$$\begin{aligned} \int_a^t \varphi^* \eta'_n(H_n \circ \eta_n) \varphi \, d\mathcal{L} &= \int_a^{\eta_n(t)} \varphi^* H_n \varphi \, d\mathcal{L} + \int_a^{\eta_n(t)} (\varphi^* \circ \eta_n^{-1} - \varphi^*) H_n \varphi \, d\mathcal{L} \\ &\quad + \int_a^{\eta_n(t)} \varphi^* \circ \eta_n^{-1} H_n (\varphi \circ \eta_n^{-1} - \varphi) \, d\mathcal{L}. \end{aligned} \quad (2.14)$$

The last two terms are bounded by

$$\|\varphi\|_\infty \|\varphi \circ \eta_n^{-1} - \varphi\|_\infty \int_{\text{supp } \varphi} \text{tr } H_n \, d\mathcal{L},$$

If either $H_n \rightarrow H$ or $\eta'_n(H_n \circ \eta_n) \rightarrow H$ vaguely on $[a, b)$, the compactness of $\text{supp } \varphi$ ensures that $\int_{\text{supp } \varphi} \text{tr } H_n \, d\mathcal{L}$ is bounded in n . Indeed, if $\psi \in \mathcal{C}_c([a, b), \mathbb{C}^2)$ has $\psi_j \geq 1$ on $\text{supp } \varphi$ for $j = 1$ or 2 and $\psi_k \equiv 0$ for the other component, then $\int_{\text{supp } \varphi} H_{n,jj} \, d\mathcal{L} \leq \int_a^b \psi^* H_n \psi \, d\mathcal{L}$, but this cannot diverge if $H_n \rightarrow H$ vaguely. The same argument applies if $\eta'_n(H_n \circ \eta_n) \rightarrow H$ vaguely since $\|\eta_n^{-1} - \text{id}_{[a,b]}\|_\infty \rightarrow 0$, as this implies that all $\eta_n^{-1}(\text{supp } \varphi)$ are contained in a given compact in $[a, b)$ for n large enough. Now, if ω_φ is a modulus of continuity for φ , then

$$\|\varphi \circ \eta_n^{-1} - \varphi\|_\infty = \sup_{t \in [a,b]} |\varphi \circ \eta_n^{-1}(t) - \varphi(t)| \leq \sup_{t \in [a,b]} \omega_\varphi(|\eta_n^{-1}(t) - t|) \leq \omega_\varphi(\|\eta_n^{-1} - \text{id}_{[a,b]}\|_\infty) \rightarrow 0,$$

and therefore the last two terms of (2.14) vanish in the limit if either $H_n \rightarrow H$ or $\eta'_n(H_n \circ \eta_n) \rightarrow H$ vaguely on $[a, b)$.

Finally, notice that for both $t = b$ and $t = b_n$, the upper bounds of the first two integrals in (2.14) can be replaced with b for all n large enough, since $\eta_n(b) \rightarrow b$ and φ is compactly supported in $[a, b)$. Given our previous conclusions, this allows us to rewrite (2.14) as

$$\int_a^b \varphi^* \eta'_n(H_n \circ \eta_n) \varphi \, d\mathcal{L} = \int_a^b \varphi^* H_n \varphi \, d\mathcal{L} + o(1)$$

whenever $H_n \rightarrow H$ or $\eta'_n(H_n \circ \eta_n) \rightarrow H$ vaguely on $[a, b)$, and it is now clear that either condition implies the other one. ■

2.3 Random canonical systems

The purpose of this section is to introduce a few properties of random canonical systems. We start by discussing the measurability of their solutions, and then we make a few basic remarks about the applicability of some convergence results from §2.2 to random canonical systems.

Although I haven't seen these results articulated anywhere else, they are quite elementary so it is very much possible that they are not new. They are given mostly for the sake of completeness.

2.3.1 Random canonical systems and their solutions

By a *random canonical system* on an interval $\mathcal{I} \subseteq \mathbb{R}$, we simply mean a canonical system whose coefficient matrix is a CS $\mathcal{I}$ -valued random variable. We saw in Theorem 2.11 that CS $\mathcal{I}$ is always a separable metric space, which ensures that it has good measurability properties.

Given a random canonical system, at every point of the probability space it is certainly possible to define its solutions, but it is not immediately obvious that these are measurable. Thus, in order to be able to say anything about a random canonical system, the first step is to verify that it can be given a transfer matrix which is a well-defined random variable.

If the trace of the coefficient matrix is bounded at every point of the probability space by a suitable locally integrable function, then the continuity of the map from a coefficient matrix to the associated transfer matrix (i.e., Theorem 2.12) implies that the transfer matrix is indeed a random variable. This criterion can easily be relaxed to the following, which should cover all of the cases we'll be interested in.

Proposition 2.22. *Let $\mathcal{I} = [a, b)$ or $[a, b]$, and let H be a random coefficient matrix on $\mathcal{I}$. Suppose that for every $\varepsilon > 0$, there is an $f \in L^1_{\text{loc}}(\mathcal{I})$ such that $\mathbb{P}[\text{tr } H \leq f] \geq 1 - \varepsilon$. Then, there is a TM $\mathcal{I}$ -valued random variable that is equal to the transfer matrix of H with probability 1.*

When the hypotheses of the above result are met, we won't distinguish between the random canonical system's transfer matrix and its measurable modification, as they only differ on an event of zero probability.

Proof. For each $n \in \mathbb{N}$, take $f_n \in L^1_{\text{loc}}(\mathcal{I})$ such that $\mathbb{P}[\text{tr } H \leq f_n] \geq 1 - 1/n$, choosing the f_n 's to form a pointwise increasing sequence with the property that $\{\text{tr } H \leq f_n\} \subseteq \{\text{tr } H \leq f_{n+1}\}$ for each n . It is clear that this is possible: having chosen f_n , take $\tilde{f}_{n+1}$ with $\mathbb{P}[\text{tr } H \leq \tilde{f}_{n+1}] \geq 1 - 1/(n+1)$, and then set $f_{n+1} := f_n \vee \tilde{f}_{n+1}$. Note that by continuity of measures, $\mathbb{P}(\bigcup_{n \in \mathbb{N}} \{\text{tr } H \leq f_n\}) = 1$.

Now, let $\mathcal{T} : \text{CS } \mathcal{I} \rightarrow \text{TM } \mathcal{I}$ denote the map sending a coefficient matrix to the associated transfer matrix, and define $T_n := (\mathcal{T} \circ H) \mathbb{1}_{\{\text{tr } H \leq f_n\}} + I_2 \mathbb{1}_{\{\text{tr } H > f_n\}}$. Because $\mathcal{T}$ is continuous on each $\text{CS}_{f_n} \mathcal{I}$, each T_n is measurable, and therefore $T := \lim_{n \rightarrow \infty} T_n$ is measurable. Because the sets

$\{\operatorname{tr} H \leq f_n\}$ form an increasing sequence, $T = (\mathcal{T} \circ H)\mathbb{1}_{\bigcup_{n \in \mathbb{N}} \{\operatorname{tr} H \leq f_n\}} + I_2 \mathbb{1}_{\bigcap_{n \in \mathbb{N}} \{\operatorname{tr} H > f_n\}}$, which is equal to the transfer matrix $\mathcal{T} \circ H$ on an event of probability 1. ■

Once we add appropriate boundary conditions to a random canonical system, we can also define a Weyl–Titchmarsh function. The boundary conditions could also be random: as before they take the form $e_\phi^* Ju(t) = 0$ at an endpoint t , but now ϕ could be a $[0, \pi)$ -valued random variable. By Theorems 2.13 and 2.14, there are continuous maps from transfer matrices and boundary conditions to Weyl–Titchmarsh functions, so Proposition 2.22 (together with the fact that changing the left boundary condition from e_0 to e_ϕ changes the Weyl–Titchmarsh function by an application of a Möbius transform $R_{-\phi}$) immediately yields the following.

Corollary 2.22.1. *In the setting of Proposition 2.22, suppose also that the system is subject to a (possibly random) boundary condition $e_\phi^* Ju(a) = 0$ at a , and either that the system is a.s. limit point at b , or that it is a.s. limit circle at b with (possibly random) boundary condition $e_\phi^* Ju(b) = 0$. Then the associated Weyl–Titchmarsh function is a $\operatorname{Hol}(\mathbb{H}, \overline{\mathbb{H}}_\infty)$ -valued random variable.*

These results also yield the measurability of other important objects. For instance, when the system is limit point at b , it can be shown from Weyl theory (this is done for example in [61, Theorem 3.14(a)]) that an integrable solution to the system for $z \in \mathbb{H}$ is given by $T_H(\cdot, z) \begin{pmatrix} m_H^0(z) \\ 1 \end{pmatrix}$, where as usual T_H and m_H^0 denote the transfer matrix and the Weyl–Titchmarsh function for the boundary condition $e_0^* Ju(a) = 0$. Therefore, the integrable solution is a well-defined random variable whenever T_H and m_H^0 are.

Moreover, whenever the system’s Weyl–Titchmarsh function is a well-defined random variable, the continuity of the Herglotz representation implies that the spectral measure is a well-defined random variable with values in the space $\mathcal{M}_+(\mathbb{R})$ of locally finite (nonnegative) measures on $\mathbb{R}$. The latter is a separable metric space under the vague topology—see e.g. [45, Theorem 4.2] for a proof.

2.3.2 Convergence of random canonical systems

A sequence of random coefficient matrices, as a sequence of random variables with values in a separable metric space, can be said to converge with the usual probabilistic meanings. The following result gives a straightforward criterion to test the vague convergence in probability of random canonical systems.

Proposition 2.23. *Let $\mathcal{I} \subseteq \mathbb{R}$ be an interval, and let $\{H_n\}_{n \in \mathbb{N}}$ be a sequence of random coefficient matrices on $\mathcal{I}$. If there is another random coefficient matrix H on $\mathcal{I}$ such that for all $\varphi \in \mathcal{C}_c(\mathcal{I}, \mathbb{C}^2)$,*

$$\int_{\mathcal{I}} \varphi^* (H_n - H) \varphi \, d\mathcal{L} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0,$$

then $H_n \xrightarrow{\mathbb{P}} H$ vaguely on $\mathcal{I}$.

Proof. Let $\varepsilon > 0$. By definition of the metric $d_{\text{CS } \mathcal{I}}$ from Theorem 2.11, if $d_{\text{CS } \mathcal{I}}(H_n, H) > \varepsilon$ and $N \in \mathbb{N}$ is large enough so that $\sum_{m=N+1}^{\infty} 2^{-m} \leq \varepsilon/2$, then there is an $m \leq N$ with $d_{\varphi_m}(H_n, H) > \varepsilon/2$; indeed, otherwise $d_{\text{CS } \mathcal{I}}(H_n, H) \leq \frac{\varepsilon}{2} \sum_{m=1}^N 2^{-m} + \sum_{m=N+1}^{\infty} 2^{-m} \leq \varepsilon$. Hence, $\mathbb{P}[d_{\text{CS } \mathcal{I}}(H_n, H) > \varepsilon] \leq \sum_{m=1}^N \mathbb{P}[d_{\varphi_m}(H_n, H) > \varepsilon/2]$, and the result follows. ■

In a similar way, unsurprisingly, it suffices to check the convergence of random transfer matrices on compact sets of times.

Proposition 2.24. *Let $\mathcal{I} \subseteq \mathbb{R}$ be an interval, and let T_n, T be random transfer matrices on $\mathcal{I} \times \mathbb{C}$. If $T_n \rightarrow T$ compactly on $K \times \mathbb{C}$ in probability for any compact $K \subseteq \mathcal{I}$, then $T_n \rightarrow T$ compactly on $\mathcal{I} \times \mathbb{C}$ in probability.*

Proof. Take a sequence of compact sets $K_k \subset \mathcal{I} \times \mathbb{C}$ such that $\bigcup_{k \in \mathbb{N}} K_k = \mathcal{I} \times \mathbb{C}$. Then a metric on $\text{TM } \mathcal{I}$ is given by

$$d_{\text{TM } \mathcal{I}}(T, T') = \sum_{k=1}^{\infty} \frac{1}{2^k} \frac{d_k(T, T')}{1 + d_k(T, T')} \quad \text{where} \quad d_k(T, T') := \sup_{(t,z) \in K_k} |T(t, z) - T'(t, z)|.$$

Let $\varepsilon, \delta > 0$, and fix N large enough so that $\sum_{k=N+1}^{\infty} 2^{-k} < \delta/2$. If $T_n \rightarrow T$ compactly on each K_k , then $d_k(T_n, T) < \delta/2N$ for all $k \in \{1, \dots, N\}$ and all n large enough with probability at least $1 - \varepsilon/N$. Hence, $d_{\text{TM } \mathcal{I}}(T_n, T) < \delta$ with probability at least $1 - \varepsilon$ for all n large enough, and $T_n \rightarrow T$ compactly on $\mathcal{I} \times \mathbb{C}$ in probability. ■

Given a vaguely convergent sequence of random canonical systems, a trace domination condition that holds a.s. automatically upgrades the convergence to the compact convergence of transfer matrices by Theorem 2.12. This criterion can be relaxed to allow the trace domination condition to hold only with high probability.

Proposition 2.25. *Let $\mathcal{I} = [a, b)$ or $[a, b]$ be a real interval. Let H_n, H be CS $\mathcal{I}$ -valued random coefficient matrices, and let T_{H_n}, T_H be their transfer matrices. Suppose that for all $\varepsilon > 0$, there are $f_\varepsilon, g_\varepsilon \in L^1_{\text{loc}}(\mathcal{I})$ such that $\mathbb{P}[\text{tr } H \leq f_\varepsilon] \geq 1 - \varepsilon$ and $\mathbb{P}[\text{tr } H_n \leq g_\varepsilon] \geq 1 - \varepsilon$ for any n large enough. If $H_n \rightarrow H$ vaguely on $\mathcal{I}$ in probability, then $T_{H_n} \rightarrow T_H$ compactly in probability.*

Proof. Let $d_{\text{TM}\mathcal{I}}$ denote the metric on $\text{TM}\mathcal{I}$. If $\varepsilon, \delta > 0$, by hypothesis there is an $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\mathbb{P}[d_{\text{TM}\mathcal{I}}(T_{H_n}, T_H) > \delta] < \mathbb{P}[d_{\text{TM}\mathcal{I}}(T_{H_n}, T_H) > \delta, \text{tr } H \leq f_\varepsilon, \text{tr } H_n \leq g_\varepsilon] + 2\varepsilon.$$

Now, pick some $\tilde{H} \in \text{CS}\mathcal{I}$ such that $\text{tr } \tilde{H} \leq f_\varepsilon + g_\varepsilon$, and set

$$H' := H \mathbb{1}_{\{\text{tr } H \leq f_\varepsilon\}} + \tilde{H} \mathbb{1}_{\{\text{tr } H > f_\varepsilon\}} \quad \text{and} \quad H'_n := H_n \mathbb{1}_{\{\text{tr } H_n \leq g_\varepsilon\}} + \tilde{H} \mathbb{1}_{\{\text{tr } H_n > g_\varepsilon\}}.$$

Then $H'_n \rightarrow H'$ vaguely on $\mathcal{I}$ in probability, and because they are $\text{CS}_{f_\varepsilon+g_\varepsilon}\mathcal{I}$ -valued, Theorem 2.12 guarantees that their transfer matrices $T_{H'_n} \rightarrow T_{H'}$ compactly so $\mathbb{P}[d_{\text{TM}\mathcal{I}}(T_{H'_n}, T_{H'}) > \delta] < \varepsilon$ for all n large enough. Because $T_{H'} = T_H$ on $\{\text{tr } H \leq f_\varepsilon\}$ and $T_{H'_n} = T_{H_n}$ on $\{\text{tr } H_n \leq g_\varepsilon\}$, it follows that $\mathbb{P}[d_{\text{TM}\mathcal{I}}(T_{H_n}, T_H) > \delta] < 3\varepsilon$ for all n large enough, and $T_{H_n} \rightarrow T_H$ compactly in probability. ■

2.4 Sturm–Liouville operators as canonical systems

In this section, we outline a general method to find a canonical system that is equivalent to a Sturm–Liouville operator, based on the analogous construction for Schrödinger operators that can be found in [61, §1.3]. To allow for more flexibility in the coefficients of the operator, we focus on *generalized* Sturm–Liouville operators, in the sense used by Eckhardt et al. in [29].

2.4.1 Generalized Sturm–Liouville operators

By a *generalized Sturm–Liouville operator*, we mean an operator of the form

$$L: f \mapsto \frac{1}{w} \left(-(f^{[1]})' + s f^{[1]} + q f \right) \quad \text{with} \quad f^{[1]} := p(f' + s f) \quad (2.15)$$

where $p, q, s, w: (a, b) \rightarrow \mathbb{R}$ satisfy $p \neq 0$ and $w > 0$ a.e. on (a, b) , and $p^{-1}, q, s, w \in L^1_{\text{loc}}(a, b)$. Here, $f^{[1]}$ is called the *(first) quasi-derivative* of f , and L is a priori assumed to act on functions from

$$\mathfrak{D}_L := \{f \in \text{AC}_{\text{loc}}(a, b) : f^{[1]} \in \text{AC}_{\text{loc}}(a, b)\}, \quad (2.16)$$

so that $L: \mathfrak{D}_L \rightarrow L^1_{\text{loc}}((a, b), w \, d\mathcal{L})$.

The spectral problem for L is studied on the Hilbert space $L^2((a, b), w \, d\mathcal{L})$ with inner product

$$\langle f, g \rangle_w := \int_a^b f \bar{g} w \, d\mathcal{L}.$$

L is first taken to act on functions $f \in \mathfrak{D}_L \cap L^2((a, b), w \, d\mathcal{L})$ such that $Lf \in L^2((a, b), w \, d\mathcal{L})$, and self-adjoint realizations can be obtained by restricting the domain with appropriate boundary

conditions, depending on whether the operator is limit point or limit circle at endpoints. We will not go into further detail about these self-adjoint realizations as we will rather adopt the point of view of studying the spectral theory of L by representing it as a canonical system, but the ideas are similar to those from §2.1.5 and the details can be found in [29, §3].

This class of generalized Sturm–Liouville operators is studied in detail by Eckhardt et al. in [29]. The point of extending classical Sturm–Liouville operators to this class of operators is essentially to allow L to have distributional potential, as s does not have to be differentiable. Still, the theory of ordinary differential equations yields the following existence and uniqueness theorem.

Theorem 2.26 [29, Theorem 2.2]. *For each $g \in L^1_{\text{loc}}((a, b), w \, d\mathcal{L})$, $z \in \mathbb{C}$, $t_0 \in (a, b)$ and $v \in \mathbb{C}^2$, there is a unique solution $f \in \mathfrak{D}_L$ to $(L - z)f = g$ such that $(f, f^{[1]})(t_0) = v$. Moreover, if g , v and z are real-valued, then f is real-valued.*

Remark. Here, we restrict $t_0 \in (a, b)$, but when L is regular at an endpoint—that is, when p^{-1} , q , s and w are integrable near that endpoint—then as long as g is also integrable near that endpoint, every solution f to $(L - z)f = g$ has a finite limit at that endpoint, and so does its quasi-derivative (this is proven in [29, Theorem 2.6]). In fact, when L is regular at an endpoint $t \in \{a, b\}$ and g is integrable near t , the whole statement of the theorem is also valid if $t_0 = t$.

2.4.2 Turning the eigenvalue equation into a canonical system

Given a generalized Sturm–Liouville operator L of the form (2.15), it is possible to convert the eigenvalue equation $Lh = zh$ to a canonical system. To do so, define $F := (h^{[1]}, h)$ for an h that solves $Lh = zh$. Then $(h^{[1]})' = sh^{[1]} + (q - zw)h$, so

$$F' = \begin{pmatrix} (h^{[1]})' \\ h' \end{pmatrix} = \begin{pmatrix} sh^{[1]} + (q - zw)h \\ p^{-1}h^{[1]} - sh \end{pmatrix} = \begin{pmatrix} s & q - zw \\ p^{-1} & -s \end{pmatrix} F.$$

For the spectral parameter z to be factored out in front of the matrix, as in a canonical system, a change of variables must be introduced. To this end, fix two linearly independent solutions f and g to $Lf = Lg = 0$, and set

$$A := \begin{pmatrix} g^{[1]} & f^{[1]} \\ g & f \end{pmatrix} \quad \text{so that} \quad A' = \begin{pmatrix} s & q \\ p^{-1} & -s \end{pmatrix} A \quad (2.17)$$

in the same way as above. Note that

$$\det A = fg^{[1]} - f^{[1]}g =: \mathcal{W}(f, g)$$

is the *Wronskian* of f and g . Because f and g solve $Lf = Lg = 0$, it is easy to see by a direct computation that $(\mathcal{W}(f, g))' \equiv 0$. This Wronskian is therefore constant, and it is nonzero since f and g are chosen to be linearly independent.

Now, set $u := A^{-1}F$. Because $0 = (A^{-1}A)' = (A^{-1})'A + A^{-1}A'$, we get $(A^{-1})' = -A^{-1}A'A^{-1}$ and

$$u' = (A^{-1})'F + A^{-1}F' = -A^{-1} \begin{pmatrix} s & q \\ p^{-1} & -s \end{pmatrix} F + A^{-1} \begin{pmatrix} s & q - zw \\ p^{-1} & -s \end{pmatrix} F = A^{-1} \begin{pmatrix} 0 & -zw \\ 0 & 0 \end{pmatrix} F.$$

Substituting back $F = Au$, we find that u solves the canonical system

$$Ju' = -zHu \quad \text{on } (a, b) \quad \text{with} \quad H := wJA^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} A. \quad (2.18)$$

Conversely, given a solution u to this system, the steps can be taken in reverse to see that the second entry of Au solves $Lf = zf$. In that sense, these two equations are equivalent.

The coefficient matrix obtained in (2.18) can easily be computed more explicitly in terms of the two solutions f and g . Indeed,

$$H = \frac{w}{\mathcal{W}(f, g)} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} f & -f^{[1]} \\ -g & g^{[1]} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} g^{[1]} & f^{[1]} \\ g & f \end{pmatrix} = \frac{w}{\mathcal{W}(f, g)} \begin{pmatrix} g & -g^{[1]} \\ f & -f^{[1]} \end{pmatrix} \begin{pmatrix} g & f \\ 0 & 0 \end{pmatrix},$$

which gives

$$H = \frac{w}{\mathcal{W}(f, g)} \begin{pmatrix} g^2 & fg \\ fg & f^2 \end{pmatrix} = \frac{w}{\mathcal{W}(f, g)} \psi \psi^* \quad \text{where} \quad \psi := \begin{pmatrix} g \\ f \end{pmatrix}. \quad (2.19)$$

2.4.3 The associated L^2 space

The factorization of the coefficient matrix H as a projector that we obtained in (2.19) yields an interesting representation of the seminorm $\|\cdot\|_H$. Indeed, if $u: (a, b) \rightarrow \mathbb{C}^2$, then

$$\|u\|_H^2 = \int_a^b u^* H u \, d\mathcal{L} = \frac{1}{\mathcal{W}(f, g)} \int_a^b u^* \psi \psi^* u w \, d\mathcal{L} = \frac{1}{\mathcal{W}(f, g)} \int_a^b |\langle u, \psi \rangle|^2 w \, d\mathcal{L}.$$

This yields an explicit relationship between the norms of the solutions to the canonical systems and those of the eigenvalue equation for the original operator. Indeed, let h denote the second entry of the vector Au . Then

$$\langle u, \psi \rangle = \psi^* A^{-1} A u = \frac{1}{\mathcal{W}(f, g)} \begin{pmatrix} g & f \end{pmatrix} \begin{pmatrix} f & -f^{[1]} \\ -g & g^{[1]} \end{pmatrix} \begin{pmatrix} \cdot \\ h \end{pmatrix} = h,$$

so in fact

$$\|u\|_H^2 = \frac{1}{\mathcal{W}(f, g)} \|h\|_w^2 \quad (2.20)$$

where $\|\cdot\|_w$ denotes the norm in the weighted L^2 space $L^2((a, b), w \, d\mathcal{L})$.

In other words, the mapping between solutions of the generalized Sturm–Liouville operator, which live in $L^2((a, b), w \, d\mathcal{L})$, and those of the canonical system, which live in $L_H^2(a, b)$, preserves norms up to a factor of $\mathcal{W}(f, g)$.

2.5 Jacobi operators as canonical systems

Canonical systems can encode not only second-order differential operators, but also *difference* operators. In this section, we describe how to make such a correspondence between a Jacobi operator and a canonical system, essentially following the method outlined by Remling in [61, §5.3]. For the general theory of Jacobi operators, we refer to Teschl’s book [68].

2.5.1 Jacobi operators and their matrix representations

Consider an $n \times n$ matrix

$$Y := \begin{pmatrix} b_1 + \alpha & a_1 & & & \\ & a_1 & b_2 & & \\ & & a_2 & b_3 & \ddots \\ & & & \ddots & \ddots & a_{n-1} \\ & & & & a_{n-1} & b_n + \omega \end{pmatrix} \quad (2.21)$$

where $b_j, \alpha, \omega \in \mathbb{R}$ and either $a_j > 0$ for all j or $a_j < 0$ for all j . This matrix can be thought of as a second-order difference operator τ . Indeed, the operator τ acting on a sequence $h \in \mathbb{C}^{\mathbb{N}_0}$ as

$$(\tau h)_j = a_{j-1} h_{j-1} + b_j h_j + a_j h_{j+1} \quad (2.22)$$

exactly reproduces the application of Y on a vector $h \in \mathbb{C}^n$, at least for the entries $j \in \{2, \dots, n-1\}$. This doesn’t directly extend to $j = 1$ and $j = n$, as these expressions involve the a priori undefined components h_0 and h_{n+1} .

To complete the correspondence, τ can be taken to act on a domain $\mathfrak{D}_\tau \subset \mathbb{C}^{1+n+1}$ which consists of finite sequences that satisfy certain boundary conditions at both ends. Then τ maps $\mathfrak{D}_\tau \rightarrow \mathbb{C}^n$ through the application of (2.22). To make this precise, add two values $a_0, a_n \in \mathbb{R} \setminus \{0\}$, and let $\hat{h} = (h_0, h, h_{n+1}) \in \mathbb{C}^{1+n+1}$ where $h \in \mathbb{C}^n$ —we think of this vector simply as h to which two extra

components are added at both ends to have $\tau\hat{h}$ well-defined. Then Yh and $\tau\hat{h}$ have the same first entry if

$$(Yh)_1 = (b_1 + \alpha)h_1 + a_1h_2 \quad \text{and} \quad (\tau\hat{h})_1 = a_0h_0 + b_1h_1 + a_1h_2$$

are equal, that is, if $\alpha h_1 = a_0h_0$. In the same way, the last entries of Yh and $\tau\hat{h}$ are equal if

$$(Yh)_n = a_{n-1}h_{n-1} + (b_n + \omega)h_n = a_{n-1}h_{n-1} + b_nh_n + a_nh_{n+1} = (\tau\hat{h})_n \iff \omega h_n = a_nh_{n+1}.$$

The appropriate domain for τ is therefore

$$\mathfrak{D}_\tau := \left\{ \hat{h} = (h_0, h, h_{n+1}) \in \mathbb{C}^{1+n+1} : h_0 = \frac{\alpha}{a_0}h_1, h_{n+1} = \frac{\omega}{a_n}h_n \right\}, \quad (2.23)$$

in the sense that now $\tau: \mathfrak{D}_\tau \rightarrow \mathbb{C}^n$ exactly reproduces the action of $Y: \mathbb{C}^n \rightarrow \mathbb{C}^n$. In particular, this makes the eigenvalue equations $Yh = zh$ and $\tau\hat{h} = zh$ completely equivalent if $\hat{h} \in \mathfrak{D}_\tau$. For the rest of the section, we will slightly abuse notation and use the same symbols for elements of $\mathfrak{D}_\tau$ and their restrictions to $\mathbb{C}^n$ obtained by removing the first and last entries.

2.5.2 Turning the eigenvalue equation into a canonical system

The eigenvalue equation $Yh = zh$, or equivalently $\tau h = zh$ with $h \in \mathfrak{D}_\tau$, can be turned into a canonical system in a similar way as what we did in §2.4 for Sturm–Liouville operators. This is done in two steps: first we build an equivalent discrete canonical system, and then we extend it to an actual one.

2.5.2.1 Building an equivalent discrete canonical system

Suppose $h \in \mathfrak{D}_\tau$ solves $\tau h = zh$, and set $F_j := (h_{j+1}, -a_jh_j)$ for $j \in \{0, \dots, n\}$. Then for $j \in \{1, \dots, n\}$,

$$F_j = \begin{pmatrix} \frac{1}{a_j}((z - b_j)h_j - a_{j-1}h_{j-1}) \\ -a_jh_j \end{pmatrix} = \begin{pmatrix} \frac{z-b_j}{a_j} & \frac{1}{a_j} \\ -a_j & 0 \end{pmatrix} \begin{pmatrix} h_j \\ -a_{j-1}h_{j-1} \end{pmatrix},$$

which means that

$$F_j = (C_j + zD_j)F_{j-1} \quad \text{where} \quad C_j := \begin{pmatrix} -b_j/a_j & 1/a_j \\ -a_j & 0 \end{pmatrix} \quad \text{and} \quad D_j := \begin{pmatrix} 1/a_j & 0 \\ 0 & 0 \end{pmatrix}. \quad (2.24)$$

Now, let $f, g \in \mathbb{C}^{1+n+1}$ be two linearly independent solutions to $\tau f = \tau g = 0$ (not necessarily in $\mathfrak{D}_\tau$), and define

$$A_j := \begin{pmatrix} g_{j+1} & f_{j+1} \\ -a_jg_j & -a_jf_j \end{pmatrix} \quad \text{so that} \quad A_j = C_jA_{j-1} \quad (2.25)$$

for $j \in \{1, \dots, n\}$. Note that

$$\det A_j = a_j(g_j f_{j+1} - f_j g_{j+1}) =: \mathcal{W}_j(g, f)$$

is the *Wronskian* of g and f . Because f and g solve $\tau f = \tau g = 0$, it is easy to see by a direct computation that this Wronskian is actually constant, so it is always nonzero since we chose f and g to be linearly independent.

Now, set $u_j := A_j^{-1} F_j$ for $j \in \{0, \dots, n\}$. Then for $j \in \{1, \dots, n\}$,

$$u_j = A_{j-1}^{-1} C_j^{-1} (C_j + z D_j) F_{j-1} = u_{j-1} + z A_{j-1}^{-1} C_j^{-1} D_j A_{j-1} u_{j-1},$$

so

$$J(u_j - u_{j-1}) = -z H_j u_{j-1} \quad \text{with} \quad H_j := -J A_{j-1}^{-1} C_j^{-1} D_j A_{j-1}. \quad (2.26)$$

This can be understood as a discrete canonical system, and it is fully equivalent to the original eigenvalue equation $\tau h = z h$ in the sense that given a solution $\{u_j\}_{j=0}^n$ to (2.26), the steps can be taken in reverse to see that a solution to $\tau h = z h$ can be extracted from the first entries of the $A_j u_j$'s.

The coefficient matrix obtained in (2.26) can be computed more explicitly in terms of f and g . Indeed, note that

$$C_j^{-1} D_j = \begin{pmatrix} 0 & -1/a_j \\ a_j & -b_j/a_j \end{pmatrix} \begin{pmatrix} 1/a_j & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

so

$$\begin{aligned} H_j &= \frac{1}{\mathcal{W}_0(g, f)} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -a_{j-1} f_{j-1} & -f_j \\ a_{j-1} g_{j-1} & g_j \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} g_j & f_j \\ -a_{j-1} f_{j-1} & -a_{j-1} f_{j-1} \end{pmatrix} \\ &= \frac{1}{\mathcal{W}_0(g, f)} \begin{pmatrix} a_{j-1} g_{j-1} & g_j \\ a_{j-1} f_{j-1} & f_j \end{pmatrix} \begin{pmatrix} 0 & 0 \\ g_j & f_j \end{pmatrix}, \end{aligned}$$

which gives

$$H_j = \frac{1}{\mathcal{W}_0(g, f)} \begin{pmatrix} g_j^2 & f_j g_j \\ f_j g_j & f_j^2 \end{pmatrix} = \frac{1}{\mathcal{W}_0(g, f)} \psi_j \psi_j^* \quad \text{where} \quad \psi_j := \begin{pmatrix} g_j \\ f_j \end{pmatrix}. \quad (2.27)$$

2.5.2.2 Extending to an actual canonical system

To turn the discrete canonical system into an actual canonical system that runs in continuous time, fix an increasing sequence $\{t_j\}_{j=0}^n$, and define $H: [t_0, t_n) \rightarrow \mathbb{R}^{2 \times 2}$ by setting

$$H \equiv \frac{1}{t_j - t_{j-1}} H_j \quad \text{on} \quad [t_{j-1}, t_j) \quad (2.28)$$

for each $j \in \{1, \dots, n\}$. This coefficient matrix yields a canonical system that is equivalent to the discrete system from (2.26). Indeed, define $u: [t_0, t_n) \rightarrow \mathbb{C}^2$ as being continuous and given by

$$u(t) := u(t_{j-1}) + z \frac{t - t_{j-1}}{t_j - t_{j-1}} JH_j u(t_{j-1}) \quad \text{for } t \in [t_{j-1}, t_j), \quad (2.29)$$

for each $j \in \{1, \dots, n\}$. The continuity of u requires that

$$J(u(t_j) - u(t_{j-1})) = -zH_j u(t_{j-1}),$$

so we already see that $\{u(t_j)\}_{j=0}^n$ solves the discrete system (2.26). Moreover, because H_j is proportional to $\psi_j \psi_j^*$, then $H_j JH_j = 0$ since $\langle J\psi_j, \psi_j \rangle = 0$. It follows that $H_j u = H_j u(t_{j-1})$ on $[t_{j-1}, t_j)$, so on that interval

$$Ju' = -z \frac{1}{t_j - t_{j-1}} H_j u(t_{j-1}) = -z \frac{1}{t_j - t_{j-1}} H_j u = -zHu$$

and u solves the canonical system on $[t_0, t_n)$ with coefficient matrix H . By uniqueness, any solution to the canonical system must have the form (2.29) for some $u(t_0)$, which means that any solution contains a solution to the discrete canonical system, and therefore a solution to the original eigenvalue equation $\tau h = zh$.

2.5.2.3 Transferring the boundary conditions

When we introduced the operator τ earlier, we restricted its domain $\mathfrak{D}_\tau$ in (2.23) with boundary conditions for the eigenvalue equation $\tau h = zh$ to be entirely equivalent to the eigenvalue equation $Yh = zh$ of the original matrix (2.21). These boundary conditions translate to boundary conditions for the canonical system with coefficient matrix (2.28), which is clearly limit circle at both endpoints.

Take an $h \in \mathfrak{D}_\tau$ that satisfies $\tau h = zh$, and let u be the associated canonical system solution. In particular, $\{u(t_j)\}_{j=0}^n$ solves the discrete canonical system, and $u(t_j) = A_j^{-1} F_j$ where $F_j = (h_{j+1}, -a_j h_j)$. Then the left boundary condition $a_0 h_0 = \alpha h_1$ implies that u must satisfy

$$u(t_0) = A_0^{-1} F_0 = \frac{1}{\mathcal{W}_0(g, f)} \begin{pmatrix} -a_0 f_0 & -f_1 \\ a_0 g_0 & g_1 \end{pmatrix} \begin{pmatrix} h_1 \\ -\alpha h_1 \end{pmatrix} = \frac{h_1}{\mathcal{W}_0(g, f)} \begin{pmatrix} -a_0 f_0 + \alpha f_1 \\ a_0 g_0 - \alpha g_1 \end{pmatrix}.$$

Likewise, the right boundary condition $a_n h_{n+1} = \omega h_n$ implies that u must satisfy

$$u(t_n) = A_n^{-1} F_n = \frac{1}{\mathcal{W}_0(g, f)} \begin{pmatrix} -a_n f_n & -f_{n+1} \\ a_n g_n & g_{n+1} \end{pmatrix} \begin{pmatrix} \omega h_n / a_n \\ -a_n h_n \end{pmatrix} = \frac{h_n}{\mathcal{W}_0(g, f)} \begin{pmatrix} -\omega f_n + a_n f_{n+1} \\ \omega g_n - a_n g_{n+1} \end{pmatrix}.$$

Conversely, if at both ends u is parallel to the prescribed vector, then because A_0 and A_n are invertible, the solution h obtained from u belongs to the domain $\mathfrak{D}_\tau$.

To summarize, solutions to $\tau h = zh$ with $h \in \mathfrak{D}_\tau$ can be extracted from canonical system solutions that satisfy

$$u_{t_0}^* J u(t_0) = u_{t_n}^* J u(t_n) = 0 \quad \text{with} \quad u_{t_0} := \begin{pmatrix} -a_0 f_0 + \alpha f_1 \\ a_0 g_0 - \alpha g_1 \end{pmatrix}, \quad u_{t_n} := \begin{pmatrix} -\omega f_n + a_n f_{n+1} \\ \omega g_n - a_n g_{n+1} \end{pmatrix}. \quad (2.30)$$

In particular, if $\alpha = 0$ then the left boundary condition becomes $\langle u(t_0), \psi_0 \rangle = 0$, and if $\omega = 0$ then the right boundary condition becomes $\langle u(t_n), \psi_{n+1} \rangle = 0$.

2.5.3 The associated L^2 space

In the same way as in the case of a Sturm–Liouville operator, the factorization of the coefficient matrix as a projector yields a useful representation of the associated seminorm $\|\cdot\|_H$. Indeed, if $u: (t_0, t_n) \rightarrow \mathbb{C}^2$, then

$$\|u\|_H^2 = \int_{t_0}^{t_n} u^* H u \, d\mathcal{L} = \sum_{j=1}^n \frac{1}{t_j - t_{j-1}} \int_{t_{j-1}}^{t_j} u^* H_j u \, d\mathcal{L},$$

and with the expression (2.27) of H_j , we get

$$\|u\|_H^2 = \frac{1}{\mathcal{W}_0(g, f)} \sum_{j=1}^n \frac{1}{t_j - t_{j-1}} \int_{t_{j-1}}^{t_j} |\langle u, \psi_j \rangle|^2 \, d\mathcal{L}. \quad (2.31)$$

This yields a relationship between the norms of the solutions of the canonical system and those of the eigenvectors of the original tridiagonal matrix. Indeed, if u is a solution to the canonical system, then it has the form (2.29) so $\langle u, \psi_j \rangle = \langle u(t_{j-1}), \psi_j \rangle$ on $[t_{j-1}, t_j]$. If h is the associated solution to $\tau h = zh$, it then follows that

$$\langle u, \psi_j \rangle = \psi_j^* A_{j-1}^{-1} \begin{pmatrix} h_j \\ -a_{j-1} h_{j-1} \end{pmatrix} = \frac{1}{\mathcal{W}_{j-1}(g, f)} \begin{pmatrix} g_j & f_j \end{pmatrix} \begin{pmatrix} -a_{j-1} f_{j-1} & -f_j \\ a_{j-1} g_{j-1} & g_j \end{pmatrix} \begin{pmatrix} h_j \\ -a_{j-1} h_{j-1} \end{pmatrix} = h_j.$$

Therefore, the inner product $\langle u, \psi_j \rangle$ is constant on $[t_{j-1}, t_j]$, and the expression (2.31) simplifies to

$$\|u\|_H^2 = \frac{1}{\mathcal{W}_0(g, f)} \|h\|_2^2 \quad (2.32)$$

where $\|\cdot\|_2$ is the usual Euclidean norm.

2.5.4 Embedding the right boundary condition into the system

An important reason why one would want to represent a Jacobi operator as a canonical system is that such an operator can correspond to a discretization of a continuum operator, which could be for example a differential operator. We will see examples of this in Chapter 3, when studying scaling limits of tridiagonal random matrix models for β -ensembles like those of the Hermite and Laguerre β -ensembles.

In that context, the convergence of the Jacobi operator to the continuum one can be described by taking a limit as the size n of the original matrix tends to infinity. To get a meaningful limit, the canonical systems must then be set up so that their domains (t_0, t_n) at least converge to a well-defined interval. In particular, one could take $t_0 := 0$ and $t_n \rightarrow \infty$, which might be the natural thing to do especially if the canonical system for the limit continuum operator is limit point at infinity.

In that case, it can be more natural to modify the systems in the sequence so that they all act on $(0, \infty)$. This can be done by building the right boundary conditions directly into the coefficient matrices, and effectively making the systems limit point on the right.

To make this precise, consider the same setting as before: we have a canonical system on $(0, t_n)$ with coefficient matrix H given by (2.28), subject to the boundary conditions given in (2.30). Now, extend the definition of H to the full interval $(0, \infty)$ by setting

$$H := -\eta' J u_{t_n} u_{t_n}^* J \quad \text{on} \quad [t_n, \infty) \quad (2.33)$$

for some increasing absolutely continuous function $\eta: [t_n, \infty) \rightarrow [0, \infty)$ such that η' is not integrable. Note that this implies that the system is always limit point at infinity. Now, because H is still a projector, the system's solution on $[t_n, \infty)$ has the same structure as the solution (2.29) on the intervals $[t_{j-1}, t_j)$. Indeed, a direct computation confirms that the solution for $t \geq t_n$ is

$$u(t) = u(t_n) + z(\eta(t) - \eta(t_n)) u_{t_n} u_{t_n}^* J u(t_n) \quad (2.34)$$

where the first term makes the solution continuous at t_n . The norm on $L_H^2(0, \infty)$ from (2.31) is extended for a general $u: (0, \infty) \rightarrow \mathbb{C}^2$ to

$$\|u\|_H^2 = \frac{1}{\mathcal{W}_0(g, f)} \sum_{j=1}^n \frac{1}{t_j - t_{j-1}} \int_{t_{j-1}}^{t_j} |\langle u, \psi_j \rangle|^2 d\mathcal{L} + \int_{t_n}^{\infty} |\langle u, J u_{t_n} \rangle|^2 \eta' d\mathcal{L}. \quad (2.35)$$

If u solves the canonical system, then it has the form (2.34) on $[t_n, \infty)$, and therefore $\langle u, J u_{t_n} \rangle = \langle u(t_n), J u_{t_n} \rangle$ on that interval. Because we chose η' to be nonintegrable, it follows that the solution has finite $L_H^2(0, \infty)$ norm precisely when $\langle u(t_n), J u_{t_n} \rangle = 0$. This integrability condition thus reproduces the boundary condition at t_n from (2.30).

Chapter 3

The stochastic Airy, Bessel and sine operators as canonical systems

This chapter is devoted to presenting the canonical system representations of the stochastic Airy, Bessel and sine operators mentioned in the introduction.

§3.1 and §3.2 are devoted to the stochastic Airy and Bessel operators, and follow similar structures. In both cases, the section is divided in three subsections. In the first one, we present the precise definition of the operator and give some of its most important properties, along with a canonical system representation. In the second one, we construct asymptotics of eigenfunctions on the negative real line, which we will need to apply later in Chapter 4. Finally, in the third one, we prove the convergence of suitably scaled tridiagonal random matrix models to the stochastic operator, at the canonical system level. This proves Theorems 1.5 to 1.8 given in the introduction. Then, §3.3 is devoted to the stochastic sine operator. Our goals in this case are more modest: we only review the operator's definition and some of its important properties in §3.3.1, and then we briefly comment on the possibility of working out a canonical system convergence from tridiagonal random matrix models in §3.3.2.

The properties of the stochastic Airy operator presented in §3.1.1 are mostly review from earlier works, with a few exceptions. A few of those of the stochastic Bessel operator that are given in §3.2.1 are also review from earlier works, but in that case most of the section is in fact taken from [54]. The asymptotics presented in §3.1.2 and §3.2.2 are original and taken respectively from [55] and [54]. The convergence results from tridiagonal random matrix models to the stochastic Airy and Bessel operators presented in §3.1.3 and §3.2.3 are original and have not (at least to my knowledge) appeared anywhere in this form before, but they are essentially reformulations, at the canonical system level, of results from [7, 60] in the Airy case and from [58, 59] in the Bessel case. Finally, the material of §3.3 is essentially review.

More specific references are given throughout the chapter.

3.1 The stochastic Airy operator

This section is devoted to the stochastic Airy operator. In §3.1.1, we review a few of its important properties. In §3.1.2, we derive asymptotics to solutions of $\mathcal{H}_\beta f = 0$ as $t \rightarrow -\infty$. Finally, in §3.1.3, we prove the convergence of suitably scaled tridiagonal random matrix models to the stochastic Airy operator, at the canonical system level.

3.1.1 Basic properties of the stochastic Airy operator

A proper definition. Recall from (1.7) in §1.2 that the stochastic Airy operator $\mathcal{H}_\beta$, heuristically, should be thought of as acting on a function $f: [0, \infty) \rightarrow \mathbb{R}$ as

$$\mathcal{H}_\beta f(t) = -f''(t) + \left(t + \frac{2}{\sqrt{\beta}} B'(t)\right) f(t) \quad (1.7)$$

where B is a standard Brownian motion. The idea of this operator was introduced by Edelman and Sutton in [30], and there are several ways to make its definition rigorous, as of course because of the white noise in the potential, this is not yet a proper definition for $\mathcal{H}_\beta$.

Originally, in order to give a precise definition, Edelman and Sutton rather defined a different operator $\tilde{\mathcal{H}}_\beta$ acting on functions $f \in L^2([0, \infty), \varphi_\beta^2(t) dt)$ where $\varphi_\beta(t) := \exp(\frac{2}{\sqrt{\beta}} \int_0^t B(s) ds)$ as

$$\tilde{\mathcal{H}}_\beta f(t) = -f''(t) - \frac{4}{\sqrt{\beta}} B(t) f'(t) + \left(t - \frac{4}{\beta} B^2(t)\right) f(t).$$

Because the sample paths of Brownian motion are a.s. continuous, this operator is well defined pathwise as a classical Sturm–Liouville operator. In particular, it is self-adjoint when given appropriate boundary conditions. Differentiating formally the expression $\varphi_\beta \tilde{\mathcal{H}}_\beta (\varphi_\beta^{-1} f)(t)$ yields (1.7), which motivated Edelman and Sutton to define the stochastic Airy operator as $\Phi_\beta \circ \tilde{\mathcal{H}}_\beta \circ \Phi_\beta^{-1}$, with $\Phi_\beta: f \mapsto \varphi_\beta f$. It can further be shown that $\Phi_\beta: L^2([0, \infty), \varphi_\beta^2(t) dt) \rightarrow L^2[0, \infty)$ is actually an isometry between Hilbert spaces [8, Appendix A], and because $\tilde{\mathcal{H}}_\beta$ is a Sturm–Liouville operator this could allow one to study the spectral theory of the stochastic Airy operator.

However, this definition is not completely satisfying, as it only defines $\mathcal{H}_\beta$ implicitly. The first direct, rigorous definition of this operator was given by Ramírez, Rider and Virág in [60], who used the theory of Schwartz distributions in order to have a well-defined white noise.

Here, we adopt yet another approach: we see the stochastic Airy operator as a generalized Sturm–Liouville operator in the sense used by Eckhardt et al. [29] that we already saw in §2.4. The fact that the stochastic Airy operator fits this framework was first pointed out by Minami in [52].

Definition 3.1. Let $\beta > 0$ and let B be a standard Brownian motion. The *stochastic Airy operator* $\mathcal{H}_\beta$ is the random generalized Sturm–Liouville operator defined pathwise by

$$\mathcal{H}_\beta f(t) = -\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)'(t) - \frac{2}{\sqrt{\beta}}B(t)\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)(t) + \left(t - \frac{4}{\beta}B^2(t)\right)f(t),$$

acting on functions $f \in \mathfrak{D}_{\mathcal{H}_\beta} := \{f \in \text{AC}_{\text{loc}}(0, \infty) : f' - \frac{2}{\sqrt{\beta}}Bf \in \text{AC}_{\text{loc}}(0, \infty)\}$.

Remark. This operator has the form (2.15) of a generalized Sturm–Liouville operator with $p \equiv 1$, $q(t) := t - \frac{4}{\beta}B^2(t)$, $s := -\frac{2}{\sqrt{\beta}}B$ and $w \equiv 1$.

Like the distribution-based approach, this gives a direct definition of the stochastic Airy operator, in the sense that differentiating formally in the above, one retrieves the heuristic (1.7).

The spectrum of the stochastic Airy operator. When proposing to see the stochastic Airy operator as a generalized Sturm–Liouville operator, Minami also proved the following.

Theorem 3.1 (Adapted from [52, Theorem 1]). *The stochastic Airy operator is limit circle at 0 and limit point at infinity. With Dirichlet boundary condition $f(0) = 0$, it is a.s. self-adjoint and a.s. has purely discrete spectrum.*

Remark. That the spectrum of the operator is a.s. discrete was also proven by Ramírez, Rider and Virág with their distribution-based characterization—see [60, Proposition 3.5].

The spectrum of the stochastic Airy operator under the Dirichlet boundary condition $f(0) = 0$, which is the negative of the so-called Airy_β point process, describes the soft edge scaling limit of β -ensembles. We have seen an example of this in Theorem 1.1 (taken from [60, Theorems 1.1 and 1.4]), which shows that this spectrum is a scaling limit of the Hermite and Laguerre β -ensembles. It has also been shown that the spectrum of the stochastic Airy operator with a different boundary condition is a scaling limit for a perturbed version of the Hermite and Laguerre β -ensembles [7, 8]. We will see a canonical system version of this in §3.1.3.

As mentioned in §1.2, the Airy_β point process arises as a universal scaling limit in the soft edge regime of β -ensembles for large classes of potentials, which go far beyond the Hermite and Laguerre cases [6, 10, 48].

An SDE characterization of solutions to the eigenvalue equation. Since the stochastic Airy operator is a random generalized Sturm–Liouville operator, a solution to its eigenvalue equation $\mathcal{H}_\beta f = zf$ for some $z \in \mathbb{C}$ can be defined pathwise. However, in order to understand the behavior of f as a stochastic process, a different characterization can be obtained through Itô calculus when the initial conditions $f(0)$ and $f'(0)$ are independent of the Brownian motion B that drives $\mathcal{H}_\beta$.

Indeed, if f solves $\mathcal{H}_\beta f = zf$, then its sample paths are absolutely continuous and have a derivative f' a.e. such that $f' - \frac{2}{\sqrt{\beta}}Bf$ is absolutely continuous. As long as $f(0)$ and $f'(0)$ are independent of B , it follows that f is a semimartingale with finite variation. Itô's formula therefore shows that

$$d\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)(t) = df'(t) - \frac{2}{\sqrt{\beta}}f(t) dB(t) - \frac{2}{\sqrt{\beta}}B(t)f'(t) dt.$$

But $f' - \frac{2}{\sqrt{\beta}}Bf$ is absolutely continuous, so its Itô differential is also equal to $(f' - \frac{2}{\sqrt{\beta}}Bf)'(t) dt$, and thus it follows from the eigenvalue equation that

$$d\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)(t) = -\frac{2}{\sqrt{\beta}}B(t)\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)(t) dt + \left(t - z - \frac{4}{\beta}B^2(t)\right)f(t) dt.$$

Comparing these two expressions for $d(f' - \frac{2}{\sqrt{\beta}}Bf)(t)$ shows that f solves

$$\begin{aligned} df(t) &= f'(t) dt, \\ df'(t) &= (t - z)f(t) dt + \frac{2}{\sqrt{\beta}}f(t) dB(t). \end{aligned} \tag{3.1}$$

This stochastic differential equation, called the *stochastic Airy equation*, was studied in detail by Lambert and Paquette in [49].

A translation property of solutions to the eigenvalue equation. We now point out a property of the stochastic Airy operator that will be essential later in §4.4.

Proposition 3.2. *Let $\mathcal{H}_\beta$ be defined on $\mathbb{R}$ from a two-sided standard Brownian motion B , and set $\mathcal{H}_{\beta,E} := 2\sqrt{E}(\mathcal{H}_\beta - E)$ for some $E > 0$. Then f solves $\mathcal{H}_{\beta,E}f = zf$ for some $z \in \mathbb{C}$ if and only if $\tilde{f}(t) := f(t + E)$ solves $\tilde{\mathcal{H}}_\beta \tilde{f} = \frac{z}{2\sqrt{E}}\tilde{f}$ where $\tilde{\mathcal{H}}_\beta$ has the law of $\mathcal{H}_\beta$, but is defined from the two-sided standard Brownian motion $B(\cdot + E) - B(E)$.*

Proof. By definition, for all $t \in \mathbb{R}$,

$$\begin{aligned} \frac{1}{2\sqrt{E}}\mathcal{H}_{\beta,E}f(t + E) &= -\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)'(t + E) - \frac{2}{\sqrt{\beta}}B(t + E)f'(t + E) + tf(t + E) \\ &= -\left(f' - \frac{2}{\sqrt{\beta}}(B - B(E))f\right)'(t + E) \\ &\quad - \frac{2}{\sqrt{\beta}}(B(t + E) - B(E))f'(t + E) + tf(t + E), \end{aligned}$$

since the extra $B(E)$ terms cancel out. But this is precisely the expression of $\tilde{\mathcal{H}}_\beta \tilde{f}(t)$, so if $\mathcal{H}_{\beta,E}f = zf$ then $\tilde{\mathcal{H}}_\beta \tilde{f}(t) = \frac{1}{2\sqrt{E}}\mathcal{H}_{\beta,E}f(t + E) = \frac{z}{2\sqrt{E}}f(t + E) = \frac{z}{2\sqrt{E}}\tilde{f}(t)$ for all $t \in \mathbb{R}$, and vice versa. ■

Canonical system representation. As a random generalized Sturm–Liouville operator, the stochastic Airy operator can be associated with a random canonical system that is equivalent to its eigenvalue equation, following the method detailed in §2.4. The equivalence can be made through the matrix

$$A_\beta := \begin{pmatrix} \mathfrak{g}_\beta^{[1]} & \mathfrak{f}_\beta^{[1]} \\ \mathfrak{g}_\beta & \mathfrak{f}_\beta \end{pmatrix} = \begin{pmatrix} \mathfrak{g}'_\beta - \frac{2}{\sqrt{\beta}} B \mathfrak{g}_\beta & \mathfrak{f}'_\beta - \frac{2}{\sqrt{\beta}} B \mathfrak{f}_\beta \\ \mathfrak{g}_\beta & \mathfrak{f}_\beta \end{pmatrix} \quad (3.2)$$

where $\mathfrak{f}_\beta$ and $\mathfrak{g}_\beta$ are a pair of fundamental solutions to $\mathcal{H}_\beta h = 0$ with initial conditions $\mathfrak{g}_\beta(0) = \mathfrak{f}'_\beta(0) = 0$ and $\mathfrak{g}'_\beta(0) = \mathfrak{f}_\beta(0) = 1$, and where we write $h^{[1]} := h' - \frac{2}{\sqrt{\beta}} B h$ for quasi-derivatives. The matrix A_β thus solves

$$A'_\beta(t) = \begin{pmatrix} -\frac{2}{\sqrt{\beta}} B(t) & t - \frac{4}{\beta} B^2(t) \\ 1 & \frac{2}{\sqrt{\beta}} B(t) \end{pmatrix} A_\beta(t) \quad \text{with} \quad A_\beta(0) = I_2. \quad (3.3)$$

The method from §2.4.2 shows that h solves $\mathcal{H}_\beta h = zh$ if and only if $u := A_\beta^{-1} \begin{pmatrix} h^{[1]} \\ h \end{pmatrix}$ solves the canonical system $Ju' = -zH_\beta u$ on $(0, \infty)$, where

$$H_\beta := JA_\beta^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} A_\beta = \begin{pmatrix} \mathfrak{g}_\beta^2 & \mathfrak{f}_\beta \mathfrak{g}_\beta \\ \mathfrak{f}_\beta \mathfrak{g}_\beta & \mathfrak{f}_\beta^2 \end{pmatrix} = \psi_\beta \psi_\beta^* \quad \text{where} \quad \psi_\beta := \begin{pmatrix} \mathfrak{g}_\beta \\ \mathfrak{f}_\beta \end{pmatrix}. \quad (3.4)$$

This canonical system is limit circle at 0, and it is limit point at infinity because $\mathcal{H}_\beta$ is. Hence, the eigenvalue problem $\mathcal{H}_\beta h = zh$ with $h(0) = 0$ corresponds to a self-adjoint realization of the canonical system's maximal relation obtained by restricting the maximal relation with the boundary condition $e_0^* Ju(0) = 0$.

As pointed out in §2.4.3, the norm of the associated $L_{H_\beta}^2(0, \infty)$ space has a specific form in terms of the solutions $\mathfrak{f}_\beta$ and $\mathfrak{g}_\beta$. Indeed, if $u: (0, \infty) \rightarrow \mathbb{C}^2$, then

$$\|u\|_{H_\beta}^2 = \int_0^\infty |\langle u(t), \psi_\beta(t) \rangle|^2 dt. \quad (3.5)$$

In particular, if $u = A_\beta^{-1} \begin{pmatrix} h^{[1]} \\ h \end{pmatrix}$ for some $h: (0, \infty) \rightarrow \mathbb{C}$, then $\|u\|_{H_\beta} = \|h\|_2$.

3.1.2 Asymptotics of solutions to the eigenvalue equation as $t \rightarrow -\infty$

In this section, we understand the stochastic Airy operator as being defined on the whole of $\mathbb{R}$ from a two-sided Brownian motion, and we derive asymptotics for solutions to the eigenvalue equation $\mathcal{H}_\beta f = 0$ as $t \rightarrow -\infty$. In addition to providing helpful insight into the behavior of these solutions on the negative real line, these asymptotics will be crucial in our spectral analysis of the soft edge to bulk transition later in §4.4.2.

The asymptotics we derive, which are essentially taken from [55, §7.1], are the following.

Theorem 3.3 [55, Theorem 3]. Let $\mathcal{H}_\beta$ be defined on the full real line from a two-sided standard Brownian motion B . If f solves $\mathcal{H}_\beta f = 0$ and if $f(-1)$ and $f'(-1)$ are independent of the restriction of B to $(-\infty, -1)$, then for $t \geq 1$,

$$f(-t) = C_f t^{-1/4+1/2\beta} e^{X(t)} \cos \xi_\beta(t) \quad \text{and} \quad f'(-t) = C_f t^{1/4+1/2\beta} e^{X(t)} \sin \xi_\beta(t)$$

where $C_f^2 := f^2(-1) + f'^2(-1)$, ξ_β is a process that satisfies $\xi_\beta(t) - 2\pi \lfloor \frac{\xi_\beta(t)}{2\pi} \rfloor \rightarrow U \sim \text{Unif}[0, 2\pi)$ in law as $t \rightarrow \infty$, and X is a process such that for any $\varepsilon, \delta > 0$, there is a $C > 0$ for which

$$\mathbb{P} \left[\forall t \geq 1, |X(t)| \leq C(1 + (\log t)^{1/2+\delta}) \right] \geq 1 - \varepsilon.$$

Remark. These asymptotics complement the $t \rightarrow \infty$ asymptotics for the same solutions that were derived in [49].

The idea of the proof is to construct a pair of polar coordinates to describe the growth and oscillation of f and f' , and then to analyse their asymptotic behavior. So, our first step is the following.

Proposition 3.4 [55, Proposition 23]. Let $\mathcal{H}_\beta$ be defined from a two-sided standard Brownian motion B , and let $\tilde{B}(t) := B(-t)$. If f solves $\mathcal{H}_\beta f = 0$ and $f(-1)$ and $f'(-1)$ are independent of the restriction of $\tilde{B}$ to $(1, \infty)$, then for $t \geq 1$,

$$f(-t) = t^{-1/4} e^{r_\beta(t)} \cos \xi_\beta(t) \quad \text{and} \quad f'(-t) = -t^{1/4} e^{r_\beta(t)} \sin \xi_\beta(t)$$

where r_β and ξ_β solve the stochastic differential equations

$$\begin{aligned} dr_\beta(t) &= \frac{1}{t} \left(\frac{1}{2\beta} + \left(\frac{1}{4} + \frac{1}{\beta} \right) \cos 2\xi_\beta(t) + \frac{1}{2\beta} \cos 4\xi_\beta(t) \right) dt - \frac{1}{\sqrt{\beta t}} \sin 2\xi_\beta(t) d\tilde{B}(t), \\ d\xi_\beta(t) &= -\sqrt{t} dt - \frac{1}{t} \left(\left(\frac{1}{4} + \frac{1}{\beta} \right) \sin 2\xi_\beta(t) + \frac{1}{2\beta} \sin 4\xi_\beta(t) \right) dt - \frac{2}{\sqrt{\beta t}} \cos^2 \xi_\beta(t) d\tilde{B}(t) \end{aligned}$$

with $r_\beta(1) = \frac{1}{2} \log(f^2(-1) + f'^2(-1))$ and $\xi_\beta(1) = \arctan(f'(-1)/f(-1))$.

Proof. Set $y(t) := f(-t)$. Reversing time in the equation $\mathcal{H}_\beta f = 0$ shows that

$$0 = -\left(y' + \frac{2}{\sqrt{\beta}} \tilde{B}y\right)'(t) + \frac{2}{\sqrt{\beta}} \tilde{B}(t) \left(y' + \frac{2}{\sqrt{\beta}} \tilde{B}y\right)(t) - \left(t + \frac{4}{\beta} \tilde{B}^2(t)\right)y(t),$$

and then applying Itô's formula in the same way as we did to derive the stochastic Airy equation (3.1) (which is justified by the independence of $y(1)$ and $y'(1)$ on the restriction of $\tilde{B}$ to $(1, \infty)$) shows

that y' satisfies

$$dy'(t) = -ty(t) dt - \frac{2}{\sqrt{\beta}} y(t) d\tilde{B}(t).$$

Now, define the real-valued processes r_β and ξ_β so that $e^{r_\beta(t) + i\xi_\beta(t)} = t^{1/4}y(t) + it^{-1/4}y'(t)$. By Itô's formula,

$$\begin{aligned} dr_\beta(t) + i d\xi_\beta(t) &= \frac{1}{t^{1/4}y(t) + it^{-1/4}y'(t)} \left(\frac{1}{4} t^{-3/4} y(t) dt - \frac{i}{4} t^{-5/4} y'(t) dt + t^{1/4} y'(t) dt + it^{-1/4} dy'(t) \right) \\ &\quad + \frac{1}{2} \frac{1}{(t^{1/4}y(t) + it^{-1/4}y'(t))^2} t^{-1/2} d\langle y' \rangle(t) \\ &= e^{-2r_\beta(t)} \left(\left(\frac{1}{4} t^{-1/2} y^2(t) - \frac{1}{4} t^{-3/2} y'^2(t) \right) dt - \frac{2}{\sqrt{\beta}t} y(t) y'(t) d\tilde{B}(t) \right. \\ &\quad \left. - i \left(\frac{1}{2t} y(t) y'(t) + y'^2(t) + t y^2(t) \right) dt - \frac{2i}{\sqrt{\beta}} y^2(t) d\tilde{B}(t) \right) \\ &\quad + \frac{2}{\beta} e^{-4r_\beta(t)} t^{-1/2} y^2(t) \left(t^{1/2} y^2(t) - t^{-1/2} y'^2(t) - 2iy(t) y'(t) \right) dt. \end{aligned}$$

As $e^{r_\beta(t)} \cos \xi_\beta(t) = t^{1/4}y(t)$ and $e^{r_\beta(t)} \sin \xi_\beta(t) = t^{-1/4}y'(t)$, this simplifies to

$$\begin{aligned} dr_\beta(t) + i d\xi_\beta(t) &= \frac{1}{4t} (\cos^2 \xi_\beta(t) - \sin^2 \xi_\beta(t)) dt - \frac{2}{\sqrt{\beta}t} \cos \xi_\beta(t) \sin \xi_\beta(t) d\tilde{B}(t) \\ &\quad - i \left(\frac{1}{2t} \cos \xi_\beta(t) \sin \xi_\beta(t) + \sqrt{t} \right) dt - \frac{2i}{\sqrt{\beta}t} \cos^2 \xi_\beta(t) d\tilde{B}(t) \\ &\quad + \frac{2}{\beta} \left(\frac{1}{t} \cos^4 \xi_\beta(t) - \frac{1}{t} \cos^2 \xi_\beta(t) \sin^2 \xi_\beta(t) - \frac{2i}{t} \cos^3 \xi_\beta(t) \sin \xi_\beta(t) \right) dt. \end{aligned}$$

The announced stochastic differential equations then follow from taking the real and imaginary parts in the above and simplifying with trigonometric identities. ■

We now analyse the asymptotic behavior of these polar coordinates r_β and ξ_β . This is done in Propositions 3.6 and 3.7 respectively, and Theorem 3.3 is a direct consequence of these two results. Before proceeding, we need a simple preliminary result.

Lemma 3.5 [55, Lemma 24]. *Let ξ_β solve the SDE from Proposition 3.4. If $k \neq 0$ and $\varepsilon > 0$, there is a $C > 0$ such that*

$$\mathbb{P} \left[\sup_{t \geq 1} \left| \int_1^t \frac{e^{ki\xi_\beta(s)}}{s} ds \right| > C \right] < \varepsilon.$$

Proof. By Itô's formula,

$$d(e^{ki\xi_\beta})(s) = e^{ki\xi_\beta(s)} \left(-ki\sqrt{s} ds - \frac{3k^2}{4\beta s} ds - \frac{1}{s} R_k(s) ds - \frac{2ki}{\sqrt{\beta}s} \cos^2 \xi_\beta(s) d\tilde{B}(s) \right) \quad (3.6)$$

where

$$R_k := ki \left(\frac{1}{4} + \frac{1}{\beta} \right) \sin 2\xi_\beta + \frac{ki}{2\beta} \sin 4\xi_\beta + \frac{k^2}{\beta} \cos 2\xi_\beta + \frac{k^2}{4\beta} \cos 4\xi_\beta. \quad (3.7)$$

Therefore,

$$e^{ki\xi_\beta(s)} ds = \frac{i}{k\sqrt{s}} d(e^{ki\xi_\beta})(s) + \frac{ie^{ki\xi_\beta(s)}}{k\sqrt{s}} \left(\frac{3k^2}{4\beta s} ds + \frac{1}{s} R_k(s) ds + \frac{2ki}{\sqrt{\beta s}} \cos^2 \xi_\beta(s) d\tilde{B}(s) \right),$$

and

$$\begin{aligned} \int_1^t \frac{e^{ki\xi_\beta(s)}}{s} ds &= \frac{i}{k} \int_1^t \frac{1}{s^{3/2}} d(e^{ki\xi_\beta})(s) + \int_1^t \left(\frac{3ki}{4\beta} + \frac{i}{k} R_k(s) \right) \frac{e^{ki\xi_\beta(s)}}{s^{5/2}} ds \\ &\quad - \frac{2}{\sqrt{\beta}} \int_1^t \frac{e^{ki\xi_\beta(s)}}{s^2} \cos^2 \xi_\beta(s) d\tilde{B}(s). \end{aligned} \quad (3.8)$$

Integrating the first term by parts gives

$$\frac{i}{k} \int_1^t \frac{1}{s^{3/2}} d(e^{ki\xi_\beta})(s) = \frac{i}{k} \frac{e^{ki\xi_\beta(s)}}{s^{3/2}} \Big|_1^t + \frac{3i}{2k} \int_1^t \frac{e^{ki\xi_\beta(s)}}{s^{5/2}} ds.$$

The first term is immediately bounded by $2/k$, and since $s \mapsto s^{-5/2}$ is integrable on $[1, \infty)$, the remaining integral is also bounded by a constant depending only on k . In the same way, by definition of R_k , the second integral of (3.8) is bounded by a constant depending only on k and β . As to the remaining stochastic integral in (3.8), its bracket is bounded by

$$\frac{4}{\beta} \int_1^t \frac{1}{s^4} \cos^4 \xi_\beta(s) ds \leq \frac{4}{3\beta},$$

and thus Bernstein's inequality for martingales (see e.g. [62, Exercise IV.3.16] for a general statement of that inequality) shows that

$$\mathbb{P} \left[\sup_{t \geq 1} \frac{2}{\beta} \left| \int_1^t \frac{e^{ki\xi_\beta(s)}}{s^2} \cos^2 \xi_\beta(s) d\tilde{B}(s) \right| > x \right] \leq 4 \exp \left(-\frac{3\beta x^2}{8} \right).$$

Since the first line of (3.8) is bounded by a constant $C' > 0$, taking x so that $4 \exp(-\frac{3\beta x^2}{8}) < \varepsilon$ yields the result with $C := C' + x$. ■

Proposition 3.6 [55, Proposition 25]. *Let r_β solve the SDE from Proposition 3.4. For all $\varepsilon, \delta > 0$, there is a $C > 0$ such that*

$$\mathbb{P} \left[\forall t \geq 1, \left| r_\beta(t) - r_\beta(1) - \frac{1}{2\beta} \log t \right| \leq C(1 + (\log t)^{1/2+\delta}) \right] \geq 1 - \varepsilon.$$

Proof. Recall from Proposition 3.4 that r_β solves

$$dr_\beta(t) = \frac{1}{t} \left(\frac{1}{2\beta} + \left(\frac{1}{4} + \frac{1}{\beta} \right) \cos 2\xi_\beta(t) + \frac{1}{2\beta} \cos 4\xi_\beta(t) \right) dt - \frac{1}{\sqrt{\beta t}} \sin 2\xi_\beta(t) d\tilde{B}(t)$$

where ξ_β solves the other SDE in Proposition 3.4. By Lemma 3.5, given $\varepsilon > 0$, there are $C_2, C_4 > 0$ such that for $k = 2$ and $k = 4$,

$$\mathbb{P}\left[\sup_{t \geq 1} \left| \int_1^t \frac{\cos k \xi_\beta(s)}{s} \right| > C_k\right] < \frac{\varepsilon}{3}. \quad (3.9)$$

Then, the martingale

$$M(t) := \frac{1}{\sqrt{\beta}} \int_1^{1+t} \frac{\sin 2\xi_\beta(s)}{\sqrt{s}} d\tilde{B}(s) \quad \text{has} \quad \langle M \rangle(t) = \frac{1}{\beta} \int_1^{1+t} \frac{\sin^2 2\xi_\beta(s)}{s} ds \leq \frac{1}{\beta} \log(1+t).$$

By properties of continuous martingales (e.g., Proposition A.1), it follows that for any $\delta > 0$ there is a $C > 0$ such that

$$\mathbb{P}\left[\sup_{t \geq 1} \frac{|M(t-1)|}{1 + (\frac{1}{\beta} \log t)^{1/2+\delta}} \geq C\right] < \frac{\varepsilon}{3},$$

and the result follows by combining this event with those of (3.9). $\blacksquare$

Proposition 3.7 [55, Proposition 26]. *Let ξ_β solve the SDE from Proposition 3.4. Then $e^{i\xi_\beta(t)} \rightarrow U$ in law as $t \rightarrow \infty$ where $U \sim \text{Unif } \mathbb{S}^1$.*

Proof. Recall that if $U \sim \text{Unif } \mathbb{S}^1$, then $\mathbb{E} U^m = 0$ for any integer $m \neq 0$. Hence, by density of trigonometric polynomials in continuous functions on the unit circle, it suffices to prove that $\mathbb{E} e^{mi\xi_\beta(t)} \rightarrow 0$ as $t \rightarrow \infty$ for all $m \neq 0$. This means that it suffices to prove that $\varphi_m(t) \rightarrow 0$ as $t \rightarrow \infty$ for all $m \neq 0$ for $\varphi_m := \mathbb{E} X_m$ with $X_m(t) := \exp(mi(\xi_\beta(t) + \frac{2}{3}t^{3/2}))$.

Now, from the expression (3.6) of the Itô differential of $e^{mi\xi_\beta}$,

$$dX_m(s) = -X_m(s) \left(\frac{3m^2}{4\beta s} + \frac{1}{s} R_m(s) \right) ds - \frac{2mi}{\sqrt{\beta s}} X_m(s) \cos^2 \xi_\beta(s) d\tilde{B}(s),$$

so

$$\varphi_m(t) = \mathbb{E} X_m(1) - \mathbb{E} \int_1^t X_m(s) \left(\frac{3m^2}{4\beta s} + \frac{1}{s} R_m(s) \right) ds,$$

and it follows that

$$\varphi'_m(s) = -\frac{3m^2}{4\beta s} \varphi_m(s) - \frac{1}{s} \mathbb{E} X_m(s) R_m(s).$$

To simplify notation, set $p := \frac{3m^2}{4\beta}$. Multiplying by s^p and integrating, we get another expression for φ_m :

$$\varphi_m(t) = t^{-p} \varphi_m(1) - \mathbb{E} t^{-p} \int_1^t s^{p-1} X_m(s) R_m(s) ds.$$

As $p > 0$, it is clear that the first term vanishes as $t \rightarrow \infty$. Then, recall that R_m was defined in (3.7) as a linear combination of $e^{\pm 2i\xi_\beta}$ and $e^{\pm 4i\xi_\beta}$. Therefore, in order to prove that $\varphi_m(t) \rightarrow 0$, it suffices to prove that

$$\psi_{m,k}(t) := \frac{1}{t^p} \mathbb{E} \int_1^t s^{p-1} X_m(s) e^{ki\xi_\beta(s)} ds \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

for $k = \pm 2$ and $k = \pm 4$.

Using again the expression (3.6) of the Itô differential of $e^{ki\xi_\beta}$, we can write

$$\psi_{m,k}(t) = \frac{i}{kt^p} \left(\mathbb{E} \int_1^t s^{p-3/2} X_m(s) d(e^{ki\xi_\beta})(s) + \mathbb{E} \int_1^t s^{p-5/2} X_m(s) e^{ki\xi_\beta(s)} \left(\frac{3k^2}{4\beta} + R_k(s) \right) ds \right).$$

Now, if $Y_m(s) := s^{p-3/2} X_m(s)$, then

$$dY_m(s) = -\frac{3}{2} s^{p-5/2} X_m(s) ds - s^{p-5/2} X_m(s) R_m(s) ds - \frac{2mi}{\sqrt{\beta}} s^{p-2} X_m(s) \cos^2 \xi_\beta(s) d\tilde{B}(s),$$

so

$$d\langle Y_m, e^{ki\xi_\beta} \rangle(s) = -\frac{4km}{\beta} s^{p-5/2} X_m(s) e^{ki\xi_\beta(s)} \cos^4 \xi_\beta(s) ds.$$

From this, we can integrate the first term of $\psi_{m,k}$ by parts:

$$\begin{aligned} \mathbb{E} \int_1^t s^{p-3/2} X_m(s) d(e^{ki\xi_\beta})(s) &= \mathbb{E} s^{p-3/2} X_m(s) e^{ki\xi_\beta(s)} \Big|_1^t + \frac{3}{2} \mathbb{E} \int_1^t s^{p-5/2} X_m(s) e^{ki\xi_\beta(s)} ds \\ &\quad + \mathbb{E} \int_1^t s^{p-5/2} X_m(s) R_m(s) e^{ki\xi_\beta(s)} ds + \frac{4km}{\beta} \mathbb{E} \int_1^t s^{p-5/2} X_m(s) e^{ki\xi_\beta(s)} \cos^4 \xi_\beta(s) ds. \end{aligned}$$

Finally, because X_m , R_k , R_m and $e^{ki\xi_\beta}$ are all bounded processes, substituting the result of this integration by parts in the expression of $\psi_{m,k}(t)$ shows that $|\psi_{m,k}(t)| \lesssim t^{-3/2}$, so that $\psi_{m,k}(t) \rightarrow 0$ as $t \rightarrow \infty$, which completes the proof. ■

3.1.3 Canonical system convergence from tridiagonal matrix models

In this section, we show convergence of tridiagonal random matrix models to the stochastic Airy operator by representing both as canonical systems.

As mentioned in §1.2, the idea that tridiagonal random matrix models can, in the right scaling, correspond to discrete-time approximations of the stochastic Airy operator comes from Edelman and Sutton [30]. This was made rigorous for the first time by Ramírez, Rider and Virág in [60], who showed that for any $k \in \mathbb{N}$, the first k eigenvalues of suitable tridiagonal random matrix models converge jointly in law to the first k points of the Airy_β point process. This was then extended by Bloemendal and Virág in [7] to perturbed versions of these tridiagonal random matrix models, whose eigenvalues converge (in the same sense) to those of the stochastic Airy operator with a different boundary condition at the origin than the Dirichlet condition $h(0) = 0$. Heuristically, the perturbations act as boundary conditions for the tridiagonal matrix models, and they yield a different boundary condition for the limit operator if they remain consistent as the matrix size grows.

Several results of this section can thus be seen as reformulations, at the canonical system level, of some of the main results of [7, 60]. The advantage of the present formulation is to upgrade the

results to the operator level, under a framework that can also be used to describe other scaling limits of β -ensembles.

The section is organized as follows. We first introduce the random matrix setup that we consider in §3.1.3.1, where we give the main results as well. The proofs are given in §3.1.3.2, and the results are applied to the Hermite and Laguerre β -ensembles in §3.1.3.3.

3.1.3.1 Setup and statement of the results

Tridiagonal random matrix model. Consider an $n \times n$ tridiagonal random matrix

$$M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^\top := \begin{pmatrix} b_{n,1} + \alpha_n/\delta_n^2 & a_{n,1} & & & \\ a_{n,1} & \ddots & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & & a_{n,n-1} & b_{n,n} \\ & & & a_{n,n-1} & b_{n,n} \end{pmatrix} \quad (3.10)$$

where $\delta_n > 0$ and $a_{n,j}, b_{n,j}, \alpha_n \in \mathbb{R}$ with all $a_{n,j}$'s of the same sign. We suppose that M_n is scaled at the soft edge of β -ensembles, meaning essentially that it should discretize the stochastic Airy operator with time step δ_n . More specifically, what we assume is the following.

Hypothesis 3.1. The entries $a_{n,j}, b_{n,j}$ of the random matrix M_n in (3.10) have the form

$$a_{n,j} = -\frac{1}{\delta_n^2} + \frac{2}{\sqrt{\beta}\delta_n} \Delta Z_{n,j}^a + V_{n,j}^a \quad \text{and} \quad b_{n,j} = \frac{2}{\delta_n^2} + \frac{2}{\sqrt{\beta}\delta_n} \Delta Z_{n,j}^b + V_{n,j}^b$$

where

- i) $\delta_n \rightarrow 0$ but $n\delta_n \rightarrow \infty$ as $n \rightarrow \infty$.
- ii) for any $\tau > 0$, both $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n,j}^a| \rightarrow 0$ and $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n,j}^b| \rightarrow 0$ as $n \rightarrow \infty$, and the process $\mathcal{Z}_n(t) := \sum_{j=1}^{\lfloor t/\delta_n \rfloor} \Delta Z_{n,j}$ with $\Delta Z_{n,j} := 2\Delta Z_{n,j}^a + \Delta Z_{n,j}^b$ converges compactly in law to a standard Brownian motion,
- iii) for any $\tau > 0$, $\max_{1 \leq j \leq \tau/\delta_n} |V_{n,j-1}^a + V_{n,j}^a + V_{n,j}^b - (j-1)\delta_n| \rightarrow 0$ in law as $n \rightarrow \infty$ and both $\{V_{n,j}^a\}_{j=1}^{\lfloor \tau/\delta_n \rfloor}$ and $\{V_{n,j}^b\}_{j=1}^{\lfloor \tau/\delta_n \rfloor}$ are tight.

These conditions ensure that the entries of M_n are consistent with the soft edge scaling. As we saw when building canonical systems out of Jacobi matrices in §2.5, the first entry of the matrix is what fixes the left boundary condition of the associated canonical system. To ensure that the boundary conditions for $M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^\top$ are consistent when n grows, we add the following hypothesis.

Hypothesis 3.2. $\delta_n^{-1}(1 + \alpha_n) \rightarrow \cot \phi$ in law as $n \rightarrow \infty$ for a deterministic $\phi \in [0, \pi]$.

Remark. In particular, this is automatically satisfied with $\phi = 0$ if $\alpha_n = 0$ for all n .

Canonical system representation. As a random Jacobi matrix, M_n (or equivalently $M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^T$) can be represented as a canonical system following the method from §2.5. To do so, we must fix two extra entries $a_{n,0} := -1/\delta_n^2$ and $a_{n,n} := -1$. Then, we introduce a pair $f_{\beta,n}, g_{\beta,n}$ of fundamental solutions to the three-term recurrence

$$zh_j = a_{n,j-1}h_{j-1} + b_{n,j}h_j + a_{n,j}h_{j+1} \quad \text{for } j \in \{1, \dots, n\} \quad (3.11)$$

for $z = 0$, with initial conditions $f_{\beta,n,0} = f_{\beta,n,1} = 1$, $g_{\beta,n,0} = 0$ and $g_{\beta,n,1} = \delta_n$. These initial conditions are chosen to reproduce those of the solutions f_β and g_β from the stochastic Airy operator's coefficient matrix (3.4). Note that these solutions have Wronskian

$$\mathcal{W}_0(g_{\beta,n}, f_{\beta,n}) = a_0(g_{\beta,n,0}f_{\beta,n,1} - g_{\beta,n,1}f_{\beta,n,0}) = \frac{1}{\delta_n}.$$

As in §2.5, we can use these solutions to define a discrete canonical system which can be extended to continuous time by setting the coefficient matrix as being constant on chosen intervals. Taking the time steps to be given by the sequence $t_j := j\delta_n$, this results in a canonical system $Ju' = -zH_{\beta,n}u$ on $[0, t_n)$ with coefficient matrix

$$H_{\beta,n} \equiv \begin{pmatrix} g_{\beta,n,j}^2 & f_{\beta,n,j}g_{\beta,n,j} \\ f_{\beta,n,j}g_{\beta,n,j} & f_{\beta,n,j}^2 \end{pmatrix} \quad \text{on } [t_{j-1}, t_j]. \quad (3.12a)$$

Note that under Hypothesis 3.1, $t_n \rightarrow \infty$. This canonical system is equivalent to the eigenvalue equation (3.11) in the sense that a solution $u: [0, t_n) \rightarrow \mathbb{C}^2$ must satisfy $u(t_j) = A_{\beta,n,j}^{-1}F_j$ for

$$A_{\beta,n,j} := \begin{pmatrix} g_{\beta,n,j+1} & f_{\beta,n,j+1} \\ -a_{n,j}g_{\beta,n,j} & -a_{n,j}f_{\beta,n,j} \end{pmatrix} \quad \text{and} \quad F_j := \begin{pmatrix} h_{j+1} \\ -a_{n,j}h_j \end{pmatrix}$$

for an h solving (3.11).

To make the canonical system fully equivalent to the eigenvalue equation for $M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^T$, it remains to implement the appropriate boundary conditions. Since the stochastic Airy operator is defined on $[0, \infty)$ and is limit point at infinity, it is slightly more convenient to extend the above canonical system's domain to $[0, \infty)$ by embedding the boundary condition at t_n into the system, effectively making it limit point at infinity as well. Following the method from §2.5.4, we extend the domain to $[0, \infty)$ by setting

$$H_{\beta,n} \equiv \begin{pmatrix} g_{\beta,n,n+1}^2 & f_{\beta,n,n+1}g_{\beta,n,n+1} \\ f_{\beta,n,n+1}g_{\beta,n,n+1} & g_{\beta,n,n+1}^2 \end{pmatrix} \quad \text{on } [t_n, \infty). \quad (3.12b)$$

As seen in §2.5.4, this makes the system limit point at infinity, and the integrability condition on solutions acts as a boundary condition that reproduces the action of the lower right corner of $M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^T$ on its eigenvectors.

At 0, the boundary condition that needs to be added is given by (2.30). The resulting boundary condition can be written as

$$\mathbf{u}_n^* J u(0) = 0 \quad \text{for} \quad \mathbf{u}_n := \begin{pmatrix} 1/\delta_n^2 + \alpha_n/\delta_n^2 \\ -\alpha_n/\delta_n \end{pmatrix} = \frac{1}{\delta_n} \begin{pmatrix} \delta_n^{-1}(1 + \alpha_n) \\ 1 - (1 + \alpha_n) \end{pmatrix}. \quad (3.13)$$

Note that any scaling of the vector $\mathbf{u}_n$ can also be used instead of $\mathbf{u}_n$ itself without changing the boundary condition. This will be convenient later to see how the boundary condition changes as n grows. In fact, we immediately see that if α_n satisfies Hypothesis 3.2, then $\delta_n \mathbf{u}_n \rightarrow (\cot \phi, 1)$ in law if $|\cot \phi| < \infty$ or $\delta_n^2(1 + \alpha_n)^{-1} \mathbf{u}_n \rightarrow (1, 0)$ in law if $\cot \phi = \pm\infty$.

Thus, by construction, if an integrable solution $u: [0, \infty) \rightarrow \mathbb{R}^2$ to the canonical system with coefficient matrix (3.12) and spectral parameter $z \in \mathbb{R}$ satisfies the left boundary condition (3.13), then one can extract an eigenvector $h \in \mathbb{R}^n$ of $M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^\top$ from u by taking h_j to be the first entry of $A_{\beta,n,j-1} u(t_{j-1})$ for each $j \in \{1, \dots, n\}$. By the computation from §2.5.3, the norms of u and h are related by $\|u\|_{H_{\beta,n}}^2 = \delta_n \|h\|_2^2$.

Convergence to the stochastic Airy operator. In this setup, we will show that the canonical system for M_n converges vaguely to that of the stochastic Airy operator.

Theorem 3.8. *Suppose that the $n \times n$ tridiagonal random matrix M_n from (3.10) satisfies Hypothesis 3.1. Let $H_{\beta,n}$ denote the coefficient matrix (3.12) of the associated canonical system, and let H_β denote the coefficient matrix (3.4) of the stochastic Airy operator's canonical system. Then $H_{\beta,n} \xrightarrow{\text{law}} H_\beta$ vaguely on $[0, \infty)$.*

As a corollary of this result, we also obtain the following.

Corollary 3.8.1. *In the setting of Theorem 3.8, let $T_{H_{\beta,n}}, T_{H_\beta}: [0, \infty) \times \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ denote the canonical systems' transfer matrices. Then $T_{H_{\beta,n}} \xrightarrow{\text{law}} T_{H_\beta}$ compactly on $[0, \infty) \times \mathbb{C}$, jointly with the convergence of coefficient matrices.*

Under Hypothesis 3.2, the boundary conditions for $M_n + \frac{\alpha_n}{\delta_n^2} \mathbf{e}_1 \mathbf{e}_1^\top$ are consistent in the limit, so this also extends to the Weyl–Titchmarsh functions and spectral measures of these canonical systems.

Corollary 3.8.2. *In the setting of Theorem 3.8, suppose additionally that Hypothesis 3.2 holds. Let $m_{H_{\beta,n}}^{\mathbf{u}_n}, m_{H_\beta}^\phi: \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ denote the Weyl–Titchmarsh functions of $H_{\beta,n}$ and H_β with boundary conditions $\mathbf{u}_n^* J u(0) = 0$ and $e_\phi^* J u(0) = 0$ respectively, with $\mathbf{u}_n$ as in (3.13). Then $m_{H_{\beta,n}}^{\mathbf{u}_n} \xrightarrow{\text{law}} m_{H_\beta}^\phi$ compactly on $\mathbb{H}$, jointly with both coefficient and transfer matrices. In particular, if $\mu_{H_{\beta,n}}^{\mathbf{u}_n}$ and $\mu_{H_\beta}^\phi$ are the spectral measures associated with $m_{H_{\beta,n}}^{\mathbf{u}_n}$ and $m_{H_\beta}^\phi$, then $\mu_{H_{\beta,n}}^{\mathbf{u}_n} \xrightarrow{\text{law}} \mu_{H_\beta}^\phi$ vaguely, also jointly with coefficient and transfer matrices and Weyl–Titchmarsh functions.*

It should be noted that while these results do give an interesting operator level picture of tridiagonal random matrix models scaled at the soft edge, the convergence of the spectral measures proven in Corollary 3.8.2 does not directly imply the convergence of the associated eigenvalues point processes, and therefore it does not recover the results of [7, 60] mentioned earlier (in particular in Theorem 1.1). Indeed, as discussed in a general setting in §2.2.2.2, even though the spectral measures $\mu_{H_{\beta,n}}^{u_n}$ converge to the spectral measure $\mu_{H_{\beta}}^{\phi}$ of the Airy system, at this point nothing rules out the possibility that atoms of $\mu_{H_{\beta,n}}^{u_n}$ merge or that their spectral weights vanish in the limit. Being able to deduce the point process convergence from Corollary 3.8.2 would require a more careful analysis of the spectral weights, possibly along the lines of what we saw in Proposition 2.18 for systems that are limit circle at both endpoints.

3.1.3.2 Proof of the canonical system convergence

In this section, we prove Theorem 3.8 and its corollaries. To do so, we actually prove that, under an appropriate coupling, the solutions that appear in the coefficient matrix of the finite- n canonical system converge compactly in probability to those of the limit canonical system. We prove this by comparing the solutions at the discrete times where the solutions of the finite- n system jump.

To carry out this proof, we need to be able to control the growth of the solutions to $\mathcal{H}_{\beta}h = 0$ in order to bound error terms easily. Thus, we start by proving the following lemma.

Lemma 3.9. *Let h solve $\mathcal{H}_{\beta}h = 0$. For any $\varepsilon, \delta > 0$, there is a $C > 0$ such that*

$$\mathbb{P}\left[\forall t \geq 0, |h(t)| \leq \exp(C(1+t)^{3/2+\delta})\right] \geq 1 - \varepsilon$$

and

$$\mathbb{P}\left[\forall t \geq 0, |h^{[1]}(t)| \leq \sqrt{1+t} \exp(C(1+t)^{3/2+\delta})\right] \geq 1 - \varepsilon.$$

Proof. Let $S(t) := (1+t)^{1/4}$, and define two real-valued processes r and ξ such that $e^{r+i\xi} = Sh + \frac{ih^{[1]}}{S}$. Then (slightly abusing notation and writing t for the map $t \mapsto t$)

$$\begin{aligned} r'e^{2r} &= \frac{1}{2} \left(S^2 h^2 + \frac{h^{[1]2}}{S^2} \right)' \\ &= S'Sh^2 + S^2 h \left(h^{[1]} + \frac{2}{\sqrt{\beta}} Bh \right) + \frac{h^{[1]}}{S^2} \left(-\frac{2}{\sqrt{\beta}} Bh^{[1]} + \left(t - \frac{4}{\beta} B^2 \right) h \right) - \frac{S'h^{[1]2}}{S^3} \\ &= \left(\frac{S'}{S} + \frac{2}{\sqrt{\beta}} B \right) \left(S^2 h^2 - \frac{h^{[1]2}}{S^2} \right) + \left(S^2 + \frac{t}{S^2} - \frac{4}{\beta S^2} B^2 \right) h h^{[1]}. \end{aligned}$$

In the rightmost factors of these two terms, we recognize the expressions of $e^{2r} \cos 2\xi$ and $\frac{1}{2}e^{2r} \sin 2\xi$

respectively. Expliciting the expression of S , this gives

$$r'(t) = \left(\frac{1}{4(1+t)} + \frac{2}{\sqrt{\beta}} B(t) \right) \cos 2\xi(t) + \left(\frac{1}{2} \sqrt{1+t} - \frac{1}{2\sqrt{1+t}} - \frac{2}{\beta\sqrt{1+t}} B^2(t) \right) \sin 2\xi(t).$$

This allows us to get the desired control over h from properties of Brownian motion. The property that we need here, which follows from Proposition A.1, is that for any $\varepsilon, \delta > 0$, there is a $C > 0$ such that

$$\mathbb{P} \left[\forall t \geq 0, |B(t)| \leq C(1+t)^{1/2+\delta} \right] \geq 1 - \varepsilon.$$

On that event, integrating the expression for $r'(t)$ yields

$$r(t) \leq r(0) + \frac{1}{4} \log(1+t) + C(1+t)^{3/2+\delta}$$

for a different $C > 0$. Since $|h| = S^{-1}e^r$ and $|h^{[1]}| = Se^r$, this implies the announced result. $\blacksquare$

We now show that under Hypothesis 3.1, the solutions to the three-term recurrence (3.11) with $z = 0$ converge to solutions to $\mathcal{H}_\beta h = 0$ with matching initial conditions. The canonical system convergence will be a consequence of this result.

Lemma 3.10. *Suppose that the matrices M_n from (3.10) satisfy Hypothesis 3.1. Then, there is a coupling of the matrices M_n with the stochastic Airy operator $\mathcal{H}_\beta$ under which if h_n solves the three-term recurrence (3.11) with deterministic initial conditions $h_{n,0}, h_{n,1} \in \mathbb{R}$ such that*

$$\mathfrak{h}_0 := \lim_{n \rightarrow \infty} h_{n,0} \quad \text{and} \quad \mathfrak{h}'_0 := \lim_{n \rightarrow \infty} \frac{h_{n,1} - h_{n,0}}{\delta_n}$$

both exist, then the function $t \mapsto h_{n, \lfloor t/\delta_n \rfloor}$ converges compactly in probability to the solution h to $\mathcal{H}_\beta h = 0$ with initial conditions $h(0) = \mathfrak{h}_0$, $h'(0) = \mathfrak{h}'_0$.

Remark. One might expect that it should not be hard to generalize this result to solutions to the eigenvalue equations for nonzero spectral parameters and to possibly random initial conditions. However, nothing would really be gained from doing so, as such generalizations will directly follow from the compact convergence of transfer matrices in Corollary 3.8.1.

Proof. We start by introducing the coupling in question. By condition *ii*) of Hypothesis 3.1, the process $\mathcal{Z}_n(t) := \sum_{j=1}^{\lfloor t/\delta_n \rfloor} \Delta Z_{n,j}$ converges compactly in law to a Brownian motion. Hence, by Skorokhod representation, there is a coupling of the $\mathcal{Z}_n$'s such that $\mathcal{Z}_n \rightarrow B$ compactly in probability. Our goal is now to show that under this coupling, $h_{n, \lfloor \cdot/\delta_n \rfloor} \rightarrow h$ compactly in probability.

Since h is continuous and $\delta_n \rightarrow 0$, by definition of the functions $h_{n, \lfloor \cdot/\delta_n \rfloor}$ it is clear that it suffices to compare $h(j\delta_n)$ with $h_{n,j}$ for $j \leq \tau/\delta_n$ where $\tau > 0$ remains arbitrary. From this observation, the

idea of the proof is simple: starting from recurrences for h and $h_{n, \lfloor \cdot / \delta_n \rfloor}$ at times $j\delta_n$, we obtain a recurrence for the difference between them and use it to propagate the error throughout the interval $[0, \tau]$, corresponding to roughly τ/δ_n iterations. From this, we show that the difference between the two solutions vanishes uniformly on compacts as $n \rightarrow \infty$.

To facilitate the handling of error terms throughout the proof, we work on a good event on which we can control the errors deterministically. To this end, let $\tau, \varepsilon, \delta > 0$. By Lemma 3.9, properties of Brownian motion, and Hypothesis 3.1, there are $C, C' > 0$ as well as $\varepsilon_n > 0$ with $\varepsilon_n \downarrow 0$ such that the event

$$\begin{aligned} \mathcal{G}_{n, \tau, \varepsilon} := & \left\{ \sup_{t \in [0, \tau]} (|h(t)| \vee |h^{[1]}(t)|) \leq C \right\} \cap \left\{ \sup_{t, s \in [0, \tau]} |B(t) - B(s)| \leq C'|t - s|^{1/2 - \delta} \right\} \\ & \cap \left\{ \sup_{t \in [0, \tau]} |\mathcal{Z}_n(t) - B(t)| \leq \varepsilon_n \right\} \cap \left\{ \max_{1 \leq j \leq \tau/\delta_n} (|V_{n, j}^a| \vee |V_{n, j}^b|) \leq C \right\} \\ & \cap \left\{ \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n, j}^a| \leq \varepsilon_n \right\} \cap \left\{ \max_{1 \leq j \leq \tau/\delta_n} |V_{n, j-1}^a + V_{n, j}^b + V_{n, j}^a - (j-1)\delta_n| \leq \varepsilon_n \right\} \end{aligned} \quad (3.14)$$

has $\mathbb{P}(\mathcal{G}_{n, \tau, \varepsilon}) \geq 1 - \varepsilon$ for all n large enough. Without loss of generality, we may assume that $\delta_n \leq \varepsilon_n$ for all n large enough, since it is clear that the ε_n 's could be replaced with a sequence that converges more slowly to zero.

We start by finding recurrences for h and $h^{[1]}$. To simplify notation, write $t_j := j\delta_n$. Our starting point is

$$h(t_j) = h(t_{j-1}) + \delta_n h'(t_{j-1}) + \int_{t_{j-1}}^{t_j} (h'(t) - h'(t_{j-1})) dt.$$

Since the quasi-derivative $h^{[1]} = h' - \frac{2}{\sqrt{\beta}} B h$ is absolutely continuous, we know that h' has modulus of continuity dominated by that of Brownian motion. Hence, on $\mathcal{G}_{n, \tau, \varepsilon}$, it is $(1/2 - \delta)$ -Hölder continuous, and it follows that for $j \leq \tau/\delta_n$,

$$\left| \int_{t_{j-1}}^{t_j} (h'(t) - h'(t_{j-1})) dt \right| \lesssim \int_{t_{j-1}}^{t_j} (t - t_{j-1})^{1/2 - \delta} dt \lesssim \delta_n^{3/2 - \delta}$$

where the implicit constants depend on τ and ε . Thus, for all $j \leq \tau/\delta_n$,

$$h(t_j) = h(t_{j-1}) + \delta_n \left(h^{[1]}(t_{j-1}) + \frac{2}{\sqrt{\beta}} B(t_{j-1}) h(t_{j-1}) \right) + O_{\tau, \varepsilon}(\delta_n^{3/2 - \delta}). \quad (3.15a)$$

We can obtain a similar recurrence for $h^{[1]}$ using that $\mathcal{H}_\beta h = 0$, which gives an expression for $h^{[1]}'$. Indeed, we can still write

$$h^{[1]}(t_j) = h^{[1]}(t_{j-1}) + \delta_n h^{[1]}'(t_{j-1}) + \int_{t_{j-1}}^{t_j} (h^{[1]}'(t) - h^{[1]}'(t_{j-1})) dt,$$

and because $h^{[1]'}(t) = -\frac{2}{\sqrt{\beta}}B(t)h^{[1]}(t) + (t - \frac{4}{\beta}B^2(t))h(t)$, the modulus of continuity of $h^{[1]}'$ is dominated by that of Brownian motion. Hence, in the same way as above, the remaining integral is again $O_{\tau,\varepsilon}(\delta_n^{3/2-\delta})$ on $\mathcal{G}_{n,\tau,\varepsilon}$, and it follows that for all $j \leq \tau/\delta_n$,

$$h^{[1]}(t_j) = h^{[1]}(t_{j-1}) - \delta_n \left(\frac{2}{\sqrt{\beta}}B(t_{j-1})h^{[1]}(t_{j-1}) - \left(t_{j-1} - \frac{4}{\beta}B^2(t_{j-1})\right)h(t_{j-1}) \right) + O_{\tau,\varepsilon}(\delta_n^{3/2-\delta}). \quad (3.15b)$$

We now want to find similar recurrences as (3.15), but for the discrete case. To do so, we introduce a discrete analog of the quasi-derivative associated with the stochastic Airy operator,

$$h_{n,j}^{[1]} := \frac{h_{n,j+1} - h_{n,j}}{\delta_n} - \frac{2}{\sqrt{\beta}}\mathcal{Z}_n(t_j)h_{n,j} + \frac{2}{\sqrt{\beta}}\Delta Z_{n,j}^a h_{n,j}. \quad (3.16)$$

The last term does not have a counterpart in the continuum setting, and is merely included here for technical convenience. It corresponds to an error that would otherwise telescope out when summed, but including it here lets us control it through our inductive argument.

On $\mathcal{G}_{n,\tau,\varepsilon}$ and with $j \leq \tau/\delta_n$, the expression of the discrete analog of the quasi-derivative can immediately be reversed into the recurrence

$$h_{n,j} = h_{n,j-1} + \delta_n \left(h_{n,j-1}^{[1]} + \frac{2}{\sqrt{\beta}}B(t_{j-1})h_{n,j-1} \right) + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)h_{n,j-1}. \quad (3.17a)$$

Now, writing the coefficients $a_{n,j}$ and $b_{n,j}$ under the form of Hypothesis 3.1, the three-term recurrence (3.11) for h_n with $z = 0$ becomes

$$\begin{aligned} 0 &= -\frac{h_{n,j+1} - 2h_{n,j} + h_{n,j-1}}{\delta_n^2} + \frac{2}{\sqrt{\beta}\delta_n} \left(\Delta Z_{n,j-1}^a h_{n,j-1} + \Delta Z_{n,j}^b h_{n,j} + \Delta Z_{n,j}^a h_{n,j+1} \right) \\ &\quad + V_{n,j-1}^a h_{n,j-1} + V_{n,j}^b h_{n,j} + V_{n,j}^a h_{n,j+1} \\ &= -\frac{h_{n,j}^{[1]} - h_{n,j-1}^{[1]}}{\delta_n} + \frac{2}{\sqrt{\beta}\delta_n} \left(\Delta Z_{n,j}^a h_{n,j} + \Delta Z_{n,j}^b h_{n,j} + \Delta Z_{n,j}^a h_{n,j+1} - \mathcal{Z}_n(t_j)h_{n,j} + \mathcal{Z}_n(t_{j-1})h_{n,j-1} \right) \\ &\quad + V_{n,j-1}^a h_{n,j-1} + V_{n,j}^b h_{n,j} + V_{n,j}^a h_{n,j+1}. \end{aligned}$$

Since $\mathcal{Z}_n(t_j)h_{n,j} - \mathcal{Z}_n(t_{j-1})h_{n,j-1} = \Delta Z_{n,j}h_j + \mathcal{Z}_n(t_{j-1})(h_{n,j} - h_{n,j-1})$, this reduces to

$$\begin{aligned} 0 &= -\frac{h_{n,j}^{[1]} - h_{n,j-1}^{[1]}}{\delta_n} - \frac{2}{\sqrt{\beta}}\mathcal{Z}_n(t_{j-1})h_{n,j-1}^{[1]} - \frac{4}{\beta}\mathcal{Z}_n^2(t_{j-1})h_{n,j-1} + V_{n,j-1}^a h_{n,j-1} + V_{n,j}^b h_{n,j} + V_{n,j}^a h_{n,j+1} \\ &\quad + \frac{2}{\sqrt{\beta}\delta_n}\Delta Z_{n,j}^a(h_{n,j+1} - h_{n,j}) + \frac{4}{\beta}\Delta Z_{n,j-1}^a\mathcal{Z}_n(t_{j-1})h_{n,j-1}. \end{aligned}$$

Using the recurrence (3.17a) for h_n to control the error terms, it follows that on $\mathcal{G}_{n,\tau,\varepsilon}$ for $j \leq \tau/\delta_n$,

$$\begin{aligned} h_{n,j}^{[1]} &= h_{n,j-1}^{[1]} - \delta_n \left(\frac{2}{\sqrt{\beta}}B(t_{j-1})h_{n,j-1}^{[1]} - \left(t_{j-1} - \frac{4}{\beta}B^2(t_{j-1})\right)h_{n,j-1} \right) \\ &\quad + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)h_{n,j-1}^{[1]} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)h_{n,j-1} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)h_{n,j}^{[1]} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)h_{n,j}. \end{aligned} \quad (3.17b)$$

To compare the solutions h_n and h , we set $\Delta_{n,j} := h(t_j) - h_{n,j}$ and $\Delta_{n,j}^{[1]} := h^{[1]}(t_j) - h_{n,j}^{[1]}$, and we take the differences of (3.15a) and (3.17a) and of (3.15b) and (3.17b) to get

$$\Delta_{n,j} = \Delta_{n,j-1} + \delta_n \left(\Delta_{n,j-1}^{[1]} + \frac{2}{\sqrt{\beta}} B(t_{j-1}) \Delta_{n,j-1} \right) + O_{\tau,\varepsilon}(\delta_n \varepsilon_n) \Delta_{n,j-1} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)$$

and

$$\begin{aligned} \Delta_{n,j}^{[1]} &= \Delta_{n,j-1}^{[1]} - \delta_n \left(\frac{2}{\sqrt{\beta}} B(t_{j-1}) \Delta_{n,j-1}^{[1]} - \left(t_{j-1} - \frac{4}{\beta} B^2(t_{j-1}) \right) \Delta_{n,j-1} \right) + O_{\tau,\varepsilon}(\delta_n \varepsilon_n) \\ &\quad + O_{\tau,\varepsilon}(\delta_n \varepsilon_n) \Delta_{n,j-1}^{[1]} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n) \Delta_{n,j-1} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n) \Delta_{n,j}^{[1]} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n) \Delta_{n,j} \end{aligned}$$

where we are assuming that $\delta_n^{3/2-\delta} = O(\delta_n \varepsilon_n)$, which we can certainly do by increasing ε_n if necessary. This means that if $\hat{\Delta}_{n,j} := \max_{0 \leq k \leq j} (|\Delta_{n,k}| \vee |\Delta_{n,k}^{[1]}|)$, then

$$(1 + O_{\tau,\varepsilon}(\delta_n \varepsilon_n)) \hat{\Delta}_{n,j} \leq (1 + O_{\tau,\varepsilon}(\delta_n)) \hat{\Delta}_{n,j-1} + O_{\tau,\varepsilon}(\delta_n \varepsilon_n),$$

and therefore there are constants $\hat{C}, \hat{C}' > 0$ (depending on τ and ε) such that on $\mathcal{G}_{\tau,\varepsilon}$ for all $j \leq \tau/\delta_n$,

$$\hat{\Delta}_{n,j} \leq (1 + \hat{C} \delta_n) \hat{\Delta}_{n,j-1} + \hat{C}' \delta_n \varepsilon_n.$$

A simple inductive argument then shows that

$$\hat{\Delta}_{n,j} \leq (1 + \hat{C} \delta_n)^j \hat{\Delta}_{n,0} + \hat{C}' \delta_n \varepsilon_n \sum_{k=0}^{j-1} (1 + \hat{C} \delta_n)^k \leq (1 + \hat{C} \delta_n)^j \left(\hat{\Delta}_{n,0} + \frac{\hat{C}' \varepsilon_n}{\hat{C}} \right).$$

Taking $j = \lfloor \tau/\delta_n \rfloor$, the prefactor remains bounded as n increases, so because $\hat{\Delta}_{n,0} \rightarrow 0$ by the hypothesis on the initial conditions, this shows that $\hat{\Delta}_{n, \lfloor \tau/\delta_n \rfloor} \rightarrow 0$ in probability as $n \rightarrow \infty$ and completes the proof. $\blacksquare$

The vague convergence of the canonical systems' coefficient matrices announced in Theorem 3.8 is really a consequence of this lemma.

Proof of Theorem 3.8. Recall from (3.4) and (3.12) that the coefficient matrices H_β and $H_{\beta,n}$ associated with $\mathcal{H}_\beta$ and M_n are respectively given by

$$H_\beta = \begin{pmatrix} g_\beta^2 & f_\beta g_\beta \\ f_\beta g_\beta & f_\beta^2 \end{pmatrix} \quad \text{and} \quad H_{\beta,n}(t) = \begin{pmatrix} g_{\beta,n,\lfloor t/\delta_n \rfloor}^2 & f_{\beta,n,\lfloor t/\delta_n \rfloor} g_{\beta,n,\lfloor t/\delta_n \rfloor} \\ f_{\beta,n,\lfloor t/\delta_n \rfloor} g_{\beta,n,\lfloor t/\delta_n \rfloor} & f_{\beta,n,\lfloor t/\delta_n \rfloor}^2 \end{pmatrix}$$

for $t \in [0, n\delta_n]$, where f_β and g_β solve $\mathcal{H}_\beta f_\beta = \mathcal{H}_\beta g_\beta = 0$ with $f'_\beta(0) = g_\beta(0) = 0$ and $f_\beta(0) = g'_\beta(0) = 1$, and where $f_{\beta,n}$ and $g_{\beta,n}$ solve the three-term recurrence (3.11) with $z = 0$ and initial

conditions $f_{\beta,n,0} = f_{\beta,n,1} = 1$, $g_{\beta,n,0} = 0$ and $g_{\beta,n,1} = \delta_n$. The initial conditions for these two pairs of solutions match: $f_{\beta,n,0} = f_\beta(0)$ and $g_{\beta,n,0} = g_\beta(0)$, and

$$\frac{f_{\beta,n,1} - f_{\beta,n,0}}{\delta_n} = f'_\beta(0) \quad \text{and} \quad \frac{g_{\beta,n,1} - g_{\beta,n,0}}{\delta_n} = g'_\beta(0).$$

Therefore, Lemma 3.10 shows that as $n \rightarrow \infty$, $f_{\beta,n,\lfloor \cdot / \delta_n \rfloor} \rightarrow f_\beta$ and $g_{\beta,n,\lfloor \cdot / \delta_n \rfloor} \rightarrow g_\beta$ compactly in probability with respect to the coupling from the lemma. Since $n\delta_n \rightarrow \infty$, any compact subset of $[0, \infty)$ is eventually contained in the domain on which the above representation of $H_{\beta,n}$ is valid, and it follows that $H_{\beta,n} \rightarrow H_\beta$ vaguely on $[0, \infty)$ in probability under this coupling. ■

It is now easy to deduce the convergence of transfer matrices and Weyl–Titchmarsh functions.

Proof of Corollary 3.8.1. By Lemma 3.9, for any $\tau, \varepsilon > 0$ there is an $f_{\tau,\varepsilon} \in L^1[0, \tau]$ such that $\text{tr } H_\beta \leq f_{\tau,\varepsilon}$ on $[0, \tau]$ with probability at least $1 - \varepsilon$. Working on the coupling from Lemma 3.10, the lemma implies that $\text{tr } H_{\beta,n} \rightarrow \text{tr } H_\beta$ compactly in probability, so it is also true that for any $\tau, \varepsilon > 0$, there is a $g_{\tau,\varepsilon} \in L^1[0, \tau]$ such that $\text{tr } H_{\beta,n} \leq g_{\tau,\varepsilon}$ on $[0, \tau]$ with probability at least $1 - \varepsilon$. Hence, for any $\tau > 0$, $H_{\beta,n} \rightarrow H_\beta$ vaguely on $[0, \tau]$ in probability, and the above control on the traces implies that $T_{H_{\beta,n}} \rightarrow T_{H_\beta}$ compactly on $[0, \tau] \times \mathbb{C}$ in probability by continuity of the map from coefficient to transfer matrices on trace-dominated sets (i.e., Proposition 2.25). Because τ is arbitrary in $(0, \infty)$, it follows from Proposition 2.24 that $T_{H_{\beta,n}} \rightarrow T_{H_\beta}$ compactly on $[0, \infty) \times \mathbb{C}$ in probability.

Since $H_{\beta,n} \rightarrow H_\beta$ and $T_{H_{\beta,n}} \rightarrow T_{H_\beta}$ in probability under the same coupling, it follows that $H_{\beta,n} \rightarrow H_\beta$ and $T_{H_{\beta,n}} \rightarrow T_{H_\beta}$ jointly in law. ■

Proof of Corollary 3.8.2. Suppose first that $\alpha_n = 0$ so that $\delta_n^2 \mathbf{u}_n = e_0$. Then e_0 can equivalently be used to define the boundary condition at 0 for $H_{\beta,n}$. Then by the continuity of the map from transfer matrices to Weyl–Titchmarsh functions from Theorem 2.13, the convergence of transfer matrices from Corollary 3.8.1 implies that $m_{H_{\beta,n}}^0 \rightarrow m_{H_\beta}^0$ in law.

Let's return to the general case of Hypothesis 3.2. Write $\phi_n \in [0, \pi]$ for the (random) angle such that $e_{\phi_n} = \pm \|\mathbf{u}_n\|_2^{-1} \mathbf{u}_n$, so that $m_{H_{\beta,n}}^{\phi_n} = m_{H_{\beta,n}}^{\mathbf{u}_n}$. Now, Hypothesis 3.2 implies that

$$e_{\phi_n} = \pm \frac{\mathbf{u}_n}{\|\mathbf{u}_n\|_2} = \frac{\pm 1}{\sqrt{\delta_n^{-2}(1 + \alpha_n)^2 + \alpha_n^2}} \begin{pmatrix} \delta_n^{-1}(1 + \alpha_n) \\ -\alpha_n \end{pmatrix} \xrightarrow{\text{law}} e_\phi.$$

Since the convergence is to a constant, it holds jointly with the convergence $m_{H_{\beta,n}}^0 \rightarrow m_{H_\beta}^0$ proven above. Since $m_{H_{\beta,n}}^{\phi_n} = R_{-\phi_n} m_{H_{\beta,n}}^0$ and $m_{H_\beta}^\phi = R_{-\phi} m_{H_\beta}^0$ where R_θ denotes the rotation matrix with columns e_θ and Je_θ (see the remark below Definition 2.11), it follows that $m_{H_{\beta,n}}^{\phi_n} \rightarrow m_{H_\beta}^\phi$ in law. The convergence of the spectral measures is then a consequence of the continuity of the Herglotz representation of the Weyl–Titchmarsh functions.

To conclude the proof, note that as we have seen in the proofs of Theorem 3.8 and Corollary 3.8.1, the convergence of coefficient and transfer matrices holds in probability under the coupling from Lemma 3.10. Under this coupling, the above argument actually shows that the convergence of Weyl–Titchmarsh functions and spectral measures holds in probability, and thus all of the distributional convergences hold jointly. ■

3.1.3.3 Application to the Hermite and Laguerre β -ensembles

The results of §3.1.3.1 can easily be applied to specific random matrix models for β -ensembles by checking whether they satisfy Hypotheses 3.1 and 3.2. In order to verify condition *ii*) of Hypothesis 3.1 in practice, one method is to verify that the increments of $\mathcal{Z}$ are independent and centered with appropriate higher moments, and use a martingale central limit theorem to deduce the convergence of the process. As this can work in several cases—and in particular for the Hermite and Laguerre β -ensembles—we state this as a separate result.

Lemma 3.11. *In the notation of Hypothesis 3.1, the process $\mathcal{Z}_n$ converges compactly in law to a standard Brownian motion whenever its increments can be decomposed as $\Delta \mathcal{Z}_{n,j} =: \Delta \hat{\mathcal{Z}}_{n,j} + \Delta \check{\mathcal{Z}}_{n,j}$ so that the $\Delta \hat{\mathcal{Z}}_{n,j}$'s are independent with $\mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j} = 0$, $\mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j}^2 = \delta_n + o_\tau(\delta_n)$ and $\mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j}^4 = o_\tau(\delta_n)$ for all $j \leq \tau/\delta_n$ with any $\tau > 0$, and so that the process $\check{\mathcal{Z}}_n(t) := \sum_{j=1}^{\lfloor t/\delta_n \rfloor} \Delta \check{\mathcal{Z}}_{n,j}$ converges compactly to 0 in law.*

Proof. Since $\check{\mathcal{Z}}_n \rightarrow 0$ compactly in law, it suffices to show that the other part of $\mathcal{Z}_n$, namely the process $\hat{\mathcal{Z}}_n(t) := \sum_{j=1}^{\lfloor t/\delta_n \rfloor} \Delta \hat{\mathcal{Z}}_{n,j}$, converges compactly in law to a standard Brownian motion.

Because the increments $\Delta \hat{\mathcal{Z}}_{n,j}$ are independent with $\mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j} = 0$ and finite variance, $\hat{\mathcal{Z}}_n$ is a martingale, and by definition it is càdlàg. By the martingale central limit theorem (the precise formulation we use here is that from Ethier and Kurtz [32, Theorem 7.1.4]) together with Lévy's characterization of Brownian motion, it suffices to show that $\langle \hat{\mathcal{Z}}_n \rangle(t) \rightarrow t$ in probability for all $t \geq 0$ and that for all $\tau > 0$, $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} \Delta \hat{\mathcal{Z}}_{n,j}^2 \rightarrow 0$.

The predictable quadratic variation of $\hat{\mathcal{Z}}_n$ is directly seen to be $\langle \hat{\mathcal{Z}}_n \rangle(t) = \sum_{j=1}^{\lfloor t/\delta_n \rfloor} \mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j}^2$ from its discrete structure. Thus, the second moment assumption on its increments implies that $\langle \hat{\mathcal{Z}}_n \rangle(t) = t + o_\tau(1)$ for any $\tau > 0$.

We can then check that $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} \Delta \hat{\mathcal{Z}}_{n,j}^2 \rightarrow 0$ for all $\tau > 0$ using Markov's inequality. Indeed, for any $x > 0$,

$$\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} \Delta \hat{\mathcal{Z}}_{n,j}^2 \leq x + \sum_{j=1}^{\lfloor \tau/\delta_n \rfloor} \int_x^\infty \mathbb{P}[\Delta \hat{\mathcal{Z}}_{n,j}^2 > y] dy \leq x + \sum_{j=1}^{\lfloor \tau/\delta_n \rfloor} \frac{\mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j}^4}{x}.$$

By the fourth moment assumption, there is an $\varepsilon_n = o_\tau(\delta_n)$ such that $\mathbb{E} \Delta \hat{\mathcal{Z}}_{n,j}^4 \leq \varepsilon_n$ for all $j \leq \tau/\delta_n$. Taking $x = (\frac{\tau \varepsilon_n}{\delta_n})^{1/2}$, the above is bounded by $2(\frac{\tau \varepsilon_n}{\delta_n})^{1/2} \rightarrow 0$. ■

Application to the Hermite β -ensemble. As we have seen in the introduction, the Hermite β -ensemble is the point process with density given by (1.1) for the potential $V(\lambda) = \lambda^4/4$, and it can be realized as the eigenvalue point process of the tridiagonal random matrix

$$H_{\beta,n} := \frac{1}{\sqrt{\beta}} \begin{pmatrix} \sqrt{2}\xi_{n,1} & X_{n,1} & & \\ X_{n,1} & \ddots & \ddots & \\ & \ddots & \ddots & X_{n,n-1} \\ & & X_{n,n-1} & \sqrt{2}\xi_{n,n} \end{pmatrix} \quad \text{where} \quad \begin{aligned} \xi_{n,j} &\sim \mathcal{N}(0, 1) \\ X_{n,j} &\sim \chi_{(n-j)\beta}, \end{aligned}$$

with all $\xi_{n,j}$'s and $X_{n,j}$'s independent [24]. Here, we consider a perturbed version of this tridiagonal model: $H_{\beta,n} - \alpha_n \sqrt{n} \mathbf{e}_1 \mathbf{e}_1^\top$ for a perturbation parameter $\alpha_n \in \mathbb{R}$. From [30, 60], we know that

$$n^{1/6} (2\sqrt{n} - H_{\beta,n} + \alpha_n \sqrt{n} \mathbf{e}_1 \mathbf{e}_1^\top) \quad (3.18)$$

has a soft edge scaling with time step $\delta_n := n^{-1/3}$, that is, it should be a discrete-time approximation of the stochastic Airy operator with this time step. This matrix model has a canonical system representation with coefficient matrix of the form (3.12), which has the same spectrum as the matrix when given the boundary condition (3.13) at 0.

It is easy to verify that this matrix model satisfies Hypothesis 3.1, and that Hypothesis 3.2 is satisfied as well when the perturbation α_n has the right form.

Proposition 3.12. *The matrix $n^{1/6} (2\sqrt{n} - H_{\beta,n} + \alpha_n \sqrt{n} \mathbf{e}_1 \mathbf{e}_1^\top)$ satisfies Hypothesis 3.1 with $\delta_n = n^{-1/3}$. Moreover, if $n^{1/3} (1 + \alpha_n) \rightarrow \cot \phi$ in law for some (deterministic) $\phi \in [0, \pi]$, then Hypothesis 3.2 is satisfied as well.*

It directly follows from Proposition 3.12 and Theorem 3.8 that the associated canonical system converges vaguely on $[0, \infty)$ in law to the canonical system for the stochastic Airy operator, and then from Corollary 3.8.1 that the canonical systems' transfer matrices converge compactly on $[0, \infty) \times \mathbb{C}$ in law. When the perturbation α_n has the right form, it also follows from Corollary 3.8.2 that the Weyl–Titchmarsh functions converge compactly on $\mathbb{H}$ in law, and that the spectral measures converge vaguely in law. In particular, this proves the parts of Theorems 1.5 and 1.7 about the Hermite β -ensemble.

To prove Proposition 3.12, we use Lemma 3.11, which reduces most of the problem to computing moments of the matrix entries. Thus, we need expressions for the moments of χ random variables. The following result is elementary, and can easily be proven using basic properties of the Gamma function together with Stirling's approximation.

Proposition 3.13. *Let X be a χ_k -distributed random variable for a $k \in \mathbb{N}$. Then*

$$\mathbb{E} X = \sqrt{k} - \frac{1}{4\sqrt{k}} + O(k^{-3/2}),$$

and $\mathbb{E} X^{2\ell} = \prod_{m=0}^{\ell-1} (k + 2m)$ and $\mathbb{E} X^{2\ell+1} = \mathbb{E} X \prod_{m=0}^{\ell-1} (k + 2m + 1)$ for $\ell \in \mathbb{N}$.

We now turn to the proof of Proposition 3.12.

Proof of Proposition 3.12. Note that the condition on α_n is just the condition of Hypothesis 3.2 with $\delta_n = n^{-1/3}$, so only Hypothesis 3.1 really needs to be verified.

The matrix entries are $a_{n,j} = -\frac{1}{\sqrt{\beta}} n^{1/6} X_{n,j}$ and $b_{n,j} = 2n^{2/3} - \sqrt{\frac{2}{\beta}} n^{1/6} \xi_{n,j}$, which can be written as in Hypothesis 3.1 with $\delta_n = n^{-1/3}$,

$$a_{n,j} = -n^{2/3} + \frac{2}{\sqrt{\beta}} n^{1/3} \Delta Z_{n,j}^a + V_{n,j}^a \quad \text{and} \quad b_{n,j} = 2n^{2/3} + \frac{2}{\sqrt{\beta}} n^{1/3} \Delta Z_{n,j}^b + V_{n,j}^b$$

for

$$\begin{aligned} \Delta Z_{n,j}^a &= -\frac{1}{2} n^{-1/6} (X_{n,j} - \mathbb{E} X_{n,j}), & \Delta Z_{n,j}^b &= -\frac{1}{\sqrt{2}} n^{-1/6} \xi_{n,j}, \\ V_{n,j}^a &= n^{2/3} - \frac{n^{1/6}}{\sqrt{\beta}} \mathbb{E} X_{n,j} & V_{n,j}^b &= 0. \end{aligned}$$

Out of the three conditions of Hypothesis 3.1, *i*) is obvious. To verify *iii*), note that by Proposition 3.13, if $\tau > 0$ and $j \leq \tau n^{1/3}$ then $\mathbb{E} X_{n,j} = \sqrt{(n-j)\beta} + O_\tau(n^{-1/2}) = \sqrt{n\beta}(1-j/2n) + O_\tau(n^{-1/2})$. Hence, $V_{n,j}^a = \frac{1}{2} j n^{-1/3} + O_\tau(n^{-1/3})$, which implies both that $\{V_{n,j}^a\}_{j=1}^{\lfloor \tau/\delta_n \rfloor}$ is tight and that $V_{n,j-1}^a + V_{n,j}^a = (j-1)n^{-1/3} + O_\tau(n^{-1/3})$ so that *iii*) holds.

To verify *ii*), we use Lemma 3.11. Write $\Delta \hat{Z}_{n,j} := \Delta Z_{n,j} = 2\Delta Z_{n,j}^a + \Delta Z_{n,j}^b$ (so that in the notation of Lemma 3.11, $\check{Z}_n \equiv 0$). Then indeed $\mathbb{E} \Delta \hat{Z}_{n,j} = 0$, and if $j \leq \tau n^{1/3}$ for a $\tau > 0$, then $\text{Var } X_{n,j} = 1/2 + O_\tau(1/n)$ by Proposition 3.13 so

$$\mathbb{E} \Delta \hat{Z}_{n,j}^2 = n^{-1/3} \text{Var } X_{n,j} + \frac{1}{2} n^{-1/3} \text{Var } \xi_{n,j} = n^{-1/3} + O_\tau(n^{-4/3}).$$

Moreover, $\mathbb{E} |\Delta Z_{n,j}^b|^4 = \frac{3}{4} n^{-2/3}$ and one can easily verify from Proposition 3.13 that $\mathbb{E} |\Delta Z_{n,j}^a|^4 = O_\tau(n^{-2/3})$ for $j \leq \tau n^{1/3}$, so $\mathbb{E} \Delta \hat{Z}_{n,j}^4 \leq 8(16 \mathbb{E} |\Delta Z_{n,j}^a|^4 + \mathbb{E} |\Delta Z_{n,j}^b|^4) = O_\tau(n^{-2/3})$ by Young's inequality. All of the conditions are thus satisfied for Lemma 3.11 to show that $\mathcal{Z}_n$ converges in law to a standard Brownian motion. To finish to check that condition *ii*) holds, it only remains to show that $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n,j}^a| \rightarrow 0$ and $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n,j}^b| \rightarrow 0$, and this follows from Markov's inequality together with the bounds $\mathbb{E} |\Delta Z_{n,j}^a|^4 = O_\tau(n^{-2/3})$ and $\mathbb{E} |\Delta Z_{n,j}^b|^4 = O_\tau(n^{-2/3})$, in a similar way as in the proof of Lemma 3.11. $\blacksquare$

Application to the Laguerre β -ensemble. As we have seen in the introduction, the Laguerre β -ensemble is the point process with density (1.1) with $V(\lambda) = \frac{\lambda}{2} - (\frac{m-n+1}{2} - \frac{1}{\beta}) \log \lambda$ where $m > n-1$ is another parameter that is implicitly a function of n . It can be realized as the eigenvalue point process of the tridiagonal random matrix $L_{\beta,n} L_{\beta,n}^\top$ with $L_{\beta,n}$ as in (1.4) [24]. Here, to allow for a little more generality, we rather consider the spiked model $L_{\beta,n,\ell_n} L_{\beta,n,\ell_n}^\top$ for a perturbation parameter $\ell_n > 0$ that we have seen before in (1.6), where

$$L_{\beta,n,\ell_n} := \frac{1}{\sqrt{\beta}} \begin{pmatrix} \sqrt{\ell_n} Y_{n,1} & X_{n,1} & & & \\ & \ddots & \ddots & & \\ & & \ddots & X_{n,n-1} & \\ & & & & Y_{n,n} \end{pmatrix} \quad \text{and} \quad \begin{aligned} Y_{n,j} &\sim \chi_{(m-j+1)\beta} \\ X_{n,j} &\sim \chi_{(n-j)\beta} \end{aligned}$$

with all $X_{n,j}$'s and $Y_{n,j}$'s independent. This reduces to the usual unspiked model when $\ell_n = 1$. From [30, 60], we know that

$$\delta_n \sqrt{nm} - \frac{1}{\delta_n^2 \sqrt{nm}} L_{\beta,n,\ell_n} L_{\beta,n,\ell_n}^\top \quad \text{where} \quad \delta_n := \frac{(\sqrt{n} + \sqrt{m})^{2/3}}{n^{1/3} m^{1/3}} = \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}} \right)^{2/3}$$

has a soft edge scaling with time step δ_n . As in the Hermite case, this matrix model has a canonical system representation with coefficient matrix of the form (3.12), which has the same spectrum as the matrix itself when given the boundary condition (3.13) at 0.

This matrix satisfies Hypothesis 3.1, and it satisfies Hypothesis 3.2 as well when the spike parameter ℓ_n has the right asymptotic behavior.

Proposition 3.14. *The matrix $\delta_n \sqrt{nm} - (\delta_n^2 \sqrt{nm})^{-1} L_{\beta,n,\ell_n} L_{\beta,n,\ell_n}^\top$ satisfies Hypothesis 3.1 with $\delta_n = (n^{-1/2} + m^{-1/2})^{2/3}$ and $\alpha_n = (\beta \sqrt{nm})^{-1} (\ell_n - 1) Y_{n,1}^2$. Moreover, if the spike parameter ℓ_n is independent of $Y_{n,1}$ with $\mathbb{E}(\ell_n - 1)^2 = o(n^{1/3})$ and is such that $\delta_n^{-1} (1 + \sqrt{m/n} (\ell_n - 1)) \rightarrow \cot \phi$ in law for some (deterministic) $\phi \in [0, \pi]$, then Hypothesis 3.2 is satisfied as well.*

As in the Hermite case, Proposition 3.14 and Theorem 3.8 imply that the associated canonical system converges vaguely on $[0, \infty)$ in law to the canonical system for the stochastic Airy operator, and the convergence extends to transfer matrices by Corollary 3.8.1. When ℓ_n has the right form for Hypothesis 3.2 to be satisfied, the convergence further extends to Weyl–Titchmarsh functions and spectral measures by Corollary 3.8.2. In particular, this proves the parts of Theorems 1.5 and 1.7 about the Laguerre β -ensemble.

Proof. Recall that $m > n-1$ so that $n^{-1/3} < \delta_n \leq 2n^{-1/3}$, which in particular implies that condition i) of Hypothesis 3.1 is satisfied. Moreover,

$$\frac{1 + \alpha_n}{\delta_n} = \frac{1}{\delta_n} \left(1 + (\ell_n - 1) \sqrt{\frac{m}{n}} \left(\frac{Y_{n,1}^2 - m\beta}{m\beta} + 1 \right) \right).$$

When the hypotheses on ℓ_n are satisfied, the term $\delta_n^{-1}(\ell_n - 1)(\beta\sqrt{nm})^{-1}(Y_{n,1}^2 - \mathbb{E} Y_{n,1}^2)$ has mean zero and its variance vanishes as $n \rightarrow \infty$ since $\mathbb{E}(\ell_n - 1)^2 = o(n^{1/3})$, so it converges to zero in probability by Markov's inequality. The other condition on ℓ_n is then exactly what is needed for the other term to converge to $\cot \phi$ so Hypothesis 3.2 holds.

Now, for $j \leq n - 1$, the matrix entries are

$$a_{n,j} = -\frac{1}{\beta\delta_n^2\sqrt{nm}}X_{n,j}Y_{n,j+1} \quad \text{and} \quad b_{n,j} = \delta_n\sqrt{nm} - \frac{1}{\beta\delta_n^2\sqrt{nm}}(X_{n,j}^2 + Y_{n,j}^2),$$

and they can be decomposed under the form of Hypothesis 3.1 with

$$\begin{aligned} \Delta Z_{n,j}^a &= -\frac{X_{n,j}Y_{n,j+1} - \mathbb{E} X_{n,j}Y_{n,j+1}}{2\delta_n\sqrt{\beta nm}}, & \Delta Z_{n,j}^b &= -\frac{X_{n,j}^2 + Y_{n,j}^2 - \mathbb{E}(X_{n,j}^2 + Y_{n,j}^2)}{2\delta_n\sqrt{\beta nm}}, \\ V_{n,j}^a &= \frac{1}{\delta_n^2} - \frac{\mathbb{E} X_{n,j}Y_{n,j+1}}{\beta\delta_n^2\sqrt{nm}}, & V_{n,j}^b &= \delta_n\sqrt{nm} - \frac{2}{\delta_n^2} - \frac{\mathbb{E}(X_{n,j}^2 + Y_{n,j}^2)}{\beta\delta_n^2\sqrt{nm}}. \end{aligned}$$

By Proposition 3.13, $\mathbb{E} X_{n,j}^2 = (n-j)\beta$ and $\mathbb{E} Y_{n,j}^2 = (m-j+1)\beta$, so because $\delta_n^3 nm = (\sqrt{n} + \sqrt{m})^2$,

$$V_{n,j}^b = \frac{1}{\delta_n^2\sqrt{nm}}(\delta_n^3 nm - 2\sqrt{nm} - n - m + 2j - 1) = \frac{2j - 1}{\delta_n^2\sqrt{nm}}.$$

Then $\mathbb{E} X_{n,j}Y_{n,j+1} = \beta\sqrt{(n-j)(m-j)} + O((n-j)^{-1/2})$, so if $\tau > 0$ then for all $j \leq \tau/\delta_n$,

$$V_{n,j}^a = \frac{1}{\delta_n^2\sqrt{nm}}\left(\sqrt{nm} - \sqrt{nm}\left(1 - \frac{j}{2n} - \frac{j}{2m}\right) + O_\tau(n^{-1/2})\right) = \frac{j}{2\delta_n^2}\left(\frac{1}{n} + \frac{1}{m}\right) + O_\tau(n^{-1/3}).$$

Therefore, both $\{V_{n,j}^a\}_{j=1}^{\lfloor \tau/\delta_n \rfloor}$ and $\{V_{n,j}^b\}_{j=1}^{\lfloor \tau/\delta_n \rfloor}$ are tight, and for all $j \leq \tau/\delta_n$,

$$V_{n,j-1}^a + V_{n,j}^a + V_{n,j}^b = \frac{j-1}{\delta_n^2}\left(\frac{1}{n} + \frac{1}{m} + \frac{2}{\sqrt{nm}}\right) + O_\tau(n^{-1/3}) = (j-1)\delta_n + O_\tau(n^{-1/3}),$$

so condition *iii*) of Hypothesis 3.1 holds.

To check that condition *ii*) holds, we use Lemma 3.11 to show the convergence of $\mathcal{Z}_n$ to a Brownian motion. While checking the conditions for the lemma, we will also verify that $\mathbb{E}|\Delta Z_{n,j}^a|^4$ and $\mathbb{E}|\Delta Z_{n,j}^b|^4$ are both $O_\tau(n^{-2/3})$ for $j \leq \tau/\delta_n$, which implies that $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n,j}^a| \rightarrow 0$ and $\mathbb{E} \max_{1 \leq j \leq \tau/\delta_n} |\Delta Z_{n,j}^b| \rightarrow 0$ by Markov's inequality, as in the proof of Lemma 3.11. Now, to use the lemma, we split the increments $\Delta Z_{n,j} := 2\Delta Z_{n,j}^a + \Delta Z_{n,j}^b$ as $\Delta Z_{n,j} = \Delta \hat{Z}_{n,j} + \Delta \check{Z}_{n,j}$ where

$$\Delta \hat{Z}_{n,j} := -\frac{(X_{n,j} + Y_{n,j+1})^2 - \mathbb{E}(X_{n,j} + Y_{n,j+1})^2}{2\delta_n\sqrt{\beta nm}} \quad \text{and} \quad \Delta \check{Z}_{n,j} := \frac{Y_{n,j+1}^2 - Y_{n,j}^2 - \mathbb{E}(Y_{n,j+1}^2 - Y_{n,j}^2)}{2\delta_n\sqrt{\beta nm}}.$$

This decomposition ensures that $N \mapsto \sum_{j=1}^N \Delta \hat{Z}_{n,j}$ is a martingale with independent increments. For Lemma 3.11 to apply, we need to show that $\mathbb{E} \Delta \hat{Z}_{n,j}^2 = \delta_n + o_\tau(\delta_n)$ and $\mathbb{E} \Delta \hat{Z}_{n,j}^4 = o_\tau(\delta_n)$ for all $j \leq \tau/\delta_n$ with $\tau > 0$. Now,

$$\begin{aligned} \mathbb{E}((X_{n,j} + Y_{n,j+1})^2 - \mathbb{E}(X_{n,j} + Y_{n,j+1})^2)^2 &= 4 \mathbb{E}(X_{n,j}Y_{n,j+1} - \mathbb{E}X_{n,j}Y_{n,j+1})^2 \\ &\quad + \mathbb{E}(X_{n,j}^2 - \mathbb{E}X_{n,j}^2)^2 + 4\mathbb{E}[(X_{n,j}^2 - \mathbb{E}X_{n,j}^2)(X_{n,j}Y_{n,j+1} - \mathbb{E}X_{n,j}Y_{n,j+1})] \\ &\quad + \mathbb{E}(Y_{n,j+1}^2 - \mathbb{E}Y_{n,j+1}^2)^2 + 4\mathbb{E}[(Y_{n,j+1}^2 - \mathbb{E}Y_{n,j+1}^2)(X_{n,j}Y_{n,j+1} - \mathbb{E}X_{n,j}Y_{n,j+1})]. \end{aligned}$$

By Proposition 3.13, a χ_k -distributed random variable X satisfies $\mathbb{E}X^2 = k$, $\mathbb{E}X^4 = k(k+2)$ and $\mathbb{E}X^3 = (k+1)\mathbb{E}X$. In particular, the variances of $X_{n,j}^2$ and $Y_{n,j+1}^2$ are $2(n-j)\beta$ and $2(m-j)\beta$, and the covariances between $X_{n,j}^2$ and $X_{n,j}Y_{n,j+1}$ and between $Y_{n,j+1}^2$ and $X_{n,j}Y_{n,j+1}$ are both equal to $\mathbb{E}X_{n,j}Y_{n,j+1}$. Hence,

$$\begin{aligned} \mathbb{E}((X_{n,j} + Y_{n,j+1})^2 - \mathbb{E}(X_{n,j} + Y_{n,j+1})^2)^2 &= 2\beta(n-j) + 2\beta(m-j) + 4\beta^2(n-j)(m-j) - 4(\mathbb{E}X_{n,j}Y_{n,j+1})^2 + 8\mathbb{E}X_{n,j}Y_{n,j+1}. \end{aligned}$$

Then

$$\mathbb{E}X_{n,j}Y_{n,j+1} = \beta\sqrt{(n-j)(m-j)} - \frac{1}{4}\sqrt{\frac{n-j}{m-j}} - \frac{1}{4}\sqrt{\frac{m-j}{n-j}} + O((m-j)^{1/2}(n-j)^{-3/2}), \quad (3.19)$$

so finally

$$\frac{1}{\beta} \mathbb{E}((X_{n,j} + Y_{n,j+1})^2 - \mathbb{E}(X_{n,j} + Y_{n,j+1})^2)^2 = 4\left(\sqrt{n-j} + \sqrt{m-j}\right)^2 + O((m-j)(n-j)^{-1}).$$

If $\tau > 0$, then with $j \leq \tau/\delta_n$, the j 's can be dropped at the cost of adding an error term of $O_\tau(m^{1/2}n^{-1/6})$, and this gives

$$\mathbb{E} \Delta \hat{Z}_{n,j}^2 = \frac{(\sqrt{n} + \sqrt{m})^2}{\delta_n^2 nm} + O_\tau(n^{-4/3}) + O_\tau(m^{-1/2}n^{-1/2}) = \delta_n + o_\tau(\delta_n).$$

To check that $\mathbb{E} \Delta \hat{Z}_{n,j}^4 = o_\tau(\delta_n)$, we first use Young's inequality to bound

$$\mathbb{E} \Delta \hat{Z}_{n,j}^4 \leq \frac{1}{2\beta^2 \delta_n^4 n^2 m^2} \left(\mathbb{E}(X_{n,j}^2 + Y_{n,j+1}^2 - \mathbb{E}[X_{n,j}^2 - Y_{n,j+1}^2])^4 + 16 \mathbb{E}(X_{n,j}Y_{n,j+1} - \mathbb{E}X_{n,j}Y_{n,j+1})^4 \right).$$

Because $X_{n,j}$ and $Y_{n,j+1}$ are independent, the first term is proportional to the fourth central moment of a $\chi_{(n+m-2j)\beta}^2$, which is equal to $12(n+m-2j)^2\beta^2 + 48(n+m-2j)\beta$. Hence, that term is $O_\tau(n^{-2/3})$

for $j \leq \tau/\delta_n$. Expanding the fourth central moment of $X_{n,j}Y_{n,j+1}$ and using the basic properties of χ random variables that we used earlier, we get

$$\begin{aligned} \mathbb{E}\left(X_{n,j}Y_{n,j+1} - \mathbb{E} X_{n,j}Y_{n,j+1}\right)^4 &= (n-j)(m-j)\beta^2((n-j)\beta+2)((m-j)\beta+2) \\ &\quad + \left(2(n-j)(m-j)\beta^2 - 4(n-j)\beta - 4(m-j)\beta - 4\right)(\mathbb{E} X_{n,j}Y_{n,j+1})^2 - 3(\mathbb{E} X_{n,j}Y_{n,j+1})^4. \end{aligned}$$

From the expansion (3.19) of the first moment, it is now immediate to see that the terms of order 4 and 3 in $n-j$ and $m-j$ cancel out. Hence, this fourth central moment is $O((m-j)^2)$, and $\mathbb{E} \Delta \hat{Z}_{n,j}^4 = O_\tau(n^{-2/3})$ for $j \leq \tau/\delta_n$. We note in passing that the same arguments also show that $\mathbb{E} |\Delta Z_{n,j}^a|^4 = O_\tau(n^{-2/3})$ and $\mathbb{E} |\Delta Z_{n,j}^b|^4 = O_\tau(n^{-2/3})$.

It remains to see that the other part of $\mathcal{Z}$, namely $\check{\mathcal{Z}}(t) := \sum_{j=1}^{\lfloor t/\delta_n \rfloor} \Delta \check{Z}_{n,j}$, converges compactly to 0 in probability. Because of its telescoping structure, for any k ,

$$\sum_{j=1}^k \Delta \check{Z}_{n,j} = \frac{1}{2\delta_n \sqrt{\beta n m}} (Y_{n,k+1}^2 - \mathbb{E} Y_{n,k+1}^2 - Y_{n,1}^2 + \mathbb{E} Y_{n,1}^2),$$

and it follows that for $k \leq \tau/\delta_n$,

$$\mathbb{E} \left(\sum_{j=1}^k \Delta \check{Z}_{n,j} \right)^4 \leq \frac{1}{2\beta^2 \delta_n^4 n^2 m^2} \left(\mathbb{E} (Y_{n,k+1}^2 - \mathbb{E} Y_{n,k+1}^2)^4 + \mathbb{E} (Y_{n,1}^2 - \mathbb{E} Y_{n,1}^2)^4 \right) = O_\tau(n^{-2/3}).$$

Hence, if $\varepsilon \in (0, 1/12)$ then $\mathbb{P} \left[\max_{1 \leq k \leq \tau/\delta_n} \left| \sum_{j=1}^k \Delta \check{Z}_{n,j} \right| > n^{-1/12+\varepsilon} \right] = O_\tau(n^{-4\varepsilon})$ by Markov's inequality, and $\check{\mathcal{Z}} \rightarrow 0$ uniformly on $[0, \tau]$ in probability. This finishes to check the conditions on $\hat{\mathcal{Z}}$ and $\check{\mathcal{Z}}$, and thus by Lemma 3.11, condition *ii*) of Hypothesis 3.1 is satisfied. $\blacksquare$

3.2 The stochastic Bessel operator

This section is devoted to the stochastic Bessel operator. In §3.2.1, we review a few of its important properties. In §3.2.2, we derive asymptotics to solutions of $\mathfrak{G}_{\beta,a} f = \lambda f$ as $t \rightarrow -\infty$ for $\lambda > 0$. Finally, in §3.2.3, we prove the convergence of suitably scaled tridiagonal random matrix models to the stochastic Bessel operator, at the canonical system level.

3.2.1 Basic properties of the stochastic Bessel operator

Recall from Definition 1.1 that the stochastic Bessel operator $\mathfrak{G}_{\beta,a}$, which was introduced by Ramírez and Rider in [58], is the random operator acting on a function $f: [0, \infty) \rightarrow \mathbb{R}$ by sending it to

$$\mathfrak{G}_{\beta,a} f = -\frac{1}{w_{\beta,a}} (p_{\beta,a} f')'$$

where

$$p_{\beta,a}(t) := \exp\left(-at - \frac{2}{\sqrt{\beta}}B(t)\right) \quad \text{and} \quad w_{\beta,a}(t) := \exp\left(-(a+1)t - \frac{2}{\sqrt{\beta}}B(t)\right)$$

for two parameters $\beta > 0$ and $a > -1$, and where B is a standard Brownian motion. The idea from this operator comes again from the work of Edelman and Sutton [30], although the formulation they used was a bit different.

The stochastic Bessel operator is a generalized Sturm–Liouville operator of the form (2.15) that we saw in §2.4—in fact, it is even a Sturm–Liouville operator in the classical sense. It has parameters $p = p_{\beta,a}$, $w = w_{\beta,a}$ and $q = s = 0$, so in particular the associated quasi-derivative is $f^{[1]} := p_{\beta,a}f'$, and it is well defined on the domain $\mathfrak{D}_{\mathfrak{G}_{\beta,a}} := \{f \in \text{AC}_{\text{loc}}(0, \infty) : p_{\beta,a}f \in \text{AC}_{\text{loc}}(0, \infty)\}$.

Characterization of the kernel of the stochastic Bessel operator. An important property of the stochastic Bessel operator is that its kernel can be computed explicitly. Indeed, a function $f \in \text{AC}_{\text{loc}}(0, \infty)$ solves $\mathfrak{G}_{\beta,a}f = 0$ if and only if $(p_{\beta,a}f)' = 0$, so the solutions are exactly the functions f that make $p_{\beta,a}f'$ constant. If that constant is zero, then f itself must be constant, and otherwise f must be a constant plus a multiple of

$$\varpi_{\beta,a}(t) := \int_0^t \frac{1}{p_{\beta,a}(s)} ds = \int_0^t \exp\left(as + \frac{2}{\sqrt{\beta}}B(s)\right) ds. \quad (3.20)$$

Hence, 1 and $\varpi_{\beta,a}$ are a pair of fundamental solutions to $\mathfrak{G}_{\beta,a}f = 0$, and the general solution to that equation is $f = f(0) + f'(0)\varpi_{\beta,a}$.

Having explicit expressions for these solutions simplifies the analysis of this operator in some aspects, at least from the canonical system point of view. In particular, we will see that it yields an explicit representation of the equivalent canonical system's coefficient matrix (3.24), as opposed to the usual implicit representation of the form (2.19) one has for the coefficient matrix of a canonical system equivalent to general Sturm–Liouville operator.

The spectrum of the stochastic Bessel operator and the Weyl alternative. The $\text{Bessel}_{\beta,a}$ point process, which is defined to be the spectrum of the stochastic Bessel operator for boundary conditions $f(0) = 0$ and $\lim_{t \rightarrow \infty} p_{\beta,a}(t)f'(t) = 0$, corresponds to the hard edge scaling limit of β -ensembles. We have seen an example of this in Theorem 1.2 (taken from [58, Theorem 1]), which shows that this spectrum is a scaling limit of the Laguerre β -ensemble, and this has been generalized to a wider class of potentials [63]. Moreover, when changing the boundary condition at 0, the spectrum of the stochastic Bessel operator corresponds to the scaling limit of spiked versions of the Laguerre β -ensemble [59].

It is clear from the definition that the stochastic Bessel operator is always limit circle at 0. However, as the following result shows, it is limit circle at infinity only if $|a| < 1$. Hence, the theory of generalized Sturm–Liouville operator shows that the boundary condition at infinity is not necessary when $a \geq 1$, as the operator is already self-adjoint. This is the content of the following result.

Proposition 3.15 [54, Proposition 4]. $\mathfrak{G}_{\beta,a}$ is a.s. limit circle at infinity if $|a| < 1$, and a.s. limit point at infinity if $a \geq 1$.

Proof. By the Weyl alternative, it suffices to check the behavior of solutions to $\mathfrak{G}_{\beta,a}f = zf$ at a single value of $z \in \mathbb{C}$. For simplicity, we take $z = 0$. As pointed out above, 1 and $\varpi_{\beta,a}$ are a pair of fundamental solutions to $\mathfrak{G}_{\beta,a}f = 0$, and by definition of $w_{\beta,a}$ it is clear that $1 \in L^2((0, \infty), w_{\beta,a} d\mathcal{L})$ a.s. in any case. Therefore, $\mathfrak{G}_{\beta,a}$ is a.s. limit circle at infinity if $\varpi_{\beta,a} \in L^2((0, \infty), w_{\beta,a} d\mathcal{L})$ a.s., and it is a.s. limit point at infinity if $\varpi_{\beta,a} \notin L^2((0, \infty), w_{\beta,a} d\mathcal{L})$ a.s.

We can analyse the behavior of $\varpi_{\beta,a}$ using the following property of Brownian motion (which follows from Proposition A.1): for any $\varepsilon > 0$, there is a $C > 0$ such that

$$\mathbb{P}\left[\forall t \geq 0, \frac{2}{\sqrt{\beta}}|B(t)| < C(1 + t^{3/4})\right] \geq 1 - \varepsilon. \quad (3.21)$$

Moreover, for any $\delta > 0$ there is a $T > 0$ such that $C(1 + t^{3/4}) \leq \delta t$ for all $t \geq T$.

On the event from (3.21), if $t \geq T$ with T as above,

$$\varpi_{\beta,a}(t) \leq \int_0^T e^{as+C(1+s^{3/4})} ds + \int_T^t e^{(a+\delta)s} ds = C_T + \frac{e^{(a+\delta)t}}{a+\delta}$$

where

$$C_T := \int_0^T e^{as+C(1+s^{3/4})} ds - \frac{e^{(a+\delta)T}}{a+\delta}.$$

Therefore

$$\begin{aligned} \|\varpi_{\beta,a}\|_{w_{\beta,a}}^2 &= \int_0^\infty \varpi_{\beta,a}^2(t) w_{\beta,a}(t) dt \\ &\leq \int_0^T \left(\int_0^t e^{as+C(1+s^{3/4})} ds \right)^2 e^{-(a+1)t+C(1+t^{3/4})} dt + \int_T^\infty \left(C_T + \frac{e^{(a+\delta)t}}{a+\delta} \right)^2 e^{-(a+1-\delta)t} dt \\ &\leq C'_T + 2C_T^2 \int_T^\infty e^{-(a+1-\delta)t} dt + \frac{2}{(a+\delta)^2} \int_T^\infty e^{-(1-a-3\delta)t} dt \end{aligned}$$

where C'_T is the first term of the second line, and where the third line follows from Young's inequality. If $|a| < 1$ and $\delta \in (0, \frac{1-a}{3})$, then this is finite, so $\|\varpi_{\beta,a}\|_{w_{\beta,a}} < \infty$ with probability at least $1 - \varepsilon$ for any $\varepsilon > 0$. It follows that $\|\varpi_{\beta,a}\|_{w_{\beta,a}} < \infty$ a.s., which shows that $\mathfrak{G}_{\beta,a}$ is a.s. limit circle at infinity.

With $a > 1$ and $\delta \in (0, \frac{a-1}{3})$, essentially the same argument as above shows that $\mathfrak{G}_{\beta,a}$ is a.s. limit point at infinity. Indeed, on the event from (3.21), we now have for $t \geq T$ that

$$\varpi_{\beta,a}(t) \geq \tilde{C}_T + \frac{e^{(a-\delta)t}}{a-\delta} \quad \text{where} \quad \tilde{C}_T := \int_0^T e^{as-C(1+s^{3/4})} ds - \frac{e^{(a-\delta)T}}{a-\delta},$$

and therefore

$$\begin{aligned} \|\varpi_{\beta,a}\|_{w_{\beta,a}}^2 &\geq \int_T^\infty \left(\tilde{C}_T + \frac{e^{(a-\delta)t}}{a-\delta} \right)^2 e^{-(a+1+\delta)t} dt \\ &= \tilde{C}_T^2 \int_T^\infty e^{-(a+1+\delta)t} dt + \frac{2\tilde{C}_T}{a-\delta} \int_T^\infty e^{-(1+2\delta)t} dt + \frac{1}{(a-\delta)^2} \int_T^\infty e^{(a-1-3\delta)t} dt, \end{aligned}$$

which is infinite because the last integral diverges while the first two converge. It follows that $\|\varpi_{\beta,a}\|_{w_{\beta,a}} = \infty$ a.s., so that $\mathfrak{G}_{\beta,a}$ is a.s. limit point at infinity.

This only leaves out the case $a = 1$, which is more delicate as in that case the bound on the Brownian motion from (3.21) is not strong enough to allow us to conclude. To replace it, we define a sequence of stopping times by setting $\tau_0 := \inf\{t \geq 1 : B(t) = 0\}$ and then recursively $\tau_n := \inf\{t \geq \tau_{n-1} + 2 : B(t) = 0\}$ for $n \in \mathbb{N}$. Then, let $A_n := \{\forall t \in [\tau_n, \tau_n + 1], |B(t)| \leq 1\}$. By the strong Markov property of Brownian motion, the A_n 's are independent, and for every n ,

$$\mathbb{P}(A_n) = \mathbb{P}\left[\sup_{t \in [\tau_n, \tau_n + 1]} |B(t)| \leq 1\right] = \mathbb{P}\left[\sup_{t \in [0, 1]} |B(t)| \leq 1\right] \geq 1 - 2\mathbb{P}\left[\sup_{t \in [0, 1]} B(t) > 1\right].$$

The reflection principle shows that $\mathbb{P}[\sup_{t \in [0, 1]} B(t) > 1] = 2\mathbb{P}[B(1) > 1] = 1 - \text{erf}(1/\sqrt{2})$, so $\mathbb{P}(A_n) \geq 2\text{erf}(1/\sqrt{2}) - 1$. This number is strictly positive, so the second Borel–Cantelli lemma implies that $\mathbb{P}(\bigcap_{n \in \mathbb{N}} \bigcup_{k \geq n} A_k) = 1$. On that event, there is always a subsequence $\{A_{n_k}\}_{k \in \mathbb{N}}$ in which each event occurs, and this gives us a sequence of intervals of length 1 on which the Brownian motion is bounded by 1. We can use this to estimate the norm of $\varpi_{\beta,a}$:

$$\begin{aligned} \|\varpi_{\beta,a}\|_{w_{\beta,a}}^2 &= \int_0^\infty \left(\int_0^t e^{s + \frac{2}{\sqrt{\beta}} B(s)} ds \right)^2 e^{-2t - \frac{2}{\sqrt{\beta}} B(t)} dt \geq \sum_{k=1}^\infty \int_{\tau_{n_k} + 1/2}^{\tau_{n_k} + 1} \left(\int_{\tau_{n_k}}^{\tau_{n_k} + 1/2} e^{s - \frac{2}{\sqrt{\beta}} B(s)} ds \right)^2 e^{-2t - \frac{2}{\sqrt{\beta}} B(t)} dt \\ &\geq \frac{1}{2} e^{-6/\sqrt{\beta}} \sum_{k=1}^\infty (e^{\tau_{n_k} + 1/2} - e^{\tau_{n_k}})^2 (e^{-2(\tau_{n_k} + 1/2)} - e^{-2(\tau_{n_k} + 1)}) \end{aligned}$$

All factors of $e^{\tau_{n_k}}$ in the right-hand side cancel out, so the summand is a positive constant and the series diverges. Therefore, $\|\varpi_{\beta,a}\|_{w_{\beta,a}} = \infty$ a.s. when $a = 1$, which shows that $\mathfrak{G}_{\beta,a}$ is a.s. limit point at infinity and concludes the proof. $\blacksquare$

We record here for future reference the following result, which is essentially a more explicit version of something we have seen in the proof of the last proposition.

Lemma 3.16 [54, Lemma 5]. *For any $\varepsilon, \delta > 0$, there is a $C_{\varepsilon, \delta} > 0$ such that*

$$\mathbb{P}[\forall t \geq 0, w_{\beta, a}(t)(1 + \varpi_{\beta, a}^2(t)) \leq C_{\varepsilon, \delta} e^{-(1-|a|-\delta)t}] \geq 1 - \varepsilon.$$

Remark. In particular, if $|a| < 1$ then we can choose $\delta < 1 - |a|$ and the upper bound is integrable.

Proof. Recall the good event from (3.21), and take $T > 0$ such that $C(1 + t^{3/4}) \leq \delta t$ for $t \geq T$. On that event, the bound $C(1 + t^{3/4})$ on the Brownian motion is itself bounded on the compact $[0, T]$ by a deterministic constant, so by definition, $w_{\beta, a}$ and $\varpi_{\beta, a}$ are also bounded on $[0, T]$ by deterministic constants. Now, for $t \geq T$, on the good event it also holds that $w_{\beta, a}(t) \leq e^{-(1+a-\delta)t}$, and therefore there is a $C' > 0$ such that $w_{\beta, a}(t) \leq C' e^{-(1+a-\delta)t}$ for all $t \geq 0$. Similarly, we have seen in the proof of Proposition 3.15 that on the event from (3.21), there is a $C_T \in \mathbb{R}$ such that $\varpi_{\beta, a}(t) \leq C_T + \frac{1}{a+\delta} e^{(a+\delta)t}$ for $t \geq T$, meaning that there is a $C'' > 0$ such that $\varpi_{\beta, a}(t) \leq C'' e^{(a+\delta)t}$ for all $t \geq 0$. Combining the bounds on $w_{\beta, a}$ and $\varpi_{\beta, a}$ shows that $\varpi_{\beta, a}^2(t)w_{\beta, a}(t) \leq C' C''^2 e^{-(1-a-3\delta)t}$ for all $t \geq 0$, and summing up the bounds on $w_{\beta, a}$ and $\varpi_{\beta, a}^2 w_{\beta, a}$ completes the proof. ■

SDE characterization of solutions to the eigenvalue equation. Let $f \in \mathfrak{D}_{\mathfrak{G}_{\beta, a}}$ solve the eigenvalue equation $\mathfrak{G}_{\beta, a} f = z f$ for some $z \in \mathbb{C}$. This solution is always defined pathwise, but it can be useful to have a characterization of f as a stochastic process, which can be obtained through Itô calculus when the initial conditions $f(0)$ and $f'(0)$ are independent of the Brownian motion B that drives $\mathfrak{G}_{\beta, a}$. Indeed, if f solves $\mathfrak{G}_{\beta, a} f = z f$, then by definition $p_{\beta, a} f'$ is absolutely continuous, so

$$d(p_{\beta, a} f')'(t) = (p_{\beta, a} f')'(t) dt = -z w_{\beta, a}(t) f(t) dt.$$

As long as $f(0)$ and $f'(0)$ are independent of B , Itô's formula also yields

$$d(p_{\beta, a} f')'(t) = f'(t) dp_{\beta, a}(t) + p_{\beta, a}(t) df'(t) + d\langle p_{\beta, a}, f' \rangle(t).$$

Comparing the two expressions for $d(p_{\beta, a} f')$, we get

$$df'(t) = -\frac{1}{p_{\beta, a}(t)} \left(z w_{\beta, a}(t) f(t) dt + f'(t) dp_{\beta, a}(t) + d\langle p_{\beta, a}, f' \rangle(t) \right).$$

The Itô differential of $p_{\beta, a}$ is

$$dp_{\beta, a}(t) = -\left(a - \frac{2}{\beta}\right) p_{\beta, a}(t) dt - \frac{2}{\sqrt{\beta}} p_{\beta, a}(t) dB(t) \quad \text{so} \quad d\langle p_{\beta, a}, f' \rangle(t) = -\frac{4}{\beta} p_{\beta, a}(t) f'(t) dt,$$

and it follows that f solves

$$\begin{aligned} df(t) &= f'(t) dt, \\ df'(t) &= -ze^{-t} f(t) dt + \left(a + \frac{2}{\beta}\right) f'(t) dt + \frac{2}{\sqrt{\beta}} f'(t) dB(t). \end{aligned} \tag{3.22}$$

This SDE first appeared in [58].

A translation property of solutions to the eigenvalue equation. We now point out a property of the stochastic Bessel operator that will be essential later in §4.4. The following result and its proof are taken from [54].

Proposition 3.17 [54, Proposition 6]. *Let $\mathfrak{G}_{\beta,a}$ be defined from a two-sided standard Brownian motion B , and set $\mathfrak{G}_{\beta,a,E} := \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E)$ for some $E > 0$. Then f solves $\mathfrak{G}_{\beta,a,E}f = zf$ for some $z \in \mathbb{C}$ if and only if $\tilde{f}(t) := f(t + \log E)$ solves $\tilde{\mathfrak{G}}_{\beta,a}\tilde{f} = (1 + \frac{z}{2\sqrt{E}})\tilde{f}$ where $\tilde{\mathfrak{G}}_{\beta,a}$ has the law of $\mathfrak{G}_{\beta,a}$, but is defined from the two-sided standard Brownian motion $B(\cdot + \log E) - B(\log E)$.*

Proof. By definition, $\tilde{\mathfrak{G}}_{\beta,a}$ is the stochastic Bessel operator defined from the Brownian motion $B(\cdot + \log E) - B(\log E)$. More explicitly, $\tilde{\mathfrak{G}}_{\beta,a}f = -\frac{1}{\tilde{w}_{\beta,a}}(\tilde{p}_{\beta,a}f')'$ where

$$\tilde{p}_{\beta,a}(t) = \exp\left(-at - \frac{2}{\sqrt{\beta}}(B(t + \log E) - B(\log E))\right) = E^a e^{\frac{2}{\sqrt{\beta}}B(\log E)} p_{\beta,a}(t + \log E)$$

and likewise $\tilde{w}_{\beta,a}(t) = E^{a+1} e^{\frac{2}{\sqrt{\beta}}B(\log E)} w_{\beta,a}(t + \log E)$. Therefore, if $t \in \mathbb{R}$, then

$$\tilde{\mathfrak{G}}_{\beta,a}\tilde{f}(t) = -\frac{1}{\tilde{w}_{\beta,a}(t)}(\tilde{p}_{\beta,a}\tilde{f}')'(t) = -\frac{1}{E w_{\beta,a}(t + \log E)}(p_{\beta,a}f')'(t + \log E) = \frac{1}{E}\mathfrak{G}_{\beta,a}f(t + \log E).$$

But if f solves $\mathfrak{G}_{\beta,a,E}f = zf$, then by definition of $\mathfrak{G}_{\beta,a,E}$ it also solves $\mathfrak{G}_{\beta,a}f = (E + \frac{z\sqrt{E}}{2})f$, and combining with the above we see that $\tilde{\mathfrak{G}}_{\beta,a}\tilde{f}(t) = (1 + \frac{z}{2\sqrt{E}})\tilde{f}(t)$ for all $t \in \mathbb{R}$. Taking these last steps in reverse yields the other implication. ■

Canonical system representation. The stochastic Bessel operator can naturally be associated with a random canonical system that is equivalent to its eigenvalue equation, following the method detailed in §2.4. As the equivalence is made from a pair of fundamental solutions to $\mathfrak{G}_{\beta,a}f = 0$, we can choose to work with the solutions 1 and $\varpi_{\beta,a}$ and build the mapping between solutions from the matrix

$$A_{\beta,a} := \begin{pmatrix} 1 & 0 \\ \varpi_{\beta,a} & 1 \end{pmatrix}. \quad (3.23)$$

By definition, this matrix solves

$$A'_{\beta,a} = \begin{pmatrix} 0 & 0 \\ p_{\beta,a}^{-1} & 0 \end{pmatrix} A_{\beta,a} \quad \text{with} \quad A_{\beta,a}(0) = I_2.$$

The method from §2.4.2 shows that f solves $\mathfrak{G}_{\beta,a}f = zf$ if and only if $u := A_{\beta,a}^{-1} \begin{pmatrix} f^{[1]} \\ f \end{pmatrix}$ solves the canonical system $Ju' = -zG_{\beta,a}u$ on $(0, \infty)$, where

$$G_{\beta,a} := w_{\beta,a} J A_{\beta,a}^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} A_{\beta,a} = w_{\beta,a} \begin{pmatrix} \varpi_{\beta,a}^2 & \varpi_{\beta,a} \\ \varpi_{\beta,a} & 1 \end{pmatrix} = w_{\beta,a} \psi_{\beta,a} \psi_{\beta,a}^*. \quad (3.24)$$

and $\psi_{\beta,a} := (\varpi_{\beta,a}, 1)$. This canonical system is limit circle at 0, and it is limit circle at infinity when $|a| < 1$ or limit point at infinity when $a \geq 1$ by Proposition 3.15. Moreover, because $A_{\beta,a}(0) = I_2$, the Dirichlet boundary condition $f(0) = 0$ at zero maps to the boundary condition $e_0^* J u(0) = 0$ for the canonical system. In the same way, because the second column of $A_{\beta,a}$ is always equal to $e_{\pi/2}$, the Neumann boundary condition at infinity $\lim_{t \rightarrow \infty} f^{[1]}(t) = 0$ maps to the boundary condition $e_{\pi/2}^* J u(\infty) = 0$.

As pointed out in §2.4.3, the norm of the associated $L_{G_{\beta,a}}^2(0, \infty)$ space has a specific form in terms on $\varpi_{\beta,a}$. Indeed, if $u: (0, \infty) \rightarrow \mathbb{C}^2$, then

$$\|u\|_{G_{\beta,a}}^2 = \int_0^\infty |\langle u(t), \psi_{\beta,a}(t) \rangle|^2 w_{\beta,a}(t) dt = \int_0^\infty |u_1(t) \varpi_{\beta,a}(t) + u_2(t)|^2 w_{\beta,a}(t) dt. \quad (3.25)$$

In particular, if $u = A_{\beta,a}^{-1} \begin{pmatrix} h \\ h^{[1]} \end{pmatrix}$ for an $h: (0, \infty) \rightarrow \mathbb{C}$, then $\|u\|_{G_{\beta,a}} = \|h\|_{w_{\beta,a}}$.

3.2.2 Asymptotics of solutions to the eigenvalue equation as $t \rightarrow -\infty$

In this section, we understand the stochastic Bessel operator as being defined on the whole of $\mathbb{R}$ from a two-sided Brownian motion, and we derive asymptotics for solutions to the eigenvalue equation $\mathfrak{G}_{\beta,a} f = \lambda f$ as $t \rightarrow -\infty$ when $\lambda > 0$. In addition to providing helpful insight into the behavior of these solutions on the negative real line, these asymptotics will be crucial in our spectral analysis of the hard edge to bulk transition later in §4.4.3.

The asymptotics we derive, which are essentially taken from [54, §5.3.1], are the following.

Theorem 3.18 [54, Theorem 3]. *Let $\mathfrak{G}_{\beta,a}$ be defined on the full real line from a two-sided Brownian motion. If f solves $\mathfrak{G}_{\beta,a} f = \lambda f$ for $\lambda > 0$ and if $f(0)$ and $f'(0)$ are independent of the restriction of B to $(-\infty, 0)$, then for $t \geq 0$,*

$$f(-t) = C_f \lambda^{-1/4} e^{(\frac{1}{2\beta} - \frac{a}{2} - \frac{1}{4})t + X(t)} \cos \xi_{\beta,a}(t) \quad \text{and} \quad f'(-t) = C_f \lambda^{1/4} e^{(\frac{1}{2\beta} - \frac{a}{2} + \frac{1}{4})t + X(t)} \sin \xi_{\beta,a}(t)$$

where $C_f^2 := f^2(0) + f'^2(0)$, $\xi_{\beta,a}$ is a process such that $\xi_{\beta,a}(t) - 2\pi \left\lfloor \frac{\xi_{\beta,a}(t)}{2\pi} \right\rfloor \rightarrow U \sim \text{Unif}[0, 2\pi)$ in law as $t \rightarrow \infty$, and X is a process such that for any $\varepsilon, \delta > 0$, there is a $C > 0$ for which

$$\mathbb{P} \left[\forall t \geq 0, |X(t)| \leq C(1 + t^{1/2+\delta}) \right] \geq 1 - \varepsilon.$$

We follow the same proof method that we used in §3.1.2 to derive the analog asymptotics for the stochastic Airy operator: we construct a pair of polar coordinates to describe the growth and oscillations of f and f' , and then we analyse their asymptotic behavior.

The first step is to construct the polar coordinates.

Proposition 3.19 [54, Proposition 18]. Let $\mathfrak{G}_{\beta,a}$ be defined from a two-sided standard Brownian motion B , and write $\tilde{B}(t) := B(-t)$. If f solves $\mathfrak{G}_{\beta,a}f = \lambda f$ with $\lambda > 0$ and if $f(0)$ and $f'(0)$ are independent of the restriction of $\tilde{B}$ to $(0, \infty)$, then for $t \geq 0$,

$$f(-t) = \lambda^{-1/4} e^{-t/4 + r_{\beta,a}(t)} \cos \xi_{\beta,a}(t) \quad \text{and} \quad f'(-t) = -\lambda^{1/4} e^{t/4 + r_{\beta,a}(t)} \sin \xi_{\beta,a}(t)$$

where $r_{\beta,a}$ and $\xi_{\beta,a}$ solve

$$\begin{aligned} dr_{\beta,a}(t) &= \left(\frac{1}{2\beta} - \frac{a}{2} \right) dt + \left(\left(\frac{1}{4} + \frac{a}{2} \right) \cos 2\xi_{\beta,a}(t) - \frac{1}{2\beta} \cos 4\xi_{\beta,a}(t) \right) dt + \frac{2}{\sqrt{\beta}} \sin^2 \xi_{\beta,a}(t) d\tilde{B}(t), \\ d\xi_{\beta,a}(t) &= -\sqrt{\lambda} e^{t/2} dt - \left(\left(\frac{1}{4} + \frac{a}{2} \right) \sin 2\xi_{\beta,a}(t) - \frac{1}{2\beta} \sin 4\xi_{\beta,a}(t) \right) dt + \frac{1}{\sqrt{\beta}} \sin 2\xi_{\beta,a}(t) d\tilde{B}(t) \end{aligned}$$

with $r_{\beta,a}(0) = \frac{1}{2} \log(f^2(0) + f'^2(0))$ and $\xi_{\beta,a}(0) = \arctan(f'(0)/f(0))$.

Proof. Let $y(t) := f(-t)$. Reversing time in the equation $\mathfrak{G}_{\beta,a}f = \lambda f$ yields $-\frac{1}{\tilde{w}_{\beta,a}(t)} (\tilde{p}_{\beta,a}y')'(t) = \lambda y(t)$ where $\tilde{p}_{\beta,a}(t) := p_{\beta,a}(-t) = \exp(at - \frac{2}{\sqrt{\beta}} \tilde{B}(t))$ and $\tilde{w}_{\beta,a}(t) := w_{\beta,a}(-t) = e^t \tilde{p}_{\beta,a}(t)$. Note that $d\tilde{p}_{\beta,a}(t) = (a + \frac{2}{\beta}) \tilde{p}_{\beta,a}(t) - \frac{2}{\sqrt{\beta}} \tilde{p}_{\beta,a}(t) d\tilde{B}(t)$, so applying Itô's formula in the same way as we did earlier to obtain the SDE (3.22) for f' , we get

$$dy'(t) = -\lambda e^t y(t) dt - \left(a - \frac{2}{\beta} \right) y'(t) dt + \frac{2}{\sqrt{\beta}} y'(t) d\tilde{B}(t).$$

Now, set r and ξ as real processes that satisfy $e^{r+i\xi} = Sy + iy'/s$ where $S(t) := \lambda^{1/4} e^{t/4}$. Then by Itô's formula, omitting the explicit time dependencies to simplify notation,

$$\begin{aligned} dr + i d\xi &= \frac{1}{Sy + iy'/s} \left(S'y dt + Sy' dt + \frac{i}{S} dy' - \frac{iS'}{S^2} y' dt \right) + \frac{1}{2(Sy + iy'/s)^2} \frac{1}{S^2} d\langle y' \rangle \\ &= e^{-2r} \left(\left(SS'y^2 + S^2yy' - \frac{S'y'^2}{S^3} \right) dt + \frac{y'}{S^2} dy' - i \left(\frac{2S'yy'}{S} + y'^2 \right) dt + iy dy' \right) \\ &\quad + \frac{e^{-4r}}{2S^2} \left(S^2y^2 - \frac{y'^2}{S^2} - 2iyy' \right) d\langle y' \rangle \\ &= e^{-2r} \left(\left(SS'y^2 + S^2yy' - \frac{S'y'^2}{S^3} - \frac{\lambda e^t yy'}{S^2} - \left(a - \frac{2}{\beta} \right) \frac{y'^2}{S^2} \right) dt + \frac{2}{\sqrt{\beta}} \frac{y'^2}{S^2} d\tilde{B} \right. \\ &\quad \left. - i \left(\frac{2S'yy'}{S} + y'^2 + \lambda e^t y^2 + \left(a - \frac{2}{\beta} \right) yy' \right) dt + \frac{2i}{\sqrt{\beta}} yy' d\tilde{B} \right) \\ &\quad + \frac{2}{\beta} e^{-4r} \left(y^2 y'^2 - \frac{y'^4}{S^4} - \frac{2iyy'^3}{S^2} \right) dt. \end{aligned}$$

With $S(t) = \lambda^{1/4} e^{t/4}$, the second and fourth terms on the first line cancel out, and the second line simplifies with $y'^2 + \lambda e^t y^2 = S^2 e^{2r}$. This also gives $S'(t) = \frac{1}{4} S(t)$, so that taking the real and

imaginary parts in the above while substituting $Sy = e^r \cos \xi$ and $y'/s = e^r \sin \xi$ yields

$$dr = \left(\frac{1}{4} \cos^2 \xi - \left(\frac{1}{4} + a - \frac{2}{\beta} \right) \sin^2 \xi + \frac{2}{\beta} \sin^2 \xi (\cos^2 \xi - \sin^2 \xi) \right) dt + \frac{2}{\sqrt{\beta}} \sin^2 \xi d\tilde{B}$$

and

$$d\xi = -\sqrt{\lambda} e^{t/2} dt - \left(\left(\frac{1}{2} + a - \frac{2}{\beta} \right) \cos \xi \sin \xi + \frac{4}{\beta} \cos \xi \sin^3 \xi \right) dt + \frac{2}{\sqrt{\beta}} \cos \xi \sin \xi d\tilde{B}.$$

Simplifying with trigonometric identities shows that $r_{\beta,a} := r$ and $\xi_{\beta,a} := \xi$ solve the announced SDEs. The representations for f and f' are then recovered using that $f(-t) = \frac{1}{S(t)} e^{r_{\beta,a}(t)} \cos \xi_{\beta,a}(t)$ and that $f'(-t) = -S(t) e^{r_{\beta,a}(t)} \sin \xi_{\beta,a}(t)$. ■

We now analyse the asymptotic behavior of these polar coordinates $r_{\beta,a}$ and $\xi_{\beta,a}$. The results are given in Proposition 3.21 and 3.22, of which Theorem 3.18 is a direct consequence. Before proceeding, we need a simple preliminary result.

Lemma 3.20. *Let $\xi_{\beta,a}$ solve the SDE from Proposition 3.19. For any $k \neq 0$ and $\varepsilon > 0$, there is a $C > 0$ such that*

$$\mathbb{P} \left[\sup_{t \geq 0} \left| \int_0^t e^{ki\xi_{\beta,a}(s)} ds \right| > C \right] < \varepsilon.$$

Proof. By Itô's formula, $e^{ki\xi_{\beta,a}}$ satisfies on $[0, \infty)$ the SDE

$$d(e^{ki\xi_{\beta,a}})(s) = e^{ki\xi_{\beta,a,E}(s)} \left(-ki\sqrt{\lambda} e^{s/2} ds + R_k(s) ds + \frac{ki}{\sqrt{\beta}} \sin 2\xi_{\beta,a,E}(s) d\tilde{B}(s) \right) \quad (3.26)$$

where

$$R_k := -\frac{k^2}{4\beta} - ki \left(\frac{1}{4} + \frac{a}{2} \right) \sin 2\xi_{\beta,a,E} + \frac{ki}{2\beta} \sin 4\xi_{\beta,a,E} + \frac{k^2}{4\beta} \cos 4\xi_{\beta,a,E}. \quad (3.27)$$

Hence,

$$\begin{aligned} ki\sqrt{\lambda} \int_0^t e^{ki\xi_{\beta,a}(s)} ds &= - \int_0^t e^{-s/2} d(e^{ki\xi_{\beta,a}})(s) + \int_0^t R_k(s) e^{ki\xi_{\beta,a}(s)-s/2} ds \\ &\quad + \frac{ki}{\sqrt{\beta}} \int_0^t e^{ki\xi_{\beta,a}(s)-s/2} \sin 2\xi_{\beta,a}(s) d\tilde{B}(s). \end{aligned}$$

Integrating by parts the first term yields

$$\int_0^t e^{-s/2} d(e^{ki\xi_{\beta,a}})(s) = e^{ki\xi_{\beta,a}(s)-s/2} \Big|_0^t + \frac{1}{2} \int_0^t e^{ki\xi_{\beta,a}(s)-s/2} ds.$$

Since R_k is bounded, it follows that

$$\begin{aligned} \left| \int_0^t e^{ki\xi_{\beta,a}(s)} ds \right| &\leq \frac{1}{|k|\sqrt{\lambda}} \left(2 + \left(\|R_k\|_\infty + \frac{1}{2} \right) \int_0^t e^{-s/2} ds \right. \\ &\quad \left. + \frac{|k|}{\sqrt{\beta}} \left| \int_0^t e^{ki\xi_{\beta,a}(s)-s/2} \sin 2\xi_{\beta,a}(s) d\tilde{B}(s) \right| \right). \end{aligned}$$

All terms on the first line are clearly bounded. As to the remaining stochastic integral, its real and imaginary parts both have quadratic variations bounded by $\frac{1}{\beta\lambda} \int_0^t e^{-s} ds = \frac{1}{\beta\lambda}$, so Bernstein's inequality for martingales shows that for all $x > 0$,

$$\mathbb{P} \left[\sup_{t \geq 0} \frac{1}{\sqrt{\beta\lambda}} \left| \int_0^t e^{ki\xi_{\beta,a}(s)-s/2} \sin 2\xi_{\beta,a}(s) d\tilde{B}(s) \right| > x \right] \leq 4 \exp\left(-\frac{\beta\lambda x^2}{2}\right).$$

Taking x large enough so that this exponential tail bound is less than ε completes the proof. $\blacksquare$

Proposition 3.21 [54, Proposition 19]. *Let $r_{\beta,a}$ solve the SDE from Proposition 3.19. For any $\varepsilon, \delta > 0$, there is a $C > 0$ such that*

$$\mathbb{P} \left[\forall t \geq 0, \left| r_{\beta,a}(t) - r_{\beta,a}(1) - \left(\frac{1}{2\beta} - \frac{a}{2} \right) t \right| \leq C(1 + t^{1/2+\delta}) \right] \geq 1 - \varepsilon.$$

Proof. Recall from Proposition 3.19 that $r_{\beta,a}$ solves

$$dr_{\beta,a}(t) = \left(\frac{1}{2\beta} - \frac{a}{2} \right) dt + \left(\left(\frac{1}{4} + \frac{a}{2} \right) \cos 2\xi_{\beta,a}(t) - \frac{1}{2\beta} \cos 4\xi_{\beta,a}(t) \right) dt + \frac{2}{\sqrt{\beta}} \sin^2 \xi_{\beta,a}(t) d\tilde{B}(t)$$

where $\xi_{\beta,a}$ solves the other SDE in Proposition 3.19. By Lemma 3.20, given $\varepsilon > 0$, there are $C_2, C_4 > 0$ such that for $k = 2$ and $k = 4$,

$$\mathbb{P} \left[\sup_{t \geq 0} \left| \int_0^t e^{ki\xi_{\beta,a}(s)} ds \right| > C_k \right] < \frac{\varepsilon}{3}. \quad (3.28)$$

Then, the martingale

$$M(t) := \frac{2}{\sqrt{\beta}} \int_0^t \sin^2 \xi_{\beta,a}(s) d\tilde{B}(s) \quad \text{has} \quad \langle M \rangle(t) = \frac{4}{\beta} \int_0^t \sin^4 \xi_{\beta,a}(s) ds \leq \frac{4t}{\beta}.$$

By properties of continuous martingales (e.g., Proposition A.1), it follows that for any $\delta > 0$ there is a $C > 0$ such that

$$\mathbb{P} \left[\sup_{t \geq 0} \frac{|M(t)|}{1 + (4t/\beta)^{1/2+\delta}} \geq C \right] \leq \frac{\varepsilon}{3},$$

and the result follows by combining this event with those of (3.28). $\blacksquare$

Proposition 3.22 [54, Proposition 20]. *Let $\xi_{\beta,a}$ solve the SDE from Proposition 3.19. Then $e^{i\xi_{\beta,a}(t)} \rightarrow U$ in law where $U \sim \text{Unif } \mathbb{S}^1$.*

Proof. In the same way as in the proof of Proposition 3.7, it suffices to prove that $\varphi_m(t) \rightarrow 0$ as $t \rightarrow \infty$ for all $m \in \mathbb{Z} \setminus \{0\}$ where $\varphi_m(t) := \mathbb{E} X_m(t)$ for $X_m(t) := \exp(mi\xi_{\beta,a}(t) + 2mi\sqrt{\lambda}e^{t/2})$.

Using the expression (3.26) of the differential of $e^{mi\xi_{\beta,a}}$, we see that

$$dX_m(s) = R_m(s)X_m(s) ds + \frac{ki}{\sqrt{\beta}} X_m(s) \sin 2\xi_{\beta,a}(s) d\tilde{B}(s).$$

This shows that $\varphi_m(t) = \varphi_m(0) + \mathbb{E} \int_0^t R_m(s) X_m(s) ds$, so that

$$\varphi'_m(t) = \mathbb{E} R_m(t) X_m(t) = -\frac{m^2}{4\beta} \varphi_m(t) + \mathbb{E} X_m(t) \left(R_m(t) + \frac{m^2}{4\beta} \right),$$

which integrates to

$$\varphi_m(t) = e^{-\frac{m^2 t}{4\beta}} \varphi_m(0) + e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{\frac{m^2 s}{4\beta}} \left(R_m(s) + \frac{m^2}{4\beta} \right) X_m(s) ds.$$

It is clear that the first term vanishes as $t \rightarrow \infty$, so by definition (3.27) of R_m , it suffices to show that

$$e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{\frac{m^2 s}{4\beta}} X_m(s) e^{ki\xi_{\beta,a}(s)} ds \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

for $k = 2$ and $k = 4$. This can be proven through integration by parts, reusing the expression (3.26) of the differential of $e^{ki\xi_{\beta,a}}$. Indeed, we get from (3.26) that

$$\begin{aligned} e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{\frac{m^2 s}{4\beta}} X_m(s) e^{ki\xi_{\beta,a}(s)} ds &= \frac{1}{ki\sqrt{\lambda}} e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{ki\xi_{\beta,a}(s) + (\frac{m^2}{4\beta} - \frac{1}{2})s} R_k(s) X_m(s) ds \\ &\quad - \frac{1}{ki\sqrt{\lambda}} e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{(\frac{m^2}{4\beta} - \frac{1}{2})s} X_m(s) d(e^{ki\xi_{\beta,a}})(s). \end{aligned}$$

Because X_m and R_k are bounded, the first term is bounded by a constant times $e^{-t/2}$, which vanishes as $t \rightarrow \infty$. Integrating by parts the second term, we get

$$\begin{aligned} e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{(\frac{m^2}{4\beta} - \frac{1}{2})s} X_m(s) d(e^{ki\xi_{\beta,a}})(s) &= e^{-\frac{m^2 t}{4\beta}} \mathbb{E} e^{ki\xi_{\beta,a}(s) + (\frac{m^2}{4\beta} - \frac{1}{2})s} X_m(s) \Big|_0^t \\ &\quad - e^{-\frac{m^2 t}{4\beta}} \mathbb{E} \int_0^t e^{ki\xi_{\beta,a}(s) + (\frac{m^2}{4\beta} - \frac{1}{2})s} \left(R_m(s) + \frac{m^2}{4\beta} - \frac{1}{2} - \frac{k^2}{\beta} \sin^2 2\xi_{\beta,a}(s) \right) X_m(s) ds. \end{aligned}$$

Again, because X_m and R_m are bounded, this is bounded by a constant times $e^{-t/2}$. This finishes to prove that $\varphi_m(t) \rightarrow 0$ as $t \rightarrow \infty$ and therefore completes the proof. $\blacksquare$

3.2.3 Canonical system convergence from tridiagonal matrix models

In this section, we show convergence of tridiagonal random matrix models to the stochastic Bessel operator, at the canonical system level.

As mentioned in §1.2, the idea that the stochastic Bessel operator should be the scaling limit of appropriate tridiagonal random matrix models comes from Edelman and Sutton [30]. This was made rigorous by Ramírez and Rider [58], who showed that for any $k \in \mathbb{N}$, the first k points of the Laguerre β -ensemble, scaled by a factor of n , converge jointly in law to the first k points of the

Bessel $_{\beta,a}$ point process, which is the spectrum of the stochastic Bessel operator with left boundary condition $f(0) = 0$. They later extended this result to spiked versions of the Laguerre β -ensemble, which converge in a similar sense to the spectrum of the stochastic Bessel operator with a different left boundary condition when the spiking parameter is suitably scaled [59].

The results of this section thus reformulate some of the main results [58, 59] at the canonical system level. In the same way as in §3.1.3 for the case of the stochastic Airy operator, the advantage of this new formulation is to upgrade everything to the operator level, under the unifying framework of canonical systems theory.

3.2.3.1 Setup and statement of the results

Tridiagonal random matrix model. Let L_n be an $n \times n$ random matrix of the form

$$L_n := \begin{pmatrix} \sqrt{\ell_n} y_{n,1} & x_{n,1} & & & \\ & \ddots & \ddots & & \\ & & \ddots & x_{n,n-1} & \\ & & & & y_{n,n} \end{pmatrix}, \quad (3.29a)$$

where $y_{n,j}, x_{n,j} > 0$ for all j and $\ell_n > 0$, and consider the matrix

$$L_n L_n^\top = \begin{pmatrix} x_{n,1}^2 + \ell_n y_{n,1}^2 & x_{n,1} y_{n,2} & & & \\ x_{n,1} y_{n,2} & x_{n,2}^2 + y_{n,2}^2 & \ddots & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & x_{n,n-1}^2 + y_{n,n-1}^2 & x_{n,n-1} y_{n,n} \\ & & & x_{n,n-1} y_{n,n} & y_{n,n}^2 \end{pmatrix}. \quad (3.29b)$$

Our goal is to have the matrix $n L_n L_n^\top$ scaled at the hard edge of β -ensembles. To this end, we assume the following.

Hypothesis 3.3. The entries $x_{n,j}$ and $y_{n,j}$ of the matrix L_n in (3.29) are all independent. Moreover, their logarithms have means

$$\mathbb{E} \log x_{n,j}^2 = \log(n-j) + \frac{\delta_1(n,j)}{n-j} + \frac{\delta_x(n,j)}{(n-j)^{1+\gamma}}$$

and

$$\mathbb{E} \log y_{n,j+1}^2 = \log(n+a-j) + \frac{\delta_1(n,j)}{n-j} + \frac{\delta_y(n,j)}{(n-j)^{1+\gamma}}$$

for some $\gamma \in (0, 1]$ and where $\delta_1, \delta_x, \delta_y: \mathbb{N}^2 \rightarrow \mathbb{R}$ are bounded, and their second and fourth central moments satisfy

$$\begin{aligned} \mathbb{E} \left(\log \frac{x_{n,j}^2}{y_{n,j+1}^2} - \mathbb{E} \log \frac{x_{n,j}^2}{y_{n,j+1}^2} \right)^2 &= \frac{4}{\beta} \frac{1}{n-j} + \frac{\delta_2(n, j)}{(n-j)^{1+\gamma}}, \\ \mathbb{E} \left(\log \frac{x_{n,j}^2}{y_{n,j+1}^2} - \mathbb{E} \log \frac{x_{n,j}^2}{y_{n,j+1}^2} \right)^4 &\leq \frac{\delta_4(n, j)}{(n-j)^{1+\gamma}} \end{aligned}$$

where $\delta_2, \delta_4: \mathbb{N}^2 \rightarrow \mathbb{R}$ are also bounded.

These conditions ensure that the matrix is scaled at the hard edge of β -ensembles (they are in fact somewhat modeled after the Laguerre β -ensemble). The canonical system that we will build from the matrix has a left boundary condition determined by the spike parameter ℓ_n . Thus, we need an additional hypothesis to make sure that these boundary conditions remain consistent as n grows.

Hypothesis 3.4. $n\ell_n \rightarrow \cot \phi$ in law as $n \rightarrow \infty$ for a deterministic $\phi \in [0, \pi]$.

Remark. In particular, this is automatically satisfied with $\phi = 0$ if $\ell_n = 1$ for all n .

Canonical system representation. $nL_n L_n^\top$ is a random Jacobi matrix, so it has an associated second-order difference operator which can be represented as a canonical system following the method from §2.5. The associated three-term recurrence with which we work is

$$\frac{z}{n} h_j = x_{n,j-1} y_{n,j} h_{j-1} + (x_{n,j}^2 + y_{n,j}^2) h_j + x_{n,j} y_{n,j+1} h_{j+1} \quad \text{for } j \in \{1, \dots, n\}. \quad (3.30)$$

To have this well defined, we need to set values for $x_{n,0}$, $x_{n,n}$ and $y_{n,n+1}$, which are undefined at this point. To match the orders of magnitude of the other matrix entries near each end, we set $x_{n,0} := \sqrt{n}$, $x_{n,n} := 1$ and $y_{n,n+1} := 1$. Note that this recurrence is different from the eigenvalue equation $nL_n L_n^\top h = zh$ not only in that we have added the usual extra terms for h_0 and h_{n+1} at both ends, but also in that the above corresponds to the case $\ell_n = 1$ and that we added an extra term of $x_{n,n}^2$ to the n, n entry of the matrix. The reason to do so is that it makes the form of the recurrence consistent for all j . It will not prevent the canonical system to reproduce the spectral data of $nL_n L_n^\top$: as we will see later, we will revert both the effect of having set $\ell_n = 1$ and the addition of the extra $x_{n,n}^2$ term through the canonical system's boundary conditions.

To build the canonical system, we need two linearly independent solutions $f_{\beta,a,n}$ and $g_{\beta,a,n}$ of (3.30) with $z = 0$. Taking the time steps to be given by j/n , these solutions yield a canonical system $Ju' = -z\tilde{G}_{\beta,a,n}u$ on $[0, 1)$ with coefficient matrix

$$\tilde{G}_{\beta,a,n} \equiv \frac{n}{\mathcal{W}_0(g_{\beta,a,n}, f_{\beta,a,n})} \begin{pmatrix} g_{\beta,a,n,j}^2 & f_{\beta,a,n,j} g_{\beta,a,n,j} \\ f_{\beta,a,n,j} g_{\beta,a,n,j} & f_{\beta,a,n,j}^2 \end{pmatrix} \quad \text{on} \quad \left[\frac{j-1}{n}, \frac{j}{n} \right). \quad (3.31)$$

This canonical system is equivalent to (3.30) in the sense that a solution $u: [0, 1) \rightarrow \mathbb{C}^2$ must satisfy $u(j/n) = A_{\beta,a,n,j}^{-1} F_j$ for

$$A_{\beta,a,n,j} := \begin{pmatrix} g_{\beta,a,n,j+1} & f_{\beta,a,n,j+1} \\ -nx_{n,j}y_{n,j+1}g_{\beta,a,n,j} & -nx_{n,j}y_{n,j+1}f_{\beta,a,n,j} \end{pmatrix} \quad \text{and} \quad F_j := \begin{pmatrix} h_{j+1} \\ -nx_{n,j}y_{n,j+1}h_j \end{pmatrix}$$

for an h solving (3.30). Of course, to have a fully specified system, we need to fully specify $f_{\beta,a,n}$ and $g_{\beta,a,n}$. We could do this as usual by choosing initial conditions, but here instead we exploit an interesting property of the recurrence (3.30): it can be solved explicitly when $z = 0$. This makes our canonical system construction analogous to the one carried out earlier for the stochastic Bessel operator, and in fact the process of solving the discrete system gives a lot of intuition on the way the discrete system approximates the continuous one.

Notice that the recurrence (3.30) can be written as

$$\frac{z}{n}h_j = x_{n,j}^2 \left(\frac{y_{n,j+1}}{x_{n,j}} h_{j+1} + h_j \right) + x_{n,j-1}y_{n,j} \left(\frac{y_{n,j}}{x_{n,j-1}} h_j + h_{j-1} \right).$$

For $j \in \{0, \dots, n\}$, define

$$p_j := \prod_{k=0}^{j-1} \frac{x_k^2}{y_{k+1}^2}, \quad (3.32)$$

understanding that $p_0 := 1$. Setting $\tilde{h}_j := (-1)^j p_j^{-1/2} h_j$, the recurrence becomes

$$\begin{aligned} \frac{z}{n} p_j^{1/2} \tilde{h}_j &= -x_{n,j}^2 \left(\frac{y_{n,j+1}}{x_{n,j}} p_{j+1}^{1/2} \tilde{h}_{j+1} - p_j^{1/2} \tilde{h}_j \right) + x_{n,j-1}y_{n,j} \left(\frac{y_{n,j}}{x_{n,j-1}} p_j^{1/2} \tilde{h}_j - p_{j-1}^{1/2} \tilde{h}_{j-1} \right) \\ &= -x_{n,j}^2 p_j^{1/2} (\tilde{h}_{j+1} - \tilde{h}_j) + x_{n,j-1}y_{n,j} p_{j-1}^{1/2} (\tilde{h}_j - \tilde{h}_{j-1}). \end{aligned}$$

Multiplying by $p_j^{1/2} = \frac{x_{n,j-1}}{y_{n,j}} p_{j-1}^{1/2}$ then yields

$$\frac{z}{n} p_j \tilde{h}_j = -x_{n,j}^2 p_j (\tilde{h}_{j+1} - \tilde{h}_j) + x_{n,j-1}^2 p_{j-1} (\tilde{h}_j - \tilde{h}_{j-1}). \quad (3.33)$$

This recurrence now essentially has the structure of the eigenvalue equation for the stochastic Bessel operator (although the time embedding, which is logarithmic, is not necessarily obvious). When $z = 0$, it is explicitly solvable, as it forces the sequence $x_{n,j-1}^2 p_{j-1} (\tilde{h}_j - \tilde{h}_{j-1})$ to be constant. If the constant is zero, then $\tilde{h}$ itself is constant. The solution for $\tilde{h} \equiv 1$ is what we choose to define $f_{\beta,a,n}$, which yields

$$f_{\beta,a,n,j} = (-1)^j p_j^{1/2}. \quad (3.34)$$

The alternative is to have the sequence $x_{n,j-1}^2 p_{j-1} (\tilde{h}_j - \tilde{h}_{j-1})$ equal to another constant. We call ϖ the solution starting at $\varpi_0 = 0$ for which this constant is 1. By induction, we directly obtain that

$$\varpi_j = \sum_{k=0}^{j-1} \frac{1}{p_k x_{n,k}^2}. \quad (3.35)$$

This is the solution we take to define $g_{\beta,a,n}$, so that

$$g_{\beta,a,n,j} = (-1)^j p_j^{1/2} \varpi_j. \quad (3.36)$$

Now, our two solutions have $f_{\beta,a,n,0} = 1$, $f_{\beta,a,n,1} = -\frac{x_{n,0}}{y_{n,1}}$, $g_{\beta,a,n,0} = 0$ and $g_{\beta,a,n,1} = -\frac{1}{x_{n,0}y_{n,1}}$, which means that $\mathcal{W}_0(g_{\beta,a,n}, f_{\beta,a,n}) = n$. Hence, the coefficient matrix (3.31) becomes

$$\tilde{G}_{\beta,a,n} \equiv p_j \begin{pmatrix} \varpi_j^2 & \varpi_j \\ \varpi_j & 1 \end{pmatrix} \quad \text{on} \quad \left[\frac{j-1}{n}, \frac{j}{n} \right). \quad (3.37)$$

This canonical system is defined on $[0, 1)$, unlike the system (3.24) for the stochastic Bessel operator, which is defined on $[0, \infty)$. Thus, we introduce a time change $\eta: [0, \infty) \rightarrow [0, 1)$ given by $\eta(t) := 1 - e^{-t}$, which is a smooth bijection with $\eta^{-1}(t) = -\log(1-t)$. From this, we build a canonical system on $[0, \infty)$ with coefficient matrix $G_{\beta,a,n} := \eta'(\tilde{G}_{\beta,a,n} \circ \eta)$. This new coefficient matrix is therefore proportional to the matrix in (3.37) on the interval $[t_{j-1}, t_j)$ where $t_j := -\log(1 - j/n)$. Because $\frac{j-1}{n} \leq \eta(t) < \frac{j}{n}$ if and only if $\lfloor n\eta(t) \rfloor = j-1$, it follows that

$$G_{\beta,a,n}(t) = p_{\lfloor n\eta(t) \rfloor + 1} e^{-t} \begin{pmatrix} \varpi_{\lfloor n\eta(t) \rfloor + 1}^2 & \varpi_{\lfloor n\eta(t) \rfloor + 1} \\ \varpi_{\lfloor n\eta(t) \rfloor + 1} & 1 \end{pmatrix}. \quad (3.38)$$

It should be noted that the time change is not arbitrarily chosen as any bijection $[0, \infty) \rightarrow [0, 1)$. This is not obvious to see at this point, but the reason to choose specifically $\eta(t) = 1 - e^{-t}$ is that (as we will see in Lemma 3.26) the map $t \mapsto p_{\lfloor nt \rfloor + 1}$ converges to $p_{\beta,a} \circ \eta^{-1}$, where $p_{\beta,a}$ is the coefficient that appears in the definition of the stochastic Bessel operator. Hence, with this time change, the prefactor $p_{\lfloor n\eta(t) \rfloor + 1} e^{-t}$ discretizes the weight factor $w_{\beta,a}(t)$ of the stochastic Bessel operator.

To make this canonical system fully equivalent to the eigenvalue equation for $nL_n L_n^\top$, it remains to add the appropriate boundary conditions. Following the general method of §2.5, at both endpoints, the boundary conditions that must be added have the form given by (2.30). To implement the appropriate boundary condition at 0, we need to account for the fact that we replaced the factor of ℓ_n with 1 when writing the recurrence (3.30). In other words, in the notation of §2.5, we took $b_1 = n(x_{n,1}^2 + y_{n,1}^2)$ and $\alpha = n(\ell_n - 1)y_{n,1}^2$. Hence, the left boundary condition is

$$\mathbf{u}_{n,0}^* J u(0) = 0 \quad \text{for} \quad \mathbf{u}_{n,0} := \begin{pmatrix} -nx_{n,0}y_{n,1} - n(\ell_n - 1)y_{n,1}^2 \cdot \frac{x_{n,0}}{y_{n,1}} \\ n(\ell_n - 1)y_{n,1}^2 \cdot \frac{1}{x_{n,0}y_{n,1}} \end{pmatrix} = -\frac{ny_{n,1}}{x_{n,0}} \begin{pmatrix} n\ell_n \\ 1 - \ell_n \end{pmatrix} \quad (3.39a)$$

where we used that $x_{n,0}^2 = n$. In particular, if Hypothesis 3.4 holds and $n\ell_n \rightarrow \cot \phi$, then $-\frac{x_{n,0}}{ny_{n,1}} \mathbf{u}_{n,0} \rightarrow (\cot \phi, 1)$. Then, for the boundary condition at infinity, we need to account for the extra term of $x_{n,n}^2$ that we added to the step $j = n$ of the recurrence (3.30). In the notation of §2.5,

this now corresponds to having taken $b_n = n(x_{n,n}^2 + y_{n,n}^2)$ and $\omega = -nx_{n,n}^2$. Following (2.30), this yields a boundary condition given by the vector

$$\mathbf{u}_{n,\infty} := \begin{pmatrix} nx_{n,n}^2(-1)^n p_n^{1/2} + nx_{n,n}y_{n,n+1}(-1)^{n+1} p_{n+1}^{1/2} \\ -nx_{n,n}^2(-1)^n p_n^{1/2} \varpi_n - nx_{n,n}y_{n,n+1}(-1)^{n+1} p_{n+1}^{1/2} \varpi_{n+1} \end{pmatrix}$$

Recall that by definition, $y_{n,n+1} p_{n+1}^{1/2} = x_{n,n} p_n^{1/2}$, and therefore the first entry cancels out. The second entry simplifies to $n(-1)^n x_{n,n}^2 p_n^{1/2} (\varpi_{n+1} - \varpi_n) = n(-1)^n p_n^{-1/2}$, which is always nonzero. Hence, $\mathbf{u}_{n,\infty}$ is always parallel to $e_{\pi/2}$, and the right boundary condition is simply

$$e_{\pi/2} J u(\infty) = 0. \quad (3.39b)$$

To summarize, by construction, if an integrable solution $u: [0, \infty) \rightarrow \mathbb{R}^2$ to the canonical system with coefficient matrix (3.24) and spectral parameter $z \in \mathbb{R}$ satisfies the boundary conditions from (3.39), then one can extract from u an eigenvector $h \in \mathbb{R}^n$ of $nL_n L_n^\top$ with eigenvalue z by taking h_j to be the first entry of $A_{\beta,a,n,j-1} u(t_{j-1})$ for each $j \in \{1, \dots, n\}$. By the computation from §2.5.3, the norms of the two solutions are related by $\|u\|_{G_{\beta,a,n}}^2 = \frac{1}{n} \|h\|_2^2$.

Convergence to the stochastic Bessel operator. The canonical system for $nL_n L_n^\top$ converges to the stochastic Bessel canonical system in the following sense.

Theorem 3.23. *Suppose that the $n \times n$ tridiagonal random matrix $L_n L_n^\top$ from (3.29) satisfies Hypothesis 3.3. Let $G_{\beta,a,n}$ denote the coefficient matrix (3.38) of the canonical system for $nL_n L_n^\top$, and let $G_{\beta,a}$ denote the coefficient matrix (3.24) of the stochastic Bessel operator's canonical system. Let $\mathcal{I}_a := [0, \infty)$ when $a \geq 1$ and $\mathcal{I}_a := [0, \infty]$ when $|a| < 1$. Then $G_{\beta,a,n} \xrightarrow{\text{law}} G_{\beta,a}$ vaguely on $\mathcal{I}_a$ as $n \rightarrow \infty$.*

As a corollary of this result, we also obtain the following.

Corollary 3.23.1. *In the setting of Theorem 3.23, let $T_{G_{\beta,a,n}}, T_{G_{\beta,a}}: \mathcal{I}_a \times \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ denote the canonical systems' transfer matrices. Then $T_{G_{\beta,a,n}} \xrightarrow{\text{law}} T_{G_{\beta,a}}$ compactly on $\mathcal{I}_a \times \mathbb{C}$, jointly with the convergence of coefficient matrices.*

Under Hypothesis 3.4, the boundary conditions remain consistent as n grows, so the convergence also transfers to the Weyl–Titchmarsh functions and spectral measures.

Corollary 3.23.2. *In the setting of Theorem 3.23, suppose additionally that Hypothesis 3.4 holds. Let $m_{G_{\beta,a,n}}^{\mathbf{u}_{n,0}}, m_{G_{\beta,a}}^\phi: \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ denote the Weyl–Titchmarsh functions of $G_{\beta,a,n}$ and $G_{\beta,a}$ with left boundary conditions $\mathbf{u}_{n,0}^* J u(0) = 0$ and $e_\phi^* J u(0) = 0$ respectively, with $\mathbf{u}_{n,0}$ as in (3.39a), and with right boundary condition $e_{\pi/2}^* J u(\infty) = 0$. Then $m_{G_{\beta,a,n}}^{\mathbf{u}_{n,0}} \xrightarrow{\text{law}} m_{G_{\beta,a}}^\phi$ compactly on $\mathbb{H}$, jointly*

with both coefficient and transfer matrices. In particular, if $\mu_{G_{\beta,a,n}}^{u_{n,0}}$ and $\mu_{G_{\beta,a}}^{\phi}$ are the spectral measures associated with $m_{G_{\beta,a,n}}^{u_{n,0}}$ and $m_{G_{\beta,a}}^{\phi}$, then $\mu_{G_{\beta,a,n}}^{u_{n,0}} \xrightarrow{\text{law}} \mu_{G_{\beta,a}}^{\phi}$ vaguely, also jointly with coefficient and transfer matrices and Weyl–Titchmarsh functions.

In the same way as in the case of the stochastic Airy operator (as discussed at the end of §3.1.3.1), the convergence of the spectral measures from Corollary 3.23.2 is unfortunately not strong enough to imply the convergence of the associated eigenvalue point processes, so it does not recover the results from [58, 59] mentioned earlier (in particular in Theorem 1.2).

However, combining Corollary 3.23.2 with results from Dumitriu and Edelman [24] leads to the following conjecture about the distribution of the spectral weights of the stochastic Bessel operator.

Conjecture 3.1. Let $\mu_{G_{\beta,a}}^0$ be the spectral measure of the stochastic Bessel canonical system with boundary conditions $e_0^* J u(0) = e_{\pi/2}^* J u(\infty) = 0$ (the latter being imposed only when $a \geq 1$). Then the spectral weight at $\lambda \in \text{supp } \mu_{G_{\beta,a}}^0$ is $\mu_{G_{\beta,a}}^0(\{\lambda\}) = \lambda \Gamma_{\lambda}$ where $\{\Gamma_{\lambda}\}_{\lambda \in \text{supp } \mu_{G_{\beta,a}}^0}$ is a collection of independent $\Gamma(\beta/2, 2/\beta)$ random variables, also independent of the eigenvalues.

We motivate this in §3.2.3.3 from the application of Theorem 3.23 to the Laguerre case.

3.2.3.2 Proof of the canonical system convergence

We now turn to the proofs of Theorem 3.23 and its corollaries. The proof of the convergence relies mainly on the analysis of the sequences $\{p_j\}_{j=0}^n$ and $\{\varpi_j\}_{j=0}^n$ introduced in (3.32) and (3.35). Indeed, we show in Lemma 3.26 that $p_{[n\eta]+1} \rightarrow p_{\beta,a}$ and $\varpi_{[n\eta]+1} \rightarrow \varpi_{\beta,a}$ compactly in law as $n \rightarrow \infty$. This is enough to deduce the vague convergence on $[0, \infty)$ from Theorem 3.23 for $a \geq 1$, as the Bessel canonical system is limit point at infinity. However, it is limit circle at infinity when $|a| < 1$, and in that case to complete the proof of Theorem 3.23 we need to upgrade the vague convergence to $[0, \infty]$. To do so, it suffices to show that the trace of the coefficient matrix $G_{\beta,a,n}$ is dominated by an integrable function independent of n , which we do in Lemma 3.28. Having proven Theorem 3.23, it is then easy to deduce the convergence of transfer matrices and Weyl–Titchmarsh functions in Corollaries 3.23.1 and 3.23.2 using general results from canonical systems theory.

Our starting point is the identification of the random walk that converges to the Brownian motion that drives the limiting stochastic Bessel operator.

Lemma 3.24. *Set*

$$\mathcal{Z}_n(t) := -\frac{\sqrt{\beta}}{2} \sum_{k=0}^{[n\eta(t)]} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right),$$

where the entries $x_{n,k}$ and $y_{n,k}$ satisfy Hypothesis 3.3. Then $\mathcal{Z}_n$ converges in law to a standard Brownian motion under the Skorokhod topology.

The proof of this result relies on the following observation, which is used repeatedly in the sequel.

Claim 3.25. *With $\eta(t) := 1 - e^{-t}$, for all $n \in \mathbb{N}$ and all $t \geq 0$,*

$$t - \log\left(1 + \frac{e^t}{n}\right) \leq \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k} \leq t + \frac{1}{n - \lfloor n\eta(t) \rfloor}.$$

In particular, if $\tau > 0$, then any $t \in [0, \tau]$ satisfies $|t - \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k}| \leq e^\tau/n$.

Proof. Note that

$$\begin{aligned} \sum_{k=0}^{\lfloor n\eta(t) \rfloor - 1} \frac{1}{n-k} &= \sum_{k=0}^{\lfloor n\eta(t) \rfloor - 1} \frac{1}{1 - k/n} \int_{k/n}^{(k+1)/n} ds \leq \sum_{k=0}^{\lfloor n\eta(t) \rfloor - 1} \int_{k/n}^{(k+1)/n} \frac{ds}{1-s} \\ &= \int_0^{\frac{1}{n} \lfloor n\eta(t) \rfloor} \frac{ds}{1-s} \leq \int_0^{\eta(t)} \frac{ds}{1-s} = -\log(1 - \eta(t)) = t, \end{aligned}$$

which proves the upper bound. Similarly,

$$\begin{aligned} \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k} &\geq \sum_{k=1}^{\lfloor n\eta(t) \rfloor} \frac{1}{1 - k/n} \int_{(k-1)/n}^{k/n} ds \geq \sum_{k=1}^{\lfloor n\eta(t) \rfloor} \int_{(k-1)/n}^{k/n} \frac{ds}{1-s} = \int_0^{\frac{1}{n} \lfloor n\eta(t) \rfloor} \frac{ds}{1-s} \\ &= -\log\left(1 - \frac{\lfloor n\eta(t) \rfloor}{n}\right) \geq -\log\left(1 - \eta(t) + \frac{1}{n}\right) = t - \log\left(1 + \frac{e^t}{n}\right), \end{aligned}$$

which proves the lower bound. The last conclusion then follows from noticing that if $\tau > 0$, then for $t \in [0, \tau]$, $\log(1 + e^t/n) \leq e^\tau/n$ and $n - \lfloor n\eta(t) \rfloor \geq n - n\eta(\tau) = ne^{-\tau}$. ■

We now prove the convergence of the random walk $\mathcal{Z}_n$ to a Brownian motion.

Proof of Lemma 3.24. The proof relies essentially on a standard martingale central limit theorem. The specific formulation we use is that from Ethier and Kurtz [32, Theorem 7.1.4].

Because the $x_{n,k}$'s and $y_{n,k}$'s are independent, $\mathcal{Z}_n$ is a càdlàg martingale with predictable quadratic variation

$$\langle \mathcal{Z}_n \rangle(t) = \frac{\beta}{4} \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \mathbb{E} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^2 = \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k} + \frac{\beta}{4} \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{\delta_2(n, k)}{(n-k)^{1+\gamma}}, \quad (3.40)$$

where the second equality simply follows from using the variance formula from Hypothesis 3.3. Now, for any $\tau > 0$, if $k \leq \lfloor n\eta(t) \rfloor$ then $n - k \geq n - n\eta(\tau) = ne^{-\tau}$, so the second term of (3.40) is bounded by $\frac{\beta}{4} \|\delta_2\|_\infty e^{(1+\gamma)\tau} n^{-\gamma}$ for all $t \in [0, \tau]$. Combining this with the last part of Claim 3.25,

we see that $\langle \mathcal{Z}_n \rangle(t) \rightarrow t$ uniformly on $[0, \tau]$. From [32, Theorem 7.1.4], the convergence of $\mathcal{Z}_n$ to a Brownian motion follows if we show that

$$\mathbb{E} \sup_{t \in [0, \tau]} |\mathcal{Z}_n(t) - \mathcal{Z}_n(t-)|^2 = \frac{\beta}{4} \mathbb{E} \max_{0 \leq k \leq \lfloor n\eta(\tau) \rfloor} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^2 \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

To show this, we use a layer cake decomposition. For $u > 0$,

$$\begin{aligned} \mathbb{E} \max_{0 \leq k \leq \lfloor n\eta(\tau) \rfloor} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^2 &\leq u + \sum_{k=0}^{\lfloor n\eta(\tau) \rfloor} \int_u^\infty \mathbb{P} \left[\left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^2 \geq v \right] dv \\ &\leq u + \sum_{k=0}^{\lfloor n\eta(\tau) \rfloor} \mathbb{E} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^4 \int_u^\infty \frac{dv}{v^2}, \end{aligned}$$

by Markov's inequality. By the fourth moment assumption in Hypothesis 3.3, if $k \leq \lfloor n\eta(\tau) \rfloor$ then

$$\mathbb{E} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^4 \leq \frac{\delta_4(n, k)}{(n-k)^{1+\gamma}} \leq \frac{\|\delta_4\|_\infty e^{(1+\gamma)\tau}}{n^{1+\gamma}},$$

so

$$\mathbb{E} \max_{0 \leq k \leq \lfloor n\eta(\tau) \rfloor} \left(\log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} \right)^2 \leq u + \frac{\|\delta_4\|_\infty e^{(1+\gamma)\tau}}{un^\gamma} \quad (3.41)$$

and the right-hand side vanishes as $n \rightarrow \infty$ if we take $u := n^{-\gamma/2}$. As noted above, it follows from [32, Theorem 7.1.4] that $\mathcal{Z}_n$ converges to a standard Brownian motion in the Skorokhod topology. ■

Lemma 3.24 can be used to prove the convergence of the sequences $\{p_j\}_{j=0}^n$ and $\{\varpi_j\}_{j=0}^n$, in a suitable continuous-time embedding, to the functions $p_{\beta,a}$ and $\varpi_{\beta,a}$.

Lemma 3.26. *Suppose that Hypothesis 3.3 is satisfied, and for $j \in \{0, \dots, n\}$, let p_j and ϖ_j be defined as in (3.32) and (3.35). Then with $p_{\beta,a}$ as in Definition 1.1 and $\varpi_{\beta,a}$ as in (3.20), $p_{\lfloor n\eta \rfloor + 1} \rightarrow p_{\beta,a}$ and $\varpi_{\lfloor n\eta \rfloor + 1} \rightarrow \varpi_{\beta,a}$ compactly and jointly in law as $n \rightarrow \infty$.*

Proof. By Lemma 3.24, $\mathcal{Z}_n$ converges in law to a Brownian motion. By Skorokhod representation, there is a coupling of the $\mathcal{Z}_n$'s and a Brownian motion B such that $\mathcal{Z}_n \rightarrow B$ compactly in probability. In this setting, we show that with $p_{\beta,a}$ defined from B and $\varpi_{\beta,a}(t) := \int_0^t p_{\beta,a}^{-1}(s) ds$, $p_{\lfloor n\eta \rfloor + 1} \rightarrow p_{\beta,a}$ and $\varpi_{\lfloor n\eta \rfloor + 1} \rightarrow \varpi_{\beta,a}$ compactly in probability, which implies the result.

By definition,

$$\log p_{\lfloor n\eta \rfloor + 1} = \sum_{k=0}^{\lfloor n\eta \rfloor} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} = \sum_{k=0}^{\lfloor n\eta \rfloor} \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} + \frac{2}{\sqrt{\beta}} \mathcal{Z}_n.$$

By Hypothesis 3.3, for each k ,

$$\mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} = -\log\left(1 + \frac{a}{n-k}\right) + \frac{\delta_x(n, k) - \delta_y(n, k)}{(n-k)^{1+\gamma}}.$$

Now, it is easy to verify that if $\xi_0 > -1$, then there is a $C > 0$ such that $-\log(1 + \xi) \leq -\xi + C\xi^2$ for all $\xi \geq \xi_0$. Here, either $\frac{a}{n-k} \geq 0$ for all k or $\frac{a}{n-k} \geq a > -1$ depending on the sign of a , so we can also write

$$\mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2} = -\frac{a}{n-k} + \frac{\delta_\Sigma(n, k)}{(n-k)^{1+\gamma}}$$

with $\delta_\Sigma := \delta_x - \delta_y + \delta_a$ and $\delta_a: \mathbb{N}^2 \rightarrow \mathbb{R}$ bounded. Thus,

$$\log p_{\lfloor n\eta(t) \rfloor + 1} = -a \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k} + \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{\delta_\Sigma(n, k)}{(n-k)^{1+\gamma}} + \frac{2}{\sqrt{\beta}} \mathcal{Z}_n(t). \quad (3.42)$$

If $\tau > 0$ and $t \in [0, \tau]$, then Claim 3.25 implies that the first term converges to $-at$, uniformly in $t \in [0, \tau]$. The second term is bounded by $\|\delta_\Sigma\|_\infty e^{(1+\gamma)\tau} n^{-\gamma} \rightarrow 0$, and the third term converges in probability to $\frac{2}{\sqrt{\beta}} B(t)$ uniformly in $t \in [0, \tau]$. Hence, $\log p_{\lfloor n\eta(t) \rfloor + 1} \rightarrow \log p_{\beta,a}(t)$ in probability, uniformly in $t \in [0, \tau]$.

Then, by definition,

$$\begin{aligned} \varpi_{\lfloor n\eta(t) \rfloor + 1} &= \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{p_k x_{n,k}^2} = \frac{1}{n} + \sum_{k=1}^{\lfloor n\eta(t) \rfloor} \frac{n}{p_k x_{n,k}^2} \int_{(k-1)/n}^{k/n} du = \frac{1}{n} + \sum_{k=1}^{\lfloor n\eta(t) \rfloor} \int_{(k-1)/n}^{k/n} \frac{n}{p_{\lfloor nu \rfloor + 1} x_{n, \lfloor nu \rfloor + 1}^2} du \\ &= \frac{1}{n} + \int_0^{\frac{1}{n} \lfloor n\eta(t) \rfloor} \frac{n}{p_{\lfloor nu \rfloor + 1} x_{n, \lfloor nu \rfloor + 1}^2} du = \frac{1}{n} + \int_0^{\eta^{-1}(\frac{1}{n} \lfloor n\eta(t) \rfloor)} \frac{ne^{-s}}{p_{\lfloor n\eta(s) \rfloor + 1} x_{n, \lfloor n\eta(s) \rfloor + 1}^2} ds. \end{aligned} \quad (3.43)$$

As we have seen in (3.41) in the proof of Lemma 3.24, $\mathbb{E} \max_{0 \leq k \leq \lfloor n\eta(\tau) \rfloor} |\log x_{n,k}^2 - \mathbb{E} \log x_{n,k}^2| \rightarrow 0$ for any fixed $\tau > 0$, so this maximum converges to zero in probability by Markov's inequality. From the expression of $\mathbb{E} \log x_{n,k}^2$ in Hypothesis 3.3, we deduce that for any $k \leq \lfloor n\eta(\tau) \rfloor$, $\log x_{n,k}^2 = \mathbb{E} \log x_{n,k}^2 + o_{\mathbb{P},\tau}(1) = \log(n-k) + o_{\mathbb{P},\tau}(1)$ where the error terms $o_{\mathbb{P},\tau}(1)$ converge to 0 in probability, uniformly for $k \leq \lfloor n\eta(\tau) \rfloor$. It follows that

$$\frac{n}{x_{\lfloor n\eta(s) \rfloor + 1}^2} = o_{\mathbb{P},\tau}(1) \frac{n}{n - \lfloor n\eta(s) \rfloor - 1} = \frac{o_{\mathbb{P},\tau}(1)}{e^{-s} + o_\tau(1)} \quad \text{so} \quad \frac{ne^{-s}}{x_{n, \lfloor n\eta(s) \rfloor + 1}^2} \xrightarrow{\mathbb{P}} 1,$$

uniformly in $s \in [0, \tau]$. Since $p_{\lfloor n\eta(t) \rfloor + 1} = p_{\beta,a}(t) + o_{\mathbb{P},\tau}(1)$ by the first part of the proof, it follows that

$$\varpi_{\lfloor n\eta(t) \rfloor + 1} = \int_0^{\eta^{-1}(\frac{1}{n} \lfloor n\eta(t) \rfloor)} \left(\frac{1}{p_{\beta,a}(s)} + o_{\mathbb{P},\tau}(1) \right) ds = \int_0^t \frac{ds}{p_{\beta,a}(s)} + o_{\mathbb{P},\tau}(1),$$

and $\varpi_{\lfloor n\eta(t) \rfloor + 1} \rightarrow \varpi_{\beta,a}(t)$ in probability, uniformly in $t \in [0, \tau]$. ■

Our next step, as announced at the beginning of this section, is to obtain a growth bound on the traces of the coefficient matrices. To obtain such a bound, we start by extending the control we have on the martingale $\mathcal{Z}_n$.

Lemma 3.27. *Let $\mathcal{Z}_n$ be as in Lemma 3.24. For any $\varepsilon, \delta > 0$, there is a $C > 0$ which does not depend on n such that $\mathbb{P}[\forall t \geq 0, |\mathcal{Z}_n(t)| \leq C(1 + t^{1/2+\delta})] \geq 1 - \varepsilon$.*

Proof. To simplify notation in the proof, write $Z_0 := 0$ and define

$$Z_j := \sum_{k=1}^j \Delta Z_k \quad \text{for} \quad \Delta Z_{k-1} := \log \frac{x_{n,k}^2}{y_{n,k+1}^2} - \mathbb{E} \log \frac{x_{n,k}^2}{y_{n,k+1}^2},$$

so that $Z_j = \mathcal{Z}_n(t_{j-1})$. To control $\mathcal{Z}_n$ at large times, we embed the discrete martingale Z into a Brownian motion using a Skorokhod embedding [44, Theorem 22.16]. More precisely, there is a Brownian motion B and a sequence of stopping times τ_j such that a.s., $Z_j = B_{\tau_j}$, $\mathbb{E}[\Delta \tau_j | \mathcal{F}_{j-1}^{Z,\tau}] = \mathbb{E}[(\Delta Z_j)^2 | \mathcal{F}_{j-1}^Z]$ and $\mathbb{E}[(\Delta \tau_j)^2 | \mathcal{F}_{j-1}^{Z,\tau}] \leq 4\mathbb{E}[(\Delta Z_j)^4 | \mathcal{F}_{j-1}^Z]$, where $\{\mathcal{F}_j^Z\}_{j=1}^n$ and $\{\mathcal{F}_j^{Z,\tau}\}_{j=1}^n$ are the filtrations generated by Z and (Z, τ) respectively, and where we write $\Delta \tau_j := \tau_j - \tau_{j-1}$. Fixing $\varepsilon > 0$, by basic properties of Brownian motion (see e.g. Proposition A.1) there is a $C > 0$ such that

$$\mathbb{P}[\forall t \geq 0, |B(t)| \leq C(1 + t^{1/2+\delta})] \geq 1 - \frac{\varepsilon}{2}. \quad (3.44)$$

Hence, in order to finish the proof we mainly need to control the growth of the stopping times τ_j .

The increments ΔZ_j are independent, and this implies the independence of the time differences $\Delta \tau_j$ as well. Indeed, $\Delta Z_2 = Z_2 - Z_1 = B_{\tau_2} - B_{\tau_1}$, whose law depends only on $\Delta \tau_2$ by the strong Markov property of Brownian motion. Because ΔZ_2 is independent of Z_1 , it follows that $\Delta \tau_2$ is independent of τ_1 . Inductively, the same idea shows that all $\Delta \tau_j$'s are independent, and in particular the relationships between moments reduce to $\mathbb{E} \Delta \tau_j = \mathbb{E}(\Delta Z_j)^2$ and $\mathbb{E}(\Delta \tau_j)^2 \leq 4\mathbb{E}(\Delta Z_j)^4$.

Set $M_0 := 0$ and $M_j := \sum_{k=1}^j (\Delta \tau_k - \mathbb{E} \Delta \tau_k)$. Then there is a $C' > 0$ such that for all $j \leq n$,

$$\tau_j = \sum_{k=1}^j \Delta \tau_k = \sum_{k=1}^j \mathbb{E} \Delta \tau_k + M_j = \sum_{k=1}^j \mathbb{E}(\Delta Z_k)^2 + M_j = \langle \mathcal{Z}_n \rangle(t_{j-1}) + M_j \leq t_{j-1} + C' + M_j \quad (3.45)$$

where the inequality follows from Claim 3.25. Because the $\Delta \tau_k$'s are independent,

$$\mathbb{E} M_n^2 = \sum_{k=1}^n \mathbb{E}(\Delta \tau_k - \mathbb{E} \Delta \tau_k)^2 \leq \sum_{k=1}^n \mathbb{E}(\Delta \tau_k)^2 \leq 4 \sum_{k=1}^n \mathbb{E}(\Delta Z_k)^4 \leq 4 \sum_{k=0}^{n-1} \frac{\delta_4(n, k)}{(n-k)^{1+\gamma}}.$$

Now, M is a martingale, so it follows from Doob's submartingale inequality [44, Theorem 9.16] that for any $\tilde{C} > 0$,

$$\mathbb{P}\left[\sup_{1 \leq j \leq n} |M_j| > \tilde{C}\right] \leq \frac{\mathbb{E} M_n^2}{\tilde{C}^2} \leq \frac{4\|\delta_4\|_\infty}{\tilde{C}^2} \sum_{k=1}^{\infty} \frac{1}{k^{1+\gamma}}.$$

Picking $\tilde{C}$ large enough so that the right-hand side is less than $\varepsilon/2$, we see from (3.45) that with probability at least $1 - \varepsilon/2$, $\tau_j \leq t_{j-1} + C' + \tilde{C}$ for all $j \in \{1, \dots, n\}$. Since $B_{\tau_j} = Z_j = \mathcal{Z}_n(t_{j-1})$, combining with the event in (3.44), we see that

$$\mathbb{P}\left[\forall j \in \{1, \dots, n\}, |\mathcal{Z}_n(t_{j-1})| \leq C(1 + (t_{j-1} + C' + \tilde{C})^{1/2+\delta})\right] \geq 1 - \varepsilon.$$

This extends to all times since $\mathcal{Z}_n(t) = \mathcal{Z}_n(t_{j-1})$ for all $t \in [t_{j-1}, t_j]$. Because C , C' and $\tilde{C}$ do not depend on n , this completes the proof. $\blacksquare$

From the control on the martingale $\mathcal{Z}_n$ that we proved in Lemma 3.27, we deduce the following uniform bound on the trace of the coefficient matrices (3.38).

Lemma 3.28. *Suppose that Hypothesis 3.3 holds and that $|a| < 1$. If $\varepsilon > 0$ and $\delta \in (0, 1 - |a|)$, then there is a $C > 0$ independent of n such that*

$$\mathbb{P}\left[\forall t \geq 0, e^{-t} p_{\lfloor n\eta(t) \rfloor + 1} (1 + \varpi_{\lfloor n\eta(t) \rfloor + 1}^2) \leq C e^{-(1-|a|-\delta)t}\right] \geq 1 - \varepsilon.$$

Proof. Note that it suffices to show the desired inequality for $t \in [0, \log n]$. Indeed, suppose that

$$\mathbb{P}\left[\forall t \in [0, \log n], e^{-t} p_{\lfloor n\eta(t) \rfloor + 1} (1 + \varpi_{\lfloor n\eta(t) \rfloor + 1}^2) \leq C e^{-(1-|a|-\delta)t}\right] \geq 1 - \varepsilon. \quad (3.46)$$

By definition of η , if $t \geq \log n$ then $\eta(t) = 1 - e^{-t} \geq 1 - 1/n$, so $\lfloor n\eta(t) \rfloor = n - 1$. This shows that $p_{\lfloor n\eta(t) \rfloor + 1} (1 + \varpi_{\lfloor n\eta(t) \rfloor + 1}^2)$ is constant for $t \geq \log n$. Hence, it also holds on the same event that for $t \geq \log n$,

$$p_{\lfloor n\eta(t) \rfloor + 1} (1 + \varpi_{\lfloor n\eta(t) \rfloor + 1}^2) = p_{\lfloor n\eta(\log n) \rfloor + 1} (1 + \varpi_{\lfloor n\eta(\log n) \rfloor + 1}^2) \leq C e^{(|a|+\delta)\log n} \leq C e^{(|a|+\delta)t},$$

which proves the result.

Now, to prove (3.46), recall from (3.42) that

$$\log p_{\lfloor n\eta(t) \rfloor + 1} = -a \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k} + \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{\delta_{\Sigma}(n, k)}{(n-k)^{1+\gamma}} + \frac{2}{\sqrt{\beta}} \mathcal{Z}_n(t).$$

Because δ_{Σ} is bounded, the second term is bounded by a constant independent of n . Moreover, the third term is controlled by Lemma 3.27, which implies that there is a $C' > 0$ such that on an event of probability at least $1 - \varepsilon/2$, $\frac{2}{\sqrt{\beta}} |\mathcal{Z}_n| \leq C'(1 + t^{3/4})$ for all $t \geq 0$. This upper bound is itself bounded by $C'' + \delta t$ for all $t \geq 0$, for another constant $C'' > 0$ still independent of n . Hence, for all $t \geq 0$,

$$\left| \log p_{\lfloor n\eta(t) \rfloor + 1} + a \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k} \right| \leq C + \delta t$$

for a $C > 0$. Claim 3.25 shows that $\sup_{t \in [0, \log n]} |t - \sum_{k=0}^{\lfloor n\eta(t) \rfloor} \frac{1}{n-k}| \leq 1$. From this, we conclude that on the event of probability at least $1 - \varepsilon/2$, for all $t \in [0, \log n]$,

$$c' e^{-(a+\delta)t} \leq p_{\lfloor n\eta(t) \rfloor + 1} \leq C' e^{-(a-\delta)t} \quad (3.47)$$

for constants $C', c' > 0$ independent of n .

To deduce a bound on $\varpi_{\lfloor n\eta \rfloor + 1}$, recall from (3.43) that

$$\varpi_{\lfloor n\eta(t) \rfloor + 1} = \frac{1}{n} + \int_0^{\eta^{-1}(\frac{1}{n} \lfloor n\eta(t) \rfloor)} \frac{ne^{-s}}{p_{\lfloor n\eta(s) \rfloor + 1} x_{n, \lfloor n\eta(s) \rfloor + 1}^2} ds.$$

To bound this, we are really only missing a uniform control on the $x_{n,k}$'s. By Markov's inequality and Hypothesis 3.3, if $\tilde{C} > 0$ then

$$\begin{aligned} \mathbb{P} \left[\max_{0 \leq k \leq n-1} |\log x_{n,k}^2 - \mathbb{E} \log x_{n,k}^2| > \tilde{C} \right] &\leq \sum_{k=0}^{n-1} \mathbb{P} \left[|\log x_{n,k}^2 - \mathbb{E} \log x_{n,k}^2|^4 > \tilde{C}^4 \right] \\ &\leq \frac{1}{\tilde{C}^4} \sum_{k=0}^{n-1} \mathbb{E} \left(\log x_{n,k}^2 - \mathbb{E} \log x_{n,k}^2 \right)^4 \leq \frac{1}{\tilde{C}^4} \sum_{k=0}^{n-1} \frac{\delta_4(n, k)}{(n-k)^{1+\gamma}} \leq \frac{\|\delta_4\|_\infty}{\tilde{C}^4} \sum_{k=0}^{\infty} \frac{1}{k^{1+\gamma}}. \end{aligned}$$

The right-hand side is then bounded by $\varepsilon/2$ for $\tilde{C}$ large enough, which can be chosen independently of n . Since $\mathbb{E} \log x_{n,k}^2 - \log(n-k)$ is bounded by a constant independent of n , it follows that, enlarging $\tilde{C}$ if needed,

$$\frac{n}{x_{n,k}^2} \leq ne^{\mathbb{E} \log x_{n,k}^2 + \tilde{C}} \leq ne^{-\log(n-k) + 2\tilde{C}} = \frac{e^{2\tilde{C}}}{1 - k/n}.$$

If $k = \lfloor n\eta(s) \rfloor + 1 \leq n-1$, then $\frac{k-1}{n} \leq \eta(s)$ so

$$e^{-s} = 1 - \eta(s) \leq 1 - \frac{k-1}{n} = \left(1 - \frac{k}{n}\right) \frac{1 - (k-1)/n}{1 - k/n} = \left(1 - \frac{k}{n}\right) \left(1 + \frac{1}{n-k}\right) \leq 2 \left(1 - \frac{k}{n}\right),$$

and therefore

$$\frac{n}{x_{n, \lfloor n\eta(s) \rfloor + 1}^2} \leq 2e^{2\tilde{C}+s}.$$

This is in fact valid for all $s \in [0, \log n]$, since when $\lfloor n\eta(s) \rfloor + 1 = n$, $x_{n, \lfloor n\eta(s) \rfloor + 1}^2 = n$ by definition. Combining this with our expression for $\varpi_{\lfloor n\eta \rfloor + 1}$, we find

$$\varpi_{\lfloor n\eta(t) \rfloor + 1} \leq 1 + 2e^{2\tilde{C}} \int_0^{\eta^{-1}(\frac{1}{n} \lfloor n\eta(t) \rfloor)} \frac{1}{p_{\lfloor n\eta(s) \rfloor + 1}} ds.$$

Using the lower bound on $p_{\lfloor n\eta \rfloor + 1}$ from (3.47), we find that for all $t \in [0, \log n]$,

$$\varpi_{\lfloor n\eta(t) \rfloor + 1} \leq 1 + \frac{2e^{2\tilde{C}}}{c'} \int_0^t e^{(a+\delta)s} ds = 1 + \frac{2e^{2\tilde{C}}}{c'} \frac{e^{(a+\delta)t} - 1}{a + \delta}.$$

Combining this with the upper bound on $p_{\lfloor n\eta \rfloor + 1}$, we get that

$$p_{\lfloor n\eta(t) \rfloor + 1} \varpi_{\lfloor n\eta(t) \rfloor + 1}^2 \leq \tilde{C}' e^{-(a-\delta)t} + \tilde{C}'' e^{(a+3\delta)t} \quad (3.48)$$

for two other constants $\tilde{C}', \tilde{C}'' > 0$, still independent of n . This holds for all $t \in [0, \log n]$, with probability at least $1 - \varepsilon$.

Summing up the upper bounds on $p_{\lfloor n\eta \rfloor + 1}$ and $p_{\lfloor n\eta \rfloor + 1} \varpi_{\lfloor n\eta \rfloor + 1}^2$ from (3.47) and (3.48), which hold on an event of probability at least $1 - \varepsilon$, yields the desired bound (3.46). ■

We are now ready to prove Theorem 3.23 and its corollaries.

Proof of Theorem 3.23. Recall from (3.38) and (3.24) that

$$G_{\beta,a,n}(t) = e^{-t} p_{\lfloor n\eta(t) \rfloor + 1} \begin{pmatrix} \varpi_{\lfloor n\eta(t) \rfloor + 1}^2 & \varpi_{\lfloor n\eta(t) \rfloor + 1} \\ \varpi_{\lfloor n\eta(t) \rfloor + 1} & 1 \end{pmatrix} \quad \text{and} \quad G_{\beta,a} = w_{\beta,a} \begin{pmatrix} \varpi_{\beta,a}^2 & \varpi_{\beta,a} \\ \varpi_{\beta,a} & 1 \end{pmatrix}.$$

Our goal is to show that $G_{\beta,a,n} \rightarrow G_{\beta,a}$ in law, vaguely on $\mathcal{I}_a$ as $n \rightarrow \infty$, where $\mathcal{I}_a := [0, \infty)$ when $a \geq 1$ but $\mathcal{I}_a := [0, \infty]$ when $|a| < 1$.

By Lemma 3.26, $p_{\lfloor n\eta \rfloor + 1} \rightarrow p_{\beta,a}$ and $\varpi_{\lfloor n\eta \rfloor + 1} \rightarrow \varpi_{\beta,a}$ compactly and jointly in law as $n \rightarrow \infty$. Hence, by Skorokhod representation, there is a coupling of the systems such that these convergences hold in probability. By Proposition 2.23, this reduces the problem to showing that for all $\varphi \in \mathcal{C}_c(\mathcal{I}_a, \mathbb{C}^2)$,

$$\int_{\mathcal{I}_a} \varphi^* (G_{\beta,a,n} - G_{\beta,a}) \varphi d\mathcal{L} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0.$$

Fix $\varphi \in \mathcal{C}_c(\mathcal{I}_a, \mathbb{C}^2)$, and for $\tau > 0$, write

$$\int_{\mathcal{I}_a} \varphi^* (G_{\beta,a,n} - G_{\beta,a}) \varphi d\mathcal{L} = \int_0^\tau \varphi^* (G_{\beta,a,n} - G_{\beta,a}) \varphi d\mathcal{L} + \int_\tau^\infty \varphi^* (G_{\beta,a,n} - G_{\beta,a}) \varphi d\mathcal{L}. \quad (3.49)$$

If $a \geq 1$, then we can choose τ large enough so that φ is supported on $[0, \tau]$, and then the second integral in (3.49) is zero. Then, because $p_{\lfloor n\eta \rfloor + 1} \rightarrow p_{\beta,a}$ and $\varpi_{\lfloor n\eta \rfloor + 1} \rightarrow \varpi_{\beta,a}$ compactly in probability, by definition $G_{\beta,a,n} \rightarrow G_{\beta,a}$ compactly in probability, so the first term of (3.49) converges to 0 in probability as $n \rightarrow \infty$.

If $|a| < 1$, then no τ can be chosen in the same way, since φ could be supported on all of $[0, \infty]$. However, by Lemma 3.28, for any $\varepsilon > 0$ and $\delta \in (0, 1 - |a|)$ there is a $C > 0$ independent of n such

that $\text{tr } G_{\beta,a,n}(t) \leq Ce^{-(1-|a|-\delta)t}$ for all $t \geq 0$ with probability at least $1 - \varepsilon/3$. A bound of the same form also holds on $\text{tr } G_{\beta,a}$ by Lemma 3.16, so

$$\left| \int_{\tau}^{\infty} \varphi^*(G_{\beta,a,n} - G_{\beta,a})\varphi \, d\mathcal{L} \right| \leq 2C\|\varphi\|_{\infty}^2 \int_{\tau}^{\infty} e^{-(1-|a|-\delta)t} \, dt$$

with probability at least $1 - 2\varepsilon/3$. Fixing $\zeta > 0$, we can choose τ large enough so that the right-hand side is bounded by $\zeta/2$. For this choice of τ , since $G_{\beta,a,n} \rightarrow G_{\beta,a}$ compactly in probability, the first term of (3.49) is bounded by $\zeta/2$ with probability at least $1 - \varepsilon/3$ for all n large enough, and it follows that $\int_{\mathcal{I}_a} \varphi^*(G_{\beta,a,n} - G_{\beta,a})\varphi \, d\mathcal{L} \rightarrow 0$ in probability as $n \rightarrow \infty$. ■

Proof of Corollary 3.23.1. Take the coupling from the proof of Theorem 3.23, under which the vague convergence $G_{\beta,a,n} \rightarrow G_{\beta,a}$ holds in probability.

When $|a| < 1$, then the high-probability dominating functions on the traces of $G_{\beta,a,n}$ and $G_{\beta,a}$ from Lemmas 3.28 and 3.16 are integrable, so the vague convergence in probability of coefficient matrices on $[0, \infty]$ implies the compact convergence in probability of the associated transfer matrices on $[0, \infty] \times \mathbb{C}$ by continuity of the map from coefficient to transfer matrices on trace-dominated sets (i.e., Proposition 2.25).

When $a \geq 1$, Lemma 3.16 still gives a high-probability dominating function on the trace of $G_{\beta,a}$, and because the coefficient matrices actually converge compactly by Lemma 3.26, this is enough for Proposition 2.25 to imply the compact convergence in probability of transfer matrices on $[0, \infty) \times \mathbb{C}$, exactly in the same way as in the proof of Corollary 3.8.1.

Because the coefficient and transfer matrices converge in probability under the same coupling, they converge jointly in law. ■

Proof of Corollary 3.23.2. Suppose first that $\ell_n = 1$ so that the boundary condition $\mathbf{u}_{n,0}$ at 0 from (3.39a) is parallel to e_0 . Then by the continuity of the maps from transfer matrices to Weyl–Titchmarsh functions from Theorems 2.13 and 2.14, the convergence of the transfer matrices implies that $m_{G_{\beta,a,n}}^0 \rightarrow m_{G_{\beta,a}}^0$ in law, since when $|a| < 1$ the boundary conditions at infinity of both systems are $e_{\pi/2}$.

To prove the result in the general case of Hypothesis 3.4, write $\phi_n \in [0, \pi]$ for the angle such that $e_{\phi_n} = \pm \|\mathbf{u}_{n,0}\|_2^{-1} \mathbf{u}_{n,0}$, so that $m_{G_{\beta,a,n}}^{\phi_n} = m_{G_{\beta,a,n}}^{\mathbf{u}_{n,0}}$. Then Hypothesis 3.4 implies that

$$e_{\phi_n} = \pm \frac{\mathbf{u}_{n,0}}{\|\mathbf{u}_{n,0}\|_2} = \frac{\pm 1}{\sqrt{n^2 \ell_n^2 + (1 - \ell_n)^2}} \begin{pmatrix} n \ell_n \\ 1 - \ell_n \end{pmatrix} \xrightarrow{\text{law}} e_{\phi}.$$

This holds jointly with the convergence $m_{G_{\beta,a,n}}^0 \rightarrow m_{G_{\beta,a}}^0$. Writing R_{θ} for the rotation matrix with columns e_{θ} and Je_{θ} , it then follows from the representations $m_{G_{\beta,a,n}}^{\phi_n} = R_{-\phi_n} m_{G_{\beta,a,n}}^0$ and $m_{G_{\beta,a}}^{\phi} =$

$R_{-\phi} m_{G_{\beta,a}}^0$ of the Weyl–Titchmarsh functions that $m_{G_{\beta,a,n}}^{\phi_n} \rightarrow m_{G_{\beta,a}}^{\phi}$ in law. The convergence of spectral measures then follows from the continuity of the Herglotz representation.

This holds jointly with the convergence of coefficient and transfer matrices, since these convergences all hold in probability under a coupling such that the compact convergence from Lemma 3.26 holds in probability (i.e., the coupling used in the proof of Theorem 3.23 and Corollary 3.23.1). ■

3.2.3.3 Application to the Laguerre β -ensemble

We now apply the convergence results of §3.2.3.1 to the Laguerre β -ensemble. To do so, it suffices to check that it satisfies Hypothesis 3.3. Then, based on these results we motivate Conjecture 3.1.

Recall from (1.6) in the introduction that the tridiagonal model $L_{\beta,n,\ell_n} L_{\beta,n,\ell_n}^T$ for the spiked Laguerre β -ensemble is built from the random bidiagonal matrix

$$L_{\beta,n,\ell_n} := \frac{1}{\sqrt{\beta}} \begin{pmatrix} \sqrt{\ell_n} Y_{n,1} & X_{n,1} & & \\ & \ddots & \ddots & \\ & & \ddots & X_{n,n-1} \\ & & & Y_{n,n} \end{pmatrix} \quad \text{and} \quad \begin{aligned} Y_{n,j} &\sim \chi_{(m-j+1)\beta} \\ X_{n,j} &\sim \chi_{(n-j)\beta} \end{aligned}$$

with all $X_{n,j}$'s and $Y_{n,j}$'s independent. From [58, 59], we know that if $m = n + a$, then $n L_{\beta,n,\ell_n} L_{\beta,n,\ell_n}^T$ has a hard edge scaling.

The matrix L_{β,n,ℓ_n} has exactly the form that we took in (3.29) with $x_{n,j} := \frac{1}{\sqrt{\beta}} X_{n,j}$ and $y_{n,j} := \frac{1}{\sqrt{\beta}} Y_{n,j}$, so it has a canonical system representation of the form we built in §3.2.3.1. A computation of the moments of the logarithms of its entries easily shows that this matrix satisfies Hypothesis 3.3.

Proposition 3.29. *The matrix L_{β,n,ℓ_n} from (1.6) with $m = n + a$ satisfies Hypothesis 3.3.*

Together with Theorem 3.23, this result implies that the associated canonical system converges vaguely to the canonical system for the stochastic Bessel operator, and the convergence extends to transfer matrices by Corollary 3.23.1. When the spike parameter ℓ_n satisfies Hypothesis 3.4, the convergence further extends to Weyl–Titchmarsh functions and spectral measures by Corollary 3.23.2. In particular, this proves Theorems 1.6 and 1.8.

Proof. Since $y_{n,j+1}^2 = \frac{1}{\beta} Y_{n,j+1}^2$ with $Y_{n,j+1} \sim \chi_{n+a-j}^2$, $y_{n,j+1}^2 \sim \Gamma(\frac{(n+a-j)\beta}{2}, \frac{2}{\beta})$ where $\Gamma(\alpha, \theta)$ is a Γ distribution with shape and scale parameters α and θ . In the same way, $x_{n,j}^2 \sim \Gamma(\frac{(n-j)\beta}{2}, \frac{2}{\beta})$. By definition, the cumulant generating function of $\log y_{n,j+1}^2$ is given by $K_{j+1}(t) := \log \mathbb{E} y_{n,j+1}^{2t}$. The formula for the t th moment of a $\Gamma(\alpha, \theta)$ random variable then yields

$$K_{j+1}(t) = t \log \frac{2}{\beta} + \log \Gamma\left(\frac{(n+a-j)\beta}{2} + t\right) - \log \Gamma\left(\frac{(n+a-j)\beta}{2}\right).$$

This allows us to compute the cumulants $\kappa_k(\log y_{n,j+1}^2) = K_{j+1}^{(k)}(0)$ of $\log y_{n,j+1}^2$. Recall that by basic properties of cumulants, $\mathbb{E} \log y_{n,j+1}^2 = \kappa_1(\log y_{n,j+1}^2)$, and writing $\mu_k(\log y_{n,j+1}^2)$ for the k th central moment of $\log y_{n,j+1}^2$, $\mu_2(\log y_{n,j+1}^2) = \kappa_2(\log y_{n,j+1}^2)$ and $\mu_4(\log y_{n,j+1}^2) = \kappa_4(\log y_{n,j+1}^2) - 3\mu_2^2(\log y_{n,j+1}^2)$. We deduce that

$$\mathbb{E} \log y_{n,j+1}^2 = \log \frac{2}{\beta} + \psi^{(0)}\left(\frac{(n+a-j)\beta}{2}\right), \quad \mu_2(\log y_{n,j+1}^2) = \psi^{(1)}\left(\frac{(n+a-j)\beta}{2}\right),$$

and

$$\mu_4(\log y_{n,j+1}^2) = \psi^{(3)}\left(\frac{(n+a-j)\beta}{2}\right) + 3\psi^{(1)}\left(\frac{(n+a-j)\beta}{2}\right)^2,$$

where $\psi^{(k)}$ is the polygamma function of order k . These functions are known to satisfy the following inequalities for all $t > 0$ (see e.g. [39, Theorems 4 and 5] and [2, Theorem 9]):

$$0 < \psi^{(0)}(t) - \log t + \frac{1}{2t} + \frac{1}{12t^2} < \frac{1}{120t^4},$$

$$-\frac{1}{30t^5} < \psi^{(1)}(t) - \frac{1}{t} - \frac{1}{2t^2} - \frac{1}{6t^3} < 0, \quad 0 < \psi^{(3)}(t) - \frac{2}{t^3} - \frac{3}{t^4} < \frac{2}{t^5}.$$

Hence,

$$\begin{aligned} \mathbb{E} \log y_{n,j+1}^2 &= \log(n+a-j) - \frac{1}{(n+a-j)\beta} + \frac{\delta_{n,j}}{(n+a-j)^2}, \\ \mu_2(\log y_{n,j+1}^2) &= \frac{2}{(n+a-j)\beta} + \frac{\delta'_{n,j}}{(n+a-j)^2}, \\ \mu_4(\log y_{n,j+1}^2) &= \frac{\delta''_{n,j}}{(n+a-j)^2} \end{aligned}$$

for some numbers $\delta_{n,j}, \delta'_{n,j}, \delta''_{n,j} \in \mathbb{R}$, which are bounded in both n and j since $a > -1$ and $n-j \geq 1$. Now,

$$\frac{1}{n+a-j} - \frac{1}{n-j} = \frac{a}{(n-j)^2} \left(1 + \frac{a}{n-j}\right)^{-1},$$

and because the factor $(1 + \frac{a}{n-j})^{-1}$ is bounded for $n-j \geq 1$, we can remove the a 's from the denominators in these expressions for the moments at the only cost of redefining $\delta_{n,j}, \delta'_{n,j}$ and $\delta''_{n,j}$.

Note that because $x_{n,j}^2 \sim \Gamma(\frac{(n-j)\beta}{2}, \frac{2}{\beta})$, the corresponding moments of $\log x_{n,j}^2$ are given by the same expressions, but without the a 's. The result thus follows from combining the moments of $\log x_{n,j}^2$ and $\log y_{n,j+1}^2$ using their independence. $\blacksquare$

To conclude this section, we motivate Conjecture 3.1. Indeed, the canonical system representation of $n\mathbf{L}_{\beta,n}\mathbf{L}_{\beta,n}^\top$ gives information from which we can guess what the spectral weights of the stochastic Bessel canonical system should be.

Take $L_n := \mathbf{L}_{\beta,n} = \mathbf{L}_{\beta,n,\ell_n}$ for $\ell_n = 1$ in the canonical system setting of §3.2.3.1. This matrix satisfies Hypothesis 3.3 by Proposition 3.29 and $\ell_n = 1$ satisfies Hypothesis 3.4 with $\phi = 0$, so the

results from Theorem 3.23 and its corollaries apply. Note that in this case, the boundary condition $u_{n,0}$ from (3.39a) can be replaced with e_0 , since these two vectors are parallel.

Because the canonical system has discrete spectrum with only eigenvalues, by Theorem 2.8, an eigenvalue λ has spectral weight $\mu_{G_{\beta,a,n}}^0(\{\lambda\})$ equal to $\|u_\lambda\|_{G_{\beta,a,n}}^{-2}$ where u_λ solves $Ju'_\lambda = -\lambda G_{\beta,a,n}u_\lambda$ with $u_\lambda(0) = e_0$. By construction of the canonical system, $\|u_\lambda\|_{G_{\beta,a,n}}^2 = \frac{1}{n}\|h_\lambda\|_2^2$ where h_λ is an eigenvector of $nL_{\beta,n}L_{\beta,n}^\top$ with eigenvalue λ . The initial condition $u_\lambda(0) = e_0$ also forces that

$$h_{\lambda,1} = (A_{\beta,a,n,0}u(0))_1 = -p_1^{1/2}\varpi_1 = -\frac{X_{n,0}}{Y_{n,1}}\frac{\beta}{p_0X_{n,0}^2} = -\frac{\sqrt{\beta}}{\sqrt{n}Y_{n,1}}.$$

Now, because h_λ is an eigenvector of $nL_{\beta,n}L_{\beta,n}^\top$, the vector $L_{\beta,n}^\top h_\lambda$ is an eigenvector of $nL_{\beta,n}^\top L_{\beta,n}$. Hence, writing $\{\lambda_j\}_{j=1}^n$ for the set of eigenvalues of $nL_{\beta,n}L_{\beta,n}^\top$, we know from the Dumitriu–Edelman theorem¹ [24, Theorem 3.4] that the vector $\{-\|L_{\beta,n}^\top h_{\lambda_j}\|_2^{-1}(L_{\beta,n}^\top h_{\lambda_j})_1\}_{j=1}^n$ can be written as $\|q\|_2^{-1}q$ where $q_j \sim \chi_\beta$ are independent of each other and of the eigenvalues. This means that the spectral weights for our problem are

$$\mu_{G_{\beta,a,n}}^0(\{\lambda_j\}) = \frac{1}{\|u_{\lambda_j}\|_{G_{\beta,a,n}}^2} = \frac{n}{\|h_{\lambda_j}\|_2^2} = \frac{n}{(L_{\beta,n}^\top h_{\lambda_j})_1^2} \frac{q_{\lambda_j}^2}{\|q\|_2^2} \frac{\|L_{\beta,n}^\top h_{\lambda_j}\|_2^2}{\|h_{\lambda_j}\|_2^2}.$$

The definition of $L_{\beta,n}$ gives $(L_{\beta,n}^\top h_{\lambda_j})_1 = \frac{1}{\sqrt{\beta}}Y_{n,1}h_{\lambda_j,1} = -1/\sqrt{n}$. Moreover, since h_{λ_j} satisfies $nL_{\beta,n}L_{\beta,n}^\top h_{\lambda_j} = \lambda_j h_{\lambda_j}$, we get $\|L_{\beta,n}^\top h_{\lambda_j}\|_2^2 = \langle L_{\beta,n}^\top h_{\lambda_j}, L_{\beta,n}^\top h_{\lambda_j} \rangle = \langle h_{\lambda_j}, L_{\beta,n}L_{\beta,n}^\top h_{\lambda_j} \rangle = \frac{\lambda_j}{n}\|h_{\lambda_j}\|_2^2$. This gives the formula

$$\mu_{G_{\beta,a,n}}^0(\{\lambda_j\}) = \frac{n\lambda_j q_{\lambda_j}^2}{\|q\|_2^2} = \frac{n\beta}{\|q\|_2^2} \lambda_j \frac{q_{\lambda_j}^2}{\beta} \quad (3.50)$$

for the spectral weights. This almost gives an asymptotic description of these spectral weights, since $\frac{1}{n\beta}\|q\|_2^2 \xrightarrow{\text{a.s.}} 1$ by the strong law of large numbers and the $\frac{1}{\beta}q_{\lambda_j}^2$'s are $\Gamma(\beta/2, 2/\beta)$ -distributed and independent of each other and of the eigenvalues. So, assuming that we knew beforehand that the eigenvalue point process converged jointly with the spectral measures, we would see that the spectral weights (3.50) converge to the points of the $\text{Bessel}_{\beta,a}$ point process multiplied by independent $\Gamma(\beta/2, 2/\beta)$ random variables also independent of the eigenvalues. Since the limits of the spectral weights (3.50) would have to be the spectral weights of the spectral measure $\mu_{G_{\beta,a}}^0$ of the stochastic Bessel canonical system, this is what motivates Conjecture 3.1.

A way to turn this idea into a proof would be to prove first the joint convergence of the spectral measures and the eigenvalue point processes. It is possible that this could be done through a more careful analysis of the spectral weights: using the idea from Proposition 2.17, the eigenvalue point

¹The correspondence between their notation and ours is that m and a , in their notation, become n and $\frac{\beta}{2}(n+a) > \frac{\beta}{2}(n-1)$ in ours, so their matrix B_β is our $L_{\beta,n}^\top$.

processes should converge jointly with the spectral measures if the spectral weights of $\mu_{G_{\beta,a}}^0$ can be proven to be limits of those of $\mu_{G_{\beta,a,n}}^0$ and if the spectral weights of the $\mu_{G_{\beta,a,n}}^0$'s are shown not to vanish in the limit in a suitable sense. Note in particular that when $|a| < 1$, the former criterion is automatically satisfied (by Proposition 2.18) since the canonical systems are all limit circle at both endpoints.

3.3 The stochastic sine operator

This short section is devoted to the stochastic sine operator. In §3.3.1, we review a few of its important properties. Then, in §3.3.2, we make a few comments about the possibility of proving the convergence of tridiagonal random matrix models to the stochastic sine operator at the canonical system level, and about the way the framework used in §3.1.3 and §3.2.3 should be expected to adapt to this case.

3.3.1 Basic properties of the stochastic sine operator

Hyperbolic Brownian motion. The stochastic sine operator is defined in terms of a complex-valued stochastic process called a hyperbolic Brownian motion. We start by reviewing some basic properties of this process. For more background, we refer to [36].

Definition 3.2. A *hyperbolic Brownian motion* with variance $\sigma^2 > 0$ started at a point $z \in \mathbb{H}$, in the upper half-plane model of the hyperbolic plane, is the strong solution to the SDE

$$dB(t) = \sigma \operatorname{Im} B(t) dW(t) \quad \text{with} \quad B(0) = z,$$

where W is a standard complex Brownian motion, with independent standard real Brownian motions as its real and imaginary parts.

This complex SDE immediately splits into two real SDEs by taking the real and imaginary parts. The simplest one is that for the imaginary part

$$d \operatorname{Im} B(t) = \sigma \operatorname{Im} B(t) d \operatorname{Im} W(t).$$

Indeed, this can immediately be recognized as the equation for a *geometric* Brownian motion, which can be solved explicitly using Itô's formula. The solution is

$$\operatorname{Im} B(t) = \operatorname{Im} B(0) \exp\left(\sigma \operatorname{Im} W(t) - \frac{\sigma^2 t}{2}\right).$$

The real part of the hyperbolic Brownian motion is thus given by the integral of this process against the independent real standard Brownian motion $\operatorname{Re} W$.

An important property of a hyperbolic Brownian motion is that it converges to a boundary point of the hyperbolic plane as $t \rightarrow \infty$ [36, Proposition 7.6.7.3]. Since we use the upper half-plane as a model of the hyperbolic plane, this means that $\mathcal{B}(\infty)$ is a well-defined *real* random variable.

The stochastic sine operator. For $\beta > 0$, let $\mathcal{B}_\beta$ be a hyperbolic Brownian motion with variance $4/\beta$ started at i in the upper half-plane, and let

$$R_\beta := \frac{1}{2 \det X_\beta} (X_\beta)^\top X_\beta \quad \text{with} \quad X_\beta := \begin{pmatrix} 1 & -\operatorname{Re} X_\beta \\ 0 & \operatorname{Im} X_\beta \end{pmatrix}.$$

Recall from Definition 1.2 in the introduction that the stochastic sine operator is the operator acting on functions $u: (0, 1) \rightarrow \mathbb{C}^2$ as $u \mapsto (R_\beta^{-1} \circ \nu)Ju'$ where $\nu(t) := -\log(1-t)$. This operator was introduced by Valkó and Virág in [76].

It should be noted that, in contrast with the stochastic Airy and Bessel operators, the stochastic sine operator is not a random generalized Sturm–Liouville operator. It is rather a random *Dirac* operator, that is, a random canonical system whose coefficient matrix (here $R_\beta \circ \nu$) is invertible at all times.

The spectrum of the stochastic sine operator and the Weyl alternative. When given the boundary conditions

$$e_0^*Ju(0) = 0 \quad \text{and} \quad u_1^*Ju(1) = 0 \quad \text{where} \quad u_1 := \begin{pmatrix} \operatorname{Re} \mathcal{B}_\beta(\infty) \\ 1 \end{pmatrix}, \quad (3.51)$$

with the boundary condition at 1 being enforced only when $\beta > 2$, the stochastic sine canonical system has a discrete spectrum called the sine_β point process. This was shown by Valkó and Virág in [76], building on their earlier work [72]. As stated in the introduction, the sine_β point process describes the bulk scaling limit of β -ensembles—we have seen an example of this in Theorem 1.3 (that is, [76, Theorem 1]), which describes precisely the scaling limit from the Hermite β -ensemble.

The reason the right boundary condition in (3.51) is not needed when $\beta \leq 2$ is, of course, that the canonical system is limit point at 1 in that case. This result was proven by Valkó and Virág in [76, Theorem 25]. It is also a consequence of the following bound on the trace of the sine system's coefficient matrix, which we will need later. We give a full proof for the sake of completeness, but it should be noted that this is also contained in the proof of [76, Theorem 25].

Lemma 3.30. *For any $\varepsilon > 0$ and $\delta \in (0, 2/\beta)$, there is a $C_{\varepsilon, \delta} > 0$ such that*

$$\mathbb{P}\left[\forall t \in [0, 1), \operatorname{tr} R_\beta \circ v(t) \leq \frac{C_{\varepsilon, \delta}}{(1-t)^{2/\beta+\delta}}\right] \geq 1 - \varepsilon.$$

Remark. In particular, if $\beta > 2$ then we can choose $\delta < 2/\beta - 1$ and the upper bound is integrable.

Proof. By definition,

$$\operatorname{tr} R_\beta \circ v = \frac{1}{2 \operatorname{Im} \mathcal{B}_\beta \circ v} (1 + |\mathcal{B}_\beta \circ v|^2).$$

Let W denote the complex Brownian motion that drives the hyperbolic Brownian motion $\mathcal{B}_\beta$. By properties of Brownian motion (for instance, applying Proposition A.1), for any $\varepsilon > 0$ there is a $\tilde{C} > 0$ such that with probability at least $1 - \varepsilon/2$, $|\operatorname{Im} W \circ v(t)| \leq \tilde{C}(1 + v(t)^{3/4})$ for all $t \in [0, 1)$. Therefore, for any $\delta > 0$, there is also a $C > 0$ such that on the same event, for all $t \in [0, 1)$,

$$\frac{1}{C}(1-t)^{2/\beta+\delta} \leq \operatorname{Im} \mathcal{B}_\beta \circ v(t) \leq C(1-t)^{2/\beta-\delta}.$$

If $\operatorname{Im} \mathcal{B}_\beta \circ v$ is bounded in that way, then for $\delta < 2/\beta$,

$$\langle \operatorname{Re} \mathcal{B}_\beta \circ v \rangle(t) \leq \frac{4C^2}{\beta} \int_0^t (1-s)^{2/\beta-\delta} d\langle \operatorname{Re} W \circ v \rangle(s) = \frac{4C^2}{\beta} \int_0^t (1-s)^{2/\beta-\delta-1} ds \leq \frac{4C^2}{2-\beta\delta}.$$

Hence, Bernstein's inequality for martingales shows that for any $x > 0$,

$$\mathbb{P}\left(\left\{|\operatorname{Im} W \circ v(t)| \leq \tilde{C}(1 + v(t)^{3/4})\right\} \cap \left\{\sup_{t \in [0, 1)} |\operatorname{Re} \mathcal{B}_\beta \circ v(t)| \geq x\right\}\right) \leq 2 \exp\left(-\frac{(2-\beta\delta)x^2}{8C^2}\right),$$

and this bound can be made smaller than $\varepsilon/2$ by taking x large enough. Combining the bounds on the real and imaginary parts of $\mathcal{B}_\beta$ yields the result with $C_{\varepsilon, \delta} := \frac{C}{2}(1 + C^2 + x^2)$. ■

3.3.2 Remarks on the convergence from tridiagonal matrix models

When building the stochastic sine operator, Valkó and Virág also built in [76] a framework to describe a family of Dirac operators that are parametrized by a path in the upper half-plane, from which a coefficient matrix can be defined, and two boundary points which serve as boundary conditions. In addition to using this framework to build the stochastic sine operator, they also used it in [73] to embed circular β -ensembles as Dirac operators, and showed that under a suitable coupling, their inverses converge a.s. to the inverse of the stochastic sine operator in Hilbert–Schmidt norm.

In a similar way, it should also be possible to obtain the stochastic sine canonical system as the limit of canonical systems built from appropriate tridiagonal random matrix models, like for

example those for the Hermite and Laguerre β -ensembles scaled in the bulk. This would upgrade to the canonical system level the results that are known for the associated eigenvalue point processes, of which we saw an example in Theorem 1.3 in the introduction.

Although we won't tackle this problem here, we point out that it could be studied using the framework we used in §3.1.3 and §3.2.3 to show convergence of tridiagonal random matrix models to the stochastic Airy and Bessel operators. However, there is an important difference between this case and the other two. Indeed, in the cases of the stochastic Airy and Bessel operators, Lemmas 3.10 and 3.26 essentially imply that the coefficient matrices of the discrete models converge compactly to the coefficient matrices of the continuum ones. In the case of the stochastic sine operator, this would certainly not be possible, because the coefficient matrices of the discrete models have a projector structure that they cannot retain in the limit, since the coefficient matrix of the sine system is invertible. Therefore, a part of the coefficient matrices must vanish in the vague limit.

A possible way to work out this convergence would be to represent the solutions that appear in the coefficient matrix of the discrete systems using pairs of polar coordinates—something similar to the so-called *Prüfer variables* used in the theory of Jacobi operators [68, §4.1]—and analyse the behavior of these polar coordinates. Having such representations of these solutions could help to split them into an “oscillating” part that would vanish in the vague limit and an “average” part that would survive in the limit. We will use a change of variables built using this kind of idea in Chapter 4 to analyse canonical system level transitions from the stochastic Airy or Bessel operator to the stochastic sine operator.

Chapter 4

Edge to bulk transitions in β -ensembles

4.1 Introduction

The point processes that describe scaling limits of β -ensembles should be expected to be related to each other. For example, one could expect that it is possible to retrieve the sine_β point process as a scaling limit of the Airy_β point process. The reason for this is that the sine_β point process describes an infinite number of points which are, on average, equally spaced on the real line, while the Airy_β point process describes a sequence of points that are spaced with increasing density on the half line. Because these two points processes are scaling limits of the same β -ensembles, one could expect that starting from the Airy_β point process and zooming in the direction with infinitely many points while rescaling the space in the right way, the randomness remains consistent so that the sine_β point process is retrieved. Intuitively, the idea here is that we start from the edge of the semicircle in Figure 1.1, and we zoom back into the bulk.

As mentioned in Theorem 1.4 in the introduction, this does happen: Valkó and Virág have shown that $2\sqrt{E}(\text{Airy}_\beta + E) \rightarrow \text{sine}_\beta$ vaguely in law as $E \rightarrow \infty$ [72, Corollary 3]. The same phenomenon also happens when starting from the hard edge instead of the soft edge (although the scaling is different)—a version of this was proven by Holcomb in [41, Theorem 1.1]. Because the Airy_β , $\text{Bessel}_{\beta,a}$ and sine_β point processes are the spectra of random differential operators, one should expect these edge to bulk transitions to hold not only at the level of the spectra, but also at the level of the operators themselves. The goal of this section is to describe such operator level transitions, using canonical systems theory.

The material of this chapter is mostly taken from [54, 55], although some of the arguments are reformulated in a generalized setting in order to combine the results of the two papers.

Canonical system representations and main results. The point process transition from the soft edge to the bulk is described by the vague limit $2\sqrt{E}(\text{Airy}_\beta + E) \rightarrow \text{sine}_\beta$. Since the Airy_β point

process is the negative of the spectrum of the stochastic Airy operator $\mathcal{H}_\beta$ and since the sine $_\beta$ point process is symmetric, one can expect from this point process transition that

$$\mathcal{H}_{\beta,E} := 2\sqrt{E}(\mathcal{H}_\beta - E) \quad (4.1)$$

converges to the stochastic sine operator as $E \rightarrow \infty$. Similarly, the spacing between the eigenvalues of the stochastic Bessel operator $\mathfrak{G}_{\beta,a}$ in the deterministic case $\beta = \infty$ can be estimated from asymptotics of Bessel functions, and this leads to expect that the rescaled operator

$$\mathfrak{G}_{\beta,a,E} := \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E) \quad (4.2)$$

should converge to the stochastic sine operator as $E \rightarrow \infty$.

Now, since $\mathcal{H}_{\beta,E}$ and $\mathfrak{G}_{\beta,a,E}$ are generalized Sturm–Liouville while the stochastic sine operator is Dirac, we need a common framework under which to represent all three operators, and for this we use (of course) canonical systems. As we will see in detail in §4.2, the general method of §2.4 yields canonical systems on $(0, \infty)$ for the operators $\mathcal{H}_{\beta,E}$ and $\mathfrak{G}_{\beta,a,E}$, with coefficient matrices

$$H_{\beta,E} := \frac{1}{2\sqrt{E}} \begin{pmatrix} \sqrt{E} g_{\beta,E}^2 & f_{\beta,E} g_{\beta,E} \\ f_{\beta,E} g_{\beta,E} & \frac{1}{\sqrt{E}} f_{\beta,E}^2 \end{pmatrix} \quad \text{and} \quad G_{\beta,a,E} := \frac{w_{\beta,a} \sqrt{E}}{2} \begin{pmatrix} \sqrt{E} g_{\beta,a,E}^2 & f_{\beta,a,E} g_{\beta,a,E} \\ f_{\beta,a,E} g_{\beta,a,E} & \frac{1}{\sqrt{E}} f_{\beta,a,E}^2 \end{pmatrix} \quad (4.3)$$

where $f_{\beta,E}$, $g_{\beta,E}$ and $f_{\beta,a,E}$, $g_{\beta,a,E}$ are pairs of fundamental solutions to $\mathcal{H}_{\beta,E} h = 0$ and $\mathfrak{G}_{\beta,a,E} h = 0$ respectively, with initial conditions

$$\begin{aligned} g_{\beta,E}(0) &= f'_{\beta,E}(0) = 0, & g'_{\beta,E}(0) &= f_{\beta,E}(0) = 1, \\ g_{\beta,a,E}(0) &= f'_{\beta,a,E}(0) = 0, & g'_{\beta,a,E}(0) &= f_{\beta,a,E}(0) = 1. \end{aligned}$$

We will then show that these canonical systems converge to the canonical system corresponding to the stochastic sine operator.

We will start by proving the vague convergence of the canonical systems, as stated in the introduction.

Theorem 1.9 [54, 55, Theorems 1]. *Let $\mathcal{I}_\beta := [0, 1)$ if $\beta \leq 2$ and $\mathcal{I}_\beta := [0, 1]$ if $\beta > 2$. For any diverging sequence $\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$, there are $\mathcal{C}^1$ bijections $\eta_{A,E_n} : [0, 1 + \varepsilon_{A,E_n}) \rightarrow [0, \infty)$ where $\varepsilon_{A,E_n} \rightarrow 0$ such that*

$$\tilde{H}_{\beta,E_n} := \eta'_{A,E_n}(H_{\beta,E_n} \circ \eta_{A,E_n}) \xrightarrow[n \rightarrow \infty]{\text{law}} \tilde{R}_\beta := R_\beta \circ v$$

vaguely on $\mathcal{I}_\beta$, where R_β and v are as in Definition 1.2. Likewise, there are $\mathcal{C}^1$ bijections

$\eta_{B,E_n} : [0, 1 + \varepsilon_{B,E_n}) \rightarrow [0, \infty)$ where $\varepsilon_{B,E_n} \rightarrow 0$ such that

$$\tilde{G}_{\beta,a,E_n} := \eta'_{B,E_n}(G_{\beta,a,E_n} \circ \eta_{B,E_n}) \xrightarrow[n \rightarrow \infty]{\text{law}} \tilde{R}_\beta$$

vaguely on $\mathcal{I}_\beta$.

Remark. The numbers $\varepsilon_{A,E}$ and $\varepsilon_{B,E}$ appear here for technical reasons related to the possible discrepancy between the behavior of the Airy or Bessel system and that of the sine system near the right endpoint. They will always vanish as $E \rightarrow \infty$ (and in fact we will take them to be exactly zero in some range of values of β and a), and at this point $\eta_{A,E}$ and $\eta_{B,E}$ should essentially be thought of as time changes between $(0, 1)$ and $(0, \infty)$. The precise definitions of $\eta_{A,E}$ and $\eta_{B,E}$ will be given in (4.11) and (4.18) respectively.

As a corollary, we will also show the convergence of the canonical systems' transfer matrices.

Corollary 1.9.1 [54, 55, Corollaries 1.1]. *Let $\{E_n\}_{n \in \mathbb{N}}$ be a diverging sequence, and let $T_{\tilde{H}_{\beta,E_n}}, T_{\tilde{G}_{\beta,a,E_n}}, T_{\tilde{R}_\beta} : \mathcal{I}_\beta \times \mathbb{C} \rightarrow \mathbb{C}^{2 \times 2}$ be the transfer matrices of the shifted and scaled Airy and Bessel systems and of the sine system. Then*

$$T_{\tilde{H}_{\beta,E_n}} \xrightarrow[n \rightarrow \infty]{\text{law}} T_{\tilde{R}_\beta} \quad \text{and} \quad T_{\tilde{G}_{\beta,a,E_n}} \xrightarrow[n \rightarrow \infty]{\text{law}} T_{\tilde{R}_\beta}$$

compactly on $\mathcal{I}_\beta \times \mathbb{C}$, jointly with the respective coefficient matrices.

Having proven the vague convergence of coefficient matrices and the compact convergence of transfer matrices, our next goal will be to show the convergence of the Weyl–Titchmarsh functions, in order to deduce the convergence of the associated spectral measures. To do so, at this point it will mostly remain to prove the convergence of the systems' boundary conditions, and this will lead us to the following.

Theorem 1.10 [54, 55, Theorems 2]. *Let $\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$ be a diverging sequence, and let $m_{\tilde{H}_{\beta,E_n}}, m_{\tilde{G}_{\beta,a,E_n}}, m_{\tilde{R}_\beta} : \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ be the Weyl–Titchmarsh functions of the shifted and scaled Airy and Bessel systems and of the sine system. Then*

$$m_{\tilde{H}_{\beta,E_n}} \xrightarrow[n \rightarrow \infty]{\text{law}} m_{\tilde{R}_\beta} \quad \text{and} \quad m_{\tilde{G}_{\beta,a,E_n}} \xrightarrow[n \rightarrow \infty]{\text{law}} m_{\tilde{R}_\beta}$$

compactly on $\mathbb{H}$, jointly with the respective coefficient and transfer matrices. In particular, the associated spectral measures converge vaguely in law, also jointly with the other objects.

The spectral measures of the Airy, Bessel and sine systems are all pure point with positive masses precisely at the (simple) eigenvalues of the corresponding operators. In general, the vague convergence of spectral measures is not strong enough to guarantee the vague convergence of the

associated eigenvalue point processes: in principle, it would be possible for spectral masses to vanish or merge with others in the limit $E \rightarrow \infty$.

However, the spectral weights of the sine operator are known to be independent $\Gamma(\beta/2, 4/\beta)$ random variables that are also independent of the eigenvalues [74, Proposition 3], and the same also holds for the shifted Airy operator $\mathcal{H}_{\beta,E}$ [55, Corollary D.2]. Because the spectral weights of $\mathcal{H}_{\beta,E}$ and of the stochastic sine operator have the same distributions and are independent of the eigenvalues, in this case the point process convergence $2\sqrt{E}(\text{Airy}_\beta + E) \xrightarrow{\text{law}} \text{sine}_\beta$ actually does follow from the convergence of the corresponding spectral measures, and Theorem 1.10 recovers Valkó and Virág's result (Theorem 1.4).

The spectral weights of the shifted Bessel operator $\mathfrak{G}_{\beta,a,E}$ are not known, but they can be easily related to those of the usual Bessel operator $\mathfrak{G}_{\beta,a}$. Their description given in Conjecture 3.1 then suggests that the spectral weights of $\mathfrak{G}_{\beta,a,E}$ converge to independent $\Gamma(\beta/2, 4/\beta)$ random variables that are also independent of the eigenvalues, and this would yield the convergence of the eigenvalue point processes. We will come back to this in more detail in §4.4.4.

Related work on operator convergence. As mentioned earlier, the two edge to bulk transitions studied here have already been studied at the point process level. Indeed, Valkó and Virág have proven in [72, Corollary 3] that $2\sqrt{E}(\text{Airy}_\beta + E) \rightarrow \text{sine}_\beta$ vaguely in law as $E \rightarrow \infty$, and Holcomb proved in [41, Theorem 1.1] that for any $a > 0$, if $\text{Bess}_{\beta,a}$ denotes the square root of the $\text{Bessel}_{\beta,a}$ point process, then $\frac{1}{4}(\text{Bess}_{\beta,a} - E) \rightarrow \text{sine}_\beta$ vaguely in law as $E \rightarrow \infty$.

These two edge to bulk transitions are not the only possible transitions between these point processes. Another important example is a transition from the hard edge to the soft edge, which occurs when taking $a \rightarrow \infty$. In that case, because both the stochastic Airy and Bessel operators are random generalized Sturm–Liouville operators, one can exploit their resolvents to describe the transition. This was done by Dumaz, Li and Valkó in [22], where it is shown that the transition occurs at the operator level, in the norm resolvent sense (see [69, §6.6] for the definition of this type of convergence).

For comparison with this type of result, note that if $S_{\tilde{H}_{\beta,E}}$, $S_{\tilde{G}_{\beta,a,E}}$ and $S_{\tilde{R}_\beta}$ denote the self-adjoint realizations of the corresponding canonical system's relations for the appropriate boundary conditions, then Theorems 1.9 and 1.10 immediately imply (by general results from canonical systems theory, see §2.2.2.3) that for any $\varphi \in \mathcal{C}_c(I, \mathbb{C}^2)$,

$$(S_{\tilde{H}_{\beta,E}} - z)^{-1} \varphi \xrightarrow[n \rightarrow \infty]{\text{law}} (S_{\tilde{R}_\beta} - z)^{-1} \varphi \quad \text{and} \quad (S_{\tilde{G}_{\beta,a,E}} - z)^{-1} \varphi \xrightarrow[n \rightarrow \infty]{\text{law}} (S_{\tilde{R}_\beta} - z)^{-1} \varphi,$$

both compactly in z . As explained in §2.2.2.3, this might extend to a strong resolvent convergence in a generalized sense where the Hilbert space on which the resolvent acts (here $L^2_{\tilde{H}_{\beta,E}}(0, 1)$ or $L^2_{\tilde{G}_{\beta,a,E}}(0, 1)$) is allowed to change along the sequence, but it is not obvious to do so.

Related work in the soft edge case. In [49], the authors give an analytic construction of the *stochastic Airy function* SAi_λ , which is a random $L^2(0, \infty)$ function, entire with respect to the parameter $\lambda \in \mathbb{C}$, that satisfies $\mathcal{H}_\beta \text{SAi}_\lambda = -\lambda \text{SAi}_\lambda$ with explicit almost sure asymptotics for $\text{SAi}_\lambda(t)$ as $t \rightarrow \infty$. The Weyl–Titchmarsh function $m_{\mathcal{H}_\beta}$ of the stochastic Airy canonical system admits a representation in terms of this function, namely

$$m_{\mathcal{H}_\beta}(z) = \frac{\text{SAi}'_{\lambda_E(z)}(0)}{\sqrt{E} \text{SAi}_{\lambda_E(z)}(0)} \quad \text{with} \quad \lambda_E(z) := -E - \frac{z}{2\sqrt{E}},$$

where the prime denotes a derivative with respect to time. Similarly, the Weyl–Titchmarsh function of the stochastic sine system can be written in terms of a random entire function ζ_β called the *stochastic zeta function*, which was introduced in [75]. Hence, Theorem 1.10 gives the following.

Corollary 1.10.1 [55, Corollary 2.1]. *There is another random analytic function $\xi: \mathbb{H} \rightarrow \mathbb{C}$ such that for any diverging sequence $\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$ and any $z \in \mathbb{H}$,*

$$\frac{\text{SAi}'_{\lambda_{E_n}(z)}(0)}{\sqrt{E_n} \text{SAi}_{\lambda_{E_n}(z)}(0)} \xrightarrow[n \rightarrow \infty]{\text{law}} \sqrt{q^2 + 1} \frac{\xi(z)}{\zeta_\beta(-z)}$$

where q is a standard Cauchy random variable independent of ξ and ζ_β . Moreover, ξ can be written in terms of q and of an independent solution to the eigenvalue equation for a differential operator τ_β that is unitarily equivalent to the stochastic sine operator.

From Theorem 1.10, the proof of this result only amounts to unraveling definitions, so we skip it in the main text. For the sake of completeness, we still provide it in Appendix A.3.

In [4, Theorem 11], an independent construction of the stochastic Airy function is given via canonical systems. The author defines the regularized Fredholm determinant for $z \in \mathbb{H}$ as

$$\mathfrak{p}(z) := \det_2(\text{id} - z\mathcal{H}_\beta^{-1}) = \lim_{t \rightarrow \infty} e^{zt(t)} (T_{\mathcal{H}_\beta}(t, z))_{11}$$

where $\mathfrak{t}(t)$ is the trace of $\mathcal{H}_\beta^{-1}$ restricted to $[0, t]$. The function $\mathfrak{p}$ can be explicitly related to $\text{SAi}_\lambda(0)$.

Finally, we mention that there also exists an independent approach from [38, 40], which constructs the semigroup of the stochastic Airy operator $e^{T\mathcal{H}_\beta}$. We do not know if there is an explicit relationship between this semigroup and the canonical system representation.

Organization of the chapter. The rest of the chapter is divided into three main sections.

First, §4.2.1 and §4.2.2 are devoted to the description of the canonical system setups for the soft edge to bulk and hard edge to bulk problems, including the construction of the time changes $\eta_{A,E}$ and $\eta_{B,E}$ that appear in Theorem 1.9 and that allow the two systems to act on $(0, 1)$ instead

of $(0, \infty)$. Now, to prove the vague convergences from Theorem 1.9, we must analyse the behavior of the solutions that appear in the coefficient matrices (4.3) of the Airy and Bessel systems. To do so, in Propositions 4.1 and 4.2, we change variables and write the (time-changed) solutions using pairs of polar coordinates. It turns out that the expressions of the coefficient matrices $H_{\beta,E}$ and $G_{\beta,a,E}$ in polar coordinates are the same, and that these coordinates are driven by SDEs with very similar structures. This allows us to infer, in §4.2.3, a general setup that combines the two vague convergence problems. In §4.2.3, we also see how the change of variables into polar coordinates gives more intuition about the convergence. Indeed, the change of variables allows us to split the coefficient matrix in two terms, and to effectively decouple a highly oscillatory part that vanishes in the vague limit, and an “average” part that converges to the sine system’s coefficient matrix. In particular, this allows us to identify the process that converges to the hyperbolic Brownian motion from the sine system’s coefficient matrix, as well as the process that converges to its driving complex Brownian motion.

Then, the goal of §4.3 is to prove Theorem 1.9, that is, to prove the soft edge to bulk and hard edge to bulk transitions at the level of the vague convergence of the canonical systems. For that part, we work in the general setup introduced in §4.2.3. Our proof is based on a coupling between the driving Brownian motions of our systems, which we build in Lemma 4.4, in §4.3.1. This allows us to deduce, in §4.3.2, quantitative estimates comparing the hyperbolic Brownian motion of the sine system’s coefficient matrix with the process that converges to it. Having done so, we complete the proof of the vague convergence in §4.3.3. At this point, we also deduce the convergence of transfer matrices from Corollary 1.9.1.

We turn to the spectral convergence in §4.4. The analysis is divided in two cases. In §4.4.1, we handle the case $\beta \leq 2$, in which the sine system is limit point on the right. Because of this, the convergence of the Weyl–Titchmarsh functions is essentially a consequence of the convergence of the transfer matrices, by general results from canonical systems theory. When $\beta > 2$, the sine system is limit circle on the right, and then to deduce the convergence of the Weyl–Titchmarsh functions we also need to prove the convergence of the boundary conditions. In that case, our analysis rests not only on the polar coordinates mentioned earlier, but also on other properties of the Airy and Bessel operators, and for this reason we handle the two cases separately in §4.4.2 and §4.4.3. In both cases, we start by a detailed sketch showing the idea of the proof, which involves using the $t \rightarrow -\infty$ asymptotics of solutions to the eigenvalue equations of $\mathcal{H}_{\beta,E}$ and $\mathfrak{G}_{\beta,a,E}$ that we obtained in §3.1.2 and §3.2.2. Finally, in §4.4.4, we also give an expression of the spectral weights of $\mathfrak{G}_{\beta,a,E}$ in terms of those of $\mathfrak{G}_{\beta,a}$, and we argue that under Conjecture 3.1, the point process level hard edge to bulk transition should follow from the convergence of the Weyl–Titchmarsh functions.

4.2 Setup and idea of the convergence

The goal of this section is to present in more detail the canonical system representations of the rescaled stochastic Airy and Bessel operators $\mathcal{H}_{\beta,E} = 2\sqrt{E}(\mathcal{H}_\beta - E)$ and $\mathfrak{G}_{\beta,a,E} = \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E)$, and to set them up for convergence to the stochastic sine canonical system.

We first describe both canonical systems independently in §4.2.1 and §4.2.2, including the construction of the time changes $\eta_{A,E}$ and $\eta_{B,E}$ under which the two systems can act on $(0, 1)$ instead of $(0, \infty)$. In both cases, we also introduce a change of variables into polar coordinates for the solutions that appear in the systems' coefficient matrices. In §4.2.3, we then see a way to unify the representations of the coefficient matrices in polar coordinates, which allows us to combine some of the arguments used in the proofs of Theorems 1.9 and 1.10. We conclude §4.2.3 with a sketch of the main arguments that we will use to prove the vague convergence in §4.3.

4.2.1 Setup for the soft edge to bulk transition

Canonical system representation. With $\mathcal{H}_{\beta,E} = 2\sqrt{E}(\mathcal{H}_\beta - E)$ for $E > 0$, by definition of the stochastic Airy operator (see Definition 3.1), $\mathcal{H}_{\beta,E}$ acts on a function $f: [0, \infty) \rightarrow \mathbb{R}$ as

$$\frac{1}{2\sqrt{E}}\mathcal{H}_{\beta,E}f(t) = -\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)'(t) - \frac{2}{\sqrt{\beta}}B(t)\left(f' - \frac{2}{\sqrt{\beta}}Bf\right)(t) + \left(t - E - \frac{4}{\beta}B^2(t)\right)f(t),$$

where B is a standard Brownian motion. Like the stochastic Airy operator itself, this operator again has the form (2.15) of a generalized Sturm–Liouville operator, but now with $p \equiv 1$, $q(t) := t - E - \frac{4}{\beta}B^2(t)$, $s := -\frac{2}{\sqrt{\beta}}B$ and $w \equiv 1/2\sqrt{E}$. Hence, the method from §2.4 yields a canonical system representation in the same way as it did for the stochastic Airy operator. We now set up the equivalence through the matrix

$$A_{\beta,E} := \begin{pmatrix} E^{1/4}g_{\beta,E}^{[1]} & E^{-1/4}f_{\beta,E}^{[1]} \\ E^{1/4}g_{\beta,E} & E^{-1/4}f_{\beta,E} \end{pmatrix} = \begin{pmatrix} E^{1/4}g'_{\beta,E} - \frac{2}{\sqrt{\beta}}E^{1/4}Bg_{\beta,E} & E^{-1/4}f'_{\beta,E} - \frac{2}{\sqrt{\beta}}E^{-1/4}Bf_{\beta,E} \\ E^{1/4}g_{\beta,E} & E^{-1/4}f_{\beta,E} \end{pmatrix} \quad (4.4)$$

where $f_{\beta,E}$ and $g_{\beta,E}$ are a pair of fundamental solutions to $\mathcal{H}_{\beta,E}h = 0$ with initial conditions

$$g_{\beta,E}(0) = f'_{\beta,E}(0) = 0, \quad g'_{\beta,E}(0) = f_{\beta,E}(0) = 1.$$

Then $A_{\beta,E}$ solves

$$A'_{\beta,E}(t) = \begin{pmatrix} -\frac{2}{\sqrt{\beta}}B(t) & t - E - \frac{4}{\beta}B^2(t) \\ 1 & \frac{2}{\sqrt{\beta}}B(t) \end{pmatrix} A_{\beta,E}(t) \quad \text{with} \quad A_{\beta,E}(0) = \begin{pmatrix} E^{1/4} & 0 \\ 0 & E^{-1/4} \end{pmatrix}.$$

The method from §2.4.2 shows that h solves $\mathcal{H}_{\beta,E}h = zh$ if and only if $u := A_{\beta,E}^{-1}\left(\begin{smallmatrix} h^{[1]} \\ h \end{smallmatrix}\right)$ solves the canonical system

$$Ju' = -zH_{\beta,E}u \quad \text{on } (0, \infty) \quad \text{with} \quad H_{\beta,E} := \frac{1}{2\sqrt{E}} \begin{pmatrix} \sqrt{E}g_{\beta,E}^2 & f_{\beta,E}g_{\beta,E} \\ f_{\beta,E}g_{\beta,E} & \frac{1}{\sqrt{E}}f_{\beta,E}^2 \end{pmatrix}. \quad (4.5)$$

This canonical system is again limit circle at 0 and limit point at infinity, and the eigenvalue problem $\mathcal{H}_{\beta,E}h = zh$ with $h(0) = 0$ corresponds to a self-adjoint realization of the canonical system's maximal relation obtained by restriction by the boundary condition $e_0^*Ju(0) = 0$.

Construction of the appropriate time change. In order to work out the vague convergence of the canonical system (4.5) to the sine canonical system, we must first modify it so that its time domain is $(0, 1)$, like the sine system. Following the time-change property of canonical systems from Theorem 2.2, we are looking for an absolutely continuous bijection $\eta_{A,E}: (0, 1 + \varepsilon_{A,E}) \rightarrow (0, \infty)$ for some $\varepsilon_{A,E} > 0$ with $\varepsilon_{A,E} \downarrow 0$.

The point of adding this parameter $\varepsilon_{A,E}$ is related to the change of behavior of the systems at the right endpoint, and to the discussion in §2.2.3. Recall that both the Airy and the sine systems are limit circle at their left endpoint 0, but the Airy system is always limit point at its right endpoint ∞ while the sine system is either limit point (when $\beta \leq 2$) or limit circle (when $\beta > 2$) at its right endpoint 1. When $\beta \leq 2$, the right endpoints of all systems are limit point and we take $\varepsilon_{A,E} = 0$. However, when $\beta > 2$, the behavior at the right endpoint changes from limit point to limit circle, so we are in the situation discussed in §2.2.3. Thus, we rather want to take $\varepsilon_{A,E} \downarrow 0$ in order to work out the vague convergence of the canonical systems on $[0, 1]$.

A good candidate for $\eta_{A,E}$ stems from the zero-temperature limit case $\beta = \infty$, in which the problem becomes deterministic. Indeed, when $\beta = \infty$ the Brownian motion disappears from $\mathcal{H}_{\beta,E}$, so the functions $f_{\infty,E}$ and $g_{\infty,E}$ are solutions of the equation $2\sqrt{E}(-h''(t) + (t - E)h(t)) = 0$. These are simply solutions of the Airy differential equation but shifted by E , so $f_{\infty,E}$ and $g_{\infty,E}$ can be written as linear combinations of $\text{Ai}(\cdot - E)$ and $\text{Bi}(\cdot - E)$, where Ai and Bi are the usual Airy functions. The Wronskian property $\text{Ai Bi}' - \text{Ai}' \text{Bi} \equiv 1/\pi$ then directly leads to

$$f_{\infty,E}(t) = \pi(\text{Bi}'(-E) \text{Ai}(t - E) - \text{Ai}'(-E) \text{Bi}(t - E)), \quad (4.6a)$$

$$g_{\infty,E}(t) = \pi(\text{Ai}(-E) \text{Bi}(t - E) - \text{Bi}(-E) \text{Ai}(t - E)). \quad (4.6b)$$

The Airy functions have well-known asymptotic expansions (see e.g. [19, 53]). For $t < E$, these lead to

$$f_{\infty,E}(t) = \frac{E^{1/4}}{(E - t)^{1/4}} \left(\cos\left(\frac{2}{3}E^{3/2} - \frac{2}{3}(E - t)^{3/2}\right) + O((E - t)^{-3/2}) \right), \quad (4.7a)$$

$$g_{\infty,E}(t) = \frac{1}{E^{1/4}(E - t)^{1/4}} \left(\sin\left(\frac{2}{3}E^{3/2} - \frac{2}{3}(E - t)^{3/2}\right) + O((E - t)^{-3/2}) \right). \quad (4.7b)$$

Dropping the errors, it follows that if $\eta_{A,E} < E$, then

$$\eta'_{A,E}(H_{\infty,E} \circ \eta_{A,E}) \approx \frac{\eta'_{A,E}}{2\sqrt{E}} \frac{1}{\sqrt{E - \eta_{A,E}}} \begin{pmatrix} \sin^2 \theta_E & \sin \theta_E \cos \theta_E \\ \sin \theta_E \cos \theta_E & \cos^2 \theta_E \end{pmatrix} \circ \eta_{A,E} \quad (4.8)$$

where $\theta_E(t) := \frac{2}{3}E^{3/2} - \frac{2}{3}(E - t)^{3/2}$. Now, $\eta_{A,E}$ should be chosen so that this coefficient matrix converges to $R_\infty \circ \nu$, which is identically equal to $\frac{1}{2}I_2$ since when $\beta = \infty$, the “hyperbolic Brownian motion” B_β has zero variance, so it is stuck at i at all times. If the function $\theta_E \circ \eta_{A,E}$ grows increasingly fast with E , we can expect the oscillations of the trigonometric functions given here to make them converge weakly to their average values, making the above matrix, without the prefactor, converge vaguely to $\frac{1}{2}I_2$ as $E \rightarrow \infty$. This leads us to choose $\eta_{A,E}$ as a function that makes the prefactor go to 1, that is, $\eta_{A,E}$ should solve the equation

$$\eta'_{A,E} = 2c_E \sqrt{E} \sqrt{E - \eta_{A,E}} \quad \text{with} \quad \eta_{A,E}(0) = 0, \quad (4.9)$$

where c_E is something that converges to 1 in an appropriate sense. The simplest choice is, of course, to force the c_E 's to be constants. In that case, the differential equation is separable and the initial value problem is solved by the function

$$\eta_{A,E}(t) := E - E(1 - c_E t)^2. \quad (4.10)$$

However, this is only a bijection $[0, 1/c_E] \rightarrow [0, E]$. When $\beta \leq 2$, we are looking for a bijection $[0, 1) \rightarrow [0, \infty)$, so we can take $c_E = 1$, use (4.10) as a bijection $[0, 1 - 1/\sqrt{E}] \rightarrow [0, E - 1]$ and complete it with a bijection $[1 - 1/\sqrt{E}, 1) \rightarrow [E - 1, \infty)$. When $\beta > 2$, we are looking for a bijection $[0, 1 + \varepsilon_{A,E}) \rightarrow [0, \infty)$, so if $c_E < 1$ we can use (4.10) as a bijection $[0, 1] \rightarrow [0, E - E(1 - c_E)^2]$. In this case, we take $c_E = 1 - 1/\sqrt{E}$ so that $\eta_{A,E}(1) = E - 1$.

In both regimes of β , the above gives a bijection $[0, \tau_E] \rightarrow [0, E - 1]$ for $\tau_E := 1 - 1/\sqrt{E}$ when $\beta \leq 2$ and $\tau_E := 1$ when $\beta > 2$. What we use to complete the bijection will turn out not to make a difference, so we leave it unspecified, assuming only that $\eta_{A,E} : [\tau_E, \tau_E + 1/\sqrt{E}) \rightarrow [E - 1, \infty)$ is a bijection so that $\eta_{A,E}$ is $\mathcal{C}^1$ on the full interval $[0, \tau_E + 1/\sqrt{E})$.

To sum up, our time change is a $\mathcal{C}^1$ bijection $\eta_{A,E} : [0, \tau_E + 1/\sqrt{E}) \rightarrow [0, \infty)$ such that

$$\eta_{A,E}(t) = E - E(1 - c_E t)^2 \quad \text{if} \quad t \leq \tau_E \quad (4.11a)$$

with

$$\begin{cases} c_E = 1, & \tau_E = 1 - 1/\sqrt{E} & \text{when } \beta \leq 2, \\ c_E = 1 - 1/\sqrt{E}, & \tau_E = 1 & \text{when } \beta > 2. \end{cases} \quad (4.11b)$$

It always satisfies $\eta_{A,E}(\tau_E) = E - 1$ and $\eta'_{A,E}(\tau_E) = 2c_E \sqrt{E}$.

Let us emphasize that the reason to take this specific time change is to counteract the natural decay of the “Airy-like” oscillations of the solutions that appear in the system’s coefficient matrix as in (4.8), in order to convert them to “sine-like” oscillations, with no decay in amplitude. Moreover, the point of taking $c_E < 1$ when $\beta > 2$ (i.e., when the sine system is limit circle at 1) is to allow us to analyse the vague convergence of the canonical systems on the closed interval $[0, 1]$, which is more natural in the canonical system setting than working on restricted intervals that would grow to $[0, 1]$. Our specific choice of $c_E = 1 - 1/\sqrt{E}$ for that case also implies that $\eta_{A,E}(1) - E = 1$ for all E , which will play an important role in our proof of the convergence of the Weyl–Titchmarsh functions in §4.4.2.

Polar coordinates. To analyse the behavior of the system and its solutions, we now introduce a change of variables with the goal of obtaining approximations of solutions analogous to those obtained from asymptotics of Airy functions in the deterministic case.

Proposition 4.1 [55, Proposition 14]. *Let $\mathcal{H}_{\beta,E}$ be defined from a standard Brownian motion B on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, and let $\eta_{A,E}$, c_E and τ_E be as in (4.11). If f solves $\mathcal{H}_{\beta,E}f = 0$ and if $f(0)$ and $f'(0)$ are independent of B , then for $t \in [0, \tau_E]$,*

$$f \circ \eta_{A,E}(t) = \frac{C_f}{E^{1/4} \sqrt{1 - c_E t}} e^{\rho_{\beta,E}(t)} \cos \xi_{\beta,E}(t)$$

and

$$f' \circ \eta_{A,E}(t) = C_f E^{1/4} \sqrt{1 - c_E t} e^{\rho_{\beta,E}(t)} \sin \xi_{\beta,E}(t)$$

where $C_f^2 := \sqrt{E} f^2(0) + \frac{1}{\sqrt{E}} f'^2(0)$ and $\rho_{\beta,E}$ and $\xi_{\beta,E}$ solve the coupled SDEs

$$\begin{aligned} d\rho_{\beta,E}(t) &= \left(\frac{1}{\beta} + \left(\frac{2}{\beta} - \frac{1}{2} \right) \cos 2\xi_{\beta,E}(t) + \frac{1}{\beta} \cos 4\xi_{\beta,E}(t) \right) \frac{c_E}{1 - c_E t} dt \\ &\quad + \sqrt{\frac{2}{\beta}} \sin 2\xi_{\beta,E}(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t), \\ d\xi_{\beta,E}(t) &= -2c_E E^{3/2} (1 - c_E t)^2 dt - \left(\left(\frac{2}{\beta} - \frac{1}{2} \right) \sin 2\xi_{\beta,E}(t) + \frac{1}{\beta} \sin 4\xi_{\beta,E}(t) \right) \frac{c_E}{1 - c_E t} dt \\ &\quad + \frac{2\sqrt{2}}{\sqrt{\beta}} \cos^2 \xi_{\beta,E}(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \end{aligned}$$

with $\rho_{\beta,E}(0) = 0$ and $\xi_{\beta,E}(0) = \arctan\left(\frac{f'(0)}{\sqrt{E} f(0)}\right)$, and where

$$B_E(t) := \frac{1}{2c_E \sqrt{E}} \int_0^{\eta_{A,E}(t)} \frac{1}{\sqrt{E - s}} dB(s)$$

is a Brownian motion and a martingale with respect to the filtration $\{\mathcal{F}_{\eta_{A,E}(t)}\}_{t \in [0, \tau_E]}$.

Proof. Let $y := f \circ \eta_{A,E}$ and define two real-valued processes r and ξ from

$$e^{r+i\xi} = Sy + \frac{iy'}{S}$$

where $S: [0, \tau_E] \rightarrow (0, \infty)$ will be specified later. Note that since $\mathcal{H}_{\beta,E}f = 0$, f and f' satisfy the coupled SDEs given in (3.1) with $z = E$. It then follows from the Dambis–Dubins–Schwarz theorem that

$$d(f' \circ \eta_{A,E})(t) = (\eta_{A,E}(t) - E)f' \circ \eta_{A,E}(t)\eta'_{A,E}(t) dt + \frac{2}{\sqrt{\beta}}f' \circ \eta_{A,E}(t)\sqrt{\eta'_{A,E}(t)} dB_E(t)$$

where B_E is the Brownian motion defined in the proposition statement. Itô's formula then shows that

$$\begin{aligned} dy'(t) &= \eta''_{A,E}(t)f' \circ \eta_{A,E}(t) dt \\ &+ \eta'_{A,E}(t) \left((\eta_{A,E}(t) - E)f' \circ \eta_{A,E}(t)\eta'_{A,E}(t) dt + \frac{2}{\sqrt{\beta}}f' \circ \eta_{A,E}(t)\sqrt{\eta'_{A,E}(t)} dB_E(t) \right). \end{aligned}$$

Omitting the explicit time dependence to simplify notation, this simplifies to

$$dy' = \frac{\eta''_{A,E}}{\eta'_{A,E}}y' dt + (\eta'_{A,E})^2(\eta_{A,E} - E)y' dt + \frac{2}{\sqrt{\beta}}(\eta'_{A,E})^{3/2}y' dB_E.$$

We now use the above expression to find the Itô differentials of r and ξ . Applying Itô's formula,

$$\begin{aligned} dr + i d\xi &= d\left(\log(Sy + iy'/S)\right) \\ &= \frac{1}{Sy + iy'/S} \left((S'y + Sy') dt + i \left(\frac{\eta''_{A,E}}{\eta'_{A,E}} \frac{y'}{S} + (\eta'_{A,E})^2(\eta_{A,E} - E) \frac{y'}{S} - \frac{S'y'}{S^2} \right) dt \right. \\ &\quad \left. + \frac{2i}{\sqrt{\beta}}(\eta'_{A,E})^{3/2} \frac{y'}{S} dB_E \right) + \frac{2}{\beta} \frac{1}{(Sy + iy'/S)^2} (\eta'_{A,E})^3 \frac{y'^2}{S^2} dt \\ &= e^{-2r} \left((SS'y^2 + S^2yy') + \frac{\eta''_{A,E}}{\eta'_{A,E}} \frac{(y')^2}{S^2} + (\eta'_{A,E})^2(\eta_{A,E} - E) \frac{yy'}{S^2} - \frac{S'(y')^2}{S^3} \right) dt \\ &\quad + i \left(\frac{\eta''_{A,E}}{\eta'_{A,E}} yy' + (\eta'_{A,E})^2(\eta_{A,E} - E)y^2 - \frac{2S'yy'}{S} - (y')^2 \right) dt \\ &\quad + \frac{2}{\sqrt{\beta}}(\eta'_{A,E})^{3/2} \frac{yy'}{S^2} dB_E + \frac{2i}{\sqrt{\beta}}(\eta'_{A,E})^{3/2} y^2 dB_E \\ &\quad + \frac{2}{\beta} e^{-4r} (\eta'_{A,E})^3 \left(y^4 - \frac{y^2(y')^2}{S^4} - 2i \frac{y^3 y'}{S^2} \right) dt. \end{aligned}$$

We thus get the Itô differentials of r and ξ by reading off the real and imaginary parts of the equation. Using the identities $y = \frac{1}{S}e^r \cos \xi$ and $y' = Se^r \sin \xi$, these differentials can be written only in terms of r and ξ as

$$\begin{aligned} dr = & \frac{2}{\sqrt{\beta}} \frac{(\eta'_{A,E})^{3/2}}{S^2} \cos \xi \sin \xi dB_E + \left(\frac{S'}{S} \cos^2 \xi + \left(S^2 + \frac{(\eta'_{A,E})^2 (\eta_{A,E} - E)}{S^2} \right) \cos \xi \sin \xi \right. \\ & \left. + \left(\frac{\eta''_{A,E}}{\eta'_{A,E}} - \frac{S'}{S} \right) \sin^2 \xi + \frac{2}{\beta} \frac{(\eta'_{A,E})^3}{S^4} \cos^2 \xi (\cos^2 \xi - \sin^2 \xi) \right) dt \end{aligned} \quad (4.12a)$$

and

$$\begin{aligned} d\xi = & \frac{2}{\sqrt{\beta}} \frac{(\eta'_{A,E})^{3/2}}{S^2} \cos^2 \xi dB_E + \left(\frac{(\eta'_{A,E})^2 (\eta_{A,E} - E)}{S^2} \cos^2 \xi + \left(\frac{\eta''_{A,E}}{\eta'_{A,E}} - \frac{2S'}{S} \right) \cos \xi \sin \xi \right. \\ & \left. - S^2 \sin^2 \xi - \frac{4}{\beta} \frac{(\eta'_{A,E})^3}{S^4} \cos^3 \xi \sin \xi \right) dt \end{aligned} \quad (4.12b)$$

We now choose S so as to simplify the equations (4.12) on $[0, \tau_E]$. To do this, note that by definition of $\eta_{A,E}$ on that interval, $\eta_{A,E}(t) - E = -E(1 - c_E t)^2 = -\frac{1}{4c_E^2 E} (\eta'_{A,E}(t))^2$. Hence, if

$$S := \frac{\eta'_{A,E}}{\sqrt{2c_E E^{1/4}}}, \quad \text{then} \quad (\eta'_{A,E})^2 (\eta_{A,E} - E) = -\frac{(\eta'_{A,E})^4}{4c_E^2 E} = -S^4,$$

which means that the coefficients of the $\cos \xi \sin \xi dt$ and $\sin^2 \xi dt$ terms in dr cancel out, and that the coefficients of the $\cos^2 \xi dt$ and $\sin^2 \xi dt$ terms in $d\xi$ are equal. Moreover, with that choice of S ,

$$\frac{S'(t)}{S(t)} = \frac{\eta''_{A,E}(t)}{\eta'_{A,E}(t)} = -\frac{c_E}{1 - c_E t} \quad \text{and} \quad \frac{(\eta'_{A,E}(t))^3}{S^4(t)} = \frac{4c_E^2 E}{\eta'_{A,E}(t)} = \frac{2c_E}{1 - c_E t},$$

so the SDEs for r and ξ become

$$\begin{aligned} dr(t) = & \left(-\cos^2 \xi(t) + \frac{4}{\beta} \cos^2 \xi(t) (\cos^2 \xi(t) - \sin^2 \xi(t)) \right) \frac{c_E}{1 - c_E t} dt \\ & + \frac{2\sqrt{2}}{\sqrt{\beta}} \cos \xi(t) \sin \xi(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \end{aligned} \quad (4.13a)$$

and

$$\begin{aligned} d\xi(t) = & -2c_E E^{3/2} (1 - c_E t)^2 dt + \left(\cos \xi(t) \sin \xi(t) - \frac{8}{\beta} \cos^3 \xi(t) \sin \xi(t) \right) \frac{c_E}{1 - c_E t} dt \\ & + \frac{2\sqrt{2}}{\sqrt{\beta}} \cos^2 \xi(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t). \end{aligned} \quad (4.13b)$$

Simplifying with trigonometric identities, we see that $\xi_{\beta,E} := \xi$ satisfies the desired SDE. For the radial part, we first rewrite (4.13a) as

$$dr(t) = \left(\frac{1}{\beta} - \frac{1}{2} + \left(\frac{2}{\beta} - \frac{1}{2} \right) \cos 2\xi(t) + \frac{1}{\beta} \cos 4\xi(t) \right) \frac{c_E}{1 - c_E t} dt + \sqrt{\frac{2}{\beta}} \sin 2\xi(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t).$$

Then we can define

$$\tilde{r}(t) := r(0) - \frac{1}{2} \int_0^t \frac{c_E}{1 - c_E s} ds = r(0) + \frac{1}{2} \log(1 - c_E t)$$

in order to integrate out explicitly the term $-\frac{1}{2} \frac{c_E}{1 - c_E t} dt$ from the equation for r , and $\rho_{\beta,E} := r - \tilde{r}$ satisfies the desired SDE. Moreover, $\rho_{\beta,E}(0) = 0$ by definition, and

$$e^{r(0)+i\xi(0)} = \sqrt{2c_E} E^{3/4} f(0) + i\sqrt{2c_E} E^{1/4} f'(0)$$

so

$$\xi(0) = \arctan\left(\frac{f'(0)}{\sqrt{E} f(0)}\right) \quad \text{and} \quad e^{r(0)} = \sqrt{2c_E E \left(\sqrt{E} f^2(0) + \frac{1}{\sqrt{E}} f'^2(0) \right)} =: \sqrt{2c_E E} C_f.$$

This finally yields the announced representations

$$f \circ \eta_{A,E}(t) = \frac{1}{S(t)} e^{r(t)} \cos \xi(t) = \frac{C_f}{E^{1/4} \sqrt{1 - c_E t}} e^{\rho_{\beta,E}(t)} \cos \xi_{\beta,E}(t)$$

and

$$f' \circ \eta_{A,E}(t) = \frac{S(t)}{\eta'_{A,E}(t)} e^{r(t)} \sin \xi(t) = C_f E^{1/4} \sqrt{1 - c_E t} e^{\rho_{\beta,E}(t)} \sin \xi_{\beta,E}(t). \quad \blacksquare$$

Now, by time-changing the coefficient matrix (4.5) of the Airy system, we get

$$\eta'_{A,E}(t) H_{\beta,E} \circ \eta_{A,E}(t) = \frac{\eta'_{A,E}(t)}{2\sqrt{E}} \begin{pmatrix} \sqrt{E} g_{\beta,E}^2 & f_{\beta,E} g_{\beta,E} \\ f_{\beta,E} g_{\beta,E} & \frac{1}{\sqrt{E}} f_{\beta,E}^2 \end{pmatrix} (\eta_{A,E}(t)). \quad (4.14)$$

The initial conditions that define $f_{\beta,E}$ and $g_{\beta,E}$ imply that the constants appearing in their representations in the polar coordinates from Proposition 4.1 are $C_{f_{\beta,E}} = E^{1/4}$ and $C_{g_{\beta,E}} = E^{-1/4}$. Because $\eta'_{A,E}(t) = 2c_E E(1 - c_E t)$, it follows that the expression of the coefficient matrix (4.14) in the polar coordinates $(\rho_{\beta,E}^f, \xi_{\beta,E}^f)$ and $(\rho_{\beta,E}^g, \xi_{\beta,E}^g)$ for $f_{\beta,E}$ and $g_{\beta,E}$ is, for $t \in [0, \tau_E]$,

$$\eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E}) = c_E \begin{pmatrix} e^{2\rho_{\beta,E}^g} \cos^2 \xi_{\beta,E}^g & e^{\rho_{\beta,E}^f + \rho_{\beta,E}^g} \cos \xi_{\beta,E}^f \cos \xi_{\beta,E}^g \\ e^{\rho_{\beta,E}^f + \rho_{\beta,E}^g} \cos \xi_{\beta,E}^f \cos \xi_{\beta,E}^g & e^{2\rho_{\beta,E}^f} \cos^2 \xi_{\beta,E}^f \end{pmatrix}. \quad (4.15)$$

This is the form of the coefficient matrix that we will use to prove the vague convergence.

4.2.2 Setup for the hard edge to bulk transition

Canonical system representation. We now turn to the canonical system representation of the rescaled stochastic Bessel operator $\mathfrak{G}_{\beta,a,E} = \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E)$ for $E > 0$. Following Definition 1.1,

$$\mathfrak{G}_{\beta,a,E} f = \frac{2}{w_{\beta,a} \sqrt{E}} (-f^{[1]'} - E w_{\beta,a} f)$$

where $f^{[1]} = p_{\beta,a} f'$ and as before

$$p_{\beta,a}(t) := \exp\left(-at - \frac{2}{\sqrt{\beta}}B(t)\right) \quad \text{and} \quad w_{\beta,a}(t) := \exp\left(-(a+1)t - \frac{2}{\sqrt{\beta}}B(t)\right).$$

This is again a generalized Sturm–Liouville operator as in (2.15), now with $p = p_{\beta,a}$, $q = -Ew_{\beta,a}$, $s \equiv 0$ and $w = w_{\beta,a}\sqrt{E}/2$. The method from §2.4 yields a canonical system representation set up through the matrix

$$A_{\beta,a,E} := \begin{pmatrix} E^{1/4} \mathfrak{g}_{\beta,a,E}^{[1]} & E^{-1/4} \mathfrak{f}_{\beta,a,E}^{[1]} \\ E^{1/4} \mathfrak{g}_{\beta,a,E} & E^{-1/4} \mathfrak{f}_{\beta,a,E} \end{pmatrix} = \begin{pmatrix} E^{1/4} p_{\beta,a} \mathfrak{g}'_{\beta,a,E} & E^{-1/4} p_{\beta,a} \mathfrak{f}'_{\beta,a,E} \\ E^{1/4} \mathfrak{g}_{\beta,a,E} & E^{-1/4} \mathfrak{f}_{\beta,a,E} \end{pmatrix}, \quad (4.16)$$

where $\mathfrak{f}_{\beta,a,E}$ and $\mathfrak{g}_{\beta,a,E}$ are a pair of fundamental solutions to $\mathfrak{G}_{\beta,a,E} h = 0$ with initial conditions

$$\mathfrak{g}_{\beta,a,E}(0) = \mathfrak{f}'_{\beta,a,E}(0) = 0, \quad \mathfrak{g}'_{\beta,a,E}(0) = \mathfrak{f}_{\beta,a,E}(0) = 1.$$

Then $A_{\beta,a,E}$ solves

$$A'_{\beta,a,E}(t) = \begin{pmatrix} 0 & -Ew_{\beta,a} \\ p_{\beta,a}^{-1} & 0 \end{pmatrix} A_{\beta,a,E}(t) \quad \text{with} \quad A_{\beta,a,E}(0) = \begin{pmatrix} E^{1/4} & 0 \\ 0 & E^{-1/4} \end{pmatrix}.$$

The method from §2.4.2 shows that h solves $\mathfrak{G}_{\beta,a,E} h = zh$ if and only if $u := A_{\beta,a,E}^{-1} \begin{pmatrix} h^{[1]} \\ h \end{pmatrix}$ solves the canonical system

$$Ju' = -zG_{\beta,a,E}u \quad \text{on} \quad (0, \infty) \quad \text{with} \quad G_{\beta,a,E} := \frac{w_{\beta,a}\sqrt{E}}{2} \begin{pmatrix} \sqrt{E} \mathfrak{g}_{\beta,a,E}^2 & \mathfrak{f}_{\beta,a,E} \mathfrak{g}_{\beta,a,E} \\ \mathfrak{f}_{\beta,a,E} \mathfrak{g}_{\beta,a,E} & \frac{1}{\sqrt{E}} \mathfrak{f}_{\beta,a,E}^2 \end{pmatrix}. \quad (4.17)$$

This canonical system is limit circle at 0, and by Proposition 3.15 it is limit circle at infinity when $|a| < 1$ and limit point at infinity when $a \geq 1$. The boundary condition $h(0) = 0$ again becomes $e_0^* Ju(0) = 0$, and the boundary condition at infinity transforms in a similar way, although less explicitly. Indeed, it can be shown by standard theory of generalized Sturm–Liouville operators (see e.g. [29, §6]) that when $|a| < 1$, there is a $\phi_E \in [0, \pi)$ such that for h in the maximal domain of $\mathfrak{G}_{\beta,a,E}$, $\lim_{t \rightarrow \infty} h^{[1]}(t) = 0$ if and only if

$$\lim_{t \rightarrow \infty} \left(\mathcal{W}_{\beta,a}(E^{1/4} \mathfrak{g}_{\beta,a,E}, h)(t) \cos \phi_E + \mathcal{W}_{\beta,a}(E^{-1/4} \mathfrak{f}_{\beta,a,E}, h)(t) \sin \phi_E \right) = 0,$$

where $\mathcal{W}_{\beta,a}(f, g) := p_{\beta,a}(fg' - f'g)$ denotes the Wronskian of f and g . Note that here, because $\mathfrak{G}_{\beta,a}$ is limit circle at infinity, the limits of both Wronskians above exist (see e.g. [29, Lemma 3.2] for a proof). Under the correspondence $u = A_{\beta,a,E}^{-1} \begin{pmatrix} p_{\beta,a} h' \\ h \end{pmatrix}$, this condition is exactly the condition that $\lim_{t \rightarrow \infty} e_{\phi_E}^* Ju(t) = 0$, which is therefore the boundary condition at infinity for the canonical system.

Construction of the appropriate time change. Just like in the Airy case, in order to work out the vague convergence of the canonical system (4.17) to the sine canonical system, we need to change its time domain to $(0, 1)$. Concretely, we are again looking for an absolutely continuous bijection $\eta_{B,E} : (0, 1 + \varepsilon_{B,E}) \rightarrow (0, \infty)$ for some $\varepsilon_{B,E} > 0$ with $\varepsilon_{B,E} \downarrow 0$.

When $\beta \leq 2$, the sine system is limit point at 1, and therefore we take $\varepsilon_{B,E} = 0$. When $\beta > 2$ and $|a| < 1$, both systems are limit circle on the right, so we also take $\varepsilon_{B,E} = 0$. However, when $\beta > 2$ and $a \geq 1$, the Bessel system is limit point at ∞ while the sine system is limit circle at 1. Because of this change of behavior, following the ideas from §2.2.3 just like in the Airy case for $\beta > 2$, we want to take $\varepsilon_{B,E} > 0$ in order to be able to analyse the vague convergence on $[0, 1]$.

Again, a candidate for the appropriate time change $\eta_{B,E}$ can be guessed from the deterministic case $\beta = \infty$. In that case, the Brownian motion disappears from the problem, and it is easy to see that a solution to $\mathfrak{G}_{\beta,a,E}h = 0$ has the form

$$h(t) = e^{at/2} \left(C J_a(2\sqrt{E} e^{-t/2}) + C' Y_a(2\sqrt{E} e^{-t/2}) \right)$$

for constants $C, C' \in \mathbb{R}$, where J_a and Y_a are Bessel functions of the first and second kind respectively. This yields explicit expressions for $f_{\infty,a,E}$ and $g_{\infty,a,E}$. Using asymptotic expansions of Bessel functions (see e.g. [19]), when E is large we can approximate these expressions by

$$f_{\infty,a,E}(t) \approx e^{at/2+t/4} \cos(2\sqrt{E}(1 - e^{-t/2})) \quad \text{and} \quad g_{\infty,a,E}(t) \approx \frac{1}{\sqrt{E}} e^{at/2+t/4} \sin(2\sqrt{E}(1 - e^{-t/2})).$$

This gives

$$\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E}) \approx \eta'_{B,E} \frac{\sqrt{E} e^{-(a+1)\eta_{B,E}}}{2} \frac{e^{(a+1/2)\eta_{B,E}}}{\sqrt{E}} \begin{pmatrix} \sin^2 \theta_E & \sin \theta_E \cos \theta_E \\ \sin \theta_E \cos \theta_E & \cos^2 \theta_E \end{pmatrix} \circ \eta_{B,E}$$

where $\theta_E(t) := 2\sqrt{E}(1 - e^{-t/2})$. As $E \rightarrow \infty$, the oscillations of the trigonometric functions should make the matrix converge vaguely to $\frac{1}{2}I_2$, which is exactly the coefficient matrix of the sine system when $\beta = \infty$. Hence, $\eta_{B,E}$ should be chosen so that the prefactor cancels out when $E \rightarrow \infty$. This suggests to take $\eta_{B,E}$ as a solution of

$$\eta'_{B,E} = 2c_E e^{\eta_{B,E}/2} \tag{4.18a}$$

where we allow for a dependence on an extra parameter $c_E \in \mathbb{R}$ chosen so that $c_E \rightarrow 1$ as $E \rightarrow \infty$. It is easy to solve this equation explicitly, and the solution with $\eta_{B,E}(0) = 0$ is

$$\eta_{B,E}(t) = -2 \log(1 - c_E t). \tag{4.18b}$$

This is a $\mathcal{C}^1$ bijection $(0, 1/c_E) \rightarrow (0, \infty)$. As explained above, we want a bijection $(0, 1) \rightarrow (0, \infty)$ when $\beta \leq 2$ and when $\beta > 2$ and $|a| < 1$, so in that case we simply take $c_E := 1$. When $\beta > 2$ and

$a \geq 1$, we want to take $c_E < 1$ in order to extend slightly the time domain; the specific value we take is $c_E := 1 - 1/\sqrt{E}$, like in the Airy case.

To motivate this choice, note that by Proposition 3.17, the time domain $[0, 1]$ for the time-changed system $\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E})$ corresponds, upon inverting the time change and reversing the shift, to the time domain $[-\log E, -2\log(1 - c_E) - \log E]$ for the original operator $\mathfrak{G}_{\beta,a}$. Taking $c_E = 1 - 1/\sqrt{E}$ therefore fixes the right boundary of this time domain to always be 0 for the original operator, which removes its dependence on the shift E . This will play a role in the analysis of the right boundary conditions in §4.4.3.

In the sequel, we denote by $\tau_E := \frac{1}{c_E}(1 - 1/\sqrt{E})$ the time that satisfies $\eta_{B,E}(\tau_E) = \log E$, and thus corresponds to time 0 for $\mathfrak{G}_{\beta,a}$ when reversing the time-change and shift.

Polar coordinates. In the same way as in the Airy case, we introduce a change of variables into polar coordinates for solutions to $\mathfrak{G}_{\beta,a,E}f = 0$. This is the tool we will use to analyse the asymptotic behavior of the solutions that appear in the coefficient matrix.

Proposition 4.2 [54, Proposition 7]. *Define $\mathfrak{G}_{\beta,a,E}$ from a standard Brownian motion B on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$, and let $\eta_{B,E}$ and c_E be as in (4.18). If f solves $\mathfrak{G}_{\beta,a,E}f = 0$ and if $f(0)$ and $f'(0)$ are independent of B , then for $t \in [0, 1/c_E)$,*

$$f \circ \eta_{B,E} = \frac{C_f E^{-1/4} e^{\eta_{B,E}/4}}{\sqrt{\rho_{\beta,a} \circ \eta_{B,E}}} e^{\rho_{\beta,a,E}} \cos \xi_{\beta,a,E} \quad \text{and} \quad f' \circ \eta_{B,E} = \frac{C_f E^{1/4} e^{-\eta_{B,E}/4}}{\sqrt{\rho_{\beta,a} \circ \eta_{B,E}}} e^{\rho_{\beta,a,E}} \sin \xi_{\beta,a,E}$$

where $C_f^2 := \sqrt{E} f^2(0) + \frac{1}{\sqrt{E}} f'^2(0)$ and $\rho_{\beta,a,E}$ and $\xi_{\beta,a,E}$ solve the coupled SDEs

$$\begin{aligned} d\rho_{\beta,a,E}(t) &= \left(\frac{1}{\beta} - \left(a + \frac{1}{2} \right) \cos 2\xi_{\beta,a,E}(t) - \frac{1}{\beta} \cos 4\xi_{\beta,a,E}(t) \right) \frac{c_E}{1 - c_E t} dt \\ &\quad - \sqrt{\frac{2}{\beta}} \cos 2\xi_{\beta,a,E}(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t), \\ d\xi_{\beta,a,E}(t) &= -2c_E \sqrt{E} dt + \left(\left(a + \frac{1}{2} \right) \sin 2\xi_{\beta,a,E}(t) + \frac{1}{\beta} \sin 4\xi_{\beta,a,E}(t) \right) \frac{c_E}{1 - c_E t} dt \\ &\quad + \sqrt{\frac{2}{\beta}} \sin 2\xi_{\beta,a,E}(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \end{aligned}$$

with $\rho_{\beta,a,E}(0) = 0$ and $\xi_{\beta,a,E}(0) = \arctan\left(\frac{f'(0)}{\sqrt{E} f(0)}\right)$, and where $B_E(t) := \frac{1}{\sqrt{2c_E}} \int_0^{\eta_{B,E}(t)} e^{-s/4} dB(s)$ is a Brownian motion and a martingale with respect to the filtration $\{\mathcal{F}_{\eta_{B,E}(t)}\}_{t \in [0, 1/c_E)}$.

Proof. Let f solve $\mathfrak{G}_{\beta,a,E}f = 0$. Then f also solves $\mathfrak{G}_{\beta,a}f = Ef$, and therefore it is also a solution

to the SDE (3.22) with $z = E$. Hence, by the Dambis–Dubins–Schwarz theorem,

$$\begin{aligned} d(f' \circ \eta_{B,E})(t) &= -E e^{-\eta_{B,E}(t)} f \circ \eta_{B,E}(t) \eta'_{B,E}(t) dt \\ &\quad + \left(a + \frac{2}{\beta}\right) f' \circ \eta_{B,E}(t) \eta'_{B,E}(t) dt + \frac{2}{\sqrt{\beta}} f' \circ \eta_{B,E}(t) \sqrt{\eta'_{B,E}(t)} dB_E(t) \end{aligned}$$

where B_E is the Brownian motion defined in the proposition statement.

Set $y := f \circ \eta_{B,E}$, so that $y' = \eta'_{B,E}(f' \circ \eta_{B,E})$. From the above expression for $d(f' \circ \eta_{B,E})$, we deduce that

$$\begin{aligned} dy'(t) &= \frac{\eta''_{B,E}(t)}{\eta'_{B,E}(t)} y'(t) dt - E \eta'_{B,E}(t)^2 e^{-\eta_{B,E}(t)} y(t) dt \\ &\quad + \left(a + \frac{2}{\beta}\right) \eta'_{B,E}(t) y'(t) dt + \frac{2}{\sqrt{\beta}} \sqrt{\eta'_{B,E}(t)} y'(t) dB_E(t). \end{aligned}$$

By definition of $\eta_{B,E}$, $(\eta'_{B,E})^2 e^{-\eta_{B,E}} = 4c_E^2$ and $\eta''_{B,E}/\eta'_{B,E} = \eta'_{B,E}/2$, so this simplifies to

$$dy'(t) = -4c_E^2 E y(t) dt + \left(a + \frac{2}{\beta} + \frac{1}{2}\right) \eta'_{B,E}(t) y'(t) dt + \frac{2}{\sqrt{\beta}} \sqrt{\eta'_{B,E}(t)} y'(t) dB_E(t).$$

Now, define two real-valued stochastic processes r and ξ such that $e^{r+i\xi} = Sy + iy'/S$ where S is a constant scaling factor that will be determined later. By Itô's formula, omitting the explicit time dependence to simplify notation,

$$\begin{aligned} dr + i d\xi &= d(\log(Sy + iy'/S)) \\ &= \frac{1}{Sy + iy'/S} \left(Sy' dt + \frac{i}{S} dy' \right) + \frac{1}{2S^2} \frac{1}{(Sy + iy'/S)^2} d\langle y' \rangle \\ &= e^{-2r} \left(S^2 yy' dt + \frac{1}{S^2} y' dy' + iy dy' - iy'^2 dt \right) + \frac{e^{-4r}}{2S^2} \left(S^2 y^2 - \frac{y'^2}{S^2} - 2iyy' \right) d\langle y' \rangle \\ &= e^{-2r} \left(S^2 yy' - \frac{4c_E^2 E}{S^2} yy' + \left(a + \frac{2}{\beta} + \frac{1}{2}\right) \frac{y'^2}{S^2} \eta'_{B,E} \right) dt + \frac{2}{\beta} e^{-4r} \frac{y'^2}{S^2} \left(S^2 y^2 - \frac{y'^2}{S^2} \right) \eta'_{B,E} dt \\ &\quad + ie^{-2r} \left(-4c_E^2 E y^2 - y'^2 + \left(a + \frac{2}{\beta} + \frac{1}{2}\right) yy' \eta'_{B,E} \right) dt - \frac{4i}{\beta} e^{-4r} \frac{yy'^3}{S^2} \eta'_{B,E} dt \\ &\quad + \frac{2}{\sqrt{\beta}} e^{-2r} \frac{y'^2}{S^2} \sqrt{\eta'_{B,E}} dB_E + \frac{2i}{\sqrt{\beta}} e^{-2r} yy' \sqrt{\eta'_{B,E}} dB_E. \end{aligned}$$

This simplifies with $S := \sqrt{2c_E} E^{1/4}$. Indeed, it balances the magnitudes of the first two terms of the first two lines so that they cancel out in the first line and combine as $4c_E^2 E y^2 + y'^2 = S^2(S^2 y^2 + y'^2/S^2) = 2c_E \sqrt{E} e^{2r}$ in the second one. Extracting the SDEs for r and ξ from the real and imaginary parts of the above and replacing y and y' with their expressions in terms of r and ξ , we get

$$dr = \left(a + \frac{2}{\beta} + \frac{1}{2}\right) \sin^2 \xi \eta'_{B,E} dt + \frac{2}{\beta} \sin^2 \xi (\cos^2 \xi - \sin^2 \xi) \eta'_{B,E} dt + \frac{2}{\sqrt{\beta}} \sin^2 \xi \sqrt{\eta'_{B,E}} dB_E$$

and

$$\begin{aligned} d\xi = & -2c_E \sqrt{E} dt + \left(a + \frac{2}{\beta} + \frac{1}{2}\right) \sin \xi \cos \xi \eta'_{B,E} dt - \frac{4}{\beta} \sin^3 \xi \cos \xi \eta'_{B,E} dt \\ & + \frac{2}{\sqrt{\beta}} \sin \xi \cos \xi \sqrt{\eta'_{B,E}} dB_E. \end{aligned}$$

It is easy to see by simplifying further with trigonometric identities that $\xi_{\beta,a,E} := \xi$ follows the announced SDE with $\xi_{\beta,a,E}(0) = \arctan\left(\frac{y'(0)}{S^2 y(0)}\right) = \arctan\left(\frac{f'(0)}{\sqrt{E} f(0)}\right)$. Performing similar simplifications in the other SDE, we get

$$dr = \left(\frac{1}{\beta} + a + \frac{1}{2} - \left(a + \frac{1}{2}\right) \cos 2\xi - \frac{1}{\beta} \cos 4\xi\right) \frac{\eta'_{B,E}}{2} dt + \frac{1}{\sqrt{\beta}} (1 - \cos 2\xi) \sqrt{\eta'_{B,E}} dB_E.$$

With

$$\tilde{r}(t) := r(0) + \left(\frac{a}{2} + \frac{1}{4}\right) \eta_{B,E}(t) + \frac{1}{\sqrt{\beta}} \int_0^t \sqrt{\eta'_{B,E}(s)} dB_E(s),$$

we see that $\rho_{\beta,a,E} := r - \tilde{r}$ follows the announced SDE with $\rho_{\beta,a,E}(0) = 0$.

It remains to check the representations of $f \circ \eta_{B,E}$ and $f' \circ \eta_{B,E}$ in terms of $\rho_{\beta,a,E}$ and $\xi_{\beta,a,E}$. To do so, note that the last term of $\tilde{r}$ is exactly $\frac{1}{\sqrt{\beta}} B(\eta_{B,E}(t))$ where B is the original Brownian motion, so in fact $\tilde{r} = r(0) + \frac{\eta_{B,E}}{4} - \frac{1}{2} \log p_{\beta,a} \circ \eta_{B,E}$. Moreover,

$$e^{2r(0)} = S^2 f^2(0) + \frac{\eta'_{B,E}(0) f'^2(0)}{S^2} = 2c_E \left(\sqrt{E} f^2(0) + \frac{f'^2(0)}{\sqrt{E}} \right) =: 2c_E C_f^2.$$

It follows that

$$f \circ \eta_{B,E} = \frac{1}{S} e^r \cos \xi = \frac{C_f E^{-1/4} e^{\eta_{B,E}/4}}{\sqrt{p_{\beta,a} \circ \eta_{B,E}}} e^{\rho_{\beta,a,E}} \cos \xi_{\beta,a,E}$$

and likewise that

$$f' \circ \eta_{B,E} = \frac{S}{\eta'_{B,E}} e^r \sin \xi = \frac{C_f E^{1/4} e^{-\eta_{B,E}/4}}{\sqrt{p_{\beta,a} \circ \eta_{B,E}}} e^{\rho_{\beta,a,E}} \sin \xi_{\beta,a,E}. \quad \blacksquare$$

Now, by time-changing the coefficient matrix (4.17) of the Bessel system, we get

$$\eta'_{B,E}(t) (G_{\beta,a,E} \circ \eta_{B,E})(t) = \frac{\eta'_{B,E}(t) w_{\beta,a} \circ \eta_{B,E}(t) \sqrt{E}}{2} \begin{pmatrix} \sqrt{E} g_{\beta,a,E}^2 & f_{\beta,a,E} g_{\beta,a,E} \\ f_{\beta,a,E} g_{\beta,a,E} & \frac{1}{\sqrt{E}} f_{\beta,a,E}^2 \end{pmatrix} (\eta_{B,E}(t)).$$

The initial conditions that define $f_{\beta,a,E}$ and $g_{\beta,a,E}$ imply that the constants appearing in their representations in the polar coordinates from Proposition 4.2 are $C_{f_{\beta,a,E}} = E^{1/4}$ and $C_{g_{\beta,a,E}} = E^{-1/4}$. Hence, switching to the polar coordinates, the coefficient matrix becomes

$$\begin{aligned} \eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E}) &= \frac{\eta'_{B,E}(w_{\beta,a} \circ \eta_{B,E}) e^{\eta_{B,E}/2}}{2 p_{\beta,a} \circ \eta_{B,E}} \\ &\times \begin{pmatrix} e^{2\rho_{\beta,a,E}^g} \cos^2 \xi_{\beta,a,E}^g & e^{\rho_{\beta,a,E}^f + \rho_{\beta,a,E}^g} \cos \xi_{\beta,a,E}^f \cos \xi_{\beta,a,E}^g \\ e^{\rho_{\beta,a,E}^f + \rho_{\beta,a,E}^g} \cos \xi_{\beta,a,E}^f \cos \xi_{\beta,a,E}^g & e^{2\rho_{\beta,a,E}^f} \cos^2 \xi_{\beta,a,E}^f \end{pmatrix}. \end{aligned}$$

We chose $\eta'_{B,E} = 2c_E e^{\eta_{B,E}/2}$ and by definition $e^{\eta_{B,E}} w_{\beta,a} \circ \eta_{B,E} = p_{\beta,a} \circ \eta_{B,E}$. Therefore, the above expression simplifies to

$$\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E}) = c_E \begin{pmatrix} e^{2\rho_{\beta,a,E}^g \cos^2 \xi_{\beta,a,E}^g} & e^{\rho_{\beta,a,E}^f + \rho_{\beta,a,E}^g \cos \xi_{\beta,a,E}^f \cos \xi_{\beta,a,E}^g} \\ e^{\rho_{\beta,a,E}^f + \rho_{\beta,a,E}^g \cos \xi_{\beta,a,E}^f \cos \xi_{\beta,a,E}^g} & e^{2\rho_{\beta,a,E}^f \cos^2 \xi_{\beta,a,E}^f} \end{pmatrix}. \quad (4.19)$$

4.2.3 Unified setup and idea of the convergence

Generalizing the polar coordinates. Notice that when written in polar coordinates as in (4.15) and (4.19), the coefficient matrices $\eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E})$ and $\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E})$ of the Airy and Bessel systems on $[0, \tau_E]$ both have the form

$$M_{\beta,E} := c_E \begin{pmatrix} e^{2\rho_{\beta,E}^g \cos^2 \xi_{\beta,E}^g} & e^{\rho_{\beta,E}^f + \rho_{\beta,E}^g \cos \xi_{\beta,E}^f \cos \xi_{\beta,E}^g} \\ e^{\rho_{\beta,E}^f + \rho_{\beta,E}^g \cos \xi_{\beta,E}^f \cos \xi_{\beta,E}^g} & e^{2\rho_{\beta,E}^f \cos^2 \xi_{\beta,E}^f} \end{pmatrix} \quad (4.20)$$

where c_E is either 1 or $1 - 1/\sqrt{E}$ depending on the values of the parameters, $\tau_E = \frac{1}{c_E}(1 - 1/\sqrt{E})$, and where the $\rho_{\beta,E}$'s and $\xi_{\beta,E}$'s follow either the SDEs of Proposition 4.1 or those of 4.2, but with $\rho_{\beta,E}^f(0) = \rho_{\beta,E}^g(0) = \xi_{\beta,E}^f(0) = 0$ and $\xi_{\beta,E}^g(0) = \pi/2$ in any case.

It seems clear that the SDEs of Propositions 4.1 and 4.2 have the same structure. In fact, they can both be written as

$$\begin{aligned} d\rho_{\beta,E}(t) &= \left(\frac{1}{\beta} + \kappa \cos 2\xi_{\beta,E}(t) + \frac{1}{\beta} \cos(4\xi_{\beta,E}(t) - 2\phi) \right) \frac{c_E}{1 - c_E t} dt \\ &\quad + \sqrt{\frac{2}{\beta}} \sin(2\xi_{\beta,E}(t) - \phi) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t), \end{aligned} \quad (4.21a)$$

$$\begin{aligned} d\xi_{\beta,E}(t) &= -2c_E E^{q/2} (1 - c_E t)^{q-1} dt - \left(\kappa \sin 2\xi_{\beta,E}(t) + \frac{1}{\beta} \sin(4\xi_{\beta,E}(t) - 2\phi) \right) \frac{c_E}{1 - c_E t} dt \\ &\quad + \frac{2\sqrt{2}}{\sqrt{\beta}} \cos \xi_{\beta,E}(t) \cos(\xi_{\beta,E}(t) - \phi) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \end{aligned} \quad (4.21b)$$

for parameters $q \geq 1$, $\phi \in [0, 2\pi)$ and $\kappa \in \mathbb{R}$. In the Airy case, $q = 3$, $\phi = 0$ and $\kappa = 2/\beta - 1/2$, while in the Bessel case, $q = 1$, $\phi = \pi/2$ and $\kappa = -a - 1/2$.

The SDEs (4.21) always make sense for $t \in [0, \tau_E]$ if $\tau_E := \frac{1}{c_E}(1 - 1/\sqrt{E})$, even if c_E is allowed to take any value in $[1 - 1/\sqrt{E}, 1]$. In that case, the matrix $M_{\beta,E}$ can be taken as a general model of coefficient matrix on $[0, \tau_E) \subseteq [0, 1)$, defined as in (4.20) with pairs $\rho_{\beta,E}^f, \xi_{\beta,E}^f$ and $\rho_{\beta,E}^g, \xi_{\beta,E}^g$ solving the SDEs (4.21) with initial conditions $\rho_{\beta,E}^f(0) = \rho_{\beta,E}^g(0) = \xi_{\beta,E}^f(0) = 0$ and $\xi_{\beta,E}^g(0) = \pi/2$ and arbitrary parameters $q \geq 1$, $\phi \in [0, 2\pi)$ and $\kappa \in \mathbb{R}$. This does not necessarily define a canonical

system on the full interval $[0, 1)$, but as we will see in §4.3, it turns out that (no matter its extension on $(\tau_E, 1)$) $M_{\beta,E}$ always converges vaguely on $[0, 1)$ to the sine system's coefficient matrix.

It should be understood that this generalization of $\eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E})$ and $\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E})$ is incomplete, in the sense that it models these systems on the “main” part of the time interval, but not their boundary behavior. This also means that their spectral information is not naturally built in $M_{\beta,E}$. In fact, even in the case of the Airy system, the polar coordinates we built in Proposition 4.1 are not valid past τ_E since the time change $\eta_{A,E}$ changes at that point, so our representation of $\eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E})$ in polar coordinates does not even contain the spectral information. Nevertheless, this general model is useful to unify the analysis of both coefficient matrices on $[0, \tau_E]$, which helps to investigate the vague convergence of both systems to the sine system.

We also note that since the polar coordinates solving (4.21) interpolate between the two models, it is natural to ask whether they could be obtained from time-changed solutions to the eigenvalue equation for another generalized Sturm–Liouville operator on $[0, \infty)$, using the method we applied in Propositions 4.1 and 4.2. Following on the remark of the previous paragraph, this operator might then complete the generalization, and be used to also unify the analysis of the spectral convergence to the sine system. We won't tackle this problem here, and will rather settle for using the unified setting given by $M_{\beta,E}$ merely as a tool to study both vague convergence problems at the same time.

Idea of the convergence. We now describe briefly why one should expect the vague convergence of $M_{\beta,E}$ to the coefficient matrix of the sine system. Notice that $M_{\beta,E}$ can be written as

$$M_{\beta,E} = \frac{c_E e^{2\rho_{\beta,E}^g}}{2} \begin{pmatrix} 1 & e^{-\Delta_{\beta,E}^\rho} \cos \Delta_{\beta,E}^\xi \\ e^{-\Delta_{\beta,E}^\rho} \cos \Delta_{\beta,E}^\xi & e^{-2\Delta_{\beta,E}^\rho} \end{pmatrix} + \frac{c_E}{2} \begin{pmatrix} e^{2\rho_{\beta,E}^g} \cos 2\xi_{\beta,E}^g & e^{\Sigma_{\beta,E}^\rho} \cos \Sigma_{\beta,E}^\xi \\ e^{\Sigma_{\beta,E}^\rho} \cos \Sigma_{\beta,E}^\xi & e^{2\rho_{\beta,E}^f} \cos 2\xi_{\beta,E}^f \end{pmatrix}$$

where we introduced the notations $\Delta_{\beta,E}^\rho := \rho_{\beta,E}^g - \rho_{\beta,E}^f$ and $\Sigma_{\beta,E}^\rho := \rho_{\beta,E}^g + \rho_{\beta,E}^f$, and likewise for phases. Now, the SDEs (4.21) have the following consequence.

Claim 4.3. *If the pairs $\rho_{\beta,E}^f, \xi_{\beta,E}^f$ and $\rho_{\beta,E}^g, \xi_{\beta,E}^g$ solve (4.21) with $\rho_{\beta,E}^f(0) = \rho_{\beta,E}^g(0) = \xi_{\beta,E}^f(0) = 0$ and $\xi_{\beta,E}^g(0) = \pi/2$, then $e^{\Sigma_{\beta,E}^\rho} \sin \Delta_{\beta,E}^\xi \equiv 1$. In particular, $e^{-2\rho_{\beta,E}^g} = e^{-\Delta_{\beta,E}^\rho} \sin \Delta_{\beta,E}^\xi$.*

Remark. When these polar coordinates are those of our solutions $f_{\beta,E}, g_{\beta,E}$ or $f_{\beta,a,E}, g_{\beta,a,E}$ to the eigenvalue equations $\mathcal{H}_{\beta,E}h = 0$ or $\mathfrak{G}_{\beta,a,E}h = 0$, this is in fact a straightforward consequence of the fact that the Wronskians of these solutions are constant (as it is the case for any two solutions to the eigenvalue equation of a generalized Sturm–Liouville operator with a given spectral parameter).

We postpone the proof of this claim until the end of the section, and rather focus on an important consequence for now. Define

$$\theta_{\beta,E} := -e^{-\Delta_{\beta,E}^\rho - i\Delta_{\beta,E}^\xi}. \quad (4.22)$$

By Claim 4.3, $e^{-2\rho_{\beta,E}^g} = \text{Im } \ell_{\beta,E}$. Hence, we can rewrite the above expression of $M_{\beta,E}$ as

$$M_{\beta,E} = \frac{c_E}{2 \text{Im } \ell_{\beta,E}} \begin{pmatrix} 1 & -\text{Re } \ell_{\beta,E} \\ -\text{Re } \ell_{\beta,E} & |\ell_{\beta,E}|^2 \end{pmatrix} + \frac{c_E}{2} \begin{pmatrix} e^{2\rho_{\beta,E}^g} \cos 2\xi_{\beta,E}^g & e^{\Sigma_{\beta,E}^\rho} \cos \Sigma_{\beta,E}^\xi \\ e^{\Sigma_{\beta,E}^\rho} \cos \Sigma_{\beta,E}^\xi & e^{2\rho_{\beta,E}^f} \cos 2\xi_{\beta,E}^f \end{pmatrix}. \quad (4.23)$$

Recall that the coefficient matrix of the stochastic sine canonical system is given by

$$R_\beta \circ \nu = \frac{1}{2 \text{Im } \mathcal{B}_\beta \circ \nu} \begin{pmatrix} 1 & -\text{Re } \mathcal{B}_\beta \circ \nu \\ -\text{Re } \mathcal{B}_\beta \circ \nu & |\mathcal{B}_\beta \circ \nu|^2 \end{pmatrix}$$

where $\mathcal{B}_\beta$ is a hyperbolic Brownian motion with variance $4/\beta$ started at i in the upper half-plane, and ν is a logarithmic time change. Comparing the expressions of these two coefficient matrices provides a clear guess on how the convergence might happen. Indeed, the matrix in the first term of (4.23) has the same structure as the coefficient matrix of the sine system, with $\ell_{\beta,E}$ in place of $\mathcal{B}_\beta \circ \nu$. Moreover, it is clear from the SDE (4.21b) that the phases $\xi_{\beta,E}^f$ and $\xi_{\beta,E}^g$ will grow increasingly fast as E becomes large, which should make the second term of (4.23) vanish in the vague limit. Hence, we expect that what makes the convergence possible is that as E becomes large, the process $\ell_{\beta,E}$ becomes close to a hyperbolic Brownian motion with variance $4/\beta$, started at i in the upper half-plane and run in logarithmic time. Notice that because $\Delta_{\beta,E}^\rho(0) = 0$ and $\Delta_{\beta,E}^\xi(0) = \pi/2$, indeed $\ell_{\beta,E}(0) = i$.

To see how $\ell_{\beta,E}$ becomes a hyperbolic Brownian motion, we compute its Itô differential. To do so, note that

$$\begin{aligned} d\Delta_{\beta,E}^\rho(t) = & -\left(2\kappa \sin \Delta_{\beta,E}^\xi(t) \sin \Sigma_{\beta,E}^\xi(t) + \frac{2}{\beta} \sin 2\Delta_{\beta,E}^\xi(t) \sin(2\Sigma_{\beta,E}^\xi(t) - 2\phi)\right) \frac{c_E}{1 - c_E t} dt \\ & + \frac{2\sqrt{2}}{\sqrt{\beta}} \sin \Delta_{\beta,E}^\xi(t) \cos(\Sigma_{\beta,E}^\xi(t) - \phi) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \end{aligned}$$

and

$$\begin{aligned} d\Delta_{\beta,E}^\xi(t) = & -\left(2\kappa \sin \Delta_{\beta,E}^\xi(t) \cos \Sigma_{\beta,E}^\xi(t) + \frac{2}{\beta} \sin 2\Delta_{\beta,E}^\xi(t) \cos(2\Sigma_{\beta,E}^\xi(t) - 2\phi)\right) \frac{c_E}{1 - c_E t} dt \\ & - \frac{2\sqrt{2}}{\sqrt{\beta}} \sin \Delta_{\beta,E}^\xi(t) \sin(\Sigma_{\beta,E}^\xi(t) - \phi) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t). \end{aligned}$$

From this, applying Itô's formula (and dropping the explicit dependence on t to lighten notation) gives

$$d\ell_{\beta,E} = -\ell_{\beta,E} d\Delta_{\beta,E}^\rho - i\ell_{\beta,E} d\Delta_{\beta,E}^\xi + \frac{1}{2}\ell_{\beta,E} (d\langle \Delta_{\beta,E}^\rho \rangle + 2i d\langle \Delta_{\beta,E}^\rho, \Delta_{\beta,E}^\xi \rangle - d\langle \Delta_{\beta,E}^\xi \rangle)$$

$$\begin{aligned}
&= -\frac{2\sqrt{2}}{\sqrt{\beta}} \ell_{\beta,E} \sin \Delta_{\beta,E}^{\xi} e^{i\phi - i\Sigma_{\beta,E}^{\xi}} \sqrt{\frac{c_E}{1 - c_E t}} dB_E \\
&\quad + \ell_{\beta,E} \left(2i\kappa \sin \Delta_{\beta,E}^{\xi} e^{-i\Sigma_{\beta,E}^{\xi}} + \frac{2i}{\beta} \sin 2\Delta_{\beta,E}^{\xi} e^{2i\phi - 2i\Sigma_{\beta,E}^{\xi}} \right) \frac{c_E}{1 - c_E t} dt \\
&\quad + \frac{4}{\beta} \ell_{\beta,E} \sin^2 \Delta_{\beta,E}^{\xi} e^{2i\phi - 2i\Sigma_{\beta,E}^{\xi}} \frac{c_E}{1 - c_E t} dt \\
&= -\frac{2\sqrt{2}}{\sqrt{\beta}} \ell_{\beta,E} \sin \Delta_{\beta,E}^{\xi} e^{i\phi - i\Sigma_{\beta,E}^{\xi}} \sqrt{\frac{c_E}{1 - c_E t}} dB_E \\
&\quad + i \ell_{\beta,E} e^{-i\Sigma_{\beta,E}^{\xi}} \sin \Delta_{\beta,E}^{\xi} \left(2\kappa + \frac{4}{\beta} e^{2i\phi - i\Sigma_{\beta,E}^{\xi} - i\Delta_{\beta,E}^{\xi}} \right) \frac{c_E}{1 - c_E t} dt.
\end{aligned}$$

Now, $\Sigma_{\beta,E}^{\xi} + \Delta_{\beta,E}^{\xi} = 2\xi_{\beta,E}^g$ so $\ell_{\beta,E} e^{-i\Sigma_{\beta,E}^{\xi}} \sin \Delta_{\beta,E}^{\xi} = -e^{-\Delta_{\beta,E}^{\rho} - 2i\xi_{\beta,E}^g} \sin \Delta_{\beta,E}^{\xi} = -e^{-2i\xi_{\beta,E}^g} \operatorname{Im} \ell_{\beta,E}$, and

$$\begin{aligned}
d\ell_{\beta,E}(t) &= \frac{2}{\sqrt{\beta}} \operatorname{Im} \ell_{\beta,E}(t) e^{i\phi - 2i\xi_{\beta,E}^g(t)} \sqrt{\frac{2c_E}{1 - c_E t}} dB_E(t) \\
&\quad - i \operatorname{Im} \ell_{\beta,E}(t) \left(2\kappa e^{-2i\xi_{\beta,E}^g(t)} + \frac{4}{\beta} e^{2i\phi - 4i\xi_{\beta,E}^g(t)} \right) \frac{c_E}{1 - c_E t} dt.
\end{aligned} \tag{4.24}$$

Again, because of the increasingly fast oscillations of $e^{-2i\xi_{\beta,E}^g}$ and $e^{-4i\xi_{\beta,E}^g}$ when E grows, we expect the second line (4.24) not to contribute in the limit. Ignoring the second line, (4.24) has the structure of the SDE for a hyperbolic Brownian motion with variance $4/\beta$ in the upper half-plane, but driven by the process

$$t \mapsto \int_0^t e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s). \tag{4.25}$$

In §4.3, we will show that indeed this process converges in distribution to a standard complex Brownian motion, and we will use this to make rigorous the heuristic idea presented above in order to prove the vague convergence of $M_{\beta,E}$ to the sine system's coefficient matrix.

We finish the section with a proof of Claim 4.3.

Proof of Claim 4.3. By Itô's formula,

$$\begin{aligned}
d(e^{-2\rho_{\beta,E}^g})(t) &= -\frac{2}{\sqrt{\beta}} e^{-2\rho_{\beta,E}^g(t)} \sin(2\xi_{\beta,E}^g(t) - \phi) \sqrt{\frac{2c_E}{1 - c_E t}} dB_E(t) \\
&\quad - e^{-2\rho_{\beta,E}^g(t)} \left(2\kappa \cos 2\xi_{\beta,E}^g(t) + \frac{4}{\beta} \cos(4\xi_{\beta,E}^g(t) - 2\phi) \right) \frac{c_E}{1 - c_E t} dt.
\end{aligned}$$

Comparing with (4.24), we notice that this is exactly the SDE solved by $\operatorname{Im} \ell_{\beta,E} = e^{-\Delta_{\beta,E}^{\rho}} \sin \Delta_{\beta,E}^{\xi}$. Hence, the difference between these two processes must be constant, and because they are both initialized at 1, they must in fact be equal. Therefore $e^{\Sigma_{\beta,E}^{\rho}} \sin \Delta_{\beta,E}^{\xi} = e^{2\rho_{\beta,E}^g - \Delta_{\beta,E}^{\rho}} \sin \Delta_{\beta,E}^{\xi} \equiv 1$. ■

4.3 Vague convergence

The goal of this section is to prove Theorem 1.9, which shows the vague convergence to the stochastic sine canonical system of the canonical systems corresponding to the shifted and scaled stochastic Airy and Bessel operators $\mathcal{H}_{\beta,E}$ and $\mathfrak{G}_{\beta,a,E}$. Throughout the section, we work with the coefficient matrix $M_{\beta,E}$ from (4.23), and therefore all results can directly be applied to both models by setting the right values for the parameters q , ϕ and κ .

We start by building in §4.3.1 a coupling between the Brownian motions driving the SDEs (4.21) for polar coordinates, which is defined in such a way that the process (4.25) converges in probability to a complex Brownian motion W . This allows us to deduce, in §4.3.2, quantitative estimates comparing the process $\mathcal{L}_{\beta,E}$ from (4.22) to a hyperbolic Brownian motion driven by W . Having done so, we complete the proof of the vague convergence in §4.3.3.

The results and the arguments in this section combine and generalize those of [55, §§4–6] and [54, §§3–4], which are specific to the Airy and Bessel cases respectively.

4.3.1 Coupling the systems

Following the ideas from the end of §4.2.3, we now build a coupling between real Brownian motions B_E and a single complex Brownian motion W so that the process (4.25) becomes pathwise close to W as $E \rightarrow \infty$.

Lemma 4.4. *Let $\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$ satisfy $E_n \rightarrow \infty$. There exists a probability space on which are defined a collection $\{B_{E_n}\}_{n \in \mathbb{N}}$ of standard real Brownian motions and a standard complex Brownian motion W such that if ξ_{β,E_n}^g is the solution with $\xi_{\beta,E_n}^g(0) = \pi/2$ to the SDE (4.21b) driven by B_{E_n} , then for any $\alpha \in (0, 1/2)$ and $\delta \in (0, \alpha q/3)$, there is a $C > 0$ such that for all $E \in \{E_n\}_{n \in \mathbb{N}}$ large enough,*

$$\mathbb{P} \left[\sup_{t \in \mathcal{I}_{E,\alpha}} \left| \int_0^t e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) - W \circ v_E(t) \right| \geq E^{-\alpha q/3 + \delta} \right] \leq 3E^{4\alpha q/3} \log^2 E e^{-CE^{2\delta/3}} \quad (4.26)$$

where $\mathcal{I}_{E,\alpha} := [0, (1 - E^{-1/2+\alpha})/c_E]$ and $v_E(t) := -\log(1 - c_E t)$.

Remark. This probability space therefore supports a sequence of coefficient matrices $M_{\beta,E}$ of the form (4.23). Indeed, each Brownian motion B_E can be taken to define two pairs of polar coordinates $(\rho_{\beta,E}^f, \xi_{\beta,E}^f)$ and $(\rho_{\beta,E}^g, \xi_{\beta,E}^g)$ as solutions to the SDEs (4.21) with $\rho_{\beta,E}^f(0) = \rho_{\beta,E}^g(0) = \xi_{\beta,E}^f(0) = 0$ and $\xi_{\beta,E}^g(0) = \pi/2$, which suffice to define $M_{\beta,E}$. In the same way, the probability space also supports a stochastic sine canonical system, since its coefficient matrix can be defined from a hyperbolic Brownian motion driven by W .

Proof. To build an appropriate probability space, we start by fixing an $E > 0$ and building a coupling between a real and a complex Brownian motions in such a way that the above property is satisfied. To do this, we first couple a discretization of the integral in (4.26) with a discrete random walk, which we then extend to a complex Brownian motion. Then, we combine all spaces together.

Step 0: Discrete time setup. Fix $E > 0$ and a standard real Brownian motion B_E , and let $\xi_{\beta,E}^g$ solve the SDE (4.21b) with $\xi_{\beta,E}^g(0) = \pi/2$. With $\tau_E := \frac{1}{c_E}(1 - 1/\sqrt{E})$, we discretize the time interval $[0, \tau_E]$ by setting $t_0 := 0$, $t_N := \tau_E$, and in between

$$c_E t_j := 1 - (1 + E^{-p})^{-j} \quad \text{so that} \quad N := \left\lceil \frac{\log E}{2 \log(1 + E^{-p})} \right\rceil \quad (4.27)$$

where $p := 2\alpha q/3$. Note that because $\log(1 + x) \geq x(1 - x/2)$ for all $x > 0$, this means that $N \leq E^p \log E$ for all E large enough.

We begin by considering the integral in (4.26) at discrete times $\{t_j\}_{j=1}^N$, that is, we focus on the discrete martingale

$$\sum_{j=1}^n \mathcal{B}_j^\xi \quad \text{where} \quad \mathcal{B}_j^\xi := \int_{t_{j-1}}^{t_j} e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) \quad (4.28)$$

for $n \in \{1, \dots, N\}$. To compare this with a discretization of a complex Brownian motion, we compare it with the discrete martingale

$$\sum_{j=1}^n e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta \quad \text{where} \quad \mathcal{B}_j^\theta := \int_{t_{j-1}}^{t_j} e^{i\phi - 2i\theta(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) \quad (4.29)$$

and where θ is the deterministic part of $\xi_{\beta,E}^g$, that is, θ solves

$$\theta'(t) = -2c_E E^{q/2} (1 - c_E t)^{q-1} \quad \text{with} \quad \theta(0) = \frac{\pi}{2}, \quad \text{so} \quad \theta(t) = \frac{\pi}{2} - \frac{2}{q} E^{q/2} (1 - (1 - c_E t)^q). \quad (4.30)$$

The $\mathcal{B}_j^\theta$'s are Gaussian, and due to the fast oscillations of θ , we expect their real and imaginary parts to be almost independent and to have variances close to

$$\sigma^2 := \int_{t_{j-1}}^{t_j} \frac{c_E}{1 - c_E s} ds = \log\left(\frac{1 - c_E t_{j-1}}{1 - c_E t_j}\right) = \log(1 + E^{-p}) \quad (4.31)$$

for $j < N$. The actual correlation structure of $\mathcal{B}_j^\theta$ is given by the correlation matrix

$$\Sigma_j := \begin{pmatrix} \mathbb{E}(\text{Re } \mathcal{B}_j^\theta)^2 & \mathbb{E} \text{Re } \mathcal{B}_j^\theta \text{Im } \mathcal{B}_j^\theta \\ \mathbb{E} \text{Re } \mathcal{B}_j^\theta \text{Im } \mathcal{B}_j^\theta & \mathbb{E}(\text{Im } \mathcal{B}_j^\theta)^2 \end{pmatrix},$$

so if we define

$$W_j := e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \sigma \Sigma_j^{-1/2} \mathcal{B}_j^\theta \quad (4.32)$$

where we identify $x + iy \in \mathbb{C}$ with $(x, y) \in \mathbb{R}^2$, then $W_j \sim \mathbb{CN}(0, \sigma^2)$ in the sense that its real and imaginary parts are independent $\mathcal{N}(0, \sigma^2)$ random variables. Indeed, multiplying $\mathcal{B}_j^\theta$ by $\Sigma_j^{-1/2}$ removes its correlation structure so $\sigma \Sigma_j^{-1/2} \mathcal{B}_j^\theta \sim \mathbb{CN}(0, \sigma^2)$, and since the multivariate normal distribution is invariant under multiplication by orthogonal matrices, the multiplication by $e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))}$ has no effect on the distribution. This also shows that the W_j 's are independent.

Our first goal is thus to study, for $n \in \{1, \dots, N\}$, the sum $\sum_{j=1}^n (\mathcal{B}_j^\xi - W_j) = \Delta_n^{\xi\theta} + \Delta_n^{\theta W}$ where

$$\Delta_n^{\xi\theta} := \sum_{j=1}^n (\mathcal{B}_j^\xi - e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta) \quad \text{and} \quad \Delta_n^{\theta W} := \sum_{j=1}^n (e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta - W_j).$$

Setting $\Delta_0^{\xi\theta} := 0 =: \Delta_0^{\theta W}$, both $\{\Delta_n^{\xi\theta}\}_{n=0}^N$ and $\{\Delta_n^{\theta W}\}_{n=0}^N$ are martingales with respect to the filtration $\{\mathcal{F}_j\}_{j=0}^N$ generated by $\{B_E(t_j)\}_{j=0}^N$.

Step 1: Replacing the phase by its deterministic approximation. We first control the martingale $\{\Delta_n^{\xi\theta}\}_{n=0}^N$. Fixing $j \in \{1, \dots, N-1\}$ and using the identity $e^{-2ix} - e^{-2iy} = 2ie^{-i(x+y)} \sin(y-x)$, we can write the martingale's j th increment as

$$\begin{aligned} \mathcal{B}_j^\xi - e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta &= 2i \int_{t_{j-1}}^{t_j} e^{i\phi - i(\xi_{\beta,E}^g(s) + \theta(s) - \theta(t_{j-1}) + \xi_{\beta,E}^g(t_{j-1}))} \\ &\quad \times \sin\left(\theta(s) - \theta(t_{j-1}) - \xi_{\beta,E}^g(s) + \xi_{\beta,E}^g(t_{j-1})\right) \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s). \end{aligned}$$

This expression already makes it obvious that the quadratic variation of the stochastic integral is bounded by $8\sigma^2$, and therefore that the increment $\mathcal{B}_j^\xi - e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta$ is $8\sigma^2$ -subgaussian, both conditionally on $\mathcal{F}_{j-1}$ and unconditionally.

This control on the subgaussian constant of the increments is not enough to obtain a good control on the whole martingale, but we can improve it by getting a better control on the conditional variances of the increments. To do so, we apply Itô's isometry to the above expression for the j th increment, and as sine is 1-Lipschitz, we get

$$\begin{aligned} \mathbb{E} \left[\left| \mathcal{B}_j^\xi - e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta \right|^2 \middle| \mathcal{F}_{j-1} \right] \\ \leq 8 \mathbb{E} \left[\int_{t_{j-1}}^{t_j} \left| \theta(s) - \theta(t_{j-1}) - \xi_{\beta,E}^g(s) + \xi_{\beta,E}^g(t_{j-1}) \right|^2 \frac{c_E}{1 - c_E s} ds \middle| \mathcal{F}_{j-1} \right]. \end{aligned}$$

Now, by definition of θ and $\xi_{\beta,E}^g$, for any $s \in [t_{j-1}, t_j]$,

$$\begin{aligned} \theta(s) - \theta(t_{j-1}) - \xi_{\beta,E}^g(s) + \xi_{\beta,E}^g(t_{j-1}) \\ = \int_{t_{j-1}}^s \left(\kappa \sin 2\xi_{\beta,E}^g(u) + \frac{1}{\beta} \sin(4\xi_{\beta,E}^g(u) - 2\phi) \right) \frac{c_E}{1 - c_E u} du \\ - \frac{2\sqrt{2}}{\sqrt{\beta}} \int_{t_{j-1}}^s \cos \xi_{\beta,E}^g(u) \cos(\xi_{\beta,E}^g(u) - \phi) \sqrt{\frac{c_E}{1 - c_E u}} dB_E(u). \end{aligned}$$

The first term is immediately bounded by $(|\kappa| + \frac{1}{\beta})\sigma^2$, while the quadratic variation of the second one is bounded by $\frac{8}{\beta}\sigma^2$ so Bernstein's inequality shows that for any $x > 0$,

$$\mathbb{P}\left[\sup_{s \in [t_{j-1}, t_j]} \left| \frac{2\sqrt{2}}{\sqrt{\beta}} \int_{t_{j-1}}^s \cos \xi_{\beta,E}^g(u) \cos(\xi_{\beta,E}^g(u) - \phi) \sqrt{\frac{c_E}{1 - c_E u}} dB_E(u) \right| \geq x \mid \mathcal{F}_{j-1} \right] \leq 2 \exp\left(-\frac{\beta x^2}{16\sigma^2}\right).$$

On the complementary event,

$$\int_{t_{j-1}}^{t_j} |\theta(s) - \theta(t_{j-1}) - \xi_{\beta,E}^g(s) + \xi_{\beta,E}^g(t_{j-1})|^2 \frac{c_E}{1 - c_E s} ds < \left(\left(|\kappa| + \frac{1}{\beta} \right) \sigma^2 + x \right)^2 \sigma^2. \quad (4.33)$$

Now, using a layer cake representation, we can write

$$\begin{aligned} \mathbb{E}\left[\left| \mathcal{B}_j^\xi - e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta \right|^2 \mid \mathcal{F}_{j-1} \right] &\leq 8 \left(|\kappa| + \frac{1}{\beta} \right)^2 \sigma^6 \\ &+ 8 \int_{(|\kappa| + 1/\beta)^2 \sigma^6}^\infty \mathbb{P}\left[\int_{t_{j-1}}^{t_j} |\theta(s) - \theta(t_{j-1}) - \xi_{\beta,E}^g(s) + \xi_{\beta,E}^g(t_{j-1})|^2 \frac{c_E}{1 - c_E s} ds \geq y \mid \mathcal{F}_{j-1} \right] dy. \end{aligned}$$

Changing variables by setting $y = ((|\kappa| + 1/\beta)\sigma^2 + x)^2 \sigma^2$, we can use the estimate (4.33) and integrate. Since $\sigma^2 = \log(1 + E^{-p}) \leq E^{-p} \leq 1$ for any $E \geq 1$, we find in that case

$$\mathbb{E}\left[\left| \mathcal{B}_j^\xi - e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta \right|^2 \mid \mathcal{F}_{j-1} \right] \leq C_{\beta,\kappa} \sigma^4 \quad (4.34)$$

for a $C_{\beta,\kappa} > 0$ depending only on β and κ (which could easily be computed explicitly).

Finally, from the estimate (4.34) on the conditional variances of increments and the fact that the increments are $8\sigma^2$ -subgaussian, both conditionally on their past and unconditionally, we can apply Corollary A.2.1 of Freedman's inequality (see Theorem A.2 in the appendix) to both the real and imaginary parts of the martingale. Because $\sigma^2 \leq E^{-p}$ and $N \leq E^p \log E$ for all $E \geq 1$, this shows that

$$\begin{aligned} \mathbb{P}\left[\sup_{1 \leq n \leq N-1} |\Delta_n^{\xi\theta}| > x \right] &\leq 2E^{2p} \log^2 E \exp\left(-\frac{1}{2} \min\left\{ \left(\frac{E^{p/2} x \log 2}{12\sqrt{2}} \right)^{2/3}, \frac{E^p x^2 \log 2}{9C_{\beta,\kappa} \log E} \right\} \right) \\ &+ 4 \exp\left(-\frac{x \log 2}{3\sqrt{2}C_{\beta,\kappa} \log E} \exp\left(\frac{1}{4} \left(\frac{E^{p/2} x \log 2}{12\sqrt{2}} \right)^{2/3} \right) \right) \end{aligned} \quad (4.35)$$

for any E large enough.

Step 2: Comparing with a Gaussian random walk. We now estimate the supremum of the martingale $\{\Delta_n^{\theta W}\}_{n=1}^N$. To do so, our plan is to estimate the conditional variances of the increments, and then apply Azuma's inequality for subgaussian martingales.

Recall that $e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta - W_j = e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} (I_2 - \sigma \Sigma_j^{-1/2}) \mathcal{B}_j^\theta$ by definition of W_j , where again we identify complex numbers with vectors in $\mathbb{R}^2$ for matrix products. This means that

$$\mathbb{E}\left[\left| e^{-2i(\xi_{\beta,E}^g(t_{j-1}) - \theta(t_{j-1}))} \mathcal{B}_j^\theta - W_j \right|^2 \mid \mathcal{F}_{j-1} \right] \leq \|I_2 - \sigma \Sigma_j^{-1/2}\|_2^2 \mathbb{E}[|\mathcal{B}_j^\theta|^2 \mid \mathcal{F}_{j-1}]. \quad (4.36)$$

The definition of $\mathcal{B}_j^\theta$ directly yields

$$\mathbb{E}[|\mathcal{B}_j^\theta|^2 \mid \mathcal{F}_{j-1}] = \int_{t_{j-1}}^{t_j} \frac{2c_E}{1 - c_E s} ds = 2\sigma^2, \quad (4.37)$$

so we only have to estimate the norm of the matrix $I_2 - \sigma \Sigma_j^{-1/2}$, which is its largest eigenvalue, in magnitude. To do so, note that by Itô isometry,

$$\begin{aligned} \mathbb{E} \operatorname{Re} \mathcal{B}_j^\theta \operatorname{Im} \mathcal{B}_j^\theta &= \int_{t_{j-1}}^{t_j} \cos(\phi - 2\theta(s)) \sin(\phi - 2\theta(s)) \frac{2c_E}{1 - c_E s} ds, \\ \mathbb{E}(\operatorname{Re} \mathcal{B}_j^\theta)^2 - \sigma^2 &= \int_{t_{j-1}}^{t_j} (2 \cos^2(\phi - 2\theta(s)) - 1) \frac{c_E}{1 - c_E s} ds, \\ \mathbb{E}(\operatorname{Im} \mathcal{B}_j^\theta)^2 - \sigma^2 &= \int_{t_{j-1}}^{t_j} (2 \sin^2(\phi - 2\theta(s)) - 1) \frac{c_E}{1 - c_E s} ds, \end{aligned}$$

so

$$\sigma^{-2} \Sigma_j - I_2 = \sigma^{-2} \begin{pmatrix} x & y \\ y & -x \end{pmatrix} \quad \text{for} \quad z = x + iy := \int_{t_{j-1}}^{t_j} e^{2i\phi - 4i\theta(s)} \frac{c_E}{1 - c_E s} ds.$$

Hence, $\sigma^{-2} \Sigma_j - I_2$ has eigenvalues $\pm \sigma^{-2} |z|$. This also means that $\sigma^{-2} \Sigma_j$ has eigenvalues $1 \pm \sigma^{-2} |z|$, so that $\sigma \Sigma_j^{-1/2}$ has eigenvalues $(1 \pm \sigma^{-2} |z|)^{-1/2}$. Writing $\sigma^{-2} \Sigma_j - I_2 = (\sigma^{-1} \Sigma_j^{1/2} - I_2)(\sigma^{-1} \Sigma_j^{1/2} + I_2)$, we can then decompose $I_2 - \sigma \Sigma_j^{-1/2} = \sigma \Sigma_j^{-1/2} (\sigma^{-2} \Sigma_j - I_2) (\sigma^{-1} \Sigma_j^{1/2} + I_2)^{-1}$. Because $\Sigma_j \succeq 0$, the eigenvalues of $\sigma^{-1} \Sigma_j^{1/2} + I_2$ must be at least 1, so combining our eigenvalue computations yields

$$\|I_2 - \sigma \Sigma_j^{-1/2}\|_2 \leq \frac{|z|}{\sigma^2 \sqrt{1 - \sigma^{-2} |z|}}. \quad (4.38)$$

To estimate $|z|$, we then use the fact that $\theta'(s) = -2c_E E^{q/2} (1 - c_E s)^{q-1}$ to integrate by parts in

$$\bar{z} = \frac{ie^{-2i\phi}}{8E^{q/2}} \int_{t_{j-1}}^{t_j} e^{4i\theta(s)} \frac{4i\theta'(s)}{(1 - c_E s)^q} ds = \frac{ie^{-2i\phi}}{8E^{q/2}} \left(\frac{e^{4i\theta(s)}}{(1 - c_E s)^q} \Big|_{t_{j-1}}^{t_j} - q \int_{t_{j-1}}^{t_j} e^{4i\theta(s)} \frac{c_E}{(1 - c_E s)^{q+1}} ds \right).$$

This can be bounded as

$$|z| \leq \frac{1}{8E^{q/2}} \left(\frac{1}{(1 - c_E t_j)^q} + \frac{1}{(1 - c_E t_{j-1})^q} + q \int_{t_{j-1}}^{t_j} \frac{c_E}{(1 - c_E s)^{q+1}} ds \right) = \frac{1}{4E^{q/2} (1 - c_E t_j)^q} =: \frac{\delta_j}{4}.$$

This can be used to control the norm in (4.38), but only if δ_j is small compared to σ^2 . To check this, note that if $c_E t_j \in [0, 1 - E^{-1/2+\alpha}]$ for some $\alpha \in (0, 1/2)$, then $(1 - c_E t_j)^{-q} \leq E^{q/2-\alpha q}$ so $\delta_j \leq E^{-\alpha q}$. Combining this with the bound $\log(1+x) \geq x(1-x/2)$ for $x > 0$ by which $2\sigma^2 \geq E^{-p}(2 - E^{-p})$,

$$\frac{\delta_j}{2\sigma^2} \leq \frac{E^{-(\alpha q - p)}}{2 - E^{-p}}.$$

With our choice $p = 2\alpha q/3 < \alpha q$, the right-hand side vanishes when $E \rightarrow \infty$, so $\delta_j \leq 2\sigma^2$ for any E large enough. Then $\sigma^{-2}|z| \leq 1/2$, and the estimate (4.38) becomes $\|I_2 - \sigma \Sigma_j^{-1/2}\|_2 \leq \delta_j/2\sqrt{2}\sigma^2$. Using this with (4.37) in (4.36), we finally conclude that

$$\mathbb{E}\left[\left|e^{-2i(\xi_{\beta,E}^g(t_{j-1})-\theta(t_{j-1}))}\mathcal{B}_j^\theta - W_j\right|^2 \middle| \mathcal{F}_{j-1}\right] \leq \frac{\delta_j^2}{4\sigma^2} = \frac{1}{4\sigma^2 E^q (1 - c_E t_j)^{2q}} \quad (4.39)$$

for any E large enough.

Now, by definition, the increments $e^{-2i(\xi_{\beta,E}^g(t_{j-1})-\theta(t_{j-1}))}\mathcal{B}_j^\theta - W_j$ are centered Gaussian conditionally on their past. Therefore, the estimate (4.39) on the conditional variances is equivalent to an estimate on the conditional subgaussian constants, and we can use this to estimate the subgaussian bracket $\langle\langle \Delta^{\theta W} \rangle\rangle$ of the full martingale (which is essentially the sum of the conditional subgaussian constants—see Appendix A.2 for the full definition). Summing up (4.39) while assuming that $c_E t_n \in [0, 1 - E^{-1/2+\alpha}]$ so that the estimate applies to all increments, we get

$$\begin{aligned} \langle\langle \Delta^{\theta W} \rangle\rangle_n &= \sum_{j=1}^n \mathbb{E}\left[\left|e^{-2i(\xi_{\beta,E}^g(t_{j-1})-\theta(t_{j-1}))}\mathcal{B}_j^\theta - W_j\right|^2 \middle| \mathcal{F}_{j-1}\right] \\ &\leq \frac{1}{4\sigma^2 E^q} \sum_{j=1}^n (1 + E^{-p})^{2qj} \leq \frac{(1 + E^{-p})^{2q}}{4\sigma^2 E^q} \frac{(1 + E^{-p})^{2qn} - 1}{E^{-p}}. \end{aligned}$$

Because $(1 + E^{-p})^n = (1 - c_E t_n)^{-1} \leq E^{1/2-\alpha}$, this yields

$$\langle\langle \Delta^{\theta W} \rangle\rangle_n \leq \frac{(1 + E^{-p})^{2q}}{2(2 - E^{-p})} E^{2p-2\alpha q}.$$

The prefactor converges to $1/4$ as $E \rightarrow \infty$, so this shows that $\langle\langle \Delta^{\theta W} \rangle\rangle_n \leq E^{-2(\alpha q - p)}$ for any E large enough. Applying Azuma's inequality (Theorem A.3) to both the real and imaginary parts of the martingale, it follows that for any E large enough and any $x > 0$,

$$\mathbb{P}\left[\sup_n |\Delta_n^{\theta W}| \geq x\right] \leq 4 \exp\left(-\frac{E^{2(\alpha q - p)} x^2}{2}\right) \quad (4.40)$$

where the supremum is taken over all n such that $c_E t_n \in [0, 1 - E^{-1/2+\alpha}]$.

Step 3: Extending to the continuum. Now, we extend the random walk $\{W_j\}_{j=1}^{N-1}$ to a complex Brownian motion. To do this, extending the probability space, we define a collection $\{X_j\}_{j=1}^{N-1}$ of independent complex Brownian bridges with $X_j(0) = X_j(1) = 0$, taken to be independent of B_E . From these, we define Brownian motions on intervals $[v_E(t_{j-1}), v_E(t_j)]$ where $v_E(t) := -\log(1 - c_E t)$. Since $W_j \sim \mathbb{C}\mathcal{N}(0, \sigma^2)$, both real and imaginary parts of $\sigma^{-1}W_j$ are real standard normals, so given an independent random variable Z , by properties of Brownian bridges, each process

$$s \mapsto Z + \sigma X_j\left(\frac{s - v_E(t_{j-1})}{\sigma^2}\right) + \frac{s - v_E(t_{j-1})}{\sigma^2} W_j$$

is a complex Brownian motion on $[\nu_E(t_{j-1}), \nu_E(t_j)]$ with the law of Z as its distribution at time $\nu_E(t_{j-1})$. To build our complex Brownian motion W , we can stitch the above Brownian motions together as follows. We set $W(0) := 0$, and then for $0 < s \leq \nu_E(t_{N-1})$ we set $k(s) := \min\{j \in \mathbb{N} : s < \nu_E(t_{j+1})\}$ to be the index j such that $\nu_E(t_j) \leq s < \nu_E(t_{j+1})$, and

$$W(s) := \sum_{j=1}^{k(s)} W_j + \sigma X_{k(s)+1} \left(\frac{s - \nu_E(t_{k(s)})}{\sigma^2} \right) + \frac{s - \nu_E(t_{k(s)})}{\sigma^2} W_{k(s)+1}.$$

Finally, for $s > \nu_E(t_{N-1})$, we simply extend again the probability space to add an independent standard complex Brownian motion $\tilde{W}$, and we define $W(s) := W(\nu_E(t_{N-1})) + \tilde{W}(s - \nu_E(t_{N-1}))$.

Since W is a complex Brownian motion on each interval $[\nu_E(t_{n-1}), \nu_E(t_n)]$ for $n < N$ as well as on $[\nu_E(t_{N-1}), \infty)$, to check that it is a complex Brownian motion on $[0, \infty)$ we only have to check that it is a.s. continuous at the endpoints $\nu_E(t_n)$ of these intervals. To do this, fix $n < N$, and note that $k(\nu_E(t_n)) = n$ by definition, so that $W(\nu_E(t_n)) = \sum_{j=1}^n W_j$ for each $n < N$. It is clear that this is a.s. the value approached from the right, since k does not change from that direction (or just because $\tilde{W}$ is a.s. continuous if $n = N - 1$). With $s \uparrow \nu_E(t_n)$, $k(s) = k(\nu_E(t_n)) - 1 = n - 1$, and as $\nu_E(t_n) - \nu_E(t_{n-1}) = \sigma^2$, we get that

$$W(s) \rightarrow \sum_{j=1}^{n-1} W_j + \sigma X_n(1) + W_n = \sum_{j=1}^n W_j.$$

Hence, W has a.s. continuous paths on all of $[0, \infty)$, and we conclude that it is a standard complex Brownian motion.

Step 4: Full comparison of the processes. We now have all of the necessary ingredients to obtain the estimate (4.26) on

$$\Delta := \sup_{t \in I_{E,\alpha}} \left| \int_0^t e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) - W \circ \nu_E(t) \right|.$$

With $S_{E,\alpha} := \{n \in \mathbb{N} : 0 \leq c_E t_{n-1} \leq 1 - E^{-1/2+\alpha}\}$, we can extend a little bit the interval over which the supremum is taken to write

$$\begin{aligned} \Delta &\leq \sup_{n \in S_{E,\alpha}} \sup_{t \in [t_{n-1}, t_n]} \left| \int_0^t e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) - W \circ \nu_E(t) \right| \\ &\leq \sup_{n \in S_{E,\alpha}} \left| \int_0^{t_{n-1}} e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) - W \circ \nu_E(t_{n-1}) \right| \\ &\quad + \sup_{n \in S_{E,\alpha}} \sup_{t \in [t_{n-1}, t_n]} \left(\left| \int_{t_{n-1}}^t e^{i\phi - 2i\xi_{\beta,E}^g(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) \right| + |W \circ \nu_E(t) - W \circ \nu_E(t_{n-1})| \right). \end{aligned}$$

The integral that appears in the first line is exactly $\sum_{j=1}^{n-1} \mathcal{B}_j^\xi$, and $W \circ \nu_E(t_{n-1}) = \sum_{j=1}^{n-1} W_j$ by construction of W . In other words, in the notation of Steps 1 and 2,

$$\Delta \leq \sup_{n \in S_{E,\alpha}} |\Delta_{n-1}^{\xi\theta}| + \sup_{n \in S_{E,\alpha}} |\Delta_{n-1}^{\theta W}| \quad (4.41a)$$

$$+ \sup_{n \in S_{E,\alpha}} \sup_{t \in [t_{n-1}, t_n]} \left| \int_{t_{n-1}}^t e^{i\phi - 2i\xi_{\beta,E}^\theta(s)} \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) \right| \quad (4.41b)$$

$$+ \sup_{n \in S_{E,\alpha}} \sup_{t \in [t_{n-1}, t_n]} |W \circ \nu_E(t) - W \circ \nu_E(t_{n-1})|. \quad (4.41c)$$

The first line is controlled by the results (4.35) and (4.40) from Steps 1 and 2. In the other two lines, for fixed n the supremum is that of a complex martingale on $[t_{n-1}, t_n]$, which we can control by bounding the brackets of their real and imaginary parts and using Bernstein's inequality on both of them. The brackets of the real and imaginary parts of the martingale on line (4.41b) are both bounded by

$$\int_{t_{n-1}}^{t_n} \frac{2c_E}{1 - c_E s} ds = 2 \log \frac{1 - c_E t_{n-1}}{1 - c_E t_n} = 2 \log(1 + E^{-p}) \leq 2E^{-p},$$

so by Bernstein's inequality, for $x > 0$, (4.41b) is bounded by x with probability at least $1 - 4N \exp(-E^p x^2/4)$. Likewise, the brackets of the real and imaginary parts of the martingale on line (4.41c) are bounded by $\nu_E(t_n) - \nu_E(t_{n-1}) \leq E^{-p}$, so Bernstein's inequality yields a similar bound. Combining this with the results (4.35) and (4.40) from Steps 1 and 2, we get from (4.41) that for E large enough and $x > 0$,

$$\begin{aligned} \mathbb{P}[\Delta \geq x] &\leq 2E^{2p} \log^2 E \exp\left(-\frac{1}{2} \min\left\{\left(\frac{E^{p/2} x \log 2}{48\sqrt{2}}\right)^{2/3}, \frac{E^p x^2 \log 2}{144C_{\beta,\kappa} \log E}\right\}\right) \\ &\quad + 4 \exp\left(-\frac{x \log 2}{12\sqrt{2}C_{\beta,\kappa} \log E} \exp\left(\frac{1}{4}\left(\frac{E^{p/2} x \log 2}{48\sqrt{2}}\right)^{2/3}\right)\right) \\ &\quad + 4 \exp\left(-\frac{E^{2(\alpha q - p)} x^2}{32}\right) + 8N \exp\left(-\frac{E^p x^2}{64}\right). \end{aligned}$$

That $p = 2\alpha q/3$ implies that $2(\alpha q - p) = p$, so the powers of E in the last two terms are balanced. If $x = E^{-\alpha q/3 + \delta}$, then all exponents are at least of order $E^{2\delta/3}$ for E large enough, and we conclude that

$$\mathbb{P}[\Delta \geq E^{-\alpha q/3 + \delta}] \leq 3E^{4\alpha/3} \log^2 E \exp(-CE^{2\delta/3}),$$

where $C > 0$ depends only on β, κ, α and δ .

Step 5: Combining all spaces together. The above construction yields, for a given $E > 0$, a probability space on which are defined a standard real Brownian motion B_E and a standard complex Brownian motion W such that (4.26) holds if E is large enough. Thus, given a sequence

$\{E_n\}_{n \in \mathbb{N}} \subset (0, \infty)$ such that $E_n \rightarrow \infty$, it remains to combine all the spaces together in such a way that the couplings between the real and complex Brownian motions are preserved. For the sake of completeness, we build the full probability space explicitly, but this construction is standard.

Let $(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ and W_n denote the probability space and the complex Brownian motion built above for $E = E_n$ and a standard real Brownian motion B_{E_n} . As the initial Brownian motion B_{E_n} is arbitrary, we can choose it to be defined as a random variable $B_{E_n} : \Omega_n \rightarrow \mathcal{C}([0, \infty), \mathbb{R})$, using the construction of Brownian motion on a Wiener space. In the same way, taking the Brownian bridges used to build W_n with continuous paths, W_n has surely continuous paths, and $W_n : \Omega_n \rightarrow \mathcal{C}([0, \infty), \mathbb{C})$ where the space $\mathcal{C}([0, \infty), \mathbb{C})$ is endowed with a product of the Wiener measure with itself. Using these constructions, on each space $(\Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ we have two Brownian motions B_{E_n} and W_n with joint law $\mathbb{P}_{W_n, B_{E_n}}$ on $\mathcal{C}([0, \infty), \mathbb{C}) \times \mathcal{C}([0, \infty), \mathbb{R})$ with the completion of its Borel σ -algebra under an appropriate metric that makes it complete and separable (and such a metric does exist, see e.g. [46, §2.4]). This measurable space is thus Borel, and there exists a regular condition distribution $\rho_{B_{E_n}|W_n} : \mathcal{C}([0, \infty), \mathbb{C}) \times \mathcal{B}(\mathcal{C}([0, \infty), \mathbb{R})) \rightarrow [0, 1]$ such that $\mathbb{P}_{W_n, B_{E_n}} = \mathbb{P}_{W_n} \otimes \rho_{B_{E_n}|W_n}$.

Now, set $\Omega := \mathcal{C}([0, \infty), \mathbb{C}) \times \mathcal{C}([0, \infty), \mathbb{R})^{\mathbb{N}}$ and let $\mathcal{F} := \mathcal{B}(\Omega)$ be its Borel σ -algebra. Then, define kernels μ_n on the partial products $\mathcal{C}([0, \infty), \mathbb{C}) \times \prod_{j=1}^n \mathcal{C}([0, \infty), \mathbb{R})$ by setting

$$\mu_n(w, b_1, \dots, b_{n-1}, A) := \rho_{B_{E_n}|W_n}(w, A).$$

By the Ionescu–Tulcea theorem, there exists a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ such that for any $A \in \mathcal{B}(\mathcal{C}([0, \infty), \mathbb{C}) \times \mathcal{C}([0, \infty), \mathbb{R})^n)$,

$$\mathbb{P}\left(A \times \prod_{j=n+1}^{\infty} \Omega_j\right) = \int_A \left(\prod_{j=1}^n \rho_{B_{E_n}|W_n}(w, db_n)\right) \mathbb{P}_{W_n}(dw),$$

and in particular if $A \in \mathcal{B}(\mathcal{C}([0, \infty), \mathbb{C}))$ and $B \in \mathcal{B}(\mathcal{C}([0, \infty), \mathbb{R}))$, then

$$\mathbb{P}(A \times \Omega_1 \times \dots \times \Omega_{n-1} \times B \times \Omega_{n+1} \times \dots) = \int_A \int_B \rho_{B_{E_n}|W_n}(w, db) \mathbb{P}_{W_n}(dw) = \mathbb{P}_{W_n, B_{E_n}}(A \times B).$$

Therefore, for any $n \in \mathbb{N}$ the projection onto coordinates 1 and $n+1$ has the desired distribution for the pair (W, B_{E_n}) , and all of the Brownian motions we need are well defined on $(\Omega, \mathcal{F}, \mathbb{P})$. ■

4.3.2 Convergence of $\mathcal{U}_{\beta, E}$ to a hyperbolic Brownian motion

On the probability space from Lemma 4.4, we can compare, pathwise, processes driven by the limit complex Brownian motion W and the processes that approximate them. In particular, we can compare a hyperbolic Brownian motion driven by W with the process $\mathcal{U}_{\beta, E}$ from (4.22). This comparison

is our main goal for this section. We start by proving a preliminary result in §4.3.2.1, and then we compare separately the imaginary and real parts of the two processes in §4.3.2.2 and §4.3.2.3 respectively. Indeed, the comparison of the imaginary parts is simpler, both because the imaginary part of a hyperbolic Brownian motion is a geometric Brownian motion, and because $\text{Im } \mathcal{E}_{\beta,E}$ can be written in terms of $\rho_{\beta,E}^g$ only, as a consequence of Claim 4.3.

4.3.2.1 Averaging of integrals with oscillatory integrands

As a preliminary step, we need to control certain integrals whose integrands oscillate quickly, with phases proportional to solutions of (4.21b), causing them to average out when E is large.

Lemma 4.5. *Let $\xi_{\beta,E}$ be a solution to the SDE (4.21b) driven by a standard Brownian motion B . Let X_E solve*

$$dX_E(t) = \mu_E(t)X_E(t) \frac{c_E}{1 - c_E t} dt + \sigma_E(t)X_E(t) \sqrt{\frac{c_E}{1 - c_E t}} dB(t) \quad (4.42)$$

on $[0, \tau_E]$ for some complex processes μ_E and σ_E . Fix $T \in (0, \tau_E]$ and a nonzero $k \in \mathbb{R}$, and suppose that μ_E and σ_E are bounded on $[0, T]$ uniformly in E . Then, there are constants $C, C' > 0$ independent of E such that for any $M, x > 0$,

$$\begin{aligned} \mathbb{P} \left(\left\{ \sup_{t \in [0, T]} |X_E| \leq M \right\} \cap \left\{ \sup_{t \in [0, T]} \left| \int_0^t X_E(s) e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds \right| \geq x + \frac{CM}{E^{q/2}(1 - c_E T)^q} \right\} \right) \\ \leq 4 \exp \left(- \frac{C'E^q(1 - c_E T)^{2q}x^2}{M^2} \right). \end{aligned}$$

Remark. The exponential structure of the SDE is not essential to this result: it would be straightforward to extend the proof given below to processes satisfying more complicated SDEs with the factors $\mu_E X_E$ and $\sigma_E X_E$ replaced by other functions that are bounded on compacts. However, the proof is a bit simpler with this exponential structure, and this is all we need here.

Proof. Recall that $\xi_{\beta,E}$ satisfies

$$d\xi_{\beta,E}(t) = -2c_E E^{q/2} (1 - c_E t)^{q-1} dt + R_{\beta,E}^\xi(t) \frac{c_E}{1 - c_E t} dt + S_{\beta,E}^\xi(t) \sqrt{\frac{c_E}{1 - c_E t}} dB(t)$$

where

$$R_{\beta,E}^\xi := -\kappa \sin 2\xi_{\beta,E} - \frac{1}{\beta} \sin(4\xi_{\beta,E} - 2\phi) \quad \text{and} \quad S_{\beta,E}^\xi := \frac{2\sqrt{2}}{\sqrt{\beta}} \cos \xi_{\beta,E} \cos(\xi_{\beta,E} - \phi).$$

An application of Itô's formula then shows that

$$\begin{aligned} d(e^{ki\xi_{\beta,E}})(t) = e^{ki\xi_{\beta,E}(t)} & \left(-2kic_E E^{q/2} (1 - c_E t)^{q-1} dt + kiS_{\beta,E}^\xi(t) \sqrt{\frac{c_E}{1 - c_E t}} dB(t) \right. \\ & \left. + \left(kiR_{\beta,E}^\xi(t) - \frac{k^2}{2} S_{\beta,E}^\xi(t)^2 \right) \frac{c_E}{1 - c_E t} dt \right), \end{aligned}$$

so for $t \in [0, T]$,

$$\begin{aligned} \int_0^t X_E(s) e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds = & -\frac{1}{2ikE^{q/2}} \int_0^t \frac{X_E(s)}{(1 - c_E s)^q} \left(d(e^{ki\xi_{\beta,E}})(s) \right. \\ & - kiS_{\beta,E}^\xi(s) e^{ki\xi_{\beta,E}(s)} \sqrt{\frac{c_E}{1 - c_E s}} dB(s) \\ & \left. - e^{ki\xi_{\beta,E}(s)} \left(kiR_{\beta,E}^\xi(s) - \frac{k^2}{2} S_{\beta,E}^\xi(s)^2 \right) \frac{c_E}{1 - c_E s} ds \right). \end{aligned} \quad (4.43)$$

If $Y_E(t) := (1 - c_E t)^{-q} X_E(t)$, then

$$dY_E(t) = X_E(t) (q + \mu_E(t)) \frac{c_E}{(1 - c_E t)^{q+1}} dt + X_E(t) \sigma_E(t) \frac{\sqrt{c_E}}{(1 - c_E t)^{q+1/2}} dB(t),$$

which also implies that

$$d\langle Y_E, e^{ki\xi_{\beta,E}} \rangle(t) = kiX_E(t) e^{ki\xi_{\beta,E}(t)} S_{\beta,E}^\xi(t) \sigma_E(t) \frac{c_E}{(1 - c_E t)^{q+1}} dt.$$

From this, the first term of (4.43) can be integrated by parts:

$$\begin{aligned} \int_0^t X_E(s) e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds = & -\frac{1}{2ikE^{q/2}} \left(\frac{X_E(s) e^{ki\xi_{\beta,E}(s)}}{(1 - c_E s)^q} \Big|_0^t \right. \\ & - \int_0^t X_E(s) e^{ki\xi_{\beta,E}(s)} \left(kiS_{\beta,E}^\xi(s) + \sigma_E(s) \right) \frac{\sqrt{c_E}}{(1 - c_E s)^{q+1/2}} dB(s) \\ & - \int_0^t X_E(s) e^{ki\xi_{\beta,E}(s)} \left(kiR_{\beta,E}^\xi(s) - \frac{k^2}{2} S_{\beta,E}^\xi(s)^2 \right. \\ & \left. \left. + q + \mu_E(s) + kiS_{\beta,E}^\xi(s) \sigma_E(s) \right) \frac{c_E}{(1 - c_E s)^{q+1}} ds \right). \end{aligned} \quad (4.44)$$

Given $M > 0$, this can easily be bounded on the event $\mathcal{G} := \{\sup_{t \in [0, T]} |X_E(t)| \leq M\}$. The first term is bounded on $\mathcal{G}$ by

$$\frac{M}{2|k|E^{q/2}} \left(\frac{1}{(1 - c_E T)^q} + 1 \right) \leq \frac{M}{|k|E^{q/2}(1 - c_E T)^q}.$$

Then, since $|R_{\beta,E}^\xi| \leq |\kappa| + 1/\beta$ and $|S_{\beta,E}^\xi| \leq 2\sqrt{2}/\sqrt{\beta}$ and since μ_E and σ_E are also bounded on $[0, T]$ uniformly in E , there is a constant $\tilde{C} > 0$ independent of E such that the time integral in (4.44) is bounded on $\mathcal{G}$ by

$$\frac{\tilde{C}M}{E^{q/2}} \int_0^t \frac{c_E}{(1 - c_E s)^{q+1}} ds \leq \frac{\tilde{C}M}{qE^{q/2}(1 - c_ET)^q}.$$

In the same way, there is another constant $\tilde{C}' > 0$ independent of E such that, still on $\mathcal{G}$, the brackets of the real and imaginary parts of the Brownian integral in (4.44) are bounded by

$$\frac{\tilde{C}'M^2}{E^q} \int_0^t \frac{c_E}{(1 - c_E s)^{2q+1}} ds \leq \frac{\tilde{C}'M^2}{2qE^q(1 - c_ET)^{2q}}.$$

As this holds for any $t \in [0, T]$, applying Bernstein's inequality for martingales to both the real and the imaginary parts of the stochastic integral shows that for any $x > 0$,

$$\begin{aligned} \mathbb{P}\left(\mathcal{G} \cap \left\{ \sup_{t \in [0, T]} \left| \frac{1}{2ikE^{q/2}} \int_0^t X_E(s) e^{ki\xi_{\beta,E}(s)} (kiS_{\beta,E}^\xi(s) + b_E(s)) \frac{\sqrt{c_E}}{(1 - c_E s)^{q+1/2}} dB(s) \right| \geq x \right\}\right) \\ \leq 4 \exp\left(-\frac{qE^q(1 - c_ET)^{2q}x^2}{\tilde{C}'M^2}\right). \end{aligned}$$

Combining the bounds on the three terms of (4.44) yields the announced result with $C := \tilde{C}/q + 1/|k|$ and $C' := q/\tilde{C}'$. $\blacksquare$

This result has two immediate corollaries.

Corollary 4.5.1. *If $k \neq 0$ and $\alpha \in (0, 1/2)$, there are $C, C' > 0$ independent of E such that for any $x > 0$,*

$$\mathbb{P}\left[\sup_{t \in \mathcal{I}_{E,\alpha}} \left| \int_0^t e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds \right| \geq x + CE^{-\alpha q}\right] \leq 4e^{-C'E^{2\alpha q}x^2}$$

where $\mathcal{I}_{E,\alpha} := [0, (1 - E^{-1/2+\alpha})/c_E]$.

Proof. We apply the lemma with $X_E \equiv 1$ so that $\mu_E = \sigma_E = 0$, and with $c_ET := 1 - E^{-1/2+\alpha}$. As in that case $\mathbb{P}[\sup_{t \in [0, T]} |X_E(t)| > 1] = 0$, we can take $M = 1$ to get that there are $C, C' > 0$ such that for any $x > 0$,

$$\mathbb{P}\left[\sup_{t \in [0, T]} \left| \int_0^t e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds \right| \geq x + \frac{C}{E^{q/2}(1 - c_ET)^q}\right] \leq 4e^{-C'E^q(1 - c_ET)^{2q}x^2}.$$

This is exactly the announced result since $E^{q/2}(1 - c_ET)^q = E^{\alpha q}$. $\blacksquare$

Corollary 4.5.2. *If $k \neq 0$, then for every $\varepsilon > 0$ there is a $C_\varepsilon > 0$ independent of E such that*

$$\mathbb{P} \left[\sup_{t \in [0, \tau_E]} \left| \int_0^t e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds \right| > C_\varepsilon \right] < \varepsilon.$$

Proof. Applying the lemma with $X_E \equiv 1$ so that $\mu_E = \sigma_E = 0$ and with $M = 1$ and $T = \tau_E$, we get that for some $C, C' > 0$ independent of E , for any $x > 0$,

$$\mathbb{P} \left[\sup_{t \in [0, \tau_E]} \left| \int_0^t e^{ki\xi_{\beta,E}(s)} \frac{c_E}{1 - c_E s} ds \right| > x + C \right] \leq 4e^{-C'x^2},$$

and taking x large enough so that $4e^{-C'x^2} < \varepsilon$ yields the result with $C_\varepsilon := x + C$. $\blacksquare$

4.3.2.2 The geometric Brownian motion

We now deduce the convergence of $\text{Im } \mathcal{C}_{\beta,E}$ to the imaginary part of the hyperbolic Brownian motion driven by W . Recall that by Claim 4.3, $\text{Im } \mathcal{C}_{\beta,E} = e^{-\Delta_{\beta,E}^\rho} \sin \Delta_{\beta,E}^\xi = e^{-2\rho_{\beta,E}^\xi}$, and that by definition the imaginary part of a hyperbolic Brownian motion is a geometric Brownian motion (see 3.3). We start by comparing the logarithms of the two processes.

Proposition 4.6. *On the probability space from Lemma 4.4, with $\alpha \in (0, 1/2)$ and $\delta \in (0, \alpha q/3)$, there is a $C > 0$ such that for all $E \in \{E_n\}_{n \in \mathbb{N}}$ large enough,*

$$\mathbb{P} \left[\sup_{t \in \mathcal{I}_{E,\alpha}} \left| 2\rho_{\beta,E}^\xi(t) + \log Z_{\beta,E}(t) \right| \geq E^{-\alpha q/3 + \delta} \right] \leq 4E^{4\alpha q/3} \log^2 E e^{-CE^{2\delta/3}}$$

where $\mathcal{I}_{E,\alpha} := [0, (1 - E^{-1/2+\alpha})/c_E]$ and $Z_{\beta,E}(t) := \exp(\frac{2}{\sqrt{\beta}} \text{Im } W \circ \nu_E(t) - \frac{2}{\beta} \nu_E(t))$.

Proof. Since $\rho_{\beta,E}^\xi$ follows (4.21a),

$$\begin{aligned} & \sup_{t \in \mathcal{I}_{E,\alpha}} \left| 2\rho_{\beta,E}^\xi(t) + \frac{2}{\sqrt{\beta}} \text{Im } W \circ \nu_E(t) - \frac{2}{\beta} \nu_E(t) \right| \\ & \leq \sup_{t \in \mathcal{I}_{E,\alpha}} \left| \frac{2}{\sqrt{\beta}} \int_0^t \sin(2\xi_{\beta,E}^\xi(s) - \phi) \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) + \frac{2}{\sqrt{\beta}} \text{Im } W \circ \nu_E(t) \right| \\ & \quad + 2|\kappa| \sup_{t \in \mathcal{I}_{E,\alpha}} \left| \int_0^t \cos 2\xi_{\beta,E}^\xi(s) \frac{c_E}{1 - c_E s} ds \right| + \frac{2}{\beta} \sup_{t \in \mathcal{I}_{E,\alpha}} \left| \int_0^t \cos(4\xi_{\beta,E}^\xi(s) - 2\phi) \frac{c_E}{1 - c_E s} ds \right|. \end{aligned}$$

The first supremum can immediately be bounded by $\frac{1}{3}E^{-\alpha q/3 + \delta}$ on a good event by Lemma 4.4, simply by taking the imaginary part in (4.26). The other two can each be bounded by $\frac{1}{3}E^{-\alpha q/3 + \delta}$ on a good event by applying Corollary 4.5.1. The result then follows by combining the tail bounds, which are dominated by that from Lemma 4.4. $\blacksquare$

Using properties of Brownian motion, we can take the exponential and compare $e^{-2\rho_{\beta,E}^g}$ and $e^{2\rho_{\beta,E}^g}$ to the geometric Brownian motion $Z_{\beta,E}$ and its reciprocal.

Corollary 4.6.1. *In the setting of Proposition 4.6, for any $\delta \in (0, \alpha q/3)$ and any E large enough,*

$$\mathbb{P} \left[\sup_{t \in I_{E,\alpha}} \left| e^{-2\rho_{\beta,E}^g(t)} - Z_{\beta,E}(t) \right| \geq E^{-\alpha q/3 + \delta} \right] \leq \frac{2}{\log E}, \quad (4.45)$$

and if $\alpha > \frac{3}{\beta q + 6}$ and $\delta < \frac{1}{3\beta}(\alpha(\beta q + 6) - 3)$, then for any E large enough,

$$\mathbb{P} \left[\sup_{t \in I_{E,\alpha}} \left| e^{2\rho_{\beta,E}^g(t)} - \frac{1}{Z_{\beta,E}(t)} \right| \geq \exp \left(-\frac{\alpha(\beta q + 6) - 3 - 3\beta\delta}{3\beta} \log E \right) \right] \leq 2 \exp \left(-\frac{\beta\delta^2}{36(1 - 2\alpha)} \log E \right).$$

Proof. Since

$$\left| e^{\mp 2\rho_{\beta,E}^g} - Z_{\beta,E}^{\pm 1} \right| \leq Z_{\beta,E}^{\pm 1} |2\rho_{\beta,E}^g + \log Z_{\beta,E}| e^{|2\rho_{\beta,E}^g + \log Z_{\beta,E}|},$$

given the proposition, what remains to control is only the supremum of the geometric Brownian motion $Z_{\beta,E}$ and that of its reciprocal. Recall that

$$Z_{\beta,E}^{\pm 1}(t) = \exp \left(\pm \frac{2}{\sqrt{\beta}} \operatorname{Im} W \circ \nu_E(t) \mp \frac{2}{\beta} \nu_E(t) \right).$$

We are interested in the supremum of this process over $c_E t \in [0, 1 - E^{-1/2+\alpha}]$, which can be understood as the supremum over $s = \nu_E(t) \in [0, (\frac{1}{2} - \alpha) \log E]$. Now, from the joint density of a Brownian motion and its running maximum, an application of Girsanov's theorem shows that for $y \geq 0$ (see e.g. [9, Part II, §2.1, formula 1.1.4] for the precise statement),

$$\begin{aligned} & \mathbb{P} \left[\sup_{s \in [0, (\frac{1}{2} - \alpha) \log E]} \left(\pm \frac{2}{\sqrt{\beta}} \operatorname{Im} W(s) \mp \frac{2s}{\beta} \right) \geq y \right] \\ &= \frac{1}{2} \left(1 - \operatorname{erf} \left(\frac{1}{2} \left(y \sqrt{\frac{\beta}{(1 - 2\alpha) \log E}} \pm \sqrt{\frac{(1 - 2\alpha) \log E}{\beta}} \right) \right) \right) \\ & \quad + \frac{e^{\mp y}}{2} \left(1 - \operatorname{erf} \left(\frac{1}{2} \left(y \sqrt{\frac{\beta}{(1 - 2\alpha) \log E}} \mp \sqrt{\frac{(1 - 2\alpha) \log E}{\beta}} \right) \right) \right). \end{aligned} \quad (4.46)$$

We can bound this in the two cases. First, for the top sign, the argument of the first error function is nonnegative so we can use the fact that $1 - \operatorname{erf} x \leq e^{-x^2}$ for $x \geq 0$, and we can simply bound the second one with the basic property $\frac{1}{2}(1 - \operatorname{erf}) \leq 1$. We thus obtain for $y = \log \log E$ and E large

enough that

$$\begin{aligned} \mathbb{P} \left[\sup_{s \in [0, (\frac{1}{2}-\alpha) \log E]} \left(\frac{2}{\sqrt{\beta}} \operatorname{Im} W(s) - \frac{2s}{\beta} \right) \geq \log \log E \right] \\ \leq \frac{1}{2} \exp \left(-\frac{1}{4} \left(\log \log E \sqrt{\frac{\beta}{(1-2\alpha) \log E}} + \sqrt{\frac{(1-2\alpha) \log E}{\beta}} \right)^2 \right) + \exp(-\log \log E). \end{aligned}$$

The dominating contribution to the first exponential is its second term so we can remove the first one, and exponentiating, we get that for E large enough,

$$\mathbb{P} \left[\sup_{t \in \mathcal{I}_{E,\alpha}} Z_{\beta,E}(t) \geq \log E \right] \leq \frac{1}{2E^{(1-2\alpha)/4\beta}} + \frac{1}{\log E}. \quad (4.47)$$

On the intersection of the complement of this event with the complement of the event in the proposition with δ replaced with $\delta/2$,

$$\sup_{t \in \mathcal{I}_{E,\alpha}} |e^{-2\rho_{\beta,E}^g(t)} - Z_{\beta,E}(t)| \leq E^{-\alpha q/3 + \delta/2} \exp(E^{-\alpha q/3 + \delta/2}) \log E \leq E^{-\alpha q/3 + \delta},$$

and combining the tail bounds (which are dominated by the $1/\log E$ term) yields (4.45).

With the bottom sign in (4.46), we use again the bound $1 - \operatorname{erf} x \leq e^{-x^2}$ valid for $x \geq 0$, but now on both error functions, so this is only valid for $y \geq \frac{1-2\alpha}{\beta} \log E$. This yields

$$\mathbb{P} \left[\sup_{s \in [0, (\frac{1}{2}-\alpha) \log E]} \left(-\frac{2}{\sqrt{\beta}} \operatorname{Im} W(s) + \frac{2s}{\beta} \right) \geq y \right] \leq \exp \left(-\frac{1}{4} \left(y \sqrt{\frac{\beta}{(1-2\alpha) \log E}} - \sqrt{\frac{(1-2\alpha) \log E}{\beta}} \right)^2 \right).$$

Taking $y = \left(\frac{1-2\alpha}{\beta} + \frac{\delta}{3} \right) \log E$, we get

$$\mathbb{P} \left[\sup_{s \in [0, (\frac{1}{2}-\alpha) \log E]} \left(-\frac{2}{\sqrt{\beta}} \operatorname{Im} W(s) + \frac{2s}{\beta} \right) \geq \left(\frac{1-2\alpha}{\beta} + \frac{\delta}{3} \right) \log E \right] \leq \exp \left(-\frac{\beta \delta^2}{36(1-2\alpha)} \log E \right).$$

On the intersection of the complement of this event with the complement of the event in the proposition with δ replaced with $\delta/3$,

$$\sup_{t \in \mathcal{I}_{E,\alpha}} \left| e^{2\rho_{\beta,E}^g(t)} - \frac{1}{Z_{\beta,E}(t)} \right| \leq \exp \left(E^{-\alpha q/3 + \delta/3} + \left(-\frac{\alpha q}{3} + \frac{\delta}{3} + \frac{1-2\alpha}{\beta} + \frac{\delta}{3} \right) \log E \right)$$

for E large enough. If $\alpha > \frac{3}{\beta q + 6}$, then taking $\delta < \frac{1}{3\beta} (\alpha(\beta q + 6) - 3)$ ensures that the exponent is negative, and combining the tail bounds (which are dominated by the negative power of E) gives the second result and completes the proof. $\blacksquare$

4.3.2.3 The real part of the hyperbolic Brownian motion

We finally turn to the comparison between $\operatorname{Re} \mathcal{C}_{\beta,E} = -e^{-\Delta_{\beta,E}^\rho} \cos \Delta_{\beta,E}^\xi$ and the real part of the hyperbolic Brownian motion driven by W .

Proposition 4.7. *On the probability space from Lemma 4.4, for any $\alpha \in (0, 1/2)$ and $\delta \in (0, \alpha q/12)$, there is a $C > 0$ such that for any $E \in \{E_n\}_{n \in \mathbb{N}}$ large enough,*

$$\mathbb{P} \left[\sup_{t \in I_{E,\alpha}} \left| \operatorname{Re} \mathcal{C}_{\beta,E}(t) - \frac{2}{\sqrt{\beta}} \int_0^t Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s) \right| \geq E^{-\alpha q/12+\delta} \right] \leq \frac{C}{\log E}$$

where $I_{E,\alpha} := [0, (1 - E^{-1/2+\alpha})/c_E]$.

Proof. We can get an SDE for $\operatorname{Re} \mathcal{C}_{\beta,E}$ by taking the real part of the SDE for $\mathcal{C}_{\beta,E}$ given in (4.24). Using that $\operatorname{Im} \mathcal{C}_{\beta,E} = e^{-2\rho_{\beta,E}^\xi}$ by Claim 4.3, this SDE can be written as

$$\begin{aligned} d(\operatorname{Re} \mathcal{C}_{\beta,E})(t) &= \frac{2}{\sqrt{\beta}} e^{-2\rho_{\beta,E}^\xi(t)} \cos(2\xi_{\beta,E}^\xi(t) - \phi) \sqrt{\frac{2c_E}{1 - c_E t}} dB_E(t) \\ &\quad - e^{-2\rho_{\beta,E}^\xi(t)} \left(2\kappa \sin 2\xi_{\beta,E}^\xi(t) + \frac{4}{\beta} \sin(4\xi_{\beta,E}^\xi(t) - 2\phi) \right) \frac{c_E}{1 - c_E t} dt. \end{aligned} \quad (4.48)$$

To prove the proposition, we proceed as follows. First, we find a good event on which the growth of $e^{-2\rho_{\beta,E}^\xi}$ can be controlled. From this, we deduce that the second line in (4.48) vanishes when $E \rightarrow \infty$, thus reducing the problem to comparing the first line in (4.48) to the real part of the hyperbolic Brownian motion driven by W . To do this, we discretize the time interval as in the proof of Lemma 4.4 in order to reduce the problem to a comparison between two discrete time martingales.

Step 1: Finding a good event on which to control the growth of integrands. Let

$$\mathcal{G} := \left\{ \sup_{t \in I_{E,\alpha}} Z_{\beta,E}(t) \leq \log E \right\} \cap \left\{ \sup_{t \in I_{E,\alpha}} |e^{-2\rho_{\beta,E}^\xi(t)} - Z_{\beta,E}(t)| \leq E^{-\alpha q/3+\delta} \right\}. \quad (4.49)$$

We have seen in (4.47) in the proof of Corollary 4.6.1 and in (4.45) in the corollary itself that the complements of both of these events have probability at most $2/\log E$ for E large enough, so $\mathbb{P}(\mathcal{G}^c) \leq 4/\log E$. Therefore, in order to prove the proposition, it suffices to prove that

$$\mathbb{P} \left(\mathcal{G} \cap \left\{ \sup_{t \in I_{E,\alpha}} \left| \operatorname{Re} \mathcal{C}_{\beta,E}(t) - \frac{2}{\sqrt{\beta}} \int_0^t Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s) \right| \geq E^{-\alpha q/12+\delta} \right\} \right) \lesssim \frac{1}{\log E}.$$

Step 2: Controlling the oscillatory terms. We now show that the second line in (4.48) does not contribute. An application of Itô's formula (or just taking the imaginary part of (4.24) and using

that $\text{Im } \theta_{\beta,E} = e^{-2\rho_{\beta,E}^g}$ shows that

$$\begin{aligned} d(e^{-2\rho_{\beta,E}^g})(t) &= -\frac{2\sqrt{2}}{\sqrt{\beta}} e^{-2\rho_{\beta,E}^g(t)} \sin(2\xi_{\beta,E}^g(t) - \phi) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \\ &\quad - e^{-2\rho_{\beta,E}^g(t)} \left(2\kappa \cos 2\xi_{\beta,E}^g(t) + \frac{4}{\beta} \cos(4\xi_{\beta,E}^g(t) - 2\phi) \right) \frac{c_E}{1 - c_E t} dt. \end{aligned}$$

This SDE has precisely the form appearing in Lemma 4.5 with

$$\mu_E = -2\kappa \cos 2\xi_{\beta,E}^g - \frac{4}{\beta} \cos(4\xi_{\beta,E}^g - 2\phi) \quad \text{and} \quad \sigma_E = -\frac{2\sqrt{2}}{\beta} \sin(2\xi_{\beta,E}^g - \phi),$$

which are both bounded processes with bounds depending only on κ and β . Applying the lemma for a $k \neq 0$ with $c_E T = 1 - E^{-1/2+\alpha}$ so that $E^{q/2}(1 - c_E T)^q = E^{\alpha q}$, we see that there are $C, C' > 0$ independent of E such that for any $x > 0$,

$$\begin{aligned} &\mathbb{P}\left(\left\{ \sup_{t \in I_{E,\alpha}} e^{-2\rho_{\beta,E}^g(t)} \leq 2 \log E \right\} \right. \\ &\quad \cap \left. \left\{ \sup_{t \in I_{E,\alpha}} \left| \int_0^t e^{-2\rho_{\beta,E}^g(s) + ki\xi_{\beta,E}^g(s)} \frac{c_E}{1 - c_E s} ds \right| \geq x + \frac{2C \log E}{E^{\alpha q}} \right\} \right) \leq 4 \exp\left(-\frac{C' E^{2\alpha q} x^2}{4 \log^2 E}\right). \end{aligned}$$

By definition, $e^{-2\rho_{\beta,E}^g} \leq 2 \log E$ on $\mathcal{G}$, so the first of the events in the above can be replaced with $\mathcal{G}$ without changing the tail bound. Taking $x = E^{-\alpha q/12+\delta}$ then yields an exponential tail bound, which vanishes faster than $1/\log E$ as $E \rightarrow \infty$. Hence, using this with $k = -2$ and $k = -4$ gives the control we need on both terms of the second line of (4.48).

Step 3: Reducing to a discrete-time problem. It remains to compare the first line of (4.48) with the real part of a hyperbolic Brownian motion driven by W and run in logarithmic time. To do so, we reduce the comparison between the two stochastic integrals to a comparison between discrete-time martingales. As in the proof of Lemma 4.4, we partition the time interval with a sequence $\{t_j\}_{j=0}^N$ defined with $t_0 := 0$, $c_E t_N := 1 - 1/\sqrt{E}$, and with

$$c_E t_j := 1 - \frac{1}{(1 + E^{-p})^j} \quad \text{for } 0 < j < N$$

where here we set $p := \alpha q/6$. As in the proof of Lemma 4.4, this means that N is as in (4.27), so that in particular $N \leq E^p \log E$ for E large enough. With $S_{E,\alpha} := \{j \in \mathbb{N} : 0 < c_E t_{j-1} \leq 1 - E^{-1/2+\alpha}\}$,

we can write

$$\begin{aligned} & \sup_{t \in \mathcal{I}_{E,\alpha}} \left| \int_0^t e^{-2\rho_{\beta,E}^g(s)} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) - \int_0^t Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s) \right| \\ & \leq \sup_{j \in S_{E,\alpha}} \left| \int_0^{t_{j-1}} e^{-2\rho_{\beta,E}^g(s)} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) - \int_0^{t_{j-1}} Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s) \right| \end{aligned} \quad (4.50a)$$

$$+ \sup_{j \in S_{E,\alpha}} \sup_{t \in [t_{j-1}, t_j] \cap \mathcal{I}_{E,\alpha}} \left| \int_{t_{j-1}}^t e^{-2\rho_{\beta,E}^g(s)} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right| \quad (4.50b)$$

$$+ \sup_{j \in S_{E,\alpha}} \sup_{t \in [t_{j-1}, t_j] \cap \mathcal{I}_{E,\alpha}} \left| \int_{t_{j-1}}^t Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s) \right|. \quad (4.50c)$$

The last two lines are easily controlled on $\mathcal{G}$. Indeed, on this event $e^{-2\rho_{\beta,E}^g} \leq 2 \log E$ uniformly for $t \in \mathcal{I}_{E,\alpha}$, so the bracket of the integral in (4.50b) is bounded for any $t \in [t_{j-1}, t_j] \cap \mathcal{I}_{E,\alpha}$ by

$$4 \log^2 E \int_{t_{j-1}}^{t_j} \frac{2c_E}{1-c_E s} ds = 8 \log^2 E \log(1 + E^{-p}) \leq 8E^{-p} \log^2 E.$$

Thus, by Bernstein's inequality, for any $x > 0$,

$$\mathbb{P} \left(\mathcal{G} \cap \left\{ \sup_{t \in [t_{j-1}, t_j] \cap \mathcal{I}_{E,\alpha}} \left| \int_{t_{j-1}}^t e^{-2\rho_{\beta,E}^g(s)} \cos 2\xi_{\beta,E}^g(s) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right| \geq x \right\} \right) \leq 2 \exp \left(-\frac{E^p x^2}{16 \log^2 E} \right).$$

Taking the supremum over $j \in S_{E,\alpha}$ adds a prefactor of N to the tail bound, so because $N \leq E^p \log E = E^{\alpha q/6} \log E$, with $x = E^{-\alpha q/12+\delta}$ the tail bound decreases exponentially in a power of E and is certainly $O(1/\log E)$.

Likewise, on $\mathcal{G}$ the geometric Brownian motion $Z_{\beta,E}$ is bounded by $\log E$, so the bracket of the integral in (4.50c) is bounded by

$$\log^2 E \int_{t_{j-1}}^{t_j} d\langle \operatorname{Re} W \circ \nu_E \rangle(s) = \log^2 E \int_{t_{j-1}}^{t_j} \frac{c_E}{1-c_E s} ds \leq E^{-p} \log^2 E.$$

In the same way as in the last case, Bernstein's inequality then yields a tail bound for (4.50c) on $\mathcal{G}$ that decreases exponentially in a power of E .

The problem is thus reduced to controlling (4.50a).

Step 4: Controlling the difference of discrete-time martingales. It remains to control (4.50a), the supremum of the difference of discrete martingales. To do this, given a process X , we let $R(X)$ denote the process X modified to be reset at every time t_k of the partition, that is, $R(X)(s) := X(s) - X(t_{k-1})$ for $s \in [t_{k-1}, t_k)$. Using this notation, we can split each difference of increments

in (4.50a) in four differences, so that the full difference takes the form

$$\begin{aligned} & \int_0^{t_{j-1}} e^{-2\rho_{\beta,E}^g(s)} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) - \int_0^{t_{j-1}} Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s) \\ &= \int_0^{t_{j-1}} R(e^{-2\rho_{\beta,E}^g}(s) \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \end{aligned} \quad (4.51a)$$

$$+ \sum_{k=1}^{j-1} \left(e^{-2\rho_{\beta,E}^g(t_{k-1})} - Z_{\beta,E}(t_{k-1}) \right) \int_{t_{k-1}}^{t_k} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \quad (4.51b)$$

$$+ \sum_{k=1}^{j-1} Z_{\beta,E}(t_{k-1}) \left(\int_{t_{k-1}}^{t_k} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right. \\ \left. - \operatorname{Re} W \circ \nu_E(t_k) + \operatorname{Re} W \circ \nu_E(t_{k-1}) \right) \quad (4.51c)$$

$$- \int_0^{t_{j-1}} R(Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s). \quad (4.51d)$$

We control each of these four lines independently.

We start with the easiest ones. First, the integral in a single term of (4.51b) has bracket

$$2 \int_{t_{k-1}}^{t_k} \cos^2(2\xi_{\beta,E}^g(s) - \phi) \frac{c_E}{1-c_E s} ds \leq 2 \log(1 + E^{-p}) \leq 2E^{-p},$$

so by Bernstein's inequality, for any $x > 0$,

$$\mathbb{P} \left[\left| \int_{t_{k-1}}^{t_k} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right| \geq x \right] \leq 2 \exp \left(-\frac{E^p x^2}{4} \right).$$

Since $|e^{-2\rho_{\beta,E}^g} - Z_{\beta,E}| \leq E^{-\alpha q/3+\delta}$ on $\mathcal{G}$, it follows that for any k and any $x > 0$,

$$\begin{aligned} \mathbb{P} \left(\mathcal{G} \cap \left\{ \left| e^{-2\rho_{\beta,E}^g(t_{k-1})} - Z_{\beta,E}(t_{k-1}) \right| \left| \int_{t_{k-1}}^{t_k} \cos 2\xi_{\beta,E}^g(s) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right| \geq x \right\} \right) \\ \leq 2 \exp \left(-\frac{E^{5\alpha q/6-2\delta} x^2}{4} \right). \end{aligned}$$

Combining the tail bounds for all of the increments and using that $N \leq E^{\alpha q/6} \log E$ for E large enough, we get

$$\begin{aligned} \mathbb{P} \left(\mathcal{G} \cap \left\{ \sup_{j \in S_{E,\alpha}} \sum_{k=1}^{j-1} \left| e^{-2\rho_{\beta,E}^g(t_{k-1})} - Z_{\beta,E}(t_{k-1}) \right| \left| \int_{t_{k-1}}^{t_k} \cos 2\xi_{\beta,E}^g(s) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right| \geq x \right\} \right) \\ \leq 2E^{\alpha q/6} \log E \exp \left(-\frac{E^{3\alpha q/6-2\delta} x^2}{4 \log^2 E} \right). \end{aligned}$$

With $x = E^{-\alpha q/12+\delta}$, this bound is $O(1/\log E)$ as needed.

Then, to control (4.51c), we use the tail bound (4.26) we got in Lemma 4.4. Indeed, as $Z_{\beta,E} \leq \log E$ on $\mathcal{G}$, for any k we get that for E large enough,

$$\frac{E^{-\alpha q/12+\delta}}{NZ_{\beta,E}(t_{k-1})} \geq \frac{E^{-\alpha q/4+\delta}}{\log^2 E} \geq E^{-\alpha q/4},$$

so by Lemma 4.4,

$$\begin{aligned} \mathbb{P}\left(\mathcal{G} \cap \left\{Z_{\beta,E}(t_{k-1}) \left| \int_{t_{k-1}}^{t_k} \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right. \right. \right. \\ \left. \left. \left. - \operatorname{Re} W \circ \nu_E(t_k) + \operatorname{Re} W \circ \nu_E(t_{k-1}) \right| \geq \frac{E^{-\alpha q/12+\delta}}{N} \right\}\right) \leq 3E^{4\alpha q/3} \log^2 E \exp(-CE^{\alpha q/18}). \end{aligned}$$

In a similar way as in the last case, it follows that the supremum of (4.51c) over $j \in S_{E,\alpha}$ only exceeds $E^{-\alpha q/12+\delta}$ on $\mathcal{G}$ with probability at most $3E^{9\alpha q/6} \log^3 E \exp(-CE^{\alpha q/18})$, which again is $O(1/\log E)$.

Now, the stochastic integral in (4.51d) has quadratic variation

$$\begin{aligned} \int_0^{t_{j-1}} R^2(Z_{\beta,E})(s) d\langle \operatorname{Re} W \circ \nu_E \rangle(s) &= \sum_{k=1}^{j-1} \int_{t_{k-1}}^{t_k} (Z_{\beta,E}(s) - Z_{\beta,E}(t_{k-1}))^2 \frac{c_E}{1-c_E s} ds \\ &= \frac{4}{\beta} \sum_{k=1}^{j-1} \int_{t_{k-1}}^{t_k} \left(\int_{t_{k-1}}^s Z_{\beta,E}(u) d(\operatorname{Im} W \circ \nu_E)(u) \right)^2 \frac{c_E}{1-c_E s} ds. \end{aligned} \quad (4.52)$$

For each k , for $s \in [t_{k-1}, t_k]$ the bracket of the remaining stochastic integral is bounded on $\mathcal{G}$ by

$$\log^2 E \int_{t_{k-1}}^{t_k} \frac{c_E}{1-c_E s} ds = \log^2 E \log(1 + E^{-p}) \leq E^{-p} \log^2 E,$$

so by Bernstein's inequality, for any $y > 0$,

$$\mathbb{P}\left(\mathcal{G} \cap \left\{ \max_{1 \leq k \leq j-1} \sup_{s \in [t_{k-1}, t_k]} \left| \int_{t_{k-1}}^s Z_{\beta,E}(u) d(\operatorname{Im} W \circ \nu_E)(u) \right| \geq y \right\}\right) \leq 2(j-1) \exp\left(-\frac{E^p y^2}{2 \log^2 E}\right).$$

On the complementary event (relative to $\mathcal{G}$), the quadratic variation of the integral in (4.51d) is bounded by

$$\frac{4y^2}{\beta} \sum_{k=1}^{j-1} \int_{t_{k-1}}^{t_k} \frac{c_E}{1-c_E s} ds = \frac{4y^2}{\beta} \int_0^{t_{j-1}} \frac{c_E}{1-c_E s} ds = -\frac{4y^2}{\beta} \log(1 - c_E t_{j-1}) \leq \frac{2y^2}{\beta} \log E.$$

Hence, applying again Bernstein's inequality, we obtain that for any $x > 0$,

$$\mathbb{P}\left(\mathcal{G} \cap \left\{ \left| \int_0^{t_{j-1}} R(Z_{\beta,E})(s) d(\operatorname{Re} W \circ \nu_E)(s) \right| \geq x \right\}\right) \leq 2(j-1) \exp\left(-\frac{E^p y^2}{2 \log^2 E}\right) + 2 \exp\left(-\frac{\beta x^2}{4y^2 \log E}\right).$$

We can get the exponents to match by setting $y^2 = \sqrt{\frac{\beta \log E}{2E^p}} x$, which yields

$$\mathbb{P}\left(\mathcal{G} \cap \left\{ \left| \int_0^{t_{j-1}} R(Z_{\beta,E})(s) d(\operatorname{Re} W \circ \nu_E)(s) \right| \geq x \right\}\right) \leq 2j \exp\left(-\frac{\sqrt{\beta} E^{p/2} x}{2\sqrt{2} \log^{3/2} E}\right).$$

Taking the supremum over $j \in S_{E,\alpha}$ replaces the prefactor with $N^2 \leq E^{\alpha q/3} \log^2 E$, and with $x = E^{-\alpha q/12+\delta}$ the power of E in the exponent remains positive, so the upper bound is $O(1/\log E)$.

It only remains to bound (4.51a), which we do in the same way as in the last case. The stochastic integral in (4.51a) quadratic variation

$$2 \sum_{k=1}^{j-1} \int_{t_{k-1}}^{t_k} \left(e^{-2\rho_{\beta,E}^g(s)} - e^{-2\rho_{\beta,E}^g(t_{k-1})} \right)^2 \cos^2(2\xi_{\beta,E}^g(s) - \phi) \frac{c_E}{1 - c_E s} ds.$$

Now,

$$\left| e^{-2\rho_{\beta,E}^g(s)} - e^{-2\rho_{\beta,E}^g(t_{k-1})} \right| \leq \left| e^{-2\rho_{\beta,E}^g(s)} - Z_{\beta,E}(s) \right| + \left| Z_{\beta,E}(s) - Z_{\beta,E}(t_{k-1}) \right| + \left| Z_{\beta,E}(t_{k-1}) - e^{-2\rho_{\beta,E}^g(t_{k-1})} \right|.$$

The square of this can be bounded with Young's inequality and the bound $|e^{-2\rho_{\beta,E}^g} - Z_{\beta,E}| \leq E^{-\alpha q/3+\delta}$ which holds on $\mathcal{G}$, and this shows that the above quadratic variation is bounded on $\mathcal{G}$ by

$$4 \sum_{k=1}^{j-1} \int_{t_{k-1}}^{t_k} \left(|Z_{\beta,E}(s) - Z_{\beta,E}(t_{k-1})|^2 + 4E^{-2\alpha q/3+2\delta} \right) \frac{c_E}{1 - c_E s} ds.$$

The second term can be bounded by $16E^{-2\alpha q/3+2\delta} \log(1 - c_E t_{j-1}) \leq 8E^{-2\alpha q/3+2\delta} \log E$ once we sum up the increments. Then, the first term is exactly four times the quadratic variation (4.52) that we bounded for the case of (4.51d). Using what we found then, we get that for any $y > 0$

$$\begin{aligned} \mathbb{P}\left(\mathcal{G} \cap \left\{ 4 \sum_{k=1}^{j-1} \int_{t_{k-1}}^{t_k} \left(|Z_{\beta,E}(s) - Z_{\beta,E}(t_{k-1})|^2 + 4E^{-2\alpha q/3+2\delta} \right) \frac{c_E}{1 - c_E s} ds \geq 8 \left(\frac{y^2}{\beta} + E^{-2\alpha q/3+2\delta} \right) \log E \right\}\right) \\ \leq 2(j-1) \exp\left(-\frac{E^p y^2}{2 \log^2 E}\right). \end{aligned}$$

This controls the quadratic variation of the stochastic integral in (4.51a), so a further application of Bernstein's inequality shows that for any $x > 0$,

$$\begin{aligned} \mathbb{P}\left(\mathcal{G} \cap \left\{ \left| \int_0^{t_{j-1}} R(e^{-2\rho_{\beta,E}^g})(s) \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) \right| \geq x \right\}\right) \\ \leq 2(j-1) \exp\left(-\frac{E^p y^2}{2 \log^2 E}\right) + 2 \exp\left(-\frac{\beta x^2}{16(y^2 + \beta E^{-2\alpha q/3+2\delta}) \log E}\right). \end{aligned}$$

Setting $y^2 = \sqrt{\frac{\beta \log E}{8E^p}} x$ would make the exponents match if the extra term $\beta E^{-2\alpha q/3+2\delta}$ did not appear in the denominator of the second exponent. Nevertheless, if the y^2 part of the quadratic variation dominates, this still gives the best order for y^2 , and taking this we obtain

$$\begin{aligned} \mathbb{P}\left(\mathcal{G} \cap \left\{ \left| \int_0^{t_{j-1}} R(e^{-2\rho_{\beta,E}^g}(s)) \cos(2\xi_{\beta,E}^g(s) - \phi) \sqrt{\frac{2c_E}{1-c_E s}} dB_E(s) \right| \geq x \right\}\right) \\ \leq 2j \exp\left(-\frac{\sqrt{\beta} E^{p/2} x}{4\sqrt{2} \log^{3/2} E} \left(1 + \frac{2\sqrt{2\beta} E^{-7\alpha q/12+2\delta}}{x\sqrt{\log E}}\right)^{-1}\right). \end{aligned}$$

Taking the supremum only converts the $2j$ factor into an N^2 factor, so with $x = E^{-\alpha q/12+\delta}$, the bound remains $O(1/\log E)$ like in the other cases.

This finishes to control the supremum of the difference between the two discrete-time martingales in (4.50a), and therefore concludes the proof. $\blacksquare$

4.3.3 Proof of the vague convergence

In the last section, we showed that the process $\mathcal{C}_{\beta,E}$, which appears in the first term of the coefficient matrix $M_{\beta,E}$ from (4.23), becomes as $E \rightarrow \infty$ a hyperbolic Brownian motion driven by W . As pointed out in §4.2.3, heuristically, this means that the first term of $M_{\beta,E}$ essentially becomes the coefficient matrix of the sine system in the limit.

In this section, we complete this argument to prove the vague convergence of the canonical systems. An important ingredient that we are missing, however, is a better control on the growth of the entries of the coefficient matrix. We start by filling this gap, and then we complete the proof of the vague convergence. Finally, we conclude this section with a proof of an important consequence of the vague convergence: the compact convergence of the canonical systems' transfer matrices, which shows that their solutions also converge.

4.3.3.1 Control on the entries of the coefficient matrix

We now prove the following.

Proposition 4.8. *Let $\rho_{\beta,E}$ solve the SDE (4.21a) with $\rho_{\beta,E}(0) = 0$. For any $\varepsilon > 0$, there are constants $C, C' > 0$ independent of E such that*

$$\mathbb{P}\left[\forall t \in [0, \tau_E], \left| \rho_{\beta,E}(t) + \frac{1}{\beta} \log(1 - c_E t) \right| \leq C + C' (-\log(1 - c_E t))^{3/4}\right] > 1 - \varepsilon.$$

Remark. This gives a good control on $e^{2\rho_{\beta,E}^g}$, $e^{\rho_{\beta,E}^g + \rho_{\beta,E}^f}$ and $e^{2\rho_{\beta,E}^g}$, which are the amplitudes of the entries of the coefficient matrix $M_{\beta,E}$ on $[0, \tau_E]$. Indeed, it shows that for $k, \ell \in \mathbb{R}$ and $\varepsilon > 0$, there

are $C, C' > 0$ such that $k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t) \leq C + C'(-\log(1 - c_E t))^{3/4} - \frac{k+\ell}{\beta} \log(1 - c_E t)$ with probability at least $1 - \varepsilon$. If $k + \ell = 2$, it follows that for any $\delta > 0$, there is a $C'' > 0$ depending only on β, δ and ε such that for all $t \in [0, \tau_E]$,

$$e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \leq \frac{C''}{(1 - c_E t)^{2/\beta + \delta}} \leq \frac{C''}{(1 - t)^{2/\beta + \delta}}. \quad (4.53)$$

When $\beta > 2$, one can choose δ so that $2/\beta + \delta < 1$, and then this last upper bound is integrable on $[0, 1)$ and does not depend on E .

Proof. By definition, $\rho_{\beta,E}$ satisfies (4.21a), that is,

$$\begin{aligned} d\rho_{\beta,E}(t) = & \left(\frac{1}{\beta} + \kappa \cos 2\xi_{\beta,E}(t) + \frac{1}{\beta} \cos(4\xi_{\beta,E}(t) - 2\phi) \right) \frac{c_E}{1 - c_E t} dt \\ & + \sqrt{\frac{2}{\beta}} \sin(2\xi_{\beta,E}(t) - \phi) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t), \end{aligned}$$

with $\rho_{\beta,E}(0) = 0$, where $\xi_{\beta,E}$ solves (4.21b). Fix $\varepsilon > 0$. By Corollary 4.5.2, the $\cos 2\xi_{\beta,E}$ and $\cos(4\xi_{\beta,E} - 2\phi)$ terms are bounded by constants with probability at least $1 - \varepsilon/3$, so it suffices to show that there is a $\tilde{C} > 0$ such that

$$\mathbb{P} \left[\sup_{t \in [0, \tau_E]} \frac{1}{1 + (-\log(1 - c_E t))^{3/4}} \left| \int_0^t \sin(2\xi_{\beta,E}(s) - \phi) \sqrt{\frac{c_E}{1 - c_E s}} dB_E(s) \right| > \tilde{C} \right] < \frac{\varepsilon}{3}. \quad (4.54)$$

But

$$M(t) := \int_0^t \sin(2\xi_{\beta,E}(s) - \phi) \sqrt{\frac{c_E}{1 - c_E s}} dB_E(s)$$

is a continuous martingale on $[0, \tau_E]$ with $M(0) = 0$ and bracket

$$\langle M \rangle(t) = \int_0^t \sin^2(2\xi_{\beta,E}(s) - \phi) \frac{c_E}{1 - c_E s} ds \leq -\log(1 - c_E t),$$

so (4.54) follows from Proposition A.1. ■

As pointed out in the remark following Proposition 4.8, this result allows us to control the entries of the coefficient matrix $M_{\beta,E}$ on $[0, \tau_E]$. Because $\tau_E \rightarrow 1$, this control will be strong enough to work out the vague convergence on $[0, 1)$. However, when $\beta > 2$, we would like to prove this vague convergence on $[0, 1]$, as this will help to deduce the convergence of Weyl–Titchmarsh functions later. If $\tau_E = 1$ then Proposition 4.8 actually gives what we need, but if $\tau_E < 1$ this is not enough: we also need a bound on $(\tau_E, 1]$.

To obtain such a bound, there is unfortunately nothing in our general model to work with. For the problems we are actually mostly interested in, the only case in which we need this stronger

control is the Bessel case for $\beta > 2$ and $|a| < 1$. Indeed, recall from §4.2.2 that in that case, both the Bessel and sine systems are limit circle on the right, so we took $c_E = 1$ and $\tau_E = 1 - 1/\sqrt{E}$. We thus obtain this stronger control on the entries of the coefficient matrix only for that case.

Proposition 4.9. *Suppose that $|a| < 1$, and let $\mathcal{I}_\beta := [0, 1)$ if $\beta \leq 2$ or $\mathcal{I}_\beta := [0, 1]$ if $\beta > 2$. For every $\varepsilon > 0$, there is an $F_\varepsilon \in L^1_{\text{loc}}(\mathcal{I}_\beta)$ such that $\mathbb{P}[\eta'_{B,E}(\text{tr } G_{\beta,a,E} \circ \eta_{B,E}) \leq F_\varepsilon] \geq 1 - \varepsilon$.*

Proof. Applying Proposition 4.8 to $\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E})$, as in the remark following the proposition we know that for each $\varepsilon, \delta > 0$, there is a $C > 0$ such that with probability at least $1 - \varepsilon/2$, for all $t \in [0, \tau_E]$,

$$\eta'_{B,E}(t) \text{tr } G_{\beta,a,E} \circ \eta_{B,E}(t) \leq e^{2\rho_{\beta,a,E}^g(t)} + e^{2\rho_{\beta,a,E}^f(t)} \leq \frac{2C}{(1-t)^{2/\beta+\delta}}.$$

When $\beta > 2$, we can take δ small enough so that $2/\beta + \delta < 1$, and then the map $t \mapsto 2C(1-t)^{-2/\beta-\delta}$ is indeed in $L^1[0, 1]$. When $\beta \leq 2$, this map is in $L^1_{\text{loc}}[0, 1)$ for any value of δ , so in both cases this bound suffices on $[0, \tau_E]$. To complete the proof, we show that with probability at least $1 - \varepsilon/2$, there is an $\tilde{F}_\varepsilon \in L^1_{\text{loc}}(\mathcal{I}_\beta)$ such that $\eta'_{B,E}(t) \text{tr } G_{\beta,a,E} \circ \eta_{B,E}(t) \leq \tilde{F}_\varepsilon(t)$ for all $t \in (\tau_E, 1)$; then $F_\varepsilon(t) := \tilde{F}_\varepsilon(t) + 2C(1-t)^{-2/\beta-1}$ satisfies the desired condition.

To obtain a bound on $(\tau_E, 1)$, we work with $\tilde{f}_{\beta,a,E}(t) := f_{\beta,a,E}(t + \log E)$ and $\tilde{g}_{\beta,a,E}(t) := g_{\beta,a,E}(t + \log E)$, which solve $\tilde{\mathfrak{G}}_{\beta,a} \tilde{h} = \tilde{h}$ by Proposition 3.17, where $\tilde{\mathfrak{G}}_{\beta,a}$ is defined from the Brownian motion $\tilde{B}_E(t) := B_E(\cdot + \log E) - B_E(\log E)$, B_E being the Brownian motion from which $\mathfrak{G}_{\beta,a,E}$ is defined. Recall from §3.2.1 that 1 and $\tilde{\omega}_{\beta,a}(t) = \int_0^t \tilde{p}_{\beta,a}^{-1}(s) ds$ are a pair of fundamental solutions to $\tilde{\mathfrak{G}}_{\beta,a} \tilde{h} = 0$. By standard theory of Sturm–Liouville operators (see e.g. [29, Lemma 2.4]), it follows that the functions φ and ψ that solve

$$\varphi(t) = 1 + \int_0^t (\tilde{\omega}_{\beta,a}(s) - \tilde{\omega}_{\beta,a}(t)) \varphi(s) \tilde{w}_{\beta,a}(s) ds$$

and

$$\psi(t) = \tilde{\omega}_{\beta,a}(t) + \int_0^t (\tilde{\omega}_{\beta,a}(s) - \tilde{\omega}_{\beta,a}(t)) \psi(s) \tilde{w}_{\beta,a}(s) ds$$

are a pair of fundamental solutions for $\tilde{\mathfrak{G}}_{\beta,a} \tilde{h} = \tilde{h}$. Since $\varphi(0) = \psi'(0) = 1$ and $\varphi'(0) = \psi(0) = 0$, it follows that for $t \geq 0$,

$$\tilde{f}_{\beta,a,E}(t) = \tilde{f}_{\beta,a,E}(0)\varphi(t) + \tilde{f}'_{\beta,a,E}(0)\psi(t) = f_{\beta,a,E}(\log E)\varphi(t) + f'_{\beta,a,E}(\log E)\psi(t)$$

and likewise $\tilde{g}_{\beta,a,E}(t) = g_{\beta,a,E}(\log E)\varphi(t) + g'_{\beta,a,E}(\log E)\psi(t)$. Because $\tilde{\mathfrak{G}}_{\beta,a}$ is limit circle at infinity by Proposition 3.15, we know that $1, \varphi, \psi \in L^2([0, \infty), \tilde{w}_{\beta,a}(t) dt)$ a.s. Hence, by the Cauchy–Schwarz inequality,

$$\int_0^t \tilde{\omega}_{\beta,a}(s) \varphi(s) \tilde{w}_{\beta,a}(s) ds \leq \|\tilde{\omega}_{\beta,a}\|_{\tilde{w}_{\beta,a}} \|\varphi\|_{\tilde{w}_{\beta,a}} \quad \text{and} \quad \int_0^t \varphi(s) \tilde{w}_{\beta,a}(s) ds \leq \|1\|_{\tilde{w}_{\beta,a}} \|\varphi\|_{\tilde{w}_{\beta,a}},$$

and likewise for ψ instead of φ . These norms are all well-defined random variables, and it follows that there is a $\tilde{C} > 0$ such that with probability at least $1 - \varepsilon/6$, for all $t \geq 0$,

$$|\tilde{f}_{\beta,a,E}(t)| \leq \tilde{C}(|f_{\beta,a,E}(\log E)| + |f'_{\beta,a,E}(\log E)|)(1 + \tilde{\omega}_{\beta,a}(t)),$$

and likewise for $\tilde{g}_{\beta,a,E}$ instead of $\tilde{f}_{\beta,a,E}$. So with probability at least $1 - \varepsilon/6$, for $t \in (\tau_E, 1)$,

$$\begin{aligned} \text{tr } G_{\beta,a,E} \circ \eta_{B,E}(t) &= \frac{w_{\beta,a}(\eta_{B,E}(t))\sqrt{E}}{2} \left(\sqrt{E} \tilde{g}_{\beta,a,E}^2(\eta_{B,E}(t) - \log E) + \frac{1}{\sqrt{E}} \tilde{f}_{\beta,a,E}^2(\eta_{B,E}(t) - \log E) \right) \\ &\leq 4\tilde{C}^2 \sqrt{E} w_{\beta,a}(\eta_{B,E}(t)) \left(\sqrt{E} |g_{\beta,a,E}(\log E)|^2 + \sqrt{E} |g'_{\beta,a,E}(\log E)|^2 \right. \\ &\quad \left. + \frac{1}{\sqrt{E}} |f_{\beta,a,E}(\log E)|^2 + \frac{1}{\sqrt{E}} |f'_{\beta,a,E}(\log E)|^2 \right) (1 + \tilde{\omega}_{\beta,a}^2(\eta_{B,E}(t) - \log E)). \end{aligned}$$

Now, since $\log E = \eta_{B,E}(\tau_E)$, the representations in polar coordinates of $f_{\beta,a,E}$ and $g_{\beta,a,E}$ give

$$\sqrt{E} |g_{\beta,a,E}(\log E)|^2 \vee \sqrt{E} |g'_{\beta,a,E}(\log E)|^2 \leq \frac{e^{2\rho_{\beta,a,E}^g(\tau_E)}}{p_{\beta,a}(\log E)}$$

and

$$\frac{1}{\sqrt{E}} |f_{\beta,a,E}(\log E)|^2 \vee \frac{1}{\sqrt{E}} |f'_{\beta,a,E}(\log E)|^2 \leq \frac{e^{2\rho_{\beta,a,E}^f(\tau_E)}}{p_{\beta,a}(\log E)},$$

and the bounds on $e^{2\rho_{\beta,a,E}^f}$ and $e^{2\rho_{\beta,a,E}^g}$ that hold on $[0, \tau_E]$ by Proposition 4.8 imply that for any $\delta > 0$, there is a $\tilde{C}' > 0$ such that $e^{2\rho_{\beta,a,E}^f(\tau_E)} \vee e^{2\rho_{\beta,a,E}^g(\tau_E)} \leq \tilde{C}' E^{1/\beta+\delta/2}$ with probability at least $1 - \varepsilon/6$. Moreover, notice that by definition of $w_{\beta,a}$ and $p_{\beta,a}$,

$$\frac{w_{\beta,a}(\eta_{B,E}(t))}{p_{\beta,a}(\log E)} = \exp\left(-(a+1)\eta_{B,E}(t) + a \log E - \frac{2}{\sqrt{\beta}}(B_E(\eta_{B,E}(t)) - B_E(\log E))\right),$$

which is exactly $E^{-1} \tilde{w}_{\beta,a}(\eta_{B,E}(t) - \log E)$. This reduces the bound on the trace for $t \in (\tau_E, 1)$ to

$$\text{tr } G_{\beta,a,E} \circ \eta_{B,E}(t) \leq 16\tilde{C}^2 \tilde{C}' E^{-1/2+1/\beta+\delta/2} \tilde{w}_{\beta,a}(\eta_{B,E}(t) - \log E) \left(1 + \tilde{\omega}_{\beta,a}^2(\eta_{B,E}(t) - \log E)\right),$$

and this holds with probability at least $1 - \varepsilon/3$. By Lemma 3.16, if δ is small enough, then there is a $C_{\varepsilon,\delta} > 0$ such that with probability at least $1 - \varepsilon/6$, $\tilde{w}_{\beta,a}(t)(1 + \tilde{\omega}_{\beta,a}^2(t)) \leq C_{\varepsilon,\delta} e^{-(1-|a|-\delta/2)t}$ for all $t \geq 0$. Since $\eta_{B,E}(t) = -2 \log(1-t)$, we deduce that on this event, for any $t \geq \tau_E$,

$$\tilde{w}_{\beta,a}(\eta_{B,E}(t) - \log E) \left(1 + \tilde{\omega}_{\beta,a}^2(\eta_{B,E}(t) - \log E)\right) \leq C_{\varepsilon,\delta} E^{1-|a|-\delta/2} (1-t)^{2(1-|a|-\delta/2)}.$$

Finally, as $\eta'_{B,E}(t) = 2(1-t)^{-1}$, this shows that with probability at least $1 - \varepsilon/2$, for all $t \in (\tau_E, 1)$,

$$\eta'_{B,E}(t) \text{tr } G_{\beta,a,E} \circ \eta_{B,E}(t) \leq C E^{1/2+1/\beta-|a|} (1-t)^{1-2|a|-\delta} \quad (4.55)$$

for $C := 32\tilde{C}^2\tilde{C}'C_{\varepsilon,\delta}$. Notice that for $t \geq \tau_E$, $1-t \leq 1/\sqrt{E}$ so $E \leq (1-t)^{-2}$. Hence, if $1/2 + 1/\beta - |a| > 0$, then (4.55) is bounded by $C(1-t)^{-2/\beta-\delta}$, which can be made integrable on $[0, 1)$ when $\beta > 2$ by taking δ small enough that $2/\beta + \delta < 1$. On the other hand, if $1/2 + 1/\beta - |a| \leq 0$, then the power of E in (4.55) is nonpositive, so (4.55) is bounded by $C(1-t)^{-(2|a|-1+\delta)}$, which is integrable on $[0, 1)$ for δ small enough since $0 < 2|a| - 1 < 1$ by the assumptions on a . This argument gives the bound we need: it is always $L^1_{\text{loc}}[0, 1)$, and can always be made $L^1[0, 1)$ when $\beta > 2$ by taking δ small enough. ■

4.3.3.2 Vague convergence of the coefficient matrices

We now have all the ingredients to prove the vague convergence.

Theorem 4.10. *Let $\{B_{E_n}\}_{n \in \mathbb{N}}$ and W be the Brownian motions from Lemma 4.4, and let R_β be the coefficient matrix of the sine system driven by W . Fix $c_{E_n} \in [1 - 1/\sqrt{E_n}, 1]$ and $\tau_{E_n} := \frac{1}{c_{E_n}}(1 - 1/\sqrt{E_n})$, and let M_{β, E_n} be a coefficient matrix on $[0, 1)$ given by (4.23) on $[0, \tau_{E_n}]$ with polar coordinates solving the SDEs (4.21) driven by B_{E_n} . Suppose that one of the following conditions hold:*

- i) $\beta \leq 2$ and $c_{E_n} = 1$,
- ii) $\beta > 2$ and for every $\varepsilon > 0$, there is an $F_\varepsilon \in L^1[0, 1]$ such that $\mathbb{P}[\text{tr } M_{\beta, E_n} \leq F_\varepsilon] \geq 1 - \varepsilon$ for all E_n large enough.

Set $\mathcal{I}_\beta := [0, 1)$ if $\beta \leq 2$ and $\mathcal{I}_\beta := [0, 1]$ if $\beta > 2$. Then for any $\varphi \in \mathcal{C}_c(\mathcal{I}_\beta, \mathbb{C}^2)$,

$$\int_{\mathcal{I}_\beta} \varphi^*(t) (M_{\beta, E_n}(t) - R_\beta \circ \nu(t)) \varphi(t) dt \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0. \quad (4.56)$$

In particular, $M_{\beta, E_n} \rightarrow R_\beta \circ \nu$ vaguely on $\mathcal{I}_\beta$ in probability and in distribution.

Given what we already know about the Airy and Bessel systems, Theorem 1.9 is a direct consequence of Theorem 4.10 for $M_{\beta, E} = \eta'_{A, E}(H_{\beta, E} \circ \eta_{A, E})$ or $M_{\beta, E} = \eta'_{B, E}(G_{\beta, a, E} \circ \eta_{B, E})$. Indeed, in the Airy case and in the Bessel case with $a \geq 1$, when $\beta > 2$ we chose $c_E = 1 - 1/\sqrt{E}$ so $\tau_E = 1$, and thus the hypothesis on the trace of the coefficient matrix is satisfied by Proposition 4.8. In the Bessel case with $|a| < 1$, the bound from Proposition 4.8 does not hold on all of $[0, 1]$, but now Proposition 4.9 shows that the hypothesis is satisfied.

Before moving on to the proof, we point out the following important corollary, which also implies Corollary 1.9.1.

Corollary 4.10.1. *In the setting of Theorem 4.10, let $T_{M_{\beta, E_n}}, T_{\tilde{R}_\beta} : \mathcal{I}_\beta \times \mathbb{C} \rightarrow \mathbb{C}^2$ be the transfer matrices of the canonical systems with coefficient matrices M_{β, E_n} and $\tilde{R}_\beta := R_\beta \circ \nu$, both defined on the probability space from Lemma 4.4. Then $T_{M_{\beta, E_n}} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} T_{\tilde{R}_\beta}$ compactly on $\mathcal{I}_\beta \times \mathbb{C}$ as $n \rightarrow \infty$.*

Proof. Given the theorem, this result is essentially a consequence of the continuity of the map from coefficient to transfer matrices on trace-dominated sets.

When $\beta > 2$, there are high-probability dominating functions on $\text{tr } M_{\beta, E_n}$ by hypothesis and on $\text{tr } \tilde{R}_\beta$ by Lemma 3.30, and therefore the result follows from Proposition 2.25.

When $\beta \leq 2$, it suffices to verify that $T_{M_{\beta, E_n}} \xrightarrow{\mathbb{P}} T_{\tilde{R}_\beta}$ compactly on $K \times \mathbb{C}$ for any compact $K \subset [0, 1)$, by Proposition 2.24. But the high-probability dominating functions on $\text{tr } M_{\beta, E_n}$ from Proposition 4.8 and on $\text{tr } \tilde{R}_\beta$ from Lemma 3.30 are both integrable on any compact $K \subset [0, 1)$, so Proposition 2.25 shows that $T_{M_{\beta, E_n}} \xrightarrow{\mathbb{P}} T_{\tilde{R}_\beta}$ compactly on $K \times \mathbb{C}$. ■

Proof of Theorem 4.10. To prove that (4.56) holds, we first simplify the problem in two ways: we argue that we can replace the time interval $\mathcal{I}_\beta$ with $\mathcal{I}_{E, \alpha} := [0, (1 - E^{-1/2+\alpha})/c_E] \subset [0, \tau_E]$ for a suitable $\alpha \in (\frac{3}{\beta q+6}, \frac{1}{2})$, and we replace the sine system's logarithmic time $\nu(t) = -\log(1-t)$ with $\nu_E(t) = -\log(1 - c_E t)$. Then, recall from (4.23) that $M_{\beta, E}$ can be written on $[0, \tau_E]$ as

$$M_{\beta, E} = \frac{c_E}{2 \text{Im } \ell_{\beta, E}} \begin{pmatrix} 1 & -\text{Re } \ell_{\beta, E} \\ -\text{Re } \ell_{\beta, E} & |\ell_{\beta, E}|^2 \end{pmatrix} + \frac{c_E}{2} \begin{pmatrix} e^{2\rho_{\beta, E}^g} \cos 2\xi_{\beta, E}^g & e^{\Sigma_{\beta, E}^p} \cos \Sigma_{\beta, E}^\xi \\ e^{\Sigma_{\beta, E}^p} \cos \Sigma_{\beta, E}^\xi & e^{2\rho_{\beta, E}^f} \cos 2\xi_{\beta, E}^f \end{pmatrix}. \quad (4.57)$$

From this, the rest of the proof is split in two main parts: first we show that (4.56) holds when $M_{\beta, E}$ is replaced with the first term of (4.57), and then we show that the second term converges vaguely to zero. Note that once (4.56) is proven, that $M_{\beta, E} \rightarrow R_\beta \circ \nu$ vaguely on $\mathcal{I}_\beta$ in probability follows directly from the general criterion for the vague convergence of random canonical systems from Proposition 2.23.

In order to simplify the notation throughout the proof, we omit the explicit indexing of the shift sequence $\{E_n\}_{n \in \mathbb{N}}$. In particular, any limit as $E \rightarrow \infty$ is taken along that sequence and should really be understood as a limit as $n \rightarrow \infty$.

Step 1: Shortening the time interval. We start by showing that instead of integrating up to time 1 in (4.56) we can also integrate only up to time $(1 - E^{-1/2+\alpha})/c_E$ for some fixed $\alpha \in (\frac{3}{\beta q+6}, \frac{1}{2})$, i.e., that

$$\int_{(1-E^{-1/2+\alpha})/c_E}^1 \varphi^*(t) (M_{\beta, E}(t) - R_\beta \circ \nu(t)) \varphi(t) dt \xrightarrow[E \rightarrow \infty]{\mathbb{P}} 0. \quad (4.58)$$

Notice that if $\beta \leq 2$, then φ is compactly supported in $[0, 1)$ so this statement is trivial, as for any E large enough, the whole interval $[(1 - E^{-1/2+\alpha})/c_E, 1]$ has left the support of φ .

To show that (4.58) also holds when $\beta > 2$, notice that for any $\varepsilon > 0$, Lemma 3.30 and the hypothesis of the theorem provide a function $F_\varepsilon \in L^1[0, 1)$ such that for all E large enough, $\text{tr } M_{\beta, E} + \text{tr } R_\beta \circ \nu \leq F_\varepsilon$ with probability at least $1 - \varepsilon$. On this event,

$$\left| \int_{(1-E^{-1/2+\alpha})/c_E}^1 \varphi^*(t) (M_{\beta, E}(t) - R_\beta \circ \nu(t)) \varphi(t) dt \right| \leq \|\varphi\|_\infty^2 \int_{(1-E^{-1/2+\alpha})/c_E}^1 F_\varepsilon(t) dt \rightarrow 0.$$

Hence, the left-hand side converges to 0 in probability as $E \rightarrow \infty$, and it suffices to prove (4.56) with $\mathcal{I}_\beta$ replaced with $\mathcal{I}_{E,\alpha} = [0, (1 - E^{-1/2+\alpha})/c_E]$.

Step 2: Changing the logarithmic time scale of the sine system. We now change the time scale of the sine system from $v(t) = -\log(1-t)$ to $v_E(t) = -\log(1-c_E t)$. Note that by hypothesis, $c_E = 1$ when $\beta \leq 2$ so $v_E = v$ in that case, and there is only something to prove when $\beta > 2$.

To see that the time scale can be changed when $\beta > 2$, note that for $\varphi \in \mathcal{C}_c([0, 1], \mathbb{C}^2)$,

$$\begin{aligned} \int_0^{(1-E^{-1/2+\alpha})/c_E} \varphi^*(t) (R_\beta \circ v(t) - R_\beta \circ v_E(t)) \varphi(t) dt &= \int_{1-E^{-1/2+\alpha}}^{(1-E^{-1/2+\alpha})/c_E} \varphi^*(t) R_\beta \circ v(t) \varphi(t) dt \\ &\quad + \int_0^{1-E^{-1/2+\alpha}} \left(\varphi^*(t) R_\beta \circ v(t) \varphi(t) - \frac{1}{c_E} \varphi^*(t/c_E) R_\beta \circ v(t) \varphi(t/c_E) \right) dt. \end{aligned}$$

where we changed variables from t to $v^{-1} \circ v_E(t) = c_E t$ in the second term of the second integral. Since $\varphi^*(R_\beta \circ v) \varphi$ is a.s. integrable, the first integral goes to zero a.s. since the size of the interval of integration is at most $(1 - E^{-1/2+\alpha})/\sqrt{E} \rightarrow 0$. Then, since R_β is real and positive semi-definite, the second integral can be written as

$$\operatorname{Re} \int_0^{1-E^{-1/2+\alpha}} \left(\varphi^*(t) + \frac{1}{\sqrt{c_E}} \varphi^*(t/c_E) \right) R_\beta \circ v(t) \left(\varphi(t) - \frac{1}{\sqrt{c_E}} \varphi(t/c_E) \right) dt.$$

The integrand is dominated by $\frac{4}{c_E} \|\varphi\|_\infty^2 \operatorname{tr}(R_\beta \circ v) \leq 8 \|\varphi\|_\infty^2 \operatorname{tr}(R_\beta \circ v)$ for E large enough, and this bound is a.s. integrable. Hence, by dominated convergence and continuity of φ , this integral vanishes a.s. in the limit.

In any case, this reduces the problem to proving that

$$\int_{\mathcal{I}_{E,\alpha}} \varphi^*(t) (M_{\beta,E}(t) - R_\beta \circ v_E(t)) \varphi(t) dt \xrightarrow[E \rightarrow \infty]{\mathbb{P}} 0.$$

Step 3: Convergence of the first term to the sine system's coefficient matrix. We now compare the first term of $M_{\beta,E}$ in the representation (4.57) to that of the sine system. Recall that by definition,

$$R_\beta = \frac{1}{2 \det X_\beta} X_\beta^\top X_\beta \quad \text{with} \quad X_\beta := \begin{pmatrix} 1 & -\operatorname{Re} B_\beta \\ 0 & \operatorname{Im} B_\beta \end{pmatrix}$$

where B_β is a hyperbolic Brownian motion with variance $4/\beta$ started at i in the upper half-plane and driven by the complex Brownian motion W . The first term of $M_{\beta,E}$ has the same structure: it is

$$\frac{c_E}{2 \operatorname{Im} \ell_{\beta,E}} \begin{pmatrix} 1 & -\operatorname{Re} \ell_{\beta,E} \\ -\operatorname{Re} \ell_{\beta,E} & |\ell_{\beta,E}|^2 \end{pmatrix} = \frac{c_E}{2 \det Y_{\beta,E}} Y_{\beta,E}^\top Y_{\beta,E} \quad \text{with} \quad Y_{\beta,E} := \begin{pmatrix} 1 & -\operatorname{Re} \ell_{\beta,E} \\ 0 & \operatorname{Im} \ell_{\beta,E} \end{pmatrix}.$$

Using these representations and setting $\tilde{X}_{\beta,E} := X_\beta \circ \nu_E$, we can write

$$\begin{aligned} \frac{c_E}{2 \det Y_{\beta,E}} Y_{\beta,E}^\top Y_{\beta,E} - \frac{1}{2 \det \tilde{X}_{\beta,E}} \tilde{X}_{\beta,E}^\top \tilde{X}_{\beta,E} &= \left(\frac{c_E}{2 \det Y_{\beta,E}} - \frac{1}{2 \det \tilde{X}_{\beta,E}} \right) \tilde{X}_{\beta,E}^\top \tilde{X}_{\beta,E} \\ &+ \frac{c_E}{2 \det Y_{\beta,E}} \left((Y_{\beta,E} - \tilde{X}_{\beta,E})^\top (Y_{\beta,E} - \tilde{X}_{\beta,E}) + (Y_{\beta,E} - \tilde{X}_{\beta,E})^\top \tilde{X}_{\beta,E} + \tilde{X}_{\beta,E}^\top (Y_{\beta,E} - \tilde{X}_{\beta,E}) \right). \end{aligned}$$

If we sandwich this matrix between φ^* and φ , we get

$$\begin{aligned} &\left| \varphi^* \left(\frac{c_E}{2 \det Y_{\beta,E}} Y_{\beta,E}^\top Y_{\beta,E} - \frac{1}{2 \det \tilde{X}_{\beta,E}} \tilde{X}_{\beta,E}^\top \tilde{X}_{\beta,E} \right) \varphi \right| \\ &\leq \left| \frac{c_E}{2 \det Y_{\beta,E}} - \frac{1}{2 \det \tilde{X}_{\beta,E}} \right| \|\tilde{X}_{\beta,E} \varphi\|_2^2 \\ &\quad + \frac{c_E}{2 \det Y_{\beta,E}} \left(\|(Y_{\beta,E} - \tilde{X}_{\beta,E}) \varphi\|_2^2 + 2 \left| \operatorname{Re} \langle \tilde{X}_{\beta,E} \varphi, (Y_{\beta,E} - \tilde{X}_{\beta,E}) \varphi \rangle \right| \right) \\ &\leq 2 \|\varphi\|_2^2 \left| \frac{1}{\det Y_{\beta,E}} - \frac{1}{\det \tilde{X}_{\beta,E}} \right| \|\tilde{X}_{\beta,E}\|_{\max}^2 + \frac{2 \|\varphi\|_2^2}{\sqrt{E} \det Y_{\beta,E}} \|\tilde{X}_{\beta,E}\|_{\max}^2 \\ &\quad + \frac{2 \|\varphi\|_2^2}{\det Y_{\beta,E}} \|Y_{\beta,E} - \tilde{X}_{\beta,E}\|_{\max}^2 + \frac{4 \|\varphi\|_2^2}{\det Y_{\beta,E}} \|\tilde{X}_{\beta,E}\|_{\max} \|Y_{\beta,E} - \tilde{X}_{\beta,E}\|_{\max} \end{aligned} \quad (4.59)$$

where $\|A\|_{\max} := \max_{j,k \in \{1,2\}} |A_{jk}|$ for $A \in \mathbb{R}^{2 \times 2}$, and where we used that $|c_E - 1| \leq 1/\sqrt{E}$ to split the first term. From the results of §4.3.2, it is not hard to find a good event on which (4.59) is small.

Indeed, recall that solving the SDE for $\operatorname{Im} \mathcal{B}_\beta$ shows that $\operatorname{Im} \mathcal{B}_\beta \circ \nu_E = Z_{\beta,E}$, with $Z_{\beta,E}$ defined as in Proposition 4.6, and then the SDE for $\operatorname{Re} \mathcal{B}_\beta$ gives

$$\operatorname{Re} \mathcal{B}_\beta \circ \nu_E(t) = \frac{2}{\sqrt{\beta}} \int_0^t Z_{\beta,E}(s) d(\operatorname{Re} W \circ \nu_E)(s).$$

Thus, Corollary 4.6.1 shows that for E large enough,

$$\mathbb{P} \left[\sup_{t \in I_{E,\alpha}} |\operatorname{Im} \mathcal{C}_{\beta,E}(t) - \operatorname{Im} \mathcal{B}_\beta \circ \nu_E(t)| \leq E^{-\alpha q/4} \right] \geq 1 - \frac{2}{\log E}. \quad (4.60a)$$

Likewise, Proposition 4.7 shows that for some absolute $C > 0$ and E large enough,

$$\mathbb{P} \left[\sup_{t \in I_{E,\alpha}} |\operatorname{Re} \mathcal{C}_{\beta,E} - \operatorname{Re} \mathcal{B}_\beta \circ \nu_E(t)| \leq E^{-\alpha q/13} \right] \geq 1 - \frac{C}{\log E}. \quad (4.60b)$$

Together, these two statements allow us to control $\|Y_{\beta,E} - \tilde{X}_{\beta,E}\|_{\max}$. Similarly, Corollary 4.6.1 also shows that for E large enough,

$$\mathbb{P} \left[\sup_{t \in I_{E,\alpha}} \left| \frac{1}{\det Y_{\beta,E}(t)} - \frac{1}{\det \tilde{X}_{\beta,E}(t)} \right| \leq E^{-\delta} \right] \geq 1 - 2 \exp \left(-\frac{\beta \delta^2}{36(1-2\alpha)} \log E \right) \quad (4.60c)$$

where $\delta := \frac{1}{6\beta} (\alpha(\beta q + 6) - 1) > 0$.

Then, recall that we have seen in (4.47) that for E large enough,

$$\mathbb{P}\left[\sup_{t \in I_{E,\alpha}} \operatorname{Im} \mathcal{B}_\beta \circ \nu_E(t) \geq \log E\right] \leq \frac{2}{\log E}.$$

On the complementary event, for any $t \in [0, \tau_E]$,

$$\langle \operatorname{Re} \mathcal{B}_\beta \circ \nu_E \rangle(t) = \frac{4}{\beta} \int_0^t |\operatorname{Im} \mathcal{B}_\beta \circ \nu_E(s)|^2 d\langle \operatorname{Re} W \circ \nu_E \rangle(s) \leq \frac{4}{\beta} \log^2 E \int_0^t \frac{c_E}{1 - c_E s} ds \leq \frac{2}{\beta} \log^3 E,$$

so by Bernstein's inequality, for any $x > 0$,

$$\mathbb{P}\left(\left\{\sup_{t \in I_{E,\alpha}} |\operatorname{Re} \mathcal{B}_\beta \circ \nu_E(t)| \geq x\right\} \cap \left\{\sup_{t \in I_{E,\alpha}} \operatorname{Im} \mathcal{B}_\beta \circ \nu_E(t) < \log E\right\}\right) \leq 2 \exp\left(-\frac{\beta x^2}{4 \log^3 E}\right).$$

Taking $x = \log^2 E$, the bound on the probability becomes a negative power of E , and it follows that for E large enough,

$$\mathbb{P}\left[\sup_{t \in I_{E,\alpha}} \|\tilde{X}_{\beta,E}(t)\|_{\max} \leq \log^2 E\right] \geq 1 - \frac{3}{\log E}. \quad (4.60d)$$

Finally, Proposition 4.8 (in particular (4.53) in the remark) shows that for any $\varepsilon, \delta > 0$, there is a $C_{\varepsilon,\delta} > 0$ such that for all E large enough,

$$\mathbb{P}\left[\forall t \in [0, \tau_E], \frac{1}{\det Y_{\beta,E}(t)} \leq \frac{C_\varepsilon}{(1-t)^{2/\beta+\delta}}\right] \geq 1 - \frac{\varepsilon}{2}. \quad (4.60e)$$

Now, given $\varepsilon, \delta > 0$, we can set $\mathcal{G}_{E,\varepsilon,\delta}$ to be the intersection of all of the good events in (4.60), and $\mathbb{P}(\mathcal{G}_{E,\varepsilon,\delta}) \geq 1 - C/\log E - \varepsilon/2$ for any E large enough, where $C > 0$ is an absolute constant. To conclude, choose $\delta > 0$ arbitrary if $\beta \leq 2$ but such that $2/\beta + \delta < 1$ if $\beta > 2$. On $\mathcal{G}_{E,\varepsilon,\delta}$, using the representation (4.59), we can bound

$$\begin{aligned} & \left| \int_{I_{E,\alpha}} \varphi^*(t) \left(\frac{c_E}{2 \det Y_{\beta,E}} Y_{\beta,E}^\top Y_{\beta,E} - \frac{1}{2 \det \tilde{X}_{\beta,E}} \tilde{X}_{\beta,E}^\top \tilde{X}_{\beta,E} \right) (t) \varphi(t) dt \right| \\ & \leq 2E^{-\delta} \log^4 E \int_{I_{E,\alpha}} \|\varphi(t)\|_2^2 dt + 2C_\varepsilon E^{-1/2} \log^4 E \int_{I_{E,\alpha}} \frac{\|\varphi(t)\|_2^2}{(1-t)^{2/\beta+\delta}} dt \\ & \quad + 2C_\varepsilon E^{-2\alpha q/13} \int_{I_{E,\alpha}} \frac{\|\varphi(t)\|_2^2}{(1-t)^{2/\beta+\delta}} dt + 4C_\varepsilon E^{-\alpha q/13} \log^2 E \int_{I_{E,\alpha}} \frac{\|\varphi(t)\|_2^2}{(1-t)^{2/\beta+\delta}} dt. \end{aligned} \quad (4.61)$$

All of these integrals are finite. Indeed, if $\beta \leq 2$, then the integrands are all bounded because the support of φ is compact in $[0, 1)$, while if $\beta > 2$, then the integrands are all dominated by $\|\varphi\|_\infty^2 (1-t)^{-2/\beta-\delta}$, which is integrable by our choice of δ . Therefore, the right-hand side here vanishes as $E \rightarrow \infty$. Since $\mathbb{P}(\mathcal{G}_{E,\varepsilon,\delta}) \geq 1 - C/\log E - \varepsilon/2$, it follows that (4.61) converges to 0 in probability as $E \rightarrow \infty$.

Step 4: Convergence of the second term to zero. It remains to prove that the second term of the coefficient matrix (4.57) does not contribute to the limit. More precisely, given the simplifications made in the first steps of the proof, it remains to show that given an arbitrary $\varphi \in \mathcal{C}_c(\mathcal{I}_\beta, \mathbb{C}^2)$, the integral on $\mathcal{I}_{E,\alpha}$ of the second term of (4.57) sandwiched between φ^* and φ vanishes as $E \rightarrow \infty$. By linearity, it suffices to show that if $\varphi \in \mathcal{C}_c(\mathcal{I}_\beta, \mathbb{R})$,

$$\int_0^{\tau_E} \varphi(t) e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \cos(k\xi_{\beta,E}^g(t) + \ell\xi_{\beta,E}^f(t)) dt \xrightarrow[E \rightarrow \infty]{\mathbb{P}} 0 \quad (4.62)$$

for $k, \ell \in \mathbb{N}_0$ with $k + \ell = 2$. To prove (4.62), we work on an event on which $e^{k\rho_{\beta,E}^g + \ell\rho_{\beta,E}^f}$ can be controlled. As in other cases, Proposition 4.8 guarantees that given $\varepsilon > 0$, there is a $C_\varepsilon > 0$ independent of E such that the event

$$\mathcal{H}_{E,\varepsilon} := \left\{ \forall t \in [0, \tau_E], e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \leq \frac{C_\varepsilon}{(1 - c_E t)^{2/\beta + \gamma}} \right\} \quad \text{where} \quad \gamma := \begin{cases} 1/4 - 1/2\beta & \text{if } \beta > 2, \\ 1/4 & \text{if } \beta \leq 2 \end{cases} \quad (4.63)$$

has $\mathbb{P}(\mathcal{H}_{E,\varepsilon}) \geq 1 - \varepsilon/2$ for all E large enough. Note that when $\beta > 2$, γ is chosen so that $2/\beta + \gamma = 1/4 + 3/2\beta < 1$ and $t \mapsto (1 - t)^{-2/\beta - \gamma}$ is integrable on $[0, 1]$.

To prove (4.62), we replace the test function φ by a piecewise constant approximation. To do so, we discretize the time interval with a partition $\{t_j\}_{j=0}^N$ of $[0, \tau_E]$ defined by setting $t_0 := 0$, and then recursively

$$c_E t_j := c_E t_{j-1} + \frac{\pi E^p}{2c_E E^{q/2} (1 - c_E t_{j-1})^{q-1}} \quad \text{where} \quad p := \begin{cases} 1/4 + 1/2\beta & \text{if } \beta > 2, \\ 1/4 & \text{if } \beta \leq 2, \end{cases} \quad (4.64)$$

until this would give $c_E t_j > 1 - 1/\sqrt{E}$, in which case we set $j = N$ and $t_N := \tau_E$ (note that this does happen in a finite number of steps because what is added to $c_E t_{j-1}$ is always at least $\frac{\pi}{2} E^{-q/2+p}$).

Then, define $\hat{\varphi}: \mathcal{I}_\beta \rightarrow \mathbb{R}$ as

$$\hat{\varphi} \equiv \hat{\varphi}_j := \frac{1}{t_j - t_{j-1}} \int_{t_{j-1}}^{t_j} \varphi(s) ds \quad \text{on} \quad [t_{j-1}, t_j] \quad (4.65)$$

for each $j \leq N$, and then $\hat{\varphi}(t_N) := \varphi(\tau_E)$ and $\hat{\varphi} \equiv 0$ on $(\tau_E, 1]$.

Since φ is continuous and compactly supported in $\mathcal{I}_\beta$, $\hat{\varphi}$ is compactly supported in $\mathcal{I}_\beta$, and it is also bounded with $\|\hat{\varphi}\|_\infty \leq \|\varphi\|_\infty$. Moreover, since $\hat{\varphi}_j$ is the average of φ in $[t_{j-1}, t_j]$, then for $t \in [t_{j-1}, t_j]$, it is clear that $|\varphi(t) - \hat{\varphi}_j| \leq \omega_\varphi(|t_j - t_{j-1}|)$ where ω_φ denotes a modulus of continuity for φ . By definition of the t_j 's, this means that

$$|\varphi(t) - \hat{\varphi}_j| \leq \omega_\varphi \left(\frac{\pi E^p}{2c_E^2 E^{q/2} (1 - c_E t_{j-1})^{q-1}} \right) \leq \omega_\varphi \left(\frac{\pi}{2c_E^2 E^{1/2-p}} \right),$$

and as this estimate does not depend on j , it is in fact a bound on $\|(\varphi - \hat{\varphi})\mathbb{1}_{[0, \tau_E]}\|_\infty$. It follows that on $\mathcal{H}_{E, \varepsilon}$,

$$\left| \int_0^{\tau_E} e^{k\rho_{\beta, E}^g + \ell\rho_{\beta, E}^f} \cos(k\xi_{\beta, E}^g + \ell\xi_{\beta, E}^f) (\varphi - \hat{\varphi}) d\mathcal{L} \right| \leq C_\varepsilon \omega_\varphi \left(\frac{\pi}{2c_E^2 E^{1/2-p}} \right) \int_0^{\tau_E} \frac{\mathbb{1}_{\text{supp } \hat{\varphi}}(t)}{(1-t)^{2/\beta+\gamma}} dt,$$

and the remaining integral is always finite, either because $\text{supp } \hat{\varphi}$ is compact (when $\beta \leq 2$) or because $2/\beta + \gamma < 1$ (when $\beta > 2$). Since $p < 1/2$ in any case, the right-hand side vanishes as $E \rightarrow \infty$ and $\mathbb{P}(\mathcal{H}_{E, \varepsilon}) \geq 1 - \varepsilon/2$, and this shows that the integral converges to zero in probability as $E \rightarrow \infty$. Therefore it suffices to prove that (4.62) holds with φ replaced with $\hat{\varphi}$.

Now, on any interval $[t_{j-1}, t_j)$, since $\hat{\varphi}$ is constant, we can integrate by parts. Recall that

$$d\rho_{\beta, E}(t) = R_{\beta, E}^\rho(t) \frac{c_E}{1 - c_E t} dt + S_{\beta, E}^\rho(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t)$$

and

$$d\xi_{\beta, E}(t) = -2c_E E^{q/2} (1 - c_E t)^{q-1} dt + R_{\beta, E}^\xi(t) \frac{c_E}{1 - c_E t} dt + S_{\beta, E}^\xi(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t)$$

where $R_{\beta, E}^\rho$, $S_{\beta, E}^\rho$, $R_{\beta, E}^\xi$ and $S_{\beta, E}^\xi$ are constants plus linear combinations of trigonometric functions of multiples of $\xi_{\beta, E}$, which are always bounded by constants. Starting from these Itô differentials, an application of Itô's formula shows that if $k + \ell = 2$, then

$$\begin{aligned} & e^{-k\rho_{\beta, E}^g(t) - \ell\rho_{\beta, E}^f(t)} d\left(e^{k\rho_{\beta, E}^g + \ell\rho_{\beta, E}^f} \sin(k\xi_{\beta, E}^g + \ell\xi_{\beta, E}^f)\right)(t) \\ &= -4c_E E^{q/2} (1 - c_E t)^{q-1} \cos(k\xi_{\beta, E}^g(t) + \ell\xi_{\beta, E}^f(t)) dt \\ & \quad + R_{\beta, E}^\Sigma(t) \frac{c_E}{1 - c_E t} dt + S_{\beta, E}^\Sigma(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \end{aligned} \quad (4.66)$$

where $R_{\beta, E}^\Sigma$ and $S_{\beta, E}^\Sigma$ are again processes which are bounded by constants uniformly in E . It follows that for each j ,

$$\begin{aligned} & \int_{t_{j-1}}^{t_j} e^{k\rho_{\beta, E}^g(t) + \ell\rho_{\beta, E}^f(t)} \cos(k\xi_{\beta, E}^g(t) + \ell\xi_{\beta, E}^f(t)) dt \\ &= -\frac{1}{4c_E E^{q/2}} \int_{t_{j-1}}^{t_j} \frac{1}{(1 - c_E t)^{q-1}} \left(d\left(e^{k\rho_{\beta, E}^g + \ell\rho_{\beta, E}^f} \sin(k\xi_{\beta, E}^g + \ell\xi_{\beta, E}^f)\right)(t) \right. \\ & \quad \left. - e^{k\rho_{\beta, E}^g(t) + \ell\rho_{\beta, E}^f(t)} R_{\beta, E}^\Sigma(t) \frac{c_E}{1 - c_E t} dt - e^{k\rho_{\beta, E}^g(t) + \ell\rho_{\beta, E}^f(t)} S_{\beta, E}^\Sigma(t) \sqrt{\frac{c_E}{1 - c_E t}} dB_E(t) \right). \end{aligned}$$

Integrating by parts and summing up the increments, we get

$$\begin{aligned} & \int_0^{\tau_E} \hat{\varphi}(t) e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \cos(k\xi_{\beta,E}^g(t) + \ell\xi_{\beta,E}^f(t)) dt \\ &= \frac{-1}{4c_E E^{q/2}} \sum_{j=1}^N \hat{\varphi}_j \frac{e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \sin(\xi_{\beta,E}^g(t) + \ell\xi_{\beta,E}^f(t))}{(1 - c_E t)^{q-1}} \Big|_{t_{j-1}}^{t_j} \end{aligned} \quad (4.67a)$$

$$\begin{aligned} &+ \frac{1}{4c_E E^{q/2}} \int_0^{\tau_E} \hat{\varphi}(t) e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \left((q-1) \sin(k\xi_{\beta,E}^g(t) + \ell\xi_{\beta,E}^f(t)) \right. \\ &\quad \left. + R_{\beta,E}^\Sigma(t) \right) \frac{c_E}{(1 - c_E t)^q} dt \end{aligned} \quad (4.67b)$$

$$+ \frac{1}{4c_E E^{q/2}} \int_0^{\tau_E} \hat{\varphi}(t) e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} S_{\beta,E}^\Sigma(t) \frac{\sqrt{c_E}}{(1 - c_E t)^{q-1/2}} dB_E(t). \quad (4.67c)$$

To control (4.67), recall that on $\mathcal{H}_{E,\varepsilon}$, for any $t \in [0, \tau_E]$, $e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \leq C_\varepsilon (1 - c_E t)^{-2/\beta - \gamma}$. By definition of τ_E , this upper bound is always bounded by $C_\varepsilon E^{1/\beta + \gamma/2}$. When $\beta > 2$, we defined $\gamma = 1/4 - 1/2\beta$ and $p = 1/4 + 1/2\beta$, so $1/\beta + \gamma/2 = p - \gamma/2$. When $\beta \leq 2$, we took $p = \gamma = 1/4$ so $p - \gamma/2 > 0$ and it is still true that $(1 - c_E t)^{-2/\beta - \gamma} \leq E^{p - \gamma/2}$ for E large enough and $t \in \text{supp } \hat{\varphi}$, as the latter set is compact in $[0, 1)$. This shows that in any case,

$$e^{k\rho_{\beta,E}^g(t) + \ell\rho_{\beta,E}^f(t)} \leq C_\varepsilon E^{p - \gamma/2} \quad \text{where} \quad p - \frac{\gamma}{2} = \begin{cases} 1/8 + 3/4\beta & \text{if } \beta > 2, \\ 1/8 & \text{if } \beta \leq 2 \end{cases} \quad (4.68)$$

for $t \in \text{supp } \hat{\varphi}$ and E large enough.

Now, on $\mathcal{H}_{E,\varepsilon}$, the estimate (4.68) shows that (4.67a) is bounded for E large enough by

$$\begin{aligned} & \frac{C_\varepsilon \|\varphi\|_\infty E^{p - \gamma/2}}{4c_E E^{q/2}} \sum_{j=1}^N \left(\frac{1}{(1 - c_E t_j)^{q-1}} + \frac{1}{(1 - c_E t_{j-1})^{q-1}} \right) \\ &= \frac{C_\varepsilon \|\varphi\|_\infty E^{p - \gamma/2}}{4c_E E^{q/2}} \left(1 + 2 \sum_{j=1}^{N-1} \frac{1}{(1 - c_E t_j)^{q-1}} + E^{(q-1)/2} \right). \end{aligned}$$

The first and last terms vanish as $E \rightarrow \infty$, since they are of order $E^{-q/2 + p - \gamma/2}$ and $E^{-1/2 + p - \gamma/2}$ respectively while $p - \gamma/2 < 1/2$ in any case. Then, by definition (4.64) of the times t_j , the second term also vanishes as $E \rightarrow \infty$:

$$\frac{C_\varepsilon \|\varphi\|_\infty}{\pi E^{\gamma/2}} \sum_{j=1}^{N-1} \frac{\pi E^p}{2c_E E^{q/2} (1 - c_E t_j)^{q-1}} \leq \frac{C_\varepsilon \|\varphi\|_\infty}{\pi E^{\gamma/2}} c_E t_N \rightarrow 0.$$

As this holds on $\mathcal{H}_{E,\varepsilon}$, this shows that (4.67a) converges to 0 in probability as $E \rightarrow \infty$.

To control (4.67b), note that the estimate (4.68) implies that this term is bounded on $\mathcal{H}_{E,\varepsilon}$ for E large enough by

$$\frac{C_\varepsilon \|\varphi\|_\infty (q - 1 + \|R_{\beta,E}^\Sigma\|_\infty) E^{p-\gamma/2}}{4c_E E^{q/2}} \int_0^{\tau_E} \frac{c_E}{(1 - c_E t)^q} dt. \quad (4.69)$$

If $q = 1$, the integral evaluates to $\frac{1}{2} \log E$, whereas if $q > 1$, it is of order $E^{(q-1)/2}$. This means that (4.69) is at most proportional to $E^{-1/2+p-\gamma/2} \log E$, which vanishes as $E \rightarrow \infty$, and it follows that (4.67b) converges to 0 in probability as $E \rightarrow \infty$.

Finally, using again the estimate (4.68), we see that the quadratic variation of (4.67c) is bounded on $\mathcal{H}_{E,\varepsilon}$ for E large enough by

$$\frac{C_\varepsilon^2 \|\varphi\|_\infty^2 \|S_{\beta,E}^\Sigma\|_\infty^2 E^{2p-\gamma}}{16c_E^2 E^q} \int_0^{\tau_E} \frac{c_E}{(1 - c_E t)^{2q-1}} dt. \quad (4.70)$$

In the same way as in the last case, this is at most proportional to $E^{-1+2p-\gamma} \log E$. Therefore, by Bernstein's inequality, for any $x > 0$, (4.67c) exceeds x with probability at most $2 \exp(-CE^{1-2p+\gamma}x^2)$ for some $C > 0$ depending only on the constants in (4.70). Since $E^{1-2p+\gamma} \rightarrow \infty$ and $\mathbb{P}(\mathcal{H}_{E,\varepsilon}) \geq 1 - \varepsilon/2$, it follows that (4.67c) converges to 0 in probability as $E \rightarrow \infty$.

As all three terms of (4.67) converge to 0 in probability as $E \rightarrow \infty$, the second term of the coefficient matrix $M_{\beta,E}$ in (4.57) converges vaguely to 0, and this concludes the proof. ■

4.4 Spectral convergence

The goal of this section is to prove Theorem 1.10, which shows the compact convergence of the Weyl–Titchmarsh functions of the shifted and scaled stochastic Airy and Bessel operators $\mathcal{H}_{\beta,E}$ and $\mathcal{G}_{\beta,a,E}$ to that of the stochastic sine operator. Our general strategy to prove this is to use the convergence of transfer matrices that we proved in §4.3, and exploit the continuity of mappings from transfer matrices to Weyl–Titchmarsh functions for canonical systems that we saw in §2.2.

Our proof of Theorem 1.10 is split in three parts (Corollary 4.10.2 and Theorems 4.11 and 4.13). We start by working out the case $\beta \leq 2$ in §4.4.1. In that case, the convergence of the Weyl–Titchmarsh functions is easy to deduce from that of the transfer matrices, since the stochastic sine operator is limit point at 1. However, there is significant work left to do when $\beta > 2$, as the stochastic sine operator is then limit circle at 1. Indeed, this means that we also need to prove the convergence of the boundary (or integrability) conditions. For this case, our proofs rely on specific properties of the stochastic Airy and Bessel operators, which prevents us from giving a common proof for the general model that we built in §4.2.3. Instead, we prove the two results independently in §4.4.2 and §4.4.3, although both proofs use the same method.

The result presented in §4.4.1 for the limit point case $\beta \leq 2$ combines the analog results for the Airy and Bessel cases given in [55, §7] and [54, §5.2]. The results and the arguments presented in §4.4.2 are reproduced from [55, §7], while those of §4.4.3 are reproduced from [54, §5.3].

4.4.1 Limit point case ($\beta \leq 2$)

As mentioned in the introduction of the section, when $\beta \leq 2$ the fact that the stochastic sine operator is limit point at 1 makes the convergence of the Weyl–Titchmarsh functions a simple consequence of the convergence of the transfer matrices, by continuity of the mapping between the two.

Corollary 4.10.2. *In the setting of Theorem 4.10, suppose that $\beta \leq 2$, and let $m_{M_{\beta,E_n}}, m_{R_\beta} : \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ be the Weyl–Titchmarsh functions associated with M_{β,E_n} and R_β . Then $m_{M_{\beta,E_n}} \rightarrow m_{R_\beta}$ compactly on $\mathbb{H}$ in probability as $n \rightarrow \infty$.*

Recall that on a separable metric space, convergence in probability is equivalent to each subsequence having a further subsequence that converges a.s. (see e.g. [44, Lemma 5.2]). This characterization can be used to prove Corollary 4.10.2 from the convergence of transfer matrices in the same way as it can be used to prove the continuous mapping theorem (as done e.g. in [44, Lemma 5.3]). Note that here, the continuous mapping theorem itself is not useful, since the mapping from transfer matrices to Weyl–Titchmarsh functions is only continuous on the set of transfer matrices of systems that are limit point on the right, whereas here $M_{\beta,E}$ could also be limit circle.

Proof. Pick an arbitrary subsequence $S \subseteq \mathbb{N}$ and write $T_{M_{\beta,E_n}}$ and $T_{\tilde{R}_\beta}$ for the transfer matrices of M_{β,E_n} and $\tilde{R}_\beta$. By Corollary 4.10.1, $T_{M_{\beta,E_n}} \xrightarrow{\mathbb{P}} T_{\tilde{R}_\beta}$ compactly on $[0, 1) \times \mathbb{C}$ as $n \rightarrow \infty$, so there is a further subsequence $S' \subseteq S$ along which $T_{M_{\beta,E_n}} \rightarrow T_{\tilde{R}_\beta}$ compactly a.s. Since $\tilde{R}_\beta$ is limit point at 1, it follows from Theorem 2.13 that $m_{M_{\beta,E_n}} \rightarrow m_{\tilde{R}_\beta}$ compactly a.s. along S' . Because the subsequence S was arbitrary, this shows that $m_{M_{\beta,E_n}} \xrightarrow{\mathbb{P}} m_{\tilde{R}_\beta}$ compactly as $n \rightarrow \infty$. ■

4.4.2 Limit circle ($\beta > 2$) case from the soft edge

The purpose of this section is to prove the following.

Theorem 4.11. *Suppose $\beta > 2$, and let $m_{H_{\beta,E_n}}, m_{R_\beta} : \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ be the Weyl–Titchmarsh functions of the shifted and scaled Airy system and of the sine system respectively, both defined on the probability space from Lemma 4.4 applied to the case where $M_{\beta,E_n} = \eta'_{A,E_n}(H_{\beta,E_n} \circ \eta_{A,E_n})$. Then $m_{H_{\beta,E_n}} \xrightarrow{\mathbb{P}} m_{R_\beta}$ compactly on $\mathbb{H}$ as $n \rightarrow \infty$.*

Our plan is to prove this from the convergence of the transfer matrices using the continuity of the map $\text{TM}_{\text{LC}}[0, 1] \times \mathcal{C}(\mathbb{H}, \mathbb{C}^2) \rightarrow \mathbb{C}_\infty^{\mathbb{H}}$ sending a transfer matrix and a right boundary condition to

a Weyl–Titchmarsh function (i.e., Theorem 2.14). Thus, what remains to be proven, essentially, is that the right boundary conditions converge. Before moving on to the proof, we describe the setup and the idea of the proof in detail.

Recall that the right boundary condition of the sine system is a vector parallel to $(\operatorname{Re} \mathcal{B}_\beta(\infty), 1)$. The Airy system, however, is limit point on the right, and because of this change of behavior, the setting that we built for that case is modelled after the ideas of §2.2.3: with $\beta > 2$, we took $c_E = 1 - 1/\sqrt{E}$ and $\tau_E = 1$, and our modified Airy system acts on $[0, 1 + 1/\sqrt{E})$. Hence, the correct right “boundary condition” to attach to its restriction to $[0, 1]$ in order to reproduce the same Weyl–Titchmarsh function is the value at 1 of an integrable solution $u_z \circ \eta_{A,E}$ to $J(u_z \circ \eta_{A,E})' = -z\eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E})(u_z \circ \eta_{A,E})$ on $[0, 1 + 1/\sqrt{E})$. By construction of the canonical system, such a solution u_z always has the form

$$u_z = A_{\beta,E}^{-1} \begin{pmatrix} h_z^{[1]} \\ h_z \end{pmatrix} = \begin{pmatrix} E^{-1/4} f_{\beta,E} & -E^{-1/4} f'_{\beta,E} \\ -E^{1/4} g_{\beta,E} & E^{1/4} g'_{\beta,E} \end{pmatrix} \begin{pmatrix} h'_z \\ h_z \end{pmatrix}$$

where h_z is an integrable solution to $\mathcal{H}_{\beta,E} h_z = z h_z$ and where $A_{\beta,E}$ is the matrix (4.4). Now, notice that using the representations of $f_{\beta,E}$ and $g_{\beta,E}$ in the polar coordinates from Proposition 4.1,

$$\begin{aligned} E^{-1/4} f_{\beta,E} \circ \eta_{A,E}(1) &= e^{\rho_{\beta,E}^f(1)} \cos \xi_{\beta,E}^f(1) \\ &= e^{-\Delta_{\beta,E}^\rho(1)} \cos \Delta_{\beta,E}^\xi(1) e^{\rho_{\beta,E}^g(1)} \cos \xi_{\beta,E}^g(1) + e^{-\Delta_{\beta,E}^\rho(1)} \sin \Delta_{\beta,E}^\xi(1) e^{\rho_{\beta,E}^g(1)} \sin \xi_{\beta,E}^g(1), \end{aligned}$$

and we retrieve $-\operatorname{Re} \ell_{\beta,E}(1) = e^{-\Delta_{\beta,E}^\rho(1)} \cos \Delta_{\beta,E}^\xi(1)$ as well as the expression in polar coordinates of $E^{1/4} g_{\beta,E} \circ \eta_{A,E}(1)$ in the first term. The identity from Claim 4.3, which gives $e^{-\Delta_{\beta,E}^\rho(1)} \sin \Delta_{\beta,E}^\xi(1) \equiv e^{-2\rho_{\beta,E}^g(1)}$, also simplifies the prefactor of the second term, and this yields

$$E^{-1/4} f_{\beta,E} \circ \eta_{A,E}(1) = -\operatorname{Re} \ell_{\beta,E}(1) E^{1/4} g_{\beta,E} \circ \eta_{A,E}(1) + e^{-\rho_{\beta,E}^g(1)} \sin \xi_{\beta,E}^g(1).$$

Similar manipulations can be applied to $E^{-1/4} f'_{\beta,E} \circ \eta_{A,E}(1)$, and yield

$$E^{-1/4} f'_{\beta,E} \circ \eta_{A,E}(1) = -\operatorname{Re} \ell_{\beta,E}(1) E^{1/4} g'_{\beta,E} \circ \eta_{A,E}(1) - e^{-\rho_{\beta,E}^g(1)} \cos \xi_{\beta,E}^g(1).$$

It follows that

$$\begin{aligned} \begin{pmatrix} E^{-1/4} f_{\beta,E} & -E^{-1/4} f'_{\beta,E} \\ -E^{1/4} g_{\beta,E} & E^{1/4} g'_{\beta,E} \end{pmatrix} (\eta_{A,E}(1)) &= \begin{pmatrix} \operatorname{Re} \ell_{\beta,E}(1) & \\ & 1 \end{pmatrix} \begin{pmatrix} -E^{1/4} g_{\beta,E} \circ \eta_{A,E}(1) & E^{1/4} g'_{\beta,E} \circ \eta_{A,E}(1) \\ 0 & 0 \end{pmatrix} \\ &\quad + e^{-\rho_{\beta,E}^g(1)} \begin{pmatrix} \sin \xi_{\beta,E}^g(1) & \cos \xi_{\beta,E}^g(1) \\ 0 & 0 \end{pmatrix}, \end{aligned}$$

and therefore that

$$u_z \circ \eta_{A,E}(1) = \mathcal{W}(h_z, E^{1/4}g_{\beta,E})(E-1) \begin{pmatrix} \operatorname{Re} \mathcal{C}_{\beta,E}(1) \\ 1 \end{pmatrix} + e^{-\rho_{\beta,E}^g(1)} \begin{pmatrix} \sin \xi_{\beta,E}^g(1) & \cos \xi_{\beta,E}^g(1) \\ 0 & 0 \end{pmatrix} \begin{pmatrix} h'_z(E-1) \\ h_z(E-1) \end{pmatrix} \quad (4.71)$$

where $\mathcal{W}(f, g) = fg' - f'g$ denotes the Wronskian of f and g , and where we used that $\eta_{A,E}(1) = E - 1$.

From the results of §4.3.2.3, we expect $\operatorname{Re} \mathcal{C}_{\beta,E}(1)$ to converge to $\operatorname{Re} \mathcal{B}_{\beta}(\infty)$. Hence, the right “boundary condition” (4.71) should indeed converge to the right boundary condition of the sine system provided its second term vanishes in the limit, and provided we can choose the integrable solution h_z to be normalized in such a way that $\mathcal{W}(h_z, E^{1/4}g_{\beta,E})(E-1) \rightarrow 1$.

To see how this can be done, recall from Proposition 3.2 that f solves $\mathcal{H}_{\beta,E}f = zf$ if and only if $\tilde{f}(t) = f(t+E)$ solves $\tilde{\mathcal{H}}_{\beta}\tilde{f} = \frac{z}{2\sqrt{E}}\tilde{f}$ where $\tilde{\mathcal{H}}_{\beta}$ has the law of $\mathcal{H}_{\beta}$, but is defined from the Brownian motion $B_E(\cdot + E) - B_E(E)$. This means that if we fix a $\gamma \in \mathbb{C} \setminus \{0\}$ and a square integrable solution Φ_{λ} to $\tilde{\mathcal{H}}_{\beta}\Phi_{\lambda} = \lambda\Phi_{\lambda}$, then we can take $h_z(t) := \gamma\Phi_{z/2\sqrt{E}}(t-E)$ as the solution appearing in (4.71). By basic properties of Sturm–Liouville operators (see e.g. [29, Theorem 2.7]), we may (and do) choose Φ_{λ} to be analytic in λ . In particular, Φ_{λ} could be chosen to be the stochastic Airy function $\operatorname{SAi}_{-\lambda}$ from [49].

This makes it clear how we would like to choose h_z : writing $\tilde{g}_{\beta,E}(t) := g_{\beta,E}(t+E)$, as long as the $\mathcal{W}(\Phi_{z/2\sqrt{E}}, E^{1/4}\tilde{g}_{\beta,E})(-1)$ is nonzero, we can use the inverse of this value as the constant γ which normalizes the integrable solution, thus removing the prefactor of the first term in (4.71). This leads us to the following.

Lemma 4.12. *On the probability space from Lemma 4.4 in the Airy case, if $\beta > 2$ and $K \subset \mathbb{C}$ is compact, then $\mathbb{P}[\forall z \in K, |\mathcal{W}(\Phi_{z/2\sqrt{E_n}}, E_n^{1/4}\tilde{g}_{\beta,E_n})(-1)| \geq 1] \rightarrow 1$ as $n \rightarrow \infty$.*

This result can be proven using the asymptotics of solutions to $\mathcal{H}_{\beta}f = 0$ as $t \rightarrow -\infty$ that we derived in §3.1.2, and allows us to complete the proof of the convergence of the Weyl–Titchmarsh functions. To avoid interrupting the argument too much, we postpone the proof of Lemma 4.12 until the end of the section.

Proof of Theorem 4.11. As we did earlier, to simplify notation throughout the proof we drop the n subscript and understand a limit as $E \rightarrow \infty$ as being taken along the sequence $\{E_n\}_{n \in \mathbb{N}}$.

By Corollary 4.10.1 applied to the case $M_{\beta,E} = \eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E})$, the transfer matrix of $\eta'_{A,E}(H_{\beta,E} \circ \eta_{A,E})$ converges in probability to that of $\tilde{R}_{\beta}$. By continuity of the map $\operatorname{TM}_{\operatorname{LC}}[0, 1] \times \mathcal{C}(\mathbb{H}, \mathbb{C}^2) \rightarrow \mathbb{C}_{\infty}^{\mathbb{H}}$ that sends a transfer matrix and a right boundary condition to the corresponding

Weyl–Titchmarsh function (i.e., Theorem 2.14), this means that it suffices to prove that the right boundary conditions converge compactly in probability. As explained above, the right “boundary condition” of the Airy system on $[0, 1]$ is the value $u_z \circ \eta_{A,E}(1)$ of a suitable integrable solution u_z to $Ju'_z = -zH_{\beta,E}u_z$. Therefore, it suffices to show that for any compact $K \subset \mathbb{C}$, $u_z \circ \eta_{A,E}(1) \rightarrow (\operatorname{Re} \mathcal{B}_\beta(\infty), 1)$ uniformly in $z \in K$ in probability.

For $z \in K$ for some compact $K \subset \mathbb{C}$, fix $u_z = A_{\beta,E}^{-1} \begin{pmatrix} h_z^{[1]} \\ h_z \end{pmatrix}$ where

$$h_z(t) := \frac{1}{\mathcal{W}(\Phi_{z/2\sqrt{E}}, E^{1/4}\tilde{g}_{\beta,E})(-1)} \Phi_{z/2\sqrt{E}}(t - E)$$

for a solution Φ_λ to $\tilde{H}_\beta \Phi_\lambda = \lambda \Phi_\lambda$ square integrable on $[0, \infty)$ with respect to time and chosen to be analytic in λ . As observed above, by the translation property of solutions from Proposition 3.2 and by construction of the canonical system, this means that u_z is an integrable solution to $Ju'_z = -zH_{\beta,E}u_z$ on $[0, 1 + 1/\sqrt{E})$. Note that in the definition of h_z , technically the Wronskian in the denominator might be zero and this would make h_z ill defined, but Lemma 4.12 ensures that this asymptotically almost surely does not happen, as it shows that the event

$$\mathcal{G}_{K,E} := \{\forall z \in K, |\mathcal{W}(\Phi_{z/2\sqrt{E}}, E^{1/4}\tilde{g}_{\beta,E})(-1)| \geq 1\}$$

has $\mathbb{P}(\mathcal{G}_{K,E}) \rightarrow 1$.

Now, h_z is chosen so that $\mathcal{W}(h_z, E^{1/4}g_{\beta,E})(E - 1) = 1$, so following (4.71),

$$u_z \circ \eta_{A,E}(1) = \begin{pmatrix} \operatorname{Re} \ell_{\beta,E}(1) \\ 1 \end{pmatrix} + e^{-\rho_{\beta,E}^g(1)} \begin{pmatrix} \sin \xi_{\beta,E}^g(1) & \cos \xi_{\beta,E}^g(1) \\ 0 & 0 \end{pmatrix} \begin{pmatrix} h'_z(E - 1) \\ h_z(E - 1) \end{pmatrix}.$$

In particular, $u_{z,2} \circ \eta_{A,E}(1) = 1$ whenever h_z is well defined and in particular on $\mathcal{G}_{K,E}$, so it suffices to compare $u_{z,1} \circ \eta_{A,E}(1)$ to $\operatorname{Re} \mathcal{B}_\beta(\infty)$. To do so, we set $\tau_{E,\alpha} := (1 - E^{-1/2+\alpha})/c_E$ for an $\alpha \in (0, 1/2)$, and we split

$$|u_{z,1} \circ \eta_{A,E}(1) - \operatorname{Re} \mathcal{B}_\beta(\infty)| \leq |\operatorname{Re} \ell_{\beta,E}(1) - \operatorname{Re} \ell_{\beta,E}(\tau_{E,\alpha})| \quad (4.72a)$$

$$+ |\operatorname{Re} \ell_{\beta,E}(\tau_{E,\alpha}) - \operatorname{Re} \mathcal{B}_\beta \circ \nu_E(\tau_{E,\alpha})| \quad (4.72b)$$

$$+ |\operatorname{Re} \mathcal{B}_\beta \circ \nu_E(\tau_{E,\alpha}) - \operatorname{Re} \mathcal{B}_\beta(\infty)| \quad (4.72c)$$

$$+ e^{-\rho_{\beta,E}^g(1)} |\sin \xi_{\beta,E}^g(1) h'_z(E - 1) + \cos \xi_{\beta,E}^g(1) h_z(E - 1)|. \quad (4.72d)$$

It is already clear that (4.72c) converges to zero a.s. since $\nu_E(\tau_{E,\alpha}) = (\frac{1}{2} - \alpha) \log E$, and that (4.72b) converges to zero in probability by Proposition 4.7. To conclude, we show that if Y_E denotes either (4.72a) or (4.72d), then for all $\varepsilon, \zeta > 0$, $\mathbb{P}(\mathcal{G}_{K,E} \cap \{|Y_E| > \zeta\}) < \varepsilon$ for all E large enough.

To control (4.72a), recall from (4.48) that

$$\begin{aligned} d(\operatorname{Re} \ell_{\beta,E})(t) &= \frac{2}{\sqrt{\beta}} e^{-2\rho_{\beta,E}^g(t)} \cos 2\xi_{\beta,E}^g(t) \sqrt{\frac{2c_E}{1-c_E t}} dB_E(t) \\ &\quad - e^{-2\rho_{\beta,E}^g(t)} \left(\left(\frac{4}{\beta} - 1 \right) \sin 2\xi_{\beta,E}^g(t) + \frac{4}{\beta} \sin 4\xi_{\beta,E}^g(t) \right) \frac{c_E}{1-c_E t} dt. \end{aligned} \quad (4.73)$$

Given $\varepsilon > 0$, we know from Proposition 4.8 that for some $C, C' > 0$,

$$\mathbb{P} \left[\forall t \in [0, 1], \left| \rho_{\beta,E}^g(t) + \frac{1}{\beta} \log(1 - c_E t) \right| \leq C + C' (-\log(1 - c_E t))^{3/4} \right] > 1 - \frac{\varepsilon}{2}.$$

On that event, if $\delta > 0$ then there is a (different) $C > 0$ such that for any $t \in [\tau_{E,\alpha}, 1]$,

$$e^{-\rho_{\beta,E}^g(t)} \leq C(1 - c_E t)^{1/\beta - \delta} \leq C E^{-(1/2 - \alpha)(1/\beta - \delta)},$$

and thus for both $k = 2$ and $k = 4$,

$$\begin{aligned} \left| \int_{\tau_{E,\alpha}}^1 e^{-2\rho_{\beta,E}^g(t)} \sin k \xi_{\beta,E}^g(t) \frac{c_E}{1 - c_E t} dt \right| \\ \leq C^2 E^{-2(1/2 - \alpha)(1/\beta - \delta)} \log \frac{1 - c_E \tau_{E,\alpha}}{1 - c_E} = \alpha C^2 E^{-2(1/2 - \alpha)(1/\beta - \delta)} \log E. \end{aligned}$$

Taking $\delta < 1/\beta$, this shows that for any $\zeta > 0$, the second line of (4.73) is bounded by $\zeta/2$ with probability at least $1 - \varepsilon/2$ for any E large enough. On the same good event of probability at least $1 - \varepsilon/2$, the bracket of the stochastic integral in (4.73) is bounded by $8\alpha C^4 \beta^{-1} E^{-4(1/2 - \alpha)(1/\beta - \delta)} \log E$.

Thus, by Bernstein's inequality,

$$\begin{aligned} \mathbb{P} \left[\sup_{t \in [\tau_{E,\alpha}, 1]} \left| \frac{2}{\sqrt{\beta}} \int_{\tau_{E,\alpha}}^t e^{-2\rho_{\beta,E}^g(s)} \cos 2\xi_{\beta,E}^g(s) \sqrt{\frac{2c_E}{1 - c_E s}} dB_E(s) \right| \geq \frac{\zeta}{2} \right] \\ \leq \frac{\varepsilon}{2} + 2 \exp \left(- \frac{\beta E^{4(1/2 - \alpha)(1/\beta - \delta)} \zeta^2}{64\alpha C^4 \log E} \right). \end{aligned}$$

Therefore, if E is also large enough that the right-hand side here is bounded by ε , then we find that $\mathbb{P}[|\operatorname{Re} \ell_{\beta,E}(1) - \operatorname{Re} \ell_{\beta,E}(\tau_{E,\alpha})| \geq \zeta] < \varepsilon$, which is the control we were looking for on (4.72a).

Finally, in order to control (4.72d), note that by definition of h_z , on $\mathcal{G}_{K,E}$,

$$|h_z(E - 1)| = \left| \frac{1}{\mathcal{W}(\Phi_{z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,E})} \Phi_{z/2\sqrt{E}}(-1) \right| \leq |\Phi_{z/2\sqrt{E}}(-1) - \Phi_0(-1)| + |\Phi_0(-1)|.$$

A similar bound in terms of Φ'_0 and $\Phi'_{z/2\sqrt{E}}$ holds for $h'_z(E - 1)$. By analyticity of $\lambda \mapsto \Phi_\lambda(-1)$, for all $z \in K$,

$$\max \left\{ |\Phi_{z/2\sqrt{E}}(-1) - \Phi_0(-1)|, |\Phi'_{z/2\sqrt{E}}(-1) - \Phi'_0(-1)| \right\} \leq \frac{X \operatorname{diam} K}{2\sqrt{E}} \quad (4.74)$$

for some positive random variable X , so there is a $C > 0$ such that this maximum is bounded by $C/\sqrt{E}$ with probability at least $1 - \varepsilon/6$. Because $\Phi_0(-1)$ is a well-defined random variable, it follows that there is an $M > 0$ large enough so that

$$\mathbb{P}(\mathcal{G}_{K,E} \cap \{|h_z(E-1)| > M\}) < \frac{\varepsilon}{3},$$

and again the same works with $h'_z(E-1)$. Then, by Proposition 4.8 as earlier, we know that for any $\delta > 0$ there is a $C > 0$ such that $e^{-\rho_{\beta,E}^g(1)} < CE^{-1/2\beta+\delta}$ with probability at least $1 - \varepsilon/3$. Taking $\delta < 1/2\beta$, this means that given $\zeta > 0$, for any E large enough that $2CME^{-1/2\beta+\delta} < \zeta$,

$$\mathbb{P}\left(\mathcal{G}_{K,E} \cap \left\{e^{-\rho_{\beta,E}^g(1)} \left| \sin \xi_{\beta,E}^g(1) h'_z(E-1) + \cos \xi_{\beta,E}^g(1) h_z(E-1) \right| > \zeta \right\}\right) < \varepsilon.$$

This finishes to prove the convergence of the first entry of the boundary conditions, and thus concludes the proof. $\blacksquare$

Proof of Lemma 4.12. Fix $E \in \{E_n\}_{n \in \mathbb{N}}$ and a compact $K \subset \mathbb{C}$. To control the Wronskian of $\Phi_{z/2\sqrt{E}}$ and $\tilde{g}_{\beta,E}$ at -1 , we first replace $\Phi_{z/2\sqrt{E}}$ with Φ_0 :

$$\mathcal{W}(\Phi_{z/2\sqrt{E}}, \tilde{g}_{\beta,E}) = \mathcal{W}(\Phi_0, \tilde{g}_{\beta,E}) + (\Phi_{z/2\sqrt{E}} - \Phi_0) \tilde{g}'_{\beta,E} - (\Phi'_{z/2\sqrt{E}} - \Phi'_0) \tilde{g}_{\beta,E}. \quad (4.75)$$

The reason for doing this is that because Φ_0 and $\tilde{g}_{\beta,E}$ both solve $\tilde{H}_\beta f = 0$, their Wronskian is constant, and because $\tilde{g}_{\beta,E}(-E) = 0$ and $\tilde{g}'_{\beta,E}(-E) = 1$, this means that $\mathcal{W}(\Phi_0, \tilde{g}_{\beta,E}) \equiv \Phi_0(-E)$.

It can be verified using arguments similar to those used in the proof of Theorem 4.11 that the last two terms of (4.75) do not contribute in the limit for $\mathcal{W}(\Phi_{z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,E})(-1)$. Indeed,

$$E^{1/4} |\tilde{g}_{\beta,E}(-1)| = E^{1/4} |g_{\beta,E} \circ \eta_{A,E}(1)| \leq e^{\rho_{\beta,E}^g(1)}, \quad E^{1/4} |\tilde{g}'_{\beta,E}(-1)| = E^{1/4} |g'_{\beta,E} \circ \eta_{A,E}(1)| \leq e^{\rho_{\beta,E}^g(1)},$$

so it follows from Proposition 4.8 that if $\varepsilon, \delta > 0$ there is a $C > 0$ such that both $E^{1/4} |\tilde{g}_{\beta,E}(-1)|$ and $E^{1/4} |\tilde{g}'_{\beta,E}(-1)|$ are bounded by $CE^{1/2\beta+\delta}$ with probability at least $1 - \varepsilon/2$. Moreover, by analyticity of $\lambda \mapsto \Phi_\lambda$, as in (4.74) we know that there is a $C' > 0$ such that with probability at least $1 - \varepsilon/2$, for all $z \in K$, both $|\Phi_{z/2\sqrt{E}}(-1) - \Phi_0(-1)|$ and $|\Phi'_{z/2\sqrt{E}}(-1) - \Phi'_0(-1)|$ are bounded by $C'/\sqrt{E}$. This means that with probability at least $1 - \varepsilon$, for all $z \in K$,

$$|\Phi_{z/2\sqrt{E}}(-1) - \Phi_0(-1)| E^{1/4} |\tilde{g}'_{\beta,E}(-1)| \leq CC' E^{-1/2+1/2\beta+\delta},$$

and the same holds with primes on the Φ 's instead of on $\tilde{g}_{\beta,E}$. With δ small enough, this vanishes as $E \rightarrow \infty$, so $(\Phi_{z/2\sqrt{E}}(-1) - \Phi_0(-1)) E^{1/4} \tilde{g}'_{\beta,E}(-1) \rightarrow 0$ in probability uniformly in $z \in K$, and likewise with primes on Φ 's instead. Combining this with our earlier observation about the Wronskian of Φ_0 and $\tilde{g}_{\beta,E}$, we conclude that

$$\mathcal{W}(\Phi_{z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,E})(-1) = E^{1/4} \Phi_0(-E) + o_{\mathbb{P},K}(1). \quad (4.76)$$

It thus remains to understand how $\Phi_0(-E)$ behaves, but this is easily done through the asymptotics we derived in §3.1.2. Indeed, by Theorem 3.3 and the Skorokhod representation theorem, we can write $\Phi_0(-E) = C_{\Phi_0} E^{-1/4+1/2\beta} e^{X(E)} \cos \xi_\beta(E)$ where $C_{\Phi_0}^2 := \Phi_0^2(-1) + \Phi_0'^2(-1)$, ξ_β satisfies $\xi_\beta(t) - 2\pi \lfloor \frac{\xi_\beta(t)}{2\pi} \rfloor \rightarrow U$ in probability for a $U \sim \text{Unif}[0, 2\pi)$, and X has the property that for any $\varepsilon, \delta > 0$, there is a $C > 0$ for which

$$\mathbb{P}\left[\forall t \geq 1, |X(t)| \leq C(1 + (\log t)^{1/2+\delta})\right] \geq 1 - \varepsilon.$$

Hence, if $\varepsilon > 0$ and $\zeta > 0$ is small enough, $|\xi_\beta(E) - 2\pi \lfloor \frac{\xi_\beta(E)}{2\pi} \rfloor| \leq \zeta$ with probability at least $1 - \varepsilon/4$ for all E large enough. Shrinking ζ further if necessary, we may assume that $|\cos x| \geq 1 - x^2/2$ for $x \in [-\zeta, \zeta]$ and that $|\cos U| \geq 2\zeta(1 - \zeta^2/2)^{-1}$ with probability at least $1 - \varepsilon/4$. This shows that

$$|\cos \xi_\beta(E)| \geq |\cos U| \left(1 - \frac{\zeta^2}{2}\right) - \zeta \geq \zeta$$

with probability at least $1 - \varepsilon/2$ for all E large enough. Moreover, shrinking again ζ if necessary, we may assume that $C_{\Phi_0} > \zeta$ with probability at least $1 - \varepsilon/4$, and we know that for all E large enough, $e^{X(E)} \geq CE^{-\delta}$ with probability at least $1 - \varepsilon/4$. Combining everything,

$$E^{1/4} |\Phi_0(-E)| = C_{\Phi_0} E^{1/2\beta} e^{X(E)} |\cos \xi_\beta(E)| \geq C \zeta^2 E^{1/2\beta-\delta} \quad (4.77)$$

with probability at least $1 - \varepsilon$ for all E large enough. With $\delta < 1/2\beta$, this diverges as $E \rightarrow \infty$, so it is certainly at least 1 for all E large enough.

Combining (4.77) with (4.76) shows that $\mathbb{P}[\forall z \in K, |\mathcal{W}(\Phi_{z/2\sqrt{E_n}}, E_n^{1/4} \tilde{g}_{\beta, E_n})(-1)| \geq 1] \rightarrow 1$ as $n \rightarrow \infty$ and concludes the proof. $\blacksquare$

4.4.3 Limit circle ($\beta > 2$) case from the hard edge

The purpose of this section is to prove the following.

Theorem 4.13. *Suppose $\beta > 2$, and let $m_{G_{\beta, a, E_n}}, m_{R_\beta}: \mathbb{H} \rightarrow \overline{\mathbb{H}}_\infty$ be the Weyl–Titchmarsh functions of the shifted and scaled Bessel system and of the sine system respectively, both defined on the probability space from Lemma 4.4 applied to the case where $M_{\beta, E_n} = \eta'_{B, E_n}(G_{\beta, a, E_n} \circ \eta_{B, E_n})$. Then $m_{G_{\beta, a, E_n}} \xrightarrow{\mathbb{P}} m_{R_\beta}$ compactly on $\mathbb{H}$ as $n \rightarrow \infty$.*

Just like in the soft edge case, our plan is to prove this from the convergence of the transfer matrices using the continuity of the map sending a transfer matrix and a right boundary condition to a Weyl–Titchmarsh function. Thus, what remains to be proven is essentially that the right boundary conditions converge. The proof is in many aspects very similar to the proof of Theorem 4.11 we

carried out in §4.4.2 for the soft edge case, and throughout this section we insist on those aspects of the proof that are different rather than on the full details. Again, before moving on to the proof we start by describing the setup and the idea of the proof in detail.

When $a \geq 1$, the Bessel system is limit point on the right. In that case, the situation is very similar to what we saw in the soft edge case: we took $c_E = 1 - 1/\sqrt{E}$ and $\tau_E = 1$ and our modified Bessel system acts on $[0, 1/c_E)$. The correct right “boundary condition” to attach to its restriction to $[0, 1]$ in order to reproduce the same Weyl–Titchmarsh function is therefore the value at 1 of an integrable solution $u_z \circ \eta_{B,E}$ to $J(u_z \circ \eta_{B,E})' = -z\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E})(u_z \circ \eta_{B,E})$ on $[0, 1/c_E)$. By construction of the canonical system, u_z has the form

$$u_z = A_{\beta,a,E}^{-1} \begin{pmatrix} h_z^{[1]} \\ h_z \end{pmatrix} = \begin{pmatrix} E^{-1/4} f_{\beta,a,E} & -E^{-1/4} p_{\beta,a} f'_{\beta,a,E} \\ -E^{1/4} g_{\beta,a,E} & E^{1/4} p_{\beta,a} g'_{\beta,a,E} \end{pmatrix} \begin{pmatrix} p_{\beta,a} h'_z \\ h_z \end{pmatrix}$$

where h_z is an $L^2([0, \infty), w_{\beta,a}(t) dt)$ solution to $\mathfrak{G}_{\beta,a,E} h_z = z h_z$ and where $A_{\beta,a,E}$ is the matrix (4.16).

When $|a| < 1$, however, the Bessel system is limit circle on the right. In that case, we rather took $c_E = 1$ so our modified Bessel system already acts on $[0, 1)$. As we have seen in §4.2.2, the right boundary condition is then $e_{\phi_E}^* J u(\infty) = 0$ where $\phi_E \in [0, \pi)$ is chosen to reproduce the boundary condition of the stochastic Bessel operator, which requires a solution h_z of $\mathfrak{G}_{\beta,a,E} h_z = z h_z$ to satisfy $h_z^{[1]}(t) \rightarrow 0$ as $t \rightarrow \infty$. Here, it will be easier to understand the boundary condition in the latter way: a canonical system solution $u_z = A_{\beta,a,E}^{-1} \begin{pmatrix} h_z^{[1]} \\ h_z \end{pmatrix}$ simply satisfies the right boundary condition if the solution h_z satisfies $h_z^{[1]}(t) \rightarrow 0$ as $t \rightarrow \infty$.

In order to uniformize the analysis of the $|a| < 1$ and $a \geq 1$ cases, we will show that when $|a| < 1$ we can effectively move the boundary condition from 1 to $1 - 1/\sqrt{E}$. Hence, in both cases, our goal is to show the convergence of $u_z \circ \eta_{B,E}(\tau_E)$ to the sine system’s boundary condition, where $\tau_E := \frac{1}{c_E}(1 - 1/\sqrt{E})$ as before and where $u_z = A_{\beta,a,E}^{-1} \begin{pmatrix} h_z^{[1]} \\ h_z \end{pmatrix}$ for a specific solution h_z to $\mathfrak{G}_{\beta,a,E} h_z = z h_z$, either an integrable one (if $a \geq 1$) or one that satisfies $h_z^{[1]}(t) \rightarrow 0$ as $t \rightarrow \infty$ (if $|a| < 1$). In any case, this defines h_z up to a multiplicative constant, so it remains to see the appropriate way to normalize it.

To understand the convergence of the boundary condition and to see how to normalize h_z , we proceed like in the soft edge case. We write

$$\begin{aligned} E^{-1/4} f_{\beta,a,E} \circ \eta_{B,E} &= \frac{E^{-1/4} e^{\eta_{B,E}/4}}{\sqrt{p_{\beta,a} \circ \eta_{B,E}}} e^{\rho_{\beta,a,E}^g - \Delta_{\beta,E}^g} \cos(\xi_{\beta,a,E}^g - \Delta_{\beta,E}^\xi) \\ &= e^{-\Delta_{\beta,E}^g} \cos \Delta_{\beta,E}^\xi E^{-1/4} g_{\beta,a,E} \circ \eta_{B,E} + e^{-\Delta_{\beta,E}^g} \sin \Delta_{\beta,E}^\xi \frac{E^{-1/4} e^{\eta_{B,E}/4}}{\sqrt{p_{\beta,a} \circ \eta_{B,E}}} e^{\rho_{\beta,a,E}^g} \sin \xi_{\beta,a,E}^g. \end{aligned}$$

Using Claim 4.3 and evaluating at $\tau_E = \eta_{B,E}^{-1}(\log E)$, we get

$$E^{-1/4}f_{\beta,a,E}(\log E) = -\operatorname{Re} \ell_{\beta,E}(\tau_E) E^{1/4} g_{\beta,a,E}(\log E) + \frac{1}{\sqrt{p_{\beta,a}(\log E)}} e^{-\rho_{\beta,a,E}^g(\tau_E)} \sin \xi_{\beta,a,E}^g(\tau_E).$$

Similar manipulations can be applied to $E^{-1/4}f'_{\beta,a,E} \circ \eta_{B,E}$, and yield

$$E^{-1/4}f'_{\beta,a,E}(\log E) = -\operatorname{Re} \ell_{\beta,E}(\tau_E) E^{1/4} g'_{\beta,a,E}(\log E) - \frac{1}{\sqrt{p_{\beta,a}(\log E)}} e^{-\rho_{\beta,a,E}^g(\tau_E)} \cos \xi_{\beta,a,E}^g(\tau_E).$$

Using these expressions for $f_{\beta,a,E}$ and $f'_{\beta,a,E}$ in $A_{\beta,a,E}^{-1}$, we can write

$$\begin{aligned} u_z(\log E) &= \mathcal{W}_{\beta,a}(h_z, E^{1/4}g_{\beta,a,E})(\log E) \begin{pmatrix} \operatorname{Re} \ell_{\beta,E}(\tau_E) \\ 1 \end{pmatrix} \\ &\quad + e^{-\rho_{\beta,a,E}^g(\tau_E)} \sqrt{p_{\beta,a}(\log E)} \begin{pmatrix} \sin \xi_{\beta,a,E}^g(\tau_E) & \cos \xi_{\beta,a,E}^g(\tau_E) \\ 0 & 0 \end{pmatrix} \begin{pmatrix} h'_z(\log E) \\ h_z(\log E) \end{pmatrix} \end{aligned} \quad (4.78)$$

where $\mathcal{W}_{\beta,a}(f, g) = p_{\beta,a}(fg' - f'g)$ is the Wronskian of f and g . Thus, in a similar way as in the soft edge case, we expect this to converge to the right boundary condition of the sine system provided the second term vanishes in the limit, and provided we can choose h_z to be normalized in such a way that the Wronskian converges to 1.

We know from Proposition 3.17 that f solves $\mathfrak{G}_{\beta,a,E}f = zf$ if and only if $\tilde{f}(t) := f(t + \log E)$ solves $\tilde{\mathfrak{G}}_{\beta,a}\tilde{f} = (1 + \frac{z}{2\sqrt{E}})\tilde{f}$ where $\tilde{\mathfrak{G}}_{\beta,a}$ has the law of $\mathfrak{G}_{\beta,a}$, but is defined from the Brownian motion $B_E(\cdot + \log E) - B_E(E)$. Thus, if we fix a $\gamma \in \mathbb{C} \setminus \{0\}$ and a solution Φ_λ to $\tilde{\mathfrak{G}}_{\beta,a}\Phi_\lambda = \lambda\Phi_\lambda$ with the right behavior at infinity (i.e., it is $L^2([0, \infty), \tilde{w}_{\beta,a} d\mathcal{L})$ if $a \geq 1$ or satisfies $\Phi_\lambda^{[1]}(t) \rightarrow 0$ as $t \rightarrow \infty$ if $|a| < 1$), then we can take $h_z(t) := \gamma\Phi_{1+z/2\sqrt{E}}(t - \log E)$ as the solution appearing in (4.78). Then with $\tilde{g}_{\beta,a,E}(t) := g_{\beta,a,E}(t + \log E)$,

$$\mathcal{W}_{\beta,a}(h_z, E^{1/4}g_{\beta,a,E})(\log E) = \gamma p_{\beta,a}(\log E) \tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E}}, E^{1/4}\tilde{g}_{\beta,a,E})(0)$$

where $\tilde{\mathcal{W}}_{\beta,a}(f, g) := \tilde{p}_{\beta,a}(fg' - f'g)$ is the Wronskian for $\tilde{\mathfrak{G}}_{\beta,a}$. Provided the Wronskian on the right-hand side doesn't vanish, we can choose γ so that this is identically 1. It is not obvious that this can be done, but we can ensure it on a good event.

Lemma 4.14. *On the probability space from Lemma 4.4 in the Bessel case, if $\beta > 2$ and $K \subset \mathbb{C}$ is compact, then $\mathbb{P}[\forall z \in K, p_{\beta,a}(\log E_n) |\tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E_n}}, E_n^{1/4}\tilde{g}_{\beta,a,E_n})(0)| \geq E_n^{-a/2}] \rightarrow 1$ as $n \rightarrow \infty$.*

This can be proven using the asymptotics of solutions to $\mathfrak{G}_{\beta,a}f = f$ as $t \rightarrow -\infty$ that we derived in §3.2.2, and allows us to complete the proof of the Weyl–Titchmarsh functions. To avoid interrupting the argument, we postpone the proof of Lemma 4.14 until the end of the section.

Proof of Theorem 4.13. As we did earlier, to simplify notation throughout the proof we drop the n subscript and understand a limit as $E \rightarrow \infty$ as being taken along the sequence $\{E_n\}_{n \in \mathbb{N}}$.

As stated before, the transfer matrix of $\eta'_{B,E}(G_{\beta,a,E} \circ \eta_{B,E})$ converges in probability to that of $\tilde{R}_\beta$ by Corollary 4.10.1, and therefore by continuity of the usual map $\text{TM}_{\text{LC}}[0, 1] \times \mathcal{C}(\mathbb{H}, \mathbb{C}^2) \rightarrow \mathbb{C}^\mathbb{H}_\infty$ (i.e., Theorem 2.14), it suffices to prove that the right boundary condition converges compactly in probability. As explained above, what we mean here by the “right boundary condition” of the Bessel system on $[0, 1]$ is the value $u_z \circ \eta_{B,E}(1)$ or a suitable solution u_z to $Ju'_z = -zG_{\beta,a,E}u_z$.

To define u_z , we fix a solution Φ_λ to $\tilde{\mathfrak{G}}_{\beta,a}\Phi_\lambda = \lambda\Phi_\lambda$ that is $L^2([0, \infty), \tilde{w}_{\beta,a} d\mathcal{L})$ if $a \geq 1$ or that satisfies $\Phi_\lambda^{[1]}(t) \rightarrow 0$ as $t \rightarrow \infty$ if $|a| < 1$, chosen to be analytic in λ . Then, we set

$$u_z := A_{\beta,a,E}^{-1} \begin{pmatrix} p_{\beta,a} h'_z \\ h_z \end{pmatrix} \quad \text{with} \quad h_z(t) := \frac{\Phi_{1+z/2\sqrt{E}}(t - \log E)}{p_{\beta,a}(\log E) \tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,a,E})(0)}. \quad (4.79)$$

Note that if we fix a compact $K \subset \mathbb{C}$, h_z is well defined for all $z \in K$ at least asymptotically almost surely by Lemma 4.14. As explained above, this u_z can be used to characterize the boundary condition at 1, so what we really want to show is that $u_z \circ \eta_{B,E}(1)$ converges in probability to the right boundary condition of the sine system, uniformly with respect to $z \in K$.

In order to uniformize the analysis of the two cases ($|a| < 1$ and $a \geq 1$), we first show that when $|a| < 1$ we can effectively move the boundary condition from 1 to τ_E , that is, we show that $u_z \circ \eta_{B,E}(1) - u_z \circ \eta_{B,E}(\tau_E) \rightarrow 0$ as $E \rightarrow \infty$. By definition,

$$u_z = \begin{pmatrix} E^{-1/4} f_{\beta,a,E} & -E^{-1/4} p_{\beta,a} f'_{\beta,a,E} \\ -E^{1/4} g_{\beta,a,E} & E^{1/4} p_{\beta,a} g'_{\beta,a,E} \end{pmatrix} \begin{pmatrix} p_{\beta,a} h'_z \\ h_z \end{pmatrix} = \begin{pmatrix} \mathcal{W}_{\beta,a}(E^{-1/4} f_{\beta,a,E}, h_z) \\ \mathcal{W}_{\beta,a}(h_z, E^{1/4} g_{\beta,a,E}) \end{pmatrix},$$

so it suffices to show that if ψ stands for either $E^{-1/4} f_{\beta,a,E}$ or $E^{1/4} g_{\beta,a,E}$, then $\mathcal{W}_{\beta,a}(h_z, \psi)(\infty) - \mathcal{W}_{\beta,a}(h_z, \psi)(\log E) \rightarrow 0$ in probability as $E \rightarrow \infty$ (this is not entirely trivial, as the solutions themselves depend on E). To prove this, note that because $\mathfrak{G}_{\beta,a,E}\psi = 0$ and $\mathfrak{G}_{\beta,a,E}h_z = zh_z$, then $(\mathcal{W}_{\beta,a}(h_z, \psi))' = \frac{z\sqrt{E}}{2} h_z \psi w_{\beta,a}$ so

$$\mathcal{W}_{\beta,a}(h_z, \psi)(\infty) - \mathcal{W}_{\beta,a}(h_z, \psi)(\log E) = \frac{z\sqrt{E}}{2} \int_{\log E}^{\infty} h_z(t) \psi(t) w_{\beta,a}(t) dt.$$

To estimate this, we can make the dependence on E clearer by shifting time and using $\tilde{h}_z(t) := h_z(t + \log E)$ and $\tilde{\psi}(t) := \psi(t + \log E)$. By Proposition 3.17, $\tilde{h}_z$ and $\tilde{\psi}$ solve $\tilde{\mathfrak{G}}_{\beta,a}\tilde{h}_z = (1 + \frac{z}{2\sqrt{E}})\tilde{h}_z$ and $\tilde{\mathfrak{G}}_{\beta,a}\tilde{\psi} = \tilde{\psi}$ where $\tilde{\mathfrak{G}}_{\beta,a}$ is defined from the Brownian motion $\tilde{B}_E := B_E(\cdot + \log E) - B_E(\log E)$. In particular, the weight of $\tilde{\mathfrak{G}}_{\beta,a}$ is related to that of $\mathfrak{G}_{\beta,a}$ by $\tilde{w}_{\beta,a}(t) = E^{a+1} e^{\frac{2}{\sqrt{\beta}} B_E(\log E)} w_{\beta,a}(t + \log E)$. Hence, shifting time by $\log E$ in the above integral yields

$$\mathcal{W}_{\beta,a}(h_z, \psi)(\infty) - \mathcal{W}_{\beta,a}(h_z, \psi)(\log E) = \frac{z}{2} E^{-1/2-a} e^{-\frac{2}{\sqrt{\beta}} B_E(\log E)} \int_0^{\infty} \tilde{h}_z(t) \tilde{\psi}(t) \tilde{w}_{\beta,a}(t) dt.$$

Recall from §3.2.1 that 1 and $\tilde{\omega}_{\beta,a}(t) := \int_0^t \tilde{p}_{\beta,a}^{-1}(s) ds$ are a pair of fundamental solutions to $\tilde{\mathfrak{G}}_{\beta,a} f = 0$. Therefore, the argument we used in the proof of Proposition 4.9 shows that for any $\varepsilon > 0$ there is a $C > 0$ such that with probability at least $1 - \varepsilon/4$, for all $t \geq 0$,

$$|\tilde{\psi}(t)| \leq C(|\psi(\log E)| + |\psi'(\log E)|)(1 + \tilde{\omega}_{\beta,a}(t))$$

and

$$|\tilde{h}_z(t)| \leq C\left(1 + \frac{|z|}{2\sqrt{E}}\right)(|\tilde{h}_z(0)| + |\tilde{h}'_z(0)|)(1 + \tilde{\omega}_{\beta,a}(t)).$$

By the representation in polar coordinates of ψ with the estimates on $e^{2\rho_{\beta,a,E}^f}$ and $e^{2\rho_{\beta,a,E}^g}$ from Proposition 4.8, for any $\delta > 0$ we know that there is a $C' > 0$ such that $|\psi(\log E)| \vee |\psi'(\log E)| \leq C'E^{1/2\beta+\delta/2}(p_{\beta,a}(\log E))^{-1/2} = C'E^{1/2\beta+a/2+\delta/2} \exp(\frac{1}{\sqrt{\beta}}B_E(\log E))$ with probability at least $1 - \varepsilon/4$. Then, on the event from Lemma 4.14, $|\tilde{h}_z(0)| \leq E^{a/2}|\Phi_{1+z/2\sqrt{E}}(0)|$ and $|\tilde{h}'_z(0)| \leq E^{a/2}|\Phi'_{1+z/2\sqrt{E}}(0)|$. Because Φ_λ is analytic in λ , $\Phi_{1+z/2\sqrt{E}}(0) = \Phi_1(0) + O_{\mathbb{P},K}(1/\sqrt{E})$ and $\Phi'_{1+z/2\sqrt{E}}(0) = \Phi'_1(0) + O_{\mathbb{P},K}(1/\sqrt{E})$, and since Φ_1 is a solution to $\tilde{\mathfrak{G}}_{\beta,a}\Phi_1 = \Phi_1$ defined by its behavior at infinity, its law does not depend on E . It follows that there is a $C'' > 0$ such that $|\tilde{h}_z(0)| \vee |\tilde{h}'_z(0)| \leq C''E^{a/2}$ for all $z \in K$ with probability at least $1 - \varepsilon/4$. Combining these estimates, we see that for a (different) $C > 0$, with probability at least $1 - 3\varepsilon/4$, for all $z \in K$,

$$|\mathcal{W}_{\beta,a}(h_z, \psi)(\infty) - \mathcal{W}_{\beta,a}(h_z, \psi)(\log E)| \leq CE^{-1/2+1/2\beta+\delta/2} e^{-\frac{1}{\sqrt{\beta}}B_E(\log E)} \int_0^\infty (1 + \tilde{\omega}_{\beta,a}(t))^2 \tilde{w}_{\beta,a}(t) dt.$$

Since $|a| < 1$ here, $1, \tilde{\omega}_{\beta,a} \in L^2([0, \infty), \tilde{w}_{\beta,a} d\mathcal{L})$, so the integral is finite. Then for E large enough $e^{-\frac{1}{\sqrt{\beta}}B_E(\log E)} \leq E^{\delta/2}$ with probability at least $1 - \varepsilon/4$ by properties of Brownian motion, so finally the difference is bounded by a constant times $E^{-1/2+1/2\beta+\delta}$ with probability at least $1 - \varepsilon$. Since $\beta > 2$, this vanishes as $E \rightarrow \infty$ for δ small enough. Therefore, the difference of the Wronskians converges to 0 in probability, and indeed $u_z(\infty) - u_z(\log E) \rightarrow 0$ compactly in probability as $E \rightarrow \infty$.

It remains to show, now for any $a > -1$, that $u_z(\log E)$ converges to the boundary condition $(\operatorname{Re} \mathcal{B}_\beta(\infty), 1)$ of the sine system. Writing $u_z(\log E)$ as in (4.78) but using the expression of h_z in terms of $\Phi_{1+z/2\sqrt{E}}$ from (4.79),

$$u_z(\log E) = \begin{pmatrix} \operatorname{Re} \ell_{\beta,E}(\tau_E) \\ 1 \end{pmatrix} + Y_E \begin{pmatrix} \sin \xi_{\beta,a,E}^g(\tau_E) & \cos \xi_{\beta,a,E}^g(\tau_E) \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \Phi'_{1+z/2\sqrt{E}}(0) \\ \Phi_{1+z/2\sqrt{E}}(0) \end{pmatrix}$$

where

$$Y_E := \frac{E^{-a/2} e^{-\rho_{\beta,a,E}^g(\tau_E) - \frac{1}{\sqrt{\beta}}B_E(\log E)}}{p_{\beta,a}(\log E) \tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,a,E})(0)}.$$

It is not hard to verify that the second term of $u_z(\log E)$ vanishes in the limit in probability. Indeed, for $\varepsilon > 0$ and $\delta \in (0, 1/2\beta)$, Proposition 4.8 shows that $e^{-\rho_{\beta,a,E}^g(\tau_E)} \leq CE^{-1/2\beta+\delta/2}$ for a $C > 0$ with

probability at least $1 - \varepsilon/3$, and $e^{-\frac{1}{\sqrt{\beta}}B_E(\log E)} \leq E^{\delta/2}$ with probability at least $1 - \varepsilon/3$ as above. Moreover, $\Phi_{1+z/2\sqrt{E}}(0) = \Phi_1(0) + O_{\mathbb{P},K}(1/\sqrt{E})$ and the law of $\Phi_1(0)$ does not depend on E , and the same goes for $\Phi'_{1+z/2\sqrt{E}}(0)$. Since the denominator of Y_E is at least $E^{-a/2}$ on the event from Lemma 4.14, it follows that for any E large enough the second term of $u_z(\log E)$ is bounded (uniformly for $z \in K$) by a constant times $E^{-1/2\beta+\delta}$ with probability at least $1 - \varepsilon$, and this means that it converges to zero 0 in probability as $E \rightarrow \infty$, uniformly in $z \in K$. In other words,

$$u_z(\log E) = \begin{pmatrix} \operatorname{Re} \ell_{\beta,E}(\tau_E) \\ 1 \end{pmatrix} + o_{\mathbb{P},K}(1).$$

This immediately implies that the second entry of $u_z(\log E)$ converges to 1, which is the desired value. To show that $\operatorname{Re} \ell_{\beta,E}(\tau_E) \rightarrow \operatorname{Re} B_\beta(\infty)$, we set $\tau_{E,\alpha} := (1 - E^{-1/2+\alpha})/c_E$, and we write

$$|\operatorname{Re} \ell_{\beta,E}(\tau_E) - \operatorname{Re} B_\beta(\infty)| \leq |\operatorname{Re} \ell_{\beta,E}(\tau_E) - \operatorname{Re} \ell_{\beta,E}(\tau_{E,\alpha})| \quad (4.80a)$$

$$+ |\operatorname{Re} \ell_{\beta,E}(\tau_{E,\alpha}) - \operatorname{Re} B_\beta \circ \nu_E(\tau_{E,\alpha})| \quad (4.80b)$$

$$+ |\operatorname{Re} B_\beta \circ \nu_E(\tau_{E,\alpha}) - \operatorname{Re} B_\beta(\infty)|. \quad (4.80c)$$

As $\nu_E(\tau_{E,\alpha}) = (\frac{1}{2} - \alpha) \log E$, (4.80c) vanishes a.s. in the limit $E \rightarrow \infty$. It is also clear from Proposition 4.7 that (4.80b) vanishes in probability as $E \rightarrow \infty$. To show that (4.80a) also vanishes, recall from (4.48) that

$$\begin{aligned} d(\operatorname{Re} \ell_{\beta,E})(t) &= \frac{2}{\sqrt{\beta}} e^{-2\rho_{\beta,a,E}^g(t)} \sin 2\xi_{\beta,a,E}^g(t) \sqrt{\frac{2c_E}{1-c_E t}} dB_E(t) \\ &\quad + e^{-2\rho_{\beta,a,E}^g(t)} \left((2a+1) \sin 2\xi_{\beta,a,E}^g(t) + \frac{4}{\beta} \sin 4\xi_{\beta,a,E}^g(t) \right) \frac{c_E}{1-c_E t} dt. \end{aligned} \quad (4.81)$$

By Proposition 4.8, we know that for any $\varepsilon > 0$ and $\delta \in (0, 2/\beta)$, there is a $C > 0$ such that $e^{-\rho_{\beta,a,E}^g(t)} \leq C(1 - c_E t)^{1/\beta-\delta/2}$ for $t \in [0, \tau_E]$ with probability at least $1 - \varepsilon$. It follows that there is a $C' > 0$ such that with probability at least $1 - \varepsilon$,

$$\begin{aligned} \left| \int_{\tau_{E,\alpha}}^{\tau_E} e^{-2\rho_{\beta,a,E}^g(t)} \left((2a+1) \sin 2\xi_{\beta,a,E}^g(t) + \frac{4}{\beta} \sin 4\xi_{\beta,a,E}^g(t) \right) \frac{c_E}{1-c_E t} dt \right| \\ \leq C' \int_{\tau_{E,\alpha}}^{\tau_E} c_E (1 - c_E t)^{-1+2/\beta-\delta} dt = \frac{C'}{2/\beta-\delta} E^{-(1/2-\alpha)(2/\beta-\delta)} (1 - E^{-\alpha(2/\beta-\delta)}), \end{aligned}$$

which vanishes as $E \rightarrow \infty$. Similarly, there is a $C'' > 0$ such that with probability at least $1 - \varepsilon$, the quadratic variation of the first line of (4.81) is bounded on $[\tau_{E,\alpha}, \tau_E]$ by

$$C'' \int_{\tau_{E,\alpha}}^{\tau_E} c_E (1 - c_E t)^{-1+4/\beta-2\delta} dt = \frac{C''}{4/\beta-2\delta} E^{-(1/2-\alpha)(4/\beta-2\delta)} (1 - E^{-\alpha(4/\beta-2\delta)}),$$

which vanishes as $E \rightarrow \infty$. Applying Bernstein's inequality, it follows that the integral of (4.81) between $\tau_{E,\alpha}$ and τ_E converges to 0 in probability as $E \rightarrow \infty$, and this concludes the proof. $\blacksquare$

Proof of Lemma 4.14. Fix $E \in \{E_n\}_{n \in \mathbb{N}}$ and a compact $K \subset \mathbb{C}$. We need to estimate, for $z \in K$, $p_{\beta,a}(\log E) \tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,a,E})(0)$.

We first replace $\Phi_{1+z/2\sqrt{E}}$ with Φ_1 . The point of doing this is that Φ_1 and $\tilde{g}_{\beta,a,E}$ both solve $\tilde{\mathfrak{G}}_{\beta,a} f = f$, so their Wronskian is constant. As $\tilde{g}_{\beta,a,E}(-\log E) = 0$ and $\tilde{g}'_{\beta,a,E}(-\log E) = 1$, this means that $\tilde{\mathcal{W}}_{\beta,a}(\Phi_1, \tilde{g}_{\beta,a,E}) \equiv \tilde{p}_{\beta,a}(-\log E) \Phi_1(-\log E)$.

Fix $\varepsilon > 0$. As we have seen before, Proposition 4.8 implies that for any $\delta > 0$, there is a $C > 0$ such that both $E^{1/4} \tilde{g}_{\beta,a,E}(0)$ and $E^{1/4} \tilde{g}'_{\beta,a,E}(0)$ are at most $CE^{1/2\beta+\delta/2} (p_{\beta,a}(\log E))^{-1/2}$ with probability at least $1 - \varepsilon/3$. Then, by definition, $(p_{\beta,a}(\log E))^{1/2} = E^{-a/2} \exp(-\frac{1}{\sqrt{\beta}} B_E(\log E))$, and by properties of Brownian motion this is bounded by $E^{-a/2+\delta/2}$ with probability at least $1 - \varepsilon/3$ for any E large enough. Finally, as Φ_λ is analytic in λ , $\Phi_{1+z/2\sqrt{E}}(0) - \Phi_1(0) = O_{\mathbb{P},K}(1/\sqrt{E})$ and $\Phi'_{1+z/2\sqrt{E}}(0) - \Phi'_1(0) = O_{\mathbb{P},K}(1/\sqrt{E})$ where the implicit constants are well-defined random variables, so with probability at least $1 - \varepsilon/3$ these differences are bounded by $C'E^{-1/2}$ for some constant $C' > 0$, uniformly in $z \in K$. It follows that with probability at least $1 - \varepsilon$, there is a $C > 0$ such that

$$E^{a/2} |p_{\beta,a}(\log E) \tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E}} - \Phi_1, E^{1/4} \tilde{g}_{\beta,a,E})(0)| \leq CE^{-1/2+1/2\beta+\delta} \quad (4.82)$$

for all $z \in K$ and all E large enough. With δ small enough, this vanishes as $E \rightarrow \infty$, so the left-hand side of (4.82) converges to 0 in probability uniformly in $z \in K$. Combining this with our earlier observation about the Wronskian of Φ_1 and $\tilde{g}_{\beta,a,E}$ and noticing that $p_{\beta,a}(\log E) \tilde{p}_{\beta,a}(-\log E) = 1$, it follows that

$$E^{a/2} p_{\beta,a}(\log E) \tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E}}, E^{1/4} \tilde{g}_{\beta,a,E})(0) = E^{1/4+a/2} \Phi_1(-\log E) + o_{\mathbb{P},K}(1). \quad (4.83)$$

We now analyse the behavior of $\Phi_1(-\log E)$ using the asymptotics derived in §3.2.2. By Theorem 3.18 and the Skorokhod representation theorem, we can write

$$\Phi_1(-\log E) = C_{\Phi_1} E^{1/2\beta-a/2-1/4} e^{X(\log E)} \cos \xi_{\beta,a}(\log E)$$

where $C_{\Phi_1}^2 = \Phi_1^2(0) + \Phi_1'^2(0)$, $\xi_{\beta,a}$ satisfies $\xi_{\beta,a}(t) - 2\pi \lfloor \frac{\xi_{\beta,a}(t)}{2\pi} \rfloor \rightarrow U \sim \text{Unif}[0, 2\pi)$ in probability as $t \rightarrow \infty$, and X has the property that for any $\varepsilon, \delta > 0$, there is a $C > 0$ such that

$$\mathbb{P} \left[\forall t \geq 0, |X(t)| \leq C(1 + t^{1/2+\delta}) \right] \geq 1 - \varepsilon.$$

Hence, in the same way as in the proof of Lemma 4.12 in the soft edge case, if $\varepsilon > 0$ and $\zeta > 0$ is small enough, then $|\cos \xi_{\beta,a}(\log E)| \geq \zeta$ with probability at least $1 - \varepsilon/3$ for all E large enough. Moreover, shrinking ζ if necessary, we may assume that $C_{\Phi_1} > \zeta$ with probability at least $1 - \varepsilon/3$, and we know that for all E large enough, $e^{X(\log E)} \geq CE^{-\delta}$ with probability at least $1 - \varepsilon/3$. Hence,

$$E^{1/4+a/2} |\Phi_1(-\log E)| = C_{\Phi_1} E^{1/2\beta} e^{X(\log E)} |\cos \xi_{\beta,a}(\log E)| \geq C\zeta^2 E^{1/2\beta-\delta} \quad (4.84)$$

with probability at least $1 - \varepsilon$ for all E large enough. With $\delta < 1/2\beta$, this diverges as $E \rightarrow \infty$, so it is certainly at least 1 for all E large enough.

Combining (4.84) with (4.83) shows that

$$\mathbb{P}[\forall z \in K, p_{\beta,a}(\log E_n) |\tilde{\mathcal{W}}_{\beta,a}(\Phi_{1+z/2\sqrt{E_n}}, E_n^{1/4} \tilde{\mathfrak{g}}_{\beta,a,E_n})(0)| \geq E_n^{-a/2}] \rightarrow 1$$

as $n \rightarrow \infty$, which concludes the proof. $\blacksquare$

4.4.4 Remarks on the spectral weights of $\mathfrak{G}_{\beta,a,E}$

The purpose of this short section is to relate the spectral weights of the shifted and scaled stochastic Bessel operator $\mathfrak{G}_{\beta,a,E}$ to those of the usual stochastic Bessel operator $\mathfrak{G}_{\beta,a}$. We obtain a relationship between these spectral weights in (4.87), and from this we see that if the expression for the spectral weights of $\mathfrak{G}_{\beta,a}$ given in Conjecture 3.1 is true, then the point process level hard edge to bulk transition should follow from the convergence of the Weyl–Titchmarsh functions.

Suppose that λ_E is an eigenvalue of the shifted problem, that is, the canonical system $Ju' = -\lambda_E G_{\beta,a,E} u$ has a solution u_E of the form $u_E = A_{\beta,a,E}^{-1} \begin{pmatrix} p_{\beta,a} h'_E \\ h_E \end{pmatrix}$ for an $h_E \in L^2_{G_{\beta,a}}(0, \infty)$ solving $\mathfrak{G}_{\beta,a,E} h_E = \lambda_E h_E$ such that $u_E(0) = e_0$ and $\lim_{t \rightarrow \infty} p_{\beta,a}(t) h'_E(t) = 0$. By Theorem 2.8, the spectral weight associated with λ_E is $\mu_{G_{\beta,a,E}}(\{\lambda_E\}) = \|u_E\|_{G_{\beta,a,E}}^{-2}$. By construction of the canonical system, since $\mathfrak{G}_{\beta,a,E}$ is a generalized Sturm–Liouville operator with weight $w_{\beta,a}\sqrt{E}/2$, this yields

$$\mu_{G_{\beta,a,E}}(\{\lambda_E\}) = \frac{1}{\|u_E\|_{G_{\beta,a,E}}^2} = \frac{1}{\|h_E\|_{w_{\beta,a}\sqrt{E}/2}^2} = \frac{2}{\sqrt{E} \|h_E\|_{w_{\beta,a}}^2}. \quad (4.85)$$

By definition of $\mathfrak{G}_{\beta,a,E}$, h_E is also an eigenfunction of $\mathfrak{G}_{\beta,a}$, since $\lambda_E h_E = \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E)h_E$ so $\mathfrak{G}_{\beta,a} h_E = (E + \frac{1}{2}\lambda_E \sqrt{E})h_E$. Set $\hat{\lambda} := E + \frac{1}{2}\lambda_E \sqrt{E}$ and $\hat{h} := E^{-1/4} h_E$, so that $\hat{h}$ solves $\mathfrak{G}_{\beta,a} \hat{h} = \hat{\lambda} \hat{h}$. By construction of the canonical system for $\mathfrak{G}_{\beta,a}$, this means that $\hat{u} := A_{\beta,a}^{-1} \begin{pmatrix} p_{\beta,a} \hat{h}' \\ \hat{h} \end{pmatrix}$ solves $J\hat{u}' = -\hat{\lambda} G_{\beta,a} \hat{u}$, where $G_{\beta,a}$ is the coefficient matrix of the stochastic Bessel canonical system as defined in (3.24) and $A_{\beta,a}$ is the matrix (3.23) that maps between solutions to $\mathfrak{G}_{\beta,a}$'s eigenvalue equation and canonical system solutions. Now, by definition,

$$\hat{u}(0) = \begin{pmatrix} \hat{h}'(0) \\ \hat{h}(0) \end{pmatrix} = E^{-1/4} \begin{pmatrix} h'_E(0) \\ h_E(0) \end{pmatrix}.$$

The initial condition $u_E(0) = e_0$ also requires that

$$\begin{pmatrix} h'_E(0) \\ h_E(0) \end{pmatrix} = A_{\beta,a,E}(0) u_E(0) = \begin{pmatrix} E^{1/4} & 0 \\ 0 & E^{-1/4} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} E^{1/4} \\ 0 \end{pmatrix},$$

and it follows that $\hat{u}(0) = e_0$. Therefore, again by Theorem 2.8, the spectral weight of $G_{\beta,a}$ at $\hat{\lambda}$ is

$$\mu_{G_{\beta,a}}(\{\hat{\lambda}\}) = \frac{1}{\|\hat{u}\|_{G_{\beta,a}}^2} = \frac{1}{\|\hat{h}\|_{w_{\beta,a}}^2}. \quad (4.86)$$

But $\hat{h} = E^{-1/4}h_E$ gives $\|\hat{h}\|_{w_{\beta,a}}^2 = \frac{1}{\sqrt{E}}\|h_E\|_{w_{\beta,a}}^2$, so comparing (4.85) and (4.86) shows that

$$\mu_{G_{\beta,a,E}}(\{\lambda_E\}) = \frac{2}{E}\mu_{G_{\beta,a}}(\{\hat{\lambda}\}) = \frac{2}{E}\mu_{G_{\beta,a}}\left(\left\{E + \frac{\lambda_E\sqrt{E}}{2}\right\}\right). \quad (4.87)$$

If Conjecture 3.1 is true, then $\mu_{G_{\beta,a}}(\{\hat{\lambda}\}) = \hat{\lambda}\Gamma_{\hat{\lambda}}$ where $\Gamma_{\hat{\lambda}} \sim \Gamma(\beta/2, 2/\beta)$ is independent of the eigenvalues and of the other $\Gamma(\beta/2, 2/\beta)$ weights. Then (4.87) gives

$$\mu_{G_{\beta,a,E}}(\{\lambda_E\}) = \frac{2}{E}\left(E + \frac{\lambda_E\sqrt{E}}{2}\right)\Gamma_{\hat{\lambda}} = \left(1 + \frac{\lambda_E}{2\sqrt{E}}\right)\Gamma'_{\lambda_E},$$

where now $\Gamma'_{\lambda_E} \sim \Gamma(\beta/2, 4/\beta)$, but it is still independent of the eigenvalues and of the other $\Gamma(\beta/2, 4/\beta)$ weights. Since the spectral weights of the stochastic sine operator are independent $\Gamma(\beta/2, 4/\beta)$ random variables also independent of the eigenvalues, the spectral weights of $\mathfrak{G}_{\beta,a,E}$ as $E \rightarrow \infty$ should match those of the stochastic sine operator. Thus, if Conjecture 3.1 holds, randomizing the argument used to deduce the point process convergence from the convergence of spectral measures in Proposition 2.17 can then be expected to yield the hard edge to bulk convergence at the point process level.

Chapter 5

Conclusion

In this thesis, we have seen several ways in which the theory of canonical systems can help in the study of scaling limits of β -ensembles. Overall, the results highlight the potential of canonical systems theory to provide a consistent, unifying framework under which to study these scaling limits. This originates from this theory's ability to represent in the same category all stochastic differential operators that arise as scaling limits of β -ensembles as well as the familiar tridiagonal models of β -ensembles from random matrix theory.

The results from §3.1.3 and §3.2.3 shed a new light on the convergences of β -ensembles in the soft edge and hard edge scaling limits to the Airy_β and $\text{Bessel}_{\beta,a}$ point processes, which were first proven in the celebrated works [60] and [58]. Indeed, the canonical system embedding allows for a more systematic method, at the operator level, to derive scaling limits of β -ensembles, and makes more obvious the similarities between the different scaling limit problems. This also leads to think that the canonical systems representation would be appropriate to derive other scaling limits of β -ensembles, beyond those that have already been described.

We have also seen in Chapter 4 that, in their canonical systems representations, the shifted and scaled stochastic Airy and Bessel operators $\mathcal{H}_{\beta,E} := 2\sqrt{E}(\mathcal{H}_\beta - E)$ and $\mathfrak{G}_{\beta,a,E} := \frac{2}{\sqrt{E}}(\mathfrak{G}_{\beta,a} - E)$ both converge to the stochastic sine operator as $E \rightarrow \infty$. These results are certainly interesting in their own right, and in addition they demonstrate how canonical systems theory can be effective at describing the relationships between the different operators that arise as scaling limits.

Several other problems that appear in the theory of β -ensembles could also possibly be approached through canonical systems theory. An important example is the one mentioned in §3.3.2, namely the canonical system level convergence of tridiagonal random matrix models for β -ensembles to the stochastic sine operator, which would reproduce results of [72, 76] at the canonical system level. A simple version of this problem is the following.

Problem 5.1. Fix $E \in (-2, 2)$, and let $M_n := \sqrt{(4 - E^2)n}(\mathbf{H}_{\beta,n} - \sqrt{n}E)$ for $n \in \mathbb{N}$ where $\mathbf{H}_{\beta,n}$ is an $n \times n$ tridiagonal random matrix for the Hermite β -ensemble, as in (1.2). Let $H_{\beta,n} \in \text{CS}[0, 1)$

denote the coefficient matrix of a canonical system on $[0, 1)$ that is equivalent to the eigenvalue problem $M_n v = z v$ under appropriate boundary conditions. With R_β and v as in Definition 1.2, prove that $H_{\beta,n} \xrightarrow{\text{law}} R_\beta \circ v$ vaguely on $\mathcal{I}_\beta$ as $n \rightarrow \infty$ where $\mathcal{I}_\beta := [0, 1)$ for $\beta \leq 2$ and $\mathcal{I}_\beta := [0, 1]$ for $\beta > 2$. Extend the convergence to transfer matrices and Weyl–Titchmarsh functions.

As the solution of this problem could be assumed to have important similarities with the proofs of Theorems 3.8 and 3.23 and their corollaries, presumably the solution would extend to more general random matrix models.

As discussed in §1.3.4, despite the success of canonical systems theory at providing a consistent approach to the study of scaling limits of β -ensembles, the method still currently lacks a good way of retrieving the eigenvalue point process convergence from a sequence of canonical systems which otherwise converge in the various senses we have seen in §2.2. In Proposition 2.17 in §2.2.2.2, we saw conditions on the spectral weights under which the convergence of spectral measures implies the convergence of the eigenvalue point processes, but they are not necessarily easy to verify in practice, especially if the canonical systems are not limit circle at both endpoints. Indeed, verifying these conditions requires a rather precise description of the spectral weights, which is not easy to obtain. This raises the following problem.

Problem 5.2. Given some interval $\mathcal{I} = [a, b)$ or $[a, b]$, take coefficient matrices $H_n, H \in \text{CS } \mathcal{I}$ with appropriate boundary conditions at endpoints. Suppose that the associated self-adjoint relations have discrete spectrum with only eigenvalues. Suppose also that $H_n \rightarrow H$ vaguely on $\mathcal{I}$ as $n \rightarrow \infty$ and that the associated transfer matrices and Weyl–Titchmarsh functions also converge. Are there simple conditions on any of these objects under which the associated eigenvalue point processes also converge vaguely?

It is possible that this problem could be approached through a more careful analysis of the eigenfunctions, as their canonical system norms are related to the spectral weights by Theorem 2.8, but maybe other avenues are also possible.

Besides the interest of being able to deduce the convergence of the eigenvalue point processes themselves, a motivation for solving Problem 5.2 is that if both the spectral measures and the eigenvalue point processes converge, then the spectral weights also have to converge, so a description of the pre-limit spectral weights can give a description of the limit spectral weights. In particular, the formula for the spectral weights of the stochastic Bessel operator from Conjecture 3.1 could be proven using this type of strategy, using the ideas from §3.2.3.3.

The results from Chapter 4 suggest another interesting problem. The whole method we used to describe both edge to bulk transitions applied very well to both problems, and we saw the stochastic sine operator appearing quite naturally as a limit once the coefficient matrix was decomposed into polar coordinates for a pair of fundamental solutions to $\mathcal{H}_{\beta,E} f = 0$ or $\mathcal{G}_{\beta,a,E} f = 0$. It is there-

fore natural to ask whether this generalizes further, and if one could find a larger class of random generalized Sturm–Liouville operators with the stochastic sine operator as a scaling limit.

Problem 5.3. Find a class of random generalized Sturm–Liouville operators L_β on $[0, \infty)$ such that for some $\sigma_E > 0$, the canonical system for the rescaled operator $L_{\beta,E} := \sigma_E(L_\beta - E)$ converges to the stochastic sine canonical system as $E \rightarrow \infty$.

Heuristically, it appears plausible that the necessary characteristics of L_β to have the stochastic sine operator as a scaling limit should mostly be to have infinitely many eigenvalues and the right sort of “noise”, so that the stochastic sine operator is retrieved when zooming in the direction of the eigenvalues while rescaling the space so that they are equally spaced on average.

A first step towards this problem is suggested by the method that we used in §4.3 to prove the vague convergence of coefficient matrices in the edge to bulk transitions. Recall that to prove this vague convergence, we started by deriving SDEs driving polar coordinates for solutions to both $\mathcal{H}_{\beta,E}f = 0$ and $\mathfrak{G}_{\beta,a,E}f = 0$. We then noticed that we could slightly generalize these SDEs to obtain polar coordinates which, in a sense, interpolate between those for the two problems, and we worked exclusively with these generalized polar coordinates to prove the vague convergence. It seems plausible to think that these polar coordinates could be used to represent (possibly time-changed) solutions to $L_{\beta,E}f = 0$ for some other random generalized Sturm–Liouville operator $L_{\beta,E}$, which would have a general form that could be specialized to describe either $\mathcal{H}_{\beta,E}$ or $\mathfrak{G}_{\beta,a,E}$. If this is true, then the operator $L_{\beta,E}$ can possibly be inferred from the polar coordinates themselves, and this would provide a first step towards solving Problem 5.3.

Problems 5.1 and 5.3 are only two examples of open problems related to β -ensembles that could be solved through the theory of canonical systems, but there are certainly many others. I hope that this thesis can convince others of the usefulness of canonical systems as tools to study problems related to β -ensembles, and more generally in random matrix theory.

Appendix

A.1 Growth bounds on paths of continuous martingales

Proposition A.1 [55, Proposition 19]. *Let M be a continuous martingale on an interval $[0, T)$ with $M(0) = 0$. For any $\varepsilon, \delta > 0$, there is a $C > 0$ such that*

$$\mathbb{P} \left[\sup_{t \in [0, T)} \frac{|M(t)|}{1 + \langle M \rangle(t)^{1/2+\delta}} \geq C \right] < \varepsilon.$$

Proof. Setting $T_t := \inf\{s \geq 0 : \langle M \rangle(s) > t\}$, by the Dambis–Dubins–Schwarz theorem, $B(t) := M(T_t)$ is a standard Brownian motion on $[0, \langle M \rangle(T))$ and $M(t) = B(\langle M \rangle(t))$ for any $t \in [0, T)$. Therefore,

$$\sup_{t \in [0, T)} \frac{|M(t)|}{1 + \langle M \rangle(t)^{1/2+\delta}} = \sup_{t \in [0, \langle M \rangle(T))} \frac{|B(t)|}{1 + t^{1/2+\delta}},$$

and the result will follow if we can prove that for any $\varepsilon, \delta > 0$, there is a $C > 0$ such that

$$\mathbb{P} \left[\sup_{t \in [0, \infty)} \frac{|B(t)|}{1 + t^{1/2+\delta}} \geq C \right] < \varepsilon.$$

Now, since

$$\frac{\sqrt{2t \log \log t}}{1 + t^{1/2+\delta}} \rightarrow 0 \quad \text{as } t \rightarrow \infty,$$

then

$$\limsup_{t \rightarrow \infty} \frac{|B(t)|}{1 + t^{1/2+\delta}} \leq \left(\limsup_{t \rightarrow \infty} \frac{|B(t)|}{\sqrt{2t \log \log t}} \right) \left(\limsup_{t \rightarrow \infty} \frac{\sqrt{2t \log \log t}}{1 + t^{1/2+\delta}} \right) = 0$$

a.s. by the law of the iterated logarithm. Since the map $t \mapsto |B(t)|(1 + t^{1/2+\delta})^{-1}$ is a.s. continuous, it follows that the random variable $\sup_{t \in [0, \infty)} |B(t)|(1 + t^{1/2+\delta})^{-1}$ is a.s. finite, and because

$$\left\{ \sup_{t \in [0, \infty)} \frac{|B(t)|}{1 + t^{1/2+\delta}} < \infty \right\} = \bigcup_{n \in \mathbb{N}} \left\{ \sup_{t \in [0, \infty)} \frac{|B(t)|}{1 + t^{1/2+\delta}} < n \right\},$$

the result follows by continuity of measures. ■

A.2 Concentration inequalities for martingales

The purpose of this appendix, which is reproduced from [55, §A], is to recall and prove some concentration inequalities for martingales that are used in the main text. The results we need are simple extensions of classical concentration inequalities (Freedman's and Azuma's) to martingales with subgaussian increments.

Throughout, for a discrete-time martingale $M = \{M_n\}_{n \in \mathbb{N}_0}$ with respect to a filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}_0}$, we denote by $\Delta M_n := M_n - M_{n-1}$ the increments of M . We define the *bracket* or *quadratic variation* process of M as

$$\langle M \rangle_n := \sum_{j=1}^n \mathbb{E}[(\Delta M_j)^2 \mid \mathcal{F}_{j-1}].$$

Moreover, given an adapted process $V = \{V_n\}_{n \in \mathbb{N}}$, we say that M is *V-conditionally subgaussian* if each increment ΔM_n is V_n -subgaussian given $\mathcal{F}_{n-1}$, that is, if for all $n \in \mathbb{N}$ and all $\lambda \in \mathbb{R}$,

$$\mathbb{E}[e^{\lambda \Delta M_n} \mid \mathcal{F}_{n-1}] \leq e^{\lambda^2 V_n / 2} \quad \text{a.s.}$$

We then define the *subgaussian bracket* $\langle\langle M \rangle\rangle$ of M as the smallest nonnegative, nondecreasing predictable process so that M is V -conditionally subgaussian for $V_n := \langle\langle M \rangle\rangle_n - \langle\langle M \rangle\rangle_{n-1}$, and we say that M is *subgaussian* if $\langle\langle M \rangle\rangle < \infty$.

The first concentration inequality that we need is an extension of Freedman's inequality [37]. Our starting point is the following formulation of Freedman's inequality.

Theorem A.2 (Freedman's inequality). *Let $M = \{M_n\}_{n \in \mathbb{N}_0}$ be a discrete-time martingale with respect to a filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}_0}$ and with $M_0 = 0$. Suppose that for some $a > 0$, the increments of M satisfy $|\Delta M_n| \leq a$ for all $n \in \mathbb{N}$. Then for any $N \in \mathbb{N}$ and any $x, y > 0$,*

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} |M_n| > x, \langle M \rangle_N \leq y\right] \leq 2 \exp\left(-\frac{\log 2}{2} \left(\frac{x}{a} \wedge \frac{x^2}{y}\right)\right).$$

Proof. Since $\langle M \rangle$ is a non-decreasing process, the event that $\sup_{1 \leq n \leq N} M_n > x$ and $\langle M \rangle_N \leq y$ is contained in the event that $M_n > x$ and $\langle M \rangle_n \leq y$ for some n . Hence, it follows from the original statement of Freedman's inequality [37, Theorem 1.6] that

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} M_n > x, \langle M \rangle_N \leq y\right] \leq e^{x/a} \left(\frac{y/a^2}{x/a + y/a^2}\right)^{x/a + y/a^2} = \exp\left(\frac{y}{a^2} \left(-\left(1 + \frac{ax}{y}\right) \log\left(1 + \frac{ax}{y}\right) + \frac{ax}{y}\right)\right). \quad (\text{A.1})$$

To get a simpler expression for the exponent, let

$$b_p(\xi) := -C\xi^p + (1 + \xi) \log(1 + \xi) - \xi$$

for $p = 1$ or $p = 2$, with $C > 0$ to be determined. Note that $b'_p(\xi) = -Cp\xi^{p-1} + \log(1 + \xi)$, so in particular for $\xi \geq 1$, $b'_1(\xi) = -C + \log(1 + \xi) \geq -C + \log 2$, and b_1 is increasing on $[1, \infty)$ if $C < \log 2$. Because $b_1(1) = -C + 2\log 2 - 1$, this guarantees that $b_1 \geq 0$ on $[1, \infty)$ if $C < 2\log 2 - 1$. Then, using the concavity of $\log(1 + \cdot)$, we see that if $\xi \in [0, 1]$, then $b'_2(\xi) \geq -2C\xi + \xi \log 2 \geq 0$ if $C \leq \frac{1}{2} \log 2$, so $b_2 \geq 0$ on $[0, 1]$ under the same condition, since $b_2(0) = 0$.

Combining the condition for b_2 on $[0, 1]$ with that for b_1 on $[1, \infty)$, we get that if $C \leq \frac{1}{2} \log 2 < 2\log 2 - 1$, then $-(1 + \xi) \log(1 + \xi) + \xi \leq -C(\xi \wedge \xi^2)$ for any $\xi \geq 0$. Taking the best constant $C = \frac{1}{2} \log 2$, this shows that the exponent of the tail bound (A.1) is bounded by

$$-\frac{\log 2}{2} \frac{y}{a^2} \left(\frac{ax}{y} \wedge \frac{a^2 x^2}{y^2} \right) = -\frac{\log 2}{2} \left(\frac{x}{a} \wedge \frac{x^2}{y} \right).$$

The theorem now follows by applying the above to both M and $-M$, and combining the results with a union bound. $\blacksquare$

We now prove the following extension of Freedman's inequality, in which the condition that increments are bounded is weakened at the price of increasing the tail bound.

Corollary A.2.1. *Let $M = \{M_n\}_{n \in \mathbb{N}_0}$ be a discrete-time martingale with respect to a filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}_0}$ and with $M_0 = 0$. Suppose there are $K, V > 0$ such that for each n , the increment ΔM_n satisfies $\mathbb{E}[(\Delta M_n)^2 \mid \mathcal{F}_{n-1}] \leq V$ and is K -subgaussian both conditionally on $\mathcal{F}_{n-1}$ and unconditionally. Then for any $N \in \mathbb{N}$ and any $x > 0$,*

$$\mathbb{P} \left[\sup_{1 \leq n \leq N} |M_n| > x \right] \leq (2 + N(N+1)) \exp \left(-\frac{1}{2} \min \left\{ \left(\frac{x \log 2}{6\sqrt{K}} \right)^{2/3}, \frac{x^2 \log 2}{9NV} \right\} \right) + 2 \exp \left(-\frac{x \log 2}{3N\sqrt{2V}} \exp \left(\frac{1}{4} \left(\frac{x \log 2}{6\sqrt{K}} \right)^{2/3} \right) \right).$$

Proof. Take $a > 0$ to be determined later. Notice that M can be decomposed as

$$M_n = \sum_{j=1}^n \left(\Delta M_j - \mathbb{E}[\Delta M_j \mid \mathcal{F}_{j-1}] \right) = M_n^{\leq} + M_n^{>}$$

where $M_n^{\leq} := \sum_{j=1}^n \Delta M_j^{\leq}$ and $M_n^{>} := \sum_{j=1}^n \Delta M_j^{>}$ are martingales with increments

$$\Delta M_j^{\leq} := \Delta M_j \mathbb{1}_{\{|\Delta M_j| \leq a\}} - \mathbb{E}[\Delta M_j \mathbb{1}_{\{|\Delta M_j| \leq a\}} \mid \mathcal{F}_{j-1}]$$

and

$$\Delta M_j^{>} := \Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}} - \mathbb{E}[\Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}} \mid \mathcal{F}_{j-1}].$$

Now, since $M^\leq$ has bounded increments and has $\langle M \rangle_N \leq NV$, Freedman's inequality (Theorem A.2) shows that

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} |M_n^\leq| > x\right] \leq 2 \exp\left(-\frac{\log 2}{2} \left(\frac{x}{2a} \wedge \frac{x^2}{NV}\right)\right). \quad (\text{A.2})$$

Thus, to control the supremum of M , we only need to control the supremum of $M^>$. Note that if $0 < x \leq a$, then

$$\mathbb{P}[|\Delta M_j| \mathbb{1}_{\{|\Delta M_j| > a\}} > x] = \mathbb{P}[|\Delta M_j| > a],$$

because for $|\Delta M_j| \mathbb{1}_{\{|\Delta M_j| > a\}} > x$, the indicator must not vanish, so it must be that $|\Delta M_j| > a$, in which case it is also true that $|\Delta M_j| > x$. If $x > a$, then when $|\Delta M_j| \mathbb{1}_{\{|\Delta M_j| > a\}} > x$ it must also be true that $|\Delta M_j| > a$, and we conclude that for any $x > 0$,

$$\mathbb{P}[|\Delta M_j| \mathbb{1}_{\{|\Delta M_j| > a\}} > x] \leq \mathbb{P}[|\Delta M_j| > a].$$

Hence, for any $x > 0$,

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} \left|\sum_{j=1}^n \Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}}\right| > x\right] \leq \sum_{n=1}^N \sum_{j=1}^n \mathbb{P}\left[|\Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}}| > \frac{x}{n}\right] \leq \sum_{n=1}^N \sum_{j=1}^n \mathbb{P}[|\Delta M_j| > a],$$

and as increments are K -subgaussian,

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} \left|\sum_{j=1}^n \Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}}\right| > x\right] \leq N(N+1) \exp\left(-\frac{a^2}{2K}\right). \quad (\text{A.3})$$

It remains to control the mean terms of the increments of $M^>$. By Cauchy–Schwarz, for each j ,

$$|\mathbb{E}[\Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}} \mid \mathcal{F}_{j-1}]| \leq \sqrt{\mathbb{E}[(\Delta M_j)^2 \mid \mathcal{F}_{j-1}] \mathbb{P}[|\Delta M_j| > a \mid \mathcal{F}_{j-1}]},$$

and as each increment ΔM_j is also K -subgaussian conditionally on $\mathcal{F}_{j-1}$, it follows that

$$\left|\sum_{j=1}^n \mathbb{E}[\Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}} \mid \mathcal{F}_{j-1}]\right| \leq n\sqrt{2V} \exp\left(-\frac{a^2}{4K}\right).$$

With the subgaussian tail bound of bounded random variables, this also gives for any $x > 0$

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} \left|\sum_{j=1}^n \mathbb{E}[\Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}} \mid \mathcal{F}_{j-1}]\right| > x\right] \leq 2 \exp\left(-\frac{x \log 2}{N\sqrt{2V}} \exp\left(\frac{a^2}{4K}\right)\right). \quad (\text{A.4})$$

To conclude, we write

$$\sup_{1 \leq n \leq N} |M_n| \leq \sup_{1 \leq n \leq N} |M_n^\leq| + \sup_{1 \leq n \leq N} \left|\sum_{j=1}^n \Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}}\right| + \sup_{1 \leq n \leq N} \left|\sum_{j=1}^n \mathbb{E}[\Delta M_j \mathbb{1}_{\{|\Delta M_j| > a\}} \mid \mathcal{F}_{j-1}]\right|.$$

Then the tail bounds we got in (A.2), (A.3) and (A.4) give

$$\begin{aligned} \mathbb{P}\left[\sup_{1 \leq n \leq N} |M_n| > x\right] &\leq 2 \exp\left(-\frac{\log 2}{2} \left(\frac{x}{6a} \wedge \frac{x^2}{9NV}\right)\right) \\ &\quad + N(N+1) \exp\left(-\frac{a^2}{2K}\right) + 2 \exp\left(-\frac{x \log 2}{3N\sqrt{2V}} \exp\left(\frac{a^2}{4K}\right)\right). \end{aligned}$$

We now choose a in such a way that the exponents in the first two tail bounds match, that is, so that

$$\frac{x \log 2}{12a} = \frac{a^2}{2K} \iff a^3 = \frac{Kx \log 2}{6}, \quad \text{which gives} \quad \frac{x \log 2}{12a} = \frac{a^2}{2K} = \frac{1}{2} \left(\frac{x \log 2}{6\sqrt{K}}\right)^{2/3}.$$

Using this value for a in the tail bounds yields the desired inequality. $\blacksquare$

Finally, we prove a version of Azuma's inequality for martingales with subgaussian increments.

Theorem A.3 (Azuma's inequality). *Let $M = \{M_n\}_{n \in \mathbb{N}_0}$ be a martingale with respect to a filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}_0}$ and with $M_0 = 0$. If M is subgaussian, then for any $N \in \mathbb{N}$ and any $x, y > 0$,*

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} |M_n| > x, \langle\langle M \rangle\rangle_N \leq y\right] \leq 2 \exp\left(-\frac{x^2}{2y}\right).$$

Proof. For $\lambda > 0$, let

$$\mathcal{E}_n := \exp\left(\lambda M_n - \frac{\lambda^2}{2} \langle\langle M \rangle\rangle_n\right).$$

Then $\{\mathcal{E}_n\}_{n \in \mathbb{N}_0}$ is a supermartingale by definition of the subgaussian bracket of M , and it satisfies $\mathcal{E}_0 = 1$. Hence, if $T := \inf\{n \in \mathbb{N} : M_n > x \text{ or } \langle\langle M \rangle\rangle_n > y\}$, the optional stopping theorem shows that $\mathbb{E} \mathcal{E}_{T \wedge N} \leq 1$ for any $N \in \mathbb{N}$.

Now, let $A_N := \{T \leq N\} \cap \{\langle\langle M \rangle\rangle_N \leq y\}$. Note that for A_N to occur, it must be that $M_T > x$, because this is the only way that $T \leq N$ can occur at the same time as $\langle\langle M \rangle\rangle_N \leq y$. It follows that

$$1 \geq \mathbb{E} \mathcal{E}_{T \wedge N} \geq \mathbb{E}[\mathcal{E}_{T \wedge N} \mid A_N] \mathbb{P}(A_N) \geq \exp\left(\lambda x - \frac{\lambda^2}{2} y\right) \mathbb{P}(A_N),$$

which gives an upper bound on $\mathbb{P}(A_N)$. Optimizing over λ , we see that the best upper bound is achieved with $\lambda = x/y$. Unraveling the definition of A_N , we get that

$$\mathbb{P}\left[\sup_{1 \leq n \leq N} M_n > x, \langle\langle M \rangle\rangle_N \leq y\right] \leq \exp\left(-\frac{x^2}{2y}\right),$$

which is half of the desired bound. The full result follows from applying the above to $-M$ and combining the results for M and $-M$ with a union bound. $\blacksquare$

A.3 Proof of Corollary 1.10.1

In this section, we detail the proof of Corollary 1.10.1. The proof, reproduced from [55, §C], simply amounts to understanding how to express the Weyl–Titchmarsh functions of the Airy and sine systems in terms of the stochastic Airy and zeta functions.

The Weyl–Titchmarsh function of the shifted and scaled Airy system is $m_{H_{\beta,E}}(z) = \mathcal{P}u(0, z)$ where $u: [0, \infty) \times \mathbb{C} \rightarrow \mathbb{C}^2$ is defined so that for each $z \in \mathbb{C}$, $u(\cdot, z)$ is an integrable solution to $J\partial_1 u(\cdot, z) = -zH_{\beta,E}u(\cdot, z)$. By construction of the canonical system,

$$u(0, z) = A_{\beta,E}^{-1}(0) \begin{pmatrix} \partial_1 h(0, z) \\ h(0, z) \end{pmatrix}$$

where $h(\cdot, z)$ is an integrable solution to $\mathcal{H}_{\beta,E}h(\cdot, z) = zh(\cdot, z)$ for each $z \in \mathbb{C}$. This means that $h(\cdot, z)$ also solves $\mathcal{H}_{\beta}h(\cdot, z) = (E + \frac{z}{2\sqrt{E}})h(\cdot, z)$, and therefore by definition of the stochastic Airy function, $h(0, z)$ has the same law as $a \text{SAi}_{-E-z/2\sqrt{E}}(0)$ for some (possibly random and z -dependent) $a \in \mathbb{C} \setminus \{0\}$. Hence,

$$m_{H_{\beta,E}}(z) \stackrel{\text{law}}{=} \frac{\text{SAi}'_{\lambda_E(z)}(0)}{\sqrt{E} \text{SAi}_{\lambda_E(z)}(0)} \quad \text{where} \quad \lambda_E(z) = -E - \frac{z}{2\sqrt{E}}, \quad (\text{A.5})$$

and where the prime denotes a time derivative.

We now recall the definition of the stochastic zeta function ζ_{β} introduced by Valkó and Virág in [75]. Let b_1 and b_2 be two independent two-sided standard Brownian motions on $\mathbb{R}$, and define the processes

$$y(t) := \exp\left(b_2(t) - \frac{t}{2}\right) \quad \text{and} \quad x(t) := \begin{cases} -\int_t^0 y(s) db_1(s) & \text{if } s \leq 0, \\ \int_0^t y(s) db_1(s) & \text{if } s \geq 0. \end{cases}$$

Then, set

$$R_{\tau_{\beta}}(t) := \frac{1}{2y(\frac{4}{\beta} \log t)} X_{\tau_{\beta}}^*\left(\frac{4}{\beta} \log t\right) X_{\tau_{\beta}}\left(\frac{4}{\beta} \log t\right) \quad \text{where} \quad X_{\tau_{\beta}} := \begin{pmatrix} 1 & -x \\ 0 & y \end{pmatrix}.$$

Following [75, Definition 43], we now define τ_{β} as the random Dirac operator mapping $u: (0, 1) \rightarrow \mathbb{C}$ to $R_{\tau_{\beta}}^{-1}Ju'$ with boundary conditions $u_0^*Ju(0) = u_1^*Ju(1) = 0$ where again $u_0 := (1, 0)$, but now $u_1 := (-q, -1)$ where q is a standard Cauchy random variable independent of b_1 and b_2 .

The stochastic zeta function ζ_{β} is the *secular function* of τ_{β} as defined in [75, Definitions 13 and 43]. For our purposes, it suffices to understand ζ_{β} through its characterization as the solution

of a differential equation: if $v: (0, 1] \times \mathbb{C} \rightarrow \mathbb{C}^2$ is defined so that for each $z \in \mathbb{C}$, $v(\cdot, z)$ is the unique solution to

$$R_{\tau_\beta}^{-1} J \partial_1 v(\cdot, z) = z v(\cdot, z) \quad \text{with} \quad \lim_{t \rightarrow 0} v(t, z) = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

then $\zeta_\beta(z) = (1, -q)v(1, z)$. This can be taken as a definition of ζ_β , since by [75, Proposition 18], this differential equation always has a unique solution (see also [75, Section 5.1]).

As explained in [75, Section 4.2], the operator τ_β is unitarily equivalent to the stochastic sine operator. Indeed, if

$$S := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad Q := \frac{1}{\sqrt{q^2 + 1}} \begin{pmatrix} q & 1 \\ -1 & q \end{pmatrix}$$

and if ρ denotes a transformation acting on a function f on $(0, 1]$ by sending it to $\rho f(t) = f(1 - t)$, then the operator $\rho^{-1} S Q \tau_\beta (S Q)^{-1} \rho$ has the same distribution as the stochastic sine operator by [75, Proposition 46]. Writing $\tau_\beta = R_{\tau_\beta}^{-1} J \partial$ with ∂ being a derivative operator, this means that the stochastic sine operator $(R_\beta^{-1} \circ \nu) J \partial$ has the same distribution as $\rho^{-1} S Q R_{\tau_\beta}^{-1} J \partial (S Q)^{-1} \rho$. Now, $\partial \rho = -\rho \partial$, so this equivalence yields the equivalence of coefficient matrices

$$R_\beta^{-1} \circ \nu \stackrel{\text{law}}{=} \rho^{-1} S Q R_{\tau_\beta}^{-1} J (S Q)^{-1} J \rho. \quad (\text{A.6})$$

With v as above, define $u: [0, 1] \times \mathbb{C} \rightarrow \mathbb{C}^2$ by setting $u(t, z) := \rho S Q v(t, -z)$. Then by definition of v , for $t \in [0, 1)$ and $z \in \mathbb{C}$, u solves

$$J \partial_1 u(t, z) = -J S Q \partial_1 v(1 - t, -z) = -z J S Q J R_{\tau_\beta} (1 - t) v(1 - t, -z) = -z J S Q J R_{\tau_\beta} (1 - t) (S Q)^{-1} u(t, z),$$

which shows that $u(\cdot, z)$ solves the sine canonical system with spectral parameter z . Moreover, by definition of v , as $t \rightarrow 1$,

$$u(t, z) \rightarrow S Q \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{q^2 + 1}} \begin{pmatrix} q \\ 1 \end{pmatrix}.$$

By [75, Lemma 45], q is the point to which converges the hyperbolic Brownian motion that drives the sine system in this setup (i.e., with coefficient matrix $\rho^{-1} J S Q J R_{\tau_\beta} (S Q)^{-1} \rho$). This shows that $u(\cdot, z)$ satisfies the sine system's boundary condition at 1 when $\beta > 2$. When $\beta \leq 2$, there is no boundary condition at 1, but it is easy to see from the definition of the sine system's norm that an integrable solution must become parallel to $(q, 1)$ as time goes to 1. Indeed,

$$\|f\|_{R_\beta \circ \nu}^2 = \frac{1}{2} \int_0^1 \frac{\|X_\beta \circ \nu(t) f(t)\|^2}{\text{Im } B_\beta \circ \nu(t)} dt$$

and

$$X_\beta \circ v(t)f(t) = \begin{pmatrix} f_1(t) - f_2(t) \operatorname{Re} \mathcal{B}_\beta \circ v(t) \\ f_2(t) \operatorname{Im} \mathcal{B}_\beta \circ v(t) \end{pmatrix}.$$

Because $\operatorname{Re} \mathcal{B}_\beta(t)$ a.s. does not vanish as $t \rightarrow \infty$ but $\operatorname{Im} \mathcal{B}_\beta(t)$ does, $f(t)$ must become parallel to $(\operatorname{Re} \mathcal{B}_\beta(\infty), 1)$ as $t \rightarrow 1$ for $\|f\|_{R_\beta \circ v}$ to be finite. As an integrable solution exists, the uniqueness statement of [75, Proposition 18] therefore implies that $u(\cdot, z)$ is (a multiple of) that integrable solution. Hence, for any $\beta > 0$, u can be taken as the solution that defines the system's Weyl–Titchmarsh function, that is, $m_{R_\beta}(z) \stackrel{\text{law}}{=} \mathcal{P}u(0, z)$.

Now, $v(1, z) = (SQ)^{-1}u(0, -z)$, and as $S^{-1} = S$ and $Q^{-1} = Q^\top$, a direct computation shows that

$$\zeta_\beta(z) = \begin{pmatrix} 1 & -q \end{pmatrix} v(1, z) = \sqrt{q^2 + 1} u_2(0, -z).$$

Therefore,

$$m_{R_\beta}(z) \stackrel{\text{law}}{=} \frac{u_1(0, z)}{u_2(0, z)} = \sqrt{q^2 + 1} \frac{u_1(0, z)}{\zeta_\beta(-z)}. \quad (\text{A.7})$$

By general canonical systems theory, the function $\xi(z) := u_1(0, z)$ is entire. Moreover, since q is independent of the Brownian motions from which τ_β is defined, it is independent of the solution u , and therefore of ξ and ζ_β .

Thus, Corollary 1.10.1 simply follows from Theorem 1.10 because of the representations (A.5) and (A.7) of the two Weyl–Titchmarsh functions.

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